

# Mono Notes

## Thermodynamics

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## Part I

# Thermodynamics

## 1 Scope, notation, and what was already written

This is the thermodynamics part of the mono notes. The source is one file. `master.tex` is US Letter, single column. `screen.tex` is the same text on a 1200 pt  $\times$  700 pt page in two columns, the widescreen size used in `DGDT_dump.tex` and in `Propulsion/LaTeXandpdfs/thermo.tex`. Build both with `documents/mono-notes/build.sh`. The originals below were not edited.

### 1.1 Where the Kittel–Kroemer notes actually are

There is no second, fuller set of reading notes on Kittel and Kroemer. The 2008 file

`mathphysics/LaTeX_and_pdfs/Kittel_Kroemer_Thermal_Physics.tex`

is a solutions manuscript with reading notes in front of some chapters. Its own abstract says the solutions are nearly complete and that notes were to be added before the problems. What is present:

- Chapter 1, States of a Model System: the heading only. Written here as §2.
- Chapter 2, Entropy and Temperature: a short derivation of thermal equilibrium, then solutions 1–4.
- Chapter 3, Boltzmann distribution and Helmholtz free energy: heat capacity, pressure, the thermodynamic identity, and the one-atom partition function, then solutions 1–11.
- Chapter 4, thermal radiation: solutions 1–15, 17, and 18. There is no solution 16 in the file.
- Chapter 5, chemical potential: notes and solutions 1–14.

- Chapter 6, ideal gas: notes and solutions. Solution 6, entropy of mixing, is a heading with no solution. The derivation is in §14. Solutions 1–4 under this heading are Fermi–Dirac and orbital-counting calculations; the next section heading is Fermi and Bose gases. They were left in the order of the manuscript.
- Chapter 7, Fermi and Bose gases: problems and solutions 1–10.
- Chapter 8, heat and work: the long geometric discussion of engines, Legendre transforms, and the Carnot cycle, then solutions 1–9.
- Chapter 9, Gibbs free energy: solutions 1–4.
- Chapter 10, phase transformations: notes and problems 1–5.
- Chapters 11–13, binary mixtures, cryogenics, and semiconductor statistics: headings only. Written here as §14, §18, and §21.
- Chapter 14, kinetic theory: the ideal-gas law, the Maxwell speed distribution, effusion, and diffusion, then the mean-speed problem.
- Chapter 15, propagation: the continuity equation written with the Hodge star, sound speed, and the start of the diffusion problems. Parts (b) and (c) of the last problem have no solution and no statement in the file, so they were not invented.

The conceptual manuscript is the 2015 file

`Propulsion/LaTeXandpdfs/thermo.tex`

already set to the same widescreen page. Its sections sit in this sequence next to the topic they extend. The part title Thermodynamics (Revisited) is not used. Overlap is kept, so nothing in either file is only in the old PDF. A preserved block begins with the year of its manuscript. Where a section title was changed so that two blocks would not share a name, the manuscript title is the italic line under the new heading.

Two neighboring files were checked and not copied. `Propulsion/thermo.py` is the SymPy scratch pad for a few numerical problems; the diesel-engine values in the 2015 notes match it, and that check is in §33.10. `Propulsion.tex` recalls Kittel–Kroemer results inside the propulsion dump. `DGDT_dump.tex` has a 2019 reminder to compare the convection form of enthalpy with Kittel–Kroemer and with Sonntag; that split is already in the 2015 manuscript, and the per-particle formula is corrected in §33.5.

The geometry books named in the 2008 bibliography are Schutz [3], Bamberg and Sternberg [4], and Frankel [5].

## 1.2 Notation and the energy identity

Kittel and Kroemer measure temperature and entropy in energy units. That convention is the one used in the new sections.

$$\tau = k_B T, \quad \sigma = \ln g, \quad S = k_B \sigma. \quad (1)$$

Here  $g$  is the multiplicity of a macrostate,  $T$  is the kelvin temperature, and  $k_B$  is Boltzmann’s constant. Partial derivatives are written as fractions. The variables held fixed are the subscripts. Einstein summation is used where a coordinate index is repeated, without a separate reminder.

The energy differential adopted for the new sections, and for every correction below, is

$$dU = \tau d\sigma - p dV + \sum_j \mu_j dN_j. \quad (2)$$

The reversible heat 1-form is  $Q = \tau d\sigma$ . The work done on the system by a slow volume change is  $-p dV$ , so the work done by the system is  $p dV$ . The engineering line  $dU = Q - W$  with that work by the system is the same statement. Equation (2) is already the identity labeled in the 2008 notes and the one used for the Helmholtz differential  $dF = -\sigma d\tau - p dV$ . The places where a preserved line uses the opposite work sign, or writes  $(\partial F / \partial \tau)_V = \sigma$ , are marked in the manuscript and collected in §33.

A correction environment in the preserved text is a later check. The surrounding sentences and formulas are the manuscript. Where they disagree, the correction is the one to use.

### 1.3 Order

Counting, entropy, and temperature come first, then heating and the dilute-gas law. The Boltzmann distribution is followed by the pressure derivation, the first law, and both heat capacities. Radiation, the chemical potential, and the Gibbs distribution come next, then the ideal gas and mixtures. Fermi and Bose gases are followed by engines and magnetic cooling, the catalogue of potentials, the Gibbs chapter, chemical equilibrium, and semiconductors. Phase coexistence, the Joule–Thomson expansion, kinetic theory, and hydrodynamics follow. Nonequilibrium transport is the next part, because it is the theory of fluxes and entropy production. The resolutions close the notes.

### 1.4 What is new

New writing is §2, §14, §18, §21, and §33. The resolutions answer the adiabatic question left open in 2015, the use of  $C_V$  on a path whose volume changes, the claim that path-dependent heat requires nontrivial topology, and the arithmetic and transcription errors listed there.

## 2 States of a model system

The 2008 manuscript begins with this heading and no text. The counting below is the input to the paramagnet solution in the entropy chapter. Nothing here is quoted from the book.

### 2.1 Two-state elements and the spin excess

Take  $N$  elements,  $N$  even, each with two states. Label the states up and down. Let  $N_\uparrow$  and  $N_\downarrow$  be the occupation numbers,

$$N_\uparrow + N_\downarrow = N. \quad (3)$$

The spin excess  $2s$  is the difference,

$$2s = N_\uparrow - N_\downarrow, \quad s \in \mathbb{Z}, \quad |s| \leq N/2. \quad (4)$$

Solving (3) and (4) gives

$$N_\uparrow = \frac{N}{2} + s, \quad N_\downarrow = \frac{N}{2} - s. \quad (5)$$

Both are nonnegative integers in the range just stated. The elements are treated as distinguishable by their positions in the model, as in a lattice of fixed sites. The multiplicity of the macrostate  $s$  is the binomial count

$$g(N, s) = \frac{N!}{N_\uparrow! N_\downarrow!} = \frac{N!}{\left(\frac{N}{2} + s\right)! \left(\frac{N}{2} - s\right)!}. \quad (6)$$

The entropy in the units (1) is  $\sigma(N, s) = \ln g(N, s)$ .

An external field  $B$  is not required for the count. When the only energy is the coupling of each element to that field, take the single-element energies to be  $-mB$  and  $+mB$ . Then

$$U(s) = -mB N_\uparrow + mB N_\downarrow = -mB (N_\uparrow - N_\downarrow) = -2mB s. \quad (7)$$

The symbol  $m$  is the magnitude of the moment of one element. Equation (7) is the energy used in the preserved paramagnet solution.

### 2.2 Stirling's approximation

For a positive integer  $n$ ,

$$\ln n! = n \ln n - n + \frac{1}{2} \ln(2\pi n) + \varepsilon_n, \quad (8)$$

with  $\varepsilon_n \rightarrow 0$  as  $n \rightarrow \infty$ . The exponent computed next uses the terms  $n \ln n - n$ . The square-root term is restored at the central multiplicity, and it is checked that it does not change the coefficient of  $s^2/N$ .

Substitute (8), without  $\varepsilon_n$  and without the square-root term, into (6):

$$\ln g(N, s) = \ln N! - \ln N_\uparrow! - \ln N_\downarrow!$$

$$\begin{aligned}
&= N \ln N - N - N_{\uparrow} \ln N_{\uparrow} + N_{\uparrow} - N_{\downarrow} \ln N_{\downarrow} + N_{\downarrow} \\
&= N \ln N - N_{\uparrow} \ln N_{\uparrow} - N_{\downarrow} \ln N_{\downarrow}.
\end{aligned} \tag{9}$$

The linear terms  $-N + N_{\uparrow} + N_{\downarrow}$  cancel by (3).

### 2.3 Gaussian peak

Set

$$x = \frac{2s}{N}, \quad |x| \leq 1, \tag{10}$$

so  $N_{\uparrow} = (N/2)(1+x)$  and  $N_{\downarrow} = (N/2)(1-x)$ . Then

$$\begin{aligned}
N_{\uparrow} \ln N_{\uparrow} &= \frac{N}{2}(1+x)(\ln(N/2) + \ln(1+x)), \\
N_{\downarrow} \ln N_{\downarrow} &= \frac{N}{2}(1-x)(\ln(N/2) + \ln(1-x)).
\end{aligned} \tag{11}$$

Add (11):

$$\begin{aligned}
N_{\uparrow} \ln N_{\uparrow} + N_{\downarrow} \ln N_{\downarrow} &= N \ln(N/2) \\
&\quad + \frac{N}{2} [\ln(1-x^2) + x \ln((1+x)/(1-x))].
\end{aligned} \tag{12}$$

The coefficient of  $\ln(N/2)$  is  $(N/2)(1+x) + (N/2)(1-x) = N$ . Insert (12) into (9):

$$\ln g(N, s) = N \ln 2 - \frac{N}{2} \ln(1-x^2) - \frac{Nx}{2} \ln\left(\frac{1+x}{1-x}\right). \tag{13}$$

The identity  $N \ln N - N \ln(N/2) = N \ln 2$  was used. Also  $Nx/2 = s$ .

For the peak, take  $|x| \ll 1$ , which is  $|s| \ll N/2$ . The Taylor expansions through order  $x^3$  in the logarithms are

$$\begin{aligned}
\ln(1-x^2) &= -x^2 + O(x^4), \\
\ln\left(\frac{1+x}{1-x}\right) &= 2(x + O(x^3)).
\end{aligned} \tag{14}$$

Hence

$$\begin{aligned}
-\frac{N}{2} \ln(1-x^2) &= \frac{N}{2} x^2 + O(Nx^4) = \frac{2s^2}{N} + O\left(\frac{s^4}{N^3}\right), \\
-\frac{Nx}{2} \ln\left(\frac{1+x}{1-x}\right) &= -Nx^2 + O(Nx^4) = -\frac{4s^2}{N} + O\left(\frac{s^4}{N^3}\right).
\end{aligned} \tag{15}$$

The first line uses  $(N/2)x^2 = (N/2)(4s^2/N^2) = 2s^2/N$ . The second uses  $Nx^2 = N(4s^2/N^2) = 4s^2/N$ . Add them:

$$\ln g(N, s) = N \ln 2 - \frac{2s^2}{N} + O\left(\frac{s^4}{N^3}\right). \tag{16}$$

At  $s = 0$ , the same truncation gives  $\ln g(N, 0) = N \ln 2$ , so

$$g(N, s) \simeq g(N, 0) \exp\left(-\frac{2s^2}{N}\right), \quad g(N, 0) \simeq 2^N, \tag{17}$$

inside the approximation that dropped both the square-root Stirling term and  $O(s^4/N^3)$ . Restoring  $\frac{1}{2} \ln(2\pi n)$  at  $s = 0$  replaces the central estimate by

$$g(N, 0) \simeq 2^N \sqrt{\frac{2}{\pi N}}. \tag{18}$$

The check is the standard one:  $\ln N! - 2 \ln((N/2)!)$  with (8) produces  $N \ln 2 + \frac{1}{2} \ln(2/(\pi N))$ . The  $s$ -dependent piece of those square-root terms is  $-\frac{1}{2} \ln(1-x^2) = O(s^2/N^2)$ , which is smaller than  $s^2/N$  by one power of  $N$ . It does not change the coefficient  $-2$  in (16).

The peak is sharp in the fractional excess. From (16),  $\sigma(s) - \sigma(0) = -2s^2/N$  at this order. At  $s = \sqrt{N}$  the multiplicity has fallen by a factor  $e^{-2}$  relative to the center, while the fractional excess is  $s/(N/2) = 2N^{-1/2}$ . For large  $N$  almost every state in the count sits at a fractional excess of order  $N^{-1/2}$  or smaller.

## 2.4 Temperature of the paramagnet in this approximation

Solve (7) for the excess,  $s = -U/(2mB)$ , and substitute into the exponent of (16):

$$\frac{2s^2}{N} = \frac{2}{N} \cdot \frac{U^2}{4m^2B^2} = \frac{U^2}{2m^2B^2N}. \quad (19)$$

Therefore

$$\sigma(N, U) = \sigma_0 - \frac{U^2}{2m^2B^2N}, \quad \sigma_0 = \ln g(N, 0), \quad (20)$$

in the same small-excess regime, and with  $N$  and  $B$  held fixed. Differentiate:

$$\frac{1}{\tau} = \left( \frac{\partial \sigma}{\partial U} \right)_N = -\frac{U}{m^2B^2N}. \quad (21)$$

Hence

$$U = -\frac{Nm^2B^2}{\tau}. \quad (22)$$

From (7),  $U = -2mB s$ , so the fractional excess in this regime is

$$\frac{2s}{N} = \frac{mB}{\tau}. \quad (23)$$

Equation (21) is the derivative written in the preserved paramagnet solution. That solution starts from the Gaussian multiplicity (17) and does not derive it. The canonical two-state free energy is in §18. Its small-field expansion reproduces (22). *2008 manuscript.*

## 3 Entropy and Temperature

### Thermal Equilibrium

EY : 20150821 Based on considering the physical setup of two systems that can only exchange energy between each other, that are in thermal contact, this is a derivation of temperature.

$U = U_1 + U_2$  is constant total energy of 2 systems 1, 2 in thermal contact  
multiplicity  $g(N, U)$  of combined system is

$$g(N, U) = \sum_{U_1 \leq U} g_1(N_1, U_1) g_2(N_2, U - U_1)$$

The “differential” of  $g(N, U)$  is

$$dg = \left( \frac{\partial g_1}{\partial U_1} \right)_{N_1} g_2 dU + g_1 \left( \frac{\partial g_2}{\partial U_2} \right)_{N_2} dU_2 = 0$$

EY : 20150821 This step can be made mathematically sensible by considering the exterior derivative  $d$  of  $g \in C^\infty(\Sigma)$ , where  $\Sigma$  is the manifold of states of the system, with local coordinates  $N, U$ , where  $U$  happens to be a global coordinate. Then, consider a curve in  $\Sigma$  s.t. it has no component in  $\frac{\partial}{\partial N}$ ,  $\frac{\partial}{\partial N_1}$ , and this curve is a “null curve” so that the vector field  $X \in \mathfrak{X}(\Sigma)$  generated by this curve is s.t.  $dg(X) = 0$ .

With  $-dU_1 = dU_2$ ,

$$\frac{1}{g_1} \left( \frac{\partial g_1}{\partial U_1} \right)_{N_1} = \frac{1}{g_2} \left( \frac{\partial g_2}{\partial U_2} \right)_{N_2} \implies \left( \frac{\partial \ln g_1}{\partial U_1} \right)_{N_1} = \left( \frac{\partial \ln g_2}{\partial U_2} \right)_{N_2}$$

Define

$$\sigma(N, U) := \ln g(N, U)$$

Then

$$\implies \left( \frac{\partial \sigma_1}{\partial U_1} \right)_{N_1} = \left( \frac{\partial \sigma_2}{\partial U_2} \right)_{N_2}$$

## Temperature

$T_1 = T_2$  - temperatures of 2 systems in thermal equilibrium are equal.

$T$  “must be a function of  $\left(\frac{\partial \sigma}{\partial U}\right)_N$  [2].

$$\implies \frac{1}{T} = k_B \left(\frac{\partial \sigma}{\partial U}\right)_N$$

Experimentally,  $k_B = 1.381 \times 10^{-23} \text{ J/K} = 1.381 \times 10^{-16} \text{ ergs/K}$ .

Now

$$\frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial U}\right)_N \tau = k_B T$$

## Problems

**Solution 1. Entropy and temperature.**

(a) Recall that  $\frac{1}{\tau} \equiv \left(\frac{\partial \sigma}{\partial U}\right)_{N,V}$  and  $\sigma(N, U) \equiv \log g(N, U)$ . Given  $g(U) = CU^{3N/2}$ ,

$$\sigma(N, U) = \log CU^{3N/2} = \log C + \frac{3N}{2} \log U$$

$$\frac{\partial \sigma}{\partial U} = \frac{3N}{2} \frac{1}{U} = \frac{1}{\tau} \implies \boxed{U = \frac{3N}{2} \tau}$$

(b)  $\left(\frac{\partial^2 \sigma}{\partial U^2}\right)_N < 0$  ?

$$\left(\frac{\partial^2 \sigma}{\partial U^2}\right)_N = -\frac{3N}{2} \left(\frac{1}{U^2}\right) < 0$$

**Solution 2. Paramagnetism.**

$$U(s) = U_1(s_1) + U_2(s_2) = -2mB(s_1 + s_2) = -2mBs \text{ or } s = \frac{U}{-2mB}$$

i.e. potential energy  $U(s) = -2s \cdot mB$ .

For  $|s| \ll N$ , then

$$g(N, s) \simeq g(N, 0) \exp(-2s^2/N) = g(N, 0) \exp\left(\frac{-U^2}{2(mB)^2 N}\right)$$

$$\sigma(N, U) = \ln g(N, U) = \sigma_0 - \frac{U^2}{2m^2 B^2} \frac{1}{N} \text{ where } \sigma_0 = \ln g(N, 0)$$

$$\frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial U}\right)_N = \frac{-U}{m^2 B^2} \frac{1}{N}$$

What is the thermal equilibrium value of this  $N$ -spin system of fractional magnetization? If  $U$  denotes  $\langle U \rangle$ , thermal average energy, we also get the thermal average spin excess.

$$\langle U \rangle = \langle -2mBs \rangle = -2mB \langle s \rangle$$

$$\implies \tau = \frac{m^2 B^2 N}{-U} = \boxed{\frac{mBN}{2\langle s \rangle}}$$

**Solution 3. Quantum harmonic oscillator.**

(a) Result from Ch. 1:  $g(N, n) = \frac{(N+n-1)!}{n!(N-1)!}$ .

Let  $N-1 \rightarrow N \implies g(N+1, n) = \frac{(N+n)!}{n!N!}$ .

$$\sigma(N+1, n) \equiv \ln g(N+1, n) = \ln \frac{(N+n)!}{n!N!} = \ln(N+n)! - \ln(n!) - \ln(N!)$$

$$\approx (N+n) \ln(N+n) - N - n - n \ln n + n - N \ln N + N = \boxed{(N+n) \ln(N+n) - n \ln n - N \ln N}$$

- (b) Let  $U$  denote total energy  $n\hbar\omega$  of oscillators.  
 $U = n\hbar\omega$  or  $n = \frac{U}{\hbar\omega}$

$$\sigma(N, U) = \left(N + \frac{U}{\omega}\right) \ln \left(N + \frac{U}{\omega}\right) - \frac{U}{\omega} \ln \frac{U}{\omega} - N \ln N$$

At  $\tau$ ,  $\frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial U}\right)_N$ .

$$\frac{1}{\tau} = \frac{1}{\omega} \ln \left(N + \frac{U}{\omega}\right) - \frac{1}{\omega} \ln \frac{U}{\omega} = \frac{1}{\omega} \ln \left(\frac{N\omega}{U} + 1\right) \text{ or } \boxed{U = \frac{N\omega}{\exp(\omega/\tau) - 1}}$$

#### Solution 4. The meaning of “never.”

Suppose  $10^{10}$  monkeys.

- (a) Hamlet represents one specific ordering of  $10^{15}$  with 44 possibilities for each character. The probability of hitting upon Hamlet from a given, random sequence is  $\left(\frac{1}{44}\right)^{100000} = \boxed{\frac{1}{44^{100000}}}$ . Given that  $\log_{10} 44 = 1.64345$ , then  $10^{1.64345} = 44$  or  $10^{-1.64345} = 44^{-1}$  so then

$$\left(\frac{1}{44}\right)^{100000} = 10^{-164345}$$

- (b)

$$(\text{age of universe}) \cdot \left(\frac{10 \text{ keys}}{\text{second}}\right) = 10^{18} \text{ s} \left(\frac{10 \text{ keys}}{\text{second}}\right) = 10^{19} \text{ keys}$$

$$10^{19} \text{ keys} \cdot 10^{10} \text{ monkeys} = 10^{29} \text{ keys typed out} \left(\frac{1 \text{ hamlet}}{10^5 \text{ characters}}\right) = 10^{24} \text{ possible “Hamlets”}$$

From part (a), the probability that a given, random sequence is Hamlet,  $10^{-164345}$

$$(10^{29} \text{ characters})(10^{-164345}) = 10^{-164316}$$

Note, I think that the probability should be  $(10^{29} \text{ characters}) \left(\frac{1 \text{ Hamlet}}{10^5 \text{ characters}}\right) (10^{-164345}) = 10^{-164321}$

Since we are considering the number of “Hamlet”,  $10^5$  character sequences.

*2015 manuscript.*

## 4 Heating and the dilute gas

*Manuscript section title: Background.* [1]

### 4.1 Heating and Temperature

Heating: keep in mind 3 different types of heating for energy exchange between two systems:

1. Heating by conduction - literal contact, molecules jiggle faster from molecules jiggling faster by bouncing off them
2. Heating by radiation - em waves from hot source strike and excite target
3. heating by convection - energy transport by flow (perhaps a fluid)

This all relates to

$Q$

## 4.2 Some dilute gas relationships

### Pressure according to kinetic theory

i.e. some kinetic theory

$F \equiv$  force on area  $A$  due to molecules

$\Delta p \equiv$  momentum transferred to wall per collision

$n \equiv$  number of collisions in time  $\Delta t$

Thus

$$F = \frac{(\Delta p)n}{\Delta t}$$

Now  $\Delta p = 2mv_x$  since  $\Delta p = mv_x - (-mv_x) = 2mv_x$  (elastic collision with momentum conservation)  
 $v_x \Delta t A$  is a volume, inside of which gas molecules can be within distance  $v_x \Delta t$  toward the wall.

$\frac{N}{V}$  number density of molecules

Assume equal distribution of velocities: Thus  $\frac{1}{2}$

$$n = (v_x \Delta t A) \frac{1}{2} \frac{N}{V}$$

$$P = \frac{F}{A} = \frac{(2mv_x)(v_x \Delta t A) \frac{1}{2} \frac{N}{V}}{A \Delta t} = mv_x^2 \frac{N}{V}$$

Suppose  $\langle v^2 \rangle = \langle v_x^2 \rangle + \langle v_y^2 \rangle + \langle v_z^2 \rangle = 3\langle v_x^2 \rangle = d\langle v_x^2 \rangle$ .

$$\implies P = \frac{mN}{dV} \langle v^2 \rangle = \frac{2}{d} \frac{1}{2} m \langle v^2 \rangle \frac{N}{V}$$

### An empirical gas law

Now

$$P = \frac{N\tau}{V} \quad (\text{empirical})$$

$$\implies \frac{1}{2} m \langle v^2 \rangle = \frac{d}{2} \tau$$

2008 manuscript.

## 5 Boltzmann Distribution and Helmholtz Free Energy

cf. **Example: Energy and heat capacity of a two state system**, pp. 62 of Kittel and Kroemer [2]. Kittel and Kroemer introduces the heat capacity very early, specific to this example.

**Definition 1.** heat capacity  $C_V$  at constant volume is defined as

$$C_V := \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_V \quad (24)$$

Recall the thermodynamic identity (which is introduced many equations later):

$$dU = \tau d\sigma - p dV \in \Omega^1(\Sigma)$$

where  $\Sigma$  is a manifold of states of all systems.

Consider local coordinates of  $\Sigma$ ,  $(\sigma, V)$ . Consider curve  $c : \mathbb{R} \rightarrow \Sigma$  s.t.  $c$  generates a vector field  $\dot{c} = \dot{\sigma} \frac{\partial}{\partial \sigma}$  i.e.

$$c(\tau) \in \Sigma$$

no component in the  $V$  direction. Notice the prescient choice of parameter  $\tau$ .

Now for internal energy  $U \in C^\infty(\Sigma)$ , taking the exterior derivative  $d$  results in

$$dU = \frac{\partial U}{\partial \sigma} d\sigma + \frac{\partial U}{\partial V} dV$$

Then applying  $dU$  onto vector field  $\dot{\sigma} \frac{\partial}{\partial \sigma}$ ,

$$dU \left( \dot{\sigma} \frac{\partial}{\partial \sigma} \right) = \frac{\partial U}{\partial \sigma} \dot{\sigma} = \dot{\sigma} \tau + 0$$

Now,

$$\left( \frac{\partial U}{\partial \sigma} \right)_V \left( \frac{\partial \sigma}{\partial \tau} \right)_V = \left( \frac{\partial U}{\partial \tau} \right)_V = \left( \frac{\partial \sigma}{\partial \tau} \right)_V \tau$$

Hence,

$$C_V := \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_V = \left( \frac{\partial U}{\partial \tau} \right)_V \quad (25)$$

EY: 20150825 Why do we need differential geometry? It's because I always wondered why you could do this:

$$C_V := \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_V \stackrel{?}{=} \left( \frac{\partial U}{\partial \tau} \right)_V \text{ with } \tau d\sigma = dU \implies \tau \partial \sigma \stackrel{?}{=} \partial U$$

and talk of “differentials.”

**Definition: Reversible process.** EY : 20150824 Mathematically, 1-forms are exact.

## Pressure

Consider coordinates  $(\sigma, V) \in \Sigma$  of manifold of thermodynamic states  $\Sigma$ .

Imagine a reversible compression of a cube system (so imagine  $dV < 0$ ; cube's volume get smaller).

$\sigma$  constant, i.e.  $d\sigma = 0$  (on this curve in  $\Sigma$ ) because as particles in cube gets squeezed, less positions particles could sit in, but they get more kinetic energy, more energetic (more momentum squared).

Now  $U = U(\sigma, V) \in C^\infty(\Sigma)$ .

$$\implies dU = \left( \frac{\partial U}{\partial \sigma} \right)_V d\sigma + \left( \frac{\partial U}{\partial V} \right)_\sigma dV$$

Again, imagine a curve  $c : \mathbb{R} \rightarrow \Sigma$ , connecting 1 state  $(\sigma, V) \in \Sigma$  to another state  $(\sigma, V + dV) \in \Sigma$  s.t.  $\dot{c} = \dot{V} \frac{\partial}{\partial V}$ .

$$\implies dU(\dot{c}) = \left( \frac{\partial U}{\partial V} \right)_\sigma \dot{V}$$

Introduce 1-form  $W \in \Omega^1(\Sigma)$  of work done on the cube system from some external agent

$$W = -pdV$$

so  $W > 0$  when  $dV < 0$ .

Then

$$\begin{aligned} W(\dot{c}) &= -p\dot{V} = dU(\dot{c}) = \left( \frac{\partial U}{\partial V} \right)_\sigma \dot{V} \\ \implies \boxed{p &= - \left( \frac{\partial U}{\partial V} \right)_\sigma} \end{aligned} \quad (26)$$

Consider another set of coordinates  $(U, V) \in \Sigma$  for manifold  $\Sigma$ . Now entropy  $\sigma$  is a function of  $U, V$ , as  $\sigma = \sigma(U, V) \in C^\infty(M)$ , so that

$$d\sigma = \left( \frac{\partial \sigma}{\partial U} \right)_V dU + \left( \frac{\partial \sigma}{\partial V} \right)_U dV$$

Consider curve  $c = (U, V) \in \Sigma$ . Then  $\dot{c} = \dot{U} \frac{\partial}{\partial U} + \dot{V} \frac{\partial}{\partial V}$ . For this curve  $c$ ,  $\sigma$  is constant, meaning  $d\sigma(\dot{c}) = 0$  (it's a “null curve” of  $d\sigma$ )

$$d\sigma(\dot{c}) = 0 = \left( \frac{\partial \sigma}{\partial U} \right)_V \dot{U} + \left( \frac{\partial \sigma}{\partial V} \right)_U \dot{V}$$

Now define

**Definition 2.**

$$\frac{1}{\tau} := \left( \frac{\partial \sigma}{\partial U} \right)_V \quad (27)$$

So then we have  $\frac{1}{\tau}\dot{U} + \left(\frac{\partial\sigma}{\partial V}\right)_U \dot{V} = 0$ . For the parameter of curve  $c$ , choose the parameter to be  $V$ , knowing that  $\sigma$  is constant on this curve, or thermodynamic process. Thus

$$\frac{1}{\tau} \left( \frac{\partial U}{\partial V} \right)_\sigma = - \left( \frac{\partial \sigma}{\partial V} \right)_U \implies \frac{-p}{\tau} = - \left( \frac{\partial \sigma}{\partial V} \right)_U \text{ or}$$

$$\boxed{p = \tau \left( \frac{\partial \sigma}{\partial V} \right)_U} \quad (28)$$

### Thermodynamic Identity

Let  $\sigma = \sigma(U, V) \in C^\infty(\Sigma)$ . Then

$$d\sigma = \left( \frac{\partial \sigma}{\partial U} \right)_V dU + \left( \frac{\partial \sigma}{\partial V} \right)_U dV \in \Omega^1(\Sigma).$$

Now recall the quantities we've recently used:  $\frac{1}{\tau} := \left( \frac{\partial \sigma}{\partial U} \right)_V$  (this is a definition) and  $\frac{p}{\tau} = \left( \frac{\partial \sigma}{\partial V} \right)_U$  (it comes from the physics, of doing work on the system, by some external agent). Then the **thermodynamic identity** is obtained:

**Theorem 1.**

$$\boxed{\tau d\sigma = dU + p dV} \quad (29)$$

### Ideal Gas: A First Look

#### One atom in a box

one atom of mass  $M$  in cubical box of volume  $V = L^3$

$$\begin{aligned} \frac{-\hbar^2}{2M} \nabla^2 \psi &= \epsilon \psi & p &= \frac{\hbar}{i} \nabla & p^2 \psi &= \epsilon \psi \\ \implies \psi(x) &= A \sin\left(\frac{n_x \pi x}{L}\right) \sin\left(\frac{n_y \pi y}{L}\right) \sin\left(\frac{n_z \pi z}{L}\right) \end{aligned}$$

$$\epsilon_n = \frac{\hbar^2}{2M} \left( \frac{\pi}{L} \right)^2 (n_x^2 + n_y^2 + n_z^2)$$

Then the partition function  $Z_1$  is

$$Z_1 = \sum_n \exp\left(\frac{-\epsilon_n}{\tau}\right) = \sum_{(n_x, n_y, n_z)} \exp\left(\frac{-\hbar^2}{2M\tau} \left(\frac{\pi}{L}\right)^2 (n_x^2 + n_y^2 + n_z^2)\right)$$

Let

$$\alpha^2 = \frac{\hbar^2 \pi^2}{2ML^2\tau} \text{ or } \alpha = \frac{\hbar\pi}{(2M\tau)^{1/2}V^{2/d}}$$

Then

$$Z_1 = \int_0^\infty dn_x \int_0^\infty dn_y \int_0^\infty dn_z \exp[-\alpha^2(n_x^2 + n_y^2 + n_z^2)] = \left( \int_0^\infty dn_x \exp(-\alpha^2 n_x^2) \right)^3 = \left( \frac{1}{\alpha} \right)^3 \left( \frac{\pi^{1/2}}{2} \right)^3 = \left( \frac{\pi^{1/2}}{2\alpha} \right)^3$$

In general,  $Z_1 = \left( \frac{\pi^{1/2}}{2\alpha} \right)^d$

Now

$$Z_1 = \left( \frac{\pi^{1/2} V^{1/d}}{2 \frac{\hbar\pi}{(2M\tau)^{1/2}}} \right)^d = \frac{V}{\left( \frac{2\hbar\pi}{M\tau} \right)^{d/2}} = n_Q V = \frac{n_Q}{n}$$

in terms of concentration  $n = 1/V$ .

$n_Q := \left( \frac{M\tau}{2\pi\hbar^2} \right)^{d/2}$  is the **quantum concentration**. “It is the concentration associated with one atom in a cube of side equal to the thermal average de Broglie wavelength.” (pp. 73, Ch. 3: Boltzmann Distribution and Helmholtz Free Energy, Section “Ideal Gas: A First Look” of Kittel and Kroemer [2])

From Kittel and Kroemer (1980) [2], Example :  $N$  atoms in a box, Chapter 3: “Boltzmann Distribution and Helmholtz Free Energy”,

state of energy  $\epsilon_\alpha(1) + \epsilon_\beta(2) + \dots + \epsilon_\xi(N)$ ,  $\alpha, \beta, \dots, \xi$  denote orbital indices of atoms in successive boxes. each entry occurs  $N!$  times in  $Z_1^N$  (EY: 20151022)  $N!$  ways to fill  $\alpha, \beta, \dots, \xi$  orbitals with  $N$  distinguishable particles. Thus,

$$Z_N = \frac{1}{N!} Z_1^N = \frac{1}{N!} (n_Q V)^N$$

## Problems

### Solution 1. Free energy of a two state system.

(a)

$$Z = 1 + e^{-\epsilon/\tau}$$

$$F = -\tau \ln Z = -\tau \ln(1 + e^{-\epsilon/\tau})$$

(b)

$$U = -\tau^2 \frac{\partial(F/\tau)}{\partial\tau} = \frac{\epsilon e^{-\epsilon/\tau}}{1 + e^{-\epsilon/\tau}}$$

$$\sigma = -\frac{\partial F}{\partial\tau} = \ln(1 + e^{-\epsilon/\tau}) + \frac{\frac{\epsilon}{\tau} e^{-\epsilon/\tau}}{(1 + e^{-\epsilon/\tau})}$$

### Solution 2. Magnetic susceptibility

(a) Remember to calculate the **multiplicity** in the  $N$ -spin system (it's not enough to sum up  $\exp(-\epsilon_s/\tau)$  factors).

$$M = 2sm \quad U_s = -MB = -2smB \quad N = N_+ + N_-$$

$$2s = N_+ - N_- = N_+ - (N - N_+) = 2N_+ - N$$

$$Z = \sum_{s=-N/2}^{N/2} \binom{N}{N_+} \exp\left(\frac{2smB}{\tau}\right) = \sum_{s=-N/2}^{N/2} \frac{N!}{(\frac{N}{2}+s)!(\frac{N}{2}-s)!} \exp\left(\frac{2mBs}{\tau}\right) = \sum_{s=0}^N \frac{N!}{s!(N-s)!} \exp\left(\frac{2mB}{\tau} \left(s - \frac{N}{2}\right)\right) =$$

$$= e^{-\frac{NmB}{\tau}} (1 + e^{\frac{2mB}{\tau}})^N = 2^N \cosh^N\left(\frac{mB}{\tau}\right)$$

where it was **crucial** to use  $(1+x)^N = \sum_{j=1}^N \binom{N}{j} x^j$ . Note, in changing the sum index, since  $N$  is large, we can neglect dropping the  $s=0$  term.

**Correction (binomial index).** The binomial theorem starts at  $j=0$ :  $(1+x)^N = \sum_{j=0}^N \binom{N}{j} x^j$ . The  $j=0$  term is 1, and the closed form on the previous line includes it.

$$\partial_\tau Z = 2^N (N) (\cosh^{N-1}\left(\frac{mB}{\tau}\right)) \sinh\left(\frac{mB}{\tau}\right) \left(\frac{-m}{\tau^2}\right)$$

$$M = -\tau^2 \frac{\partial}{\partial\tau} \ln Z = Nm \tanh\left(\frac{mB}{\tau}\right)$$

$$\chi = \frac{\partial M}{\partial B} = \frac{Nm^2}{\tau} \operatorname{sech}^2\left(\frac{mB}{\tau}\right)$$

(b)

$$F = -\tau \ln Z = -\tau \ln \left( (2 \cosh\left(\frac{mB}{\tau}\right))^N \right) = -N\tau \ln \left( 2 \cosh\left(\frac{mB}{\tau}\right) \right)$$

For  $x \equiv \frac{M}{nm} = \tanh\left(\frac{mB}{\tau}\right)$ . Now  $1 - \tanh^2 y = \operatorname{sech}^2 y$ .  $F = -N\tau \ln \left( \frac{2}{\sqrt{1-x^2}} \right) = \frac{-N\tau}{2} \ln \left( \frac{4}{1-x^2} \right)$ .

(c) For  $\frac{mB}{\tau} \ll 1$ ,  $\cosh^2\left(\frac{mB}{\tau}\right) \rightarrow 1$ .  $\chi = \frac{m^2 N}{\tau}$

### Solution 3. Free energy of a harmonic oscillator

(a)

$$Z = \sum_{s=0}^{\infty} \exp\left(\frac{-s\hbar\omega_0}{\tau}\right) = \frac{1}{1 - e^{-\hbar\omega_0/\tau}} = (1 - e^{-\hbar\omega_0/\tau})^{-1}$$

$$F = -\tau \ln Z = \tau \ln(1 - e^{-\hbar\omega_0/\tau}) \simeq \tau \ln\left(\frac{\hbar\omega_0}{\tau}\right) \text{ for } 1 \gg \frac{\hbar\omega_0}{\tau}$$

(b)

$$\sigma = -\frac{\partial F}{\partial \tau} = -\{\ln(1 - e^{-\hbar\omega_0/\tau}) + \frac{\tau}{1 - e^{-\hbar\omega_0/\tau}}(-e^{-\hbar\omega_0/\tau})\left(\frac{\hbar\omega_0}{\tau}\right)\} = \frac{-\hbar\omega_0/\tau}{e^{\hbar\omega_0/\tau} - 1} - \ln(1 - e^{-\frac{\hbar\omega_0}{\tau}})$$

**Solution 4. Energy fluctuations.**

$$\begin{aligned} Z &= \sum_s e^{-\epsilon_s \beta} \\ \frac{1}{\tau} &= \beta \\ \frac{\partial}{\partial \tau} &= \frac{\partial \beta}{\partial \tau} \frac{\partial}{\partial \beta} = -\frac{1}{\tau^2} \frac{\partial}{\partial \beta} \end{aligned} \quad \begin{aligned} \partial_\beta Z &= \sum_s -\epsilon_s e^{-\epsilon_s \beta} & U &= \frac{\sum_s \epsilon_s e^{-\epsilon_s \beta}}{Z} = -\partial_\beta \ln Z \\ \partial_\beta^2 Z &= \sum_s \epsilon_s^2 e^{-\epsilon_s \beta} \end{aligned}$$

$$\begin{aligned} \frac{\partial U}{\partial \tau} &= -\beta^2 \frac{\partial}{\partial \beta} U = \beta^2 \left( \partial_\beta \left( \frac{\partial_\beta Z}{Z} \right) \right) = \beta^2 \left( \frac{(\partial_\beta^2 Z)Z - (\partial_\beta Z)^2}{Z^2} \right) = \beta^2 \left( \frac{\partial_\beta^2 Z}{Z} - \left( \frac{\partial_\beta Z}{Z} \right)^2 \right) = \\ &\implies \tau^2 \frac{\partial U}{\partial \tau} = \langle \epsilon^2 \rangle - \langle \epsilon \rangle^2 \end{aligned}$$

**Solution 5. Overhauser effect.** System  $\mathcal{S}$  in energy eigenstate  $E_n = n\epsilon$ .

$$P(E) = (1)g_{\mathcal{R}}(E)$$

Note  $\Delta U_{\mathcal{R}} = (\alpha - 1)\epsilon$ .  $\Delta U_{\mathcal{S}} = \epsilon$ .  $\frac{dU_{\mathcal{S}}}{d\epsilon} + \frac{dU_{\mathcal{R}}}{d\epsilon} = 1 + (\alpha - 1) = \alpha = \frac{dU_{\text{tot}}}{d\epsilon}$  in a specific energy eigenstate;  $g_{\mathcal{S}}(n\epsilon) = 1$

While  $g_{\mathcal{R}}(U_{\mathcal{R}})$  = multiplicity of reservoir  $\mathcal{R}$  with  $U_{\mathcal{R}}$  energy.

Now

$$\frac{\partial \sigma_{\mathcal{R}}}{\partial E_s} = \frac{1}{\tau}$$

and

$$g_{\mathcal{R}}(U_{\mathcal{R}}) = \exp(\sigma_{\mathcal{R}}(U_{\mathcal{R}}))$$

If  $\epsilon \frac{dU_{\mathcal{R}}}{d\epsilon} = (\alpha - 1)\epsilon = \Delta U_{\mathcal{R}}$  small compared to  $U_{\mathcal{R}}$ .

$$\begin{aligned} U_{\mathcal{R}}(E_{\mathcal{S}} = (n+1)\epsilon) &= U_{\mathcal{R}}(n\epsilon) + \frac{dU_{\mathcal{R}}}{d\epsilon} \epsilon = U_{\mathcal{R}}(n\epsilon) + (\alpha - 1)\epsilon \\ \sigma_{\mathcal{R}}(U_{\mathcal{R}}((n+1)\epsilon)) &\simeq \sigma_{\mathcal{R}}(U_{\mathcal{R}}(n\epsilon)) + \frac{1}{\tau}(\alpha - 1)\epsilon \\ \frac{P(E_{\mathcal{S}} = (n+1)\epsilon)}{P(E_{\mathcal{S}} = n\epsilon)} &= \frac{\exp(\sigma_{\mathcal{R}}(U_{\mathcal{R}}(n\epsilon)) + \frac{1}{\tau}(\alpha - 1)\epsilon)}{\exp(\sigma_{\mathcal{R}}(U_{\mathcal{R}}(n\epsilon)))} = \boxed{\exp\left(-\frac{\epsilon}{\tau}(1 - \alpha)\right)} \end{aligned}$$

**Solution 6. Rotation of diatomic molecules.**

(a)  $\epsilon(j) = j(j+1)\epsilon_0$ .  $g(j) = 2j+1$

Remember that  $Z$  is a sum over all states, not over all levels.

$$Z = \sum_{j=0}^{\infty} (2j+1) e^{-j(j+1)\epsilon_0/\tau} = \sum_{j=0}^{\infty} \frac{d}{dj} (e^{-(j^2+j)\epsilon_0/\tau}) \left( \frac{-\tau}{\epsilon_0} \right) = \frac{-\tau}{\epsilon_0} \sum_{j=0}^{\infty} \frac{d}{dj} \left( e^{-\epsilon_0/\tau} \right)^{j^2+j}$$

(b) For  $1 \gg \frac{\epsilon_0}{\tau}$

$$Z_R(\tau) = -\frac{\tau}{\epsilon_0} \int_0^{\infty} \frac{d}{dx} \left( e^{-\epsilon_0/\tau} \right)^{x^2+x} dx = -\frac{\tau}{\epsilon_0} \left( (e^{-\epsilon_0/\tau})^{x^2+x} - (e^{-\epsilon_0/\tau})^0 \right) = \frac{\tau}{\epsilon_0}$$

(c) For  $\frac{\tau}{\epsilon_0} \ll 1$

$$Z_R(\tau) = 1 + 3e^{-2\epsilon_0/\tau}$$

(d)

$$\begin{aligned}
U &= \tau^2 \frac{\partial \ln Z}{\partial \tau} \quad \text{for } 1 \gg \frac{\epsilon_0}{\tau} \quad U = \tau^2 \partial_\tau \left( \ln \frac{\tau}{\epsilon_0} \right) = \tau \\
\text{for } \frac{\tau}{\epsilon_0} \ll 1, U &= \tau^2 \left( \frac{1}{1 + 3e^{-2\epsilon_0/\tau}} \right) (3e^{-2\epsilon_0/\tau}) \left( \frac{2\epsilon_0}{\tau^2} \right) = \frac{6\epsilon_0 e^{-2\epsilon_0/\tau}}{1 + 3e^{-2\epsilon_0/\tau}} \\
C_V &= \left( \frac{\partial U}{\partial \tau} \right)_V = 1 \text{ when } 1 \gg \frac{\epsilon_0}{\tau} \\
C_V &= 6\epsilon_0 \left( \frac{e^{-2\epsilon_0/\tau} \left( \frac{2\epsilon_0}{\tau^2} \right) (1 + 3e^{-2\epsilon_0/\tau}) - (3e^{-2\epsilon_0/\tau}) \left( \frac{2\epsilon_0}{\tau^2} \right) e^{-2\epsilon_0/\tau}}{(1 + 3e^{-2\epsilon_0/\tau})^2} \right) = 12\epsilon_0^2 \frac{e^{-2\epsilon_0/\tau}}{\tau^2} \left( \frac{1}{(1 + 3e^{-2\epsilon_0/\tau})^2} \right) \\
\text{For very small } \frac{\tau}{\epsilon_0} \ll 1, & \quad C_V \approx 12 \left( \frac{e^{-2\epsilon_0/\tau}}{(\tau/\epsilon_0)^2} \right)
\end{aligned}$$

(e) See sketch.

**Solution 7. Zipper problem.**

(a)  $N$  links.  $\epsilon_s = 0$  closed,  $\epsilon$  open.

$$Z = \sum_{s=0}^N \exp(-s\epsilon/\tau) = \frac{1 - e^{-(N+1)\epsilon/\tau}}{1 - e^{-\epsilon/\tau}}$$

(b)  $1 \gg \frac{\tau}{\epsilon}$ .

$$Z = \sum_{s=0}^N \exp(-s\epsilon B)$$

$$\begin{aligned}
\frac{-1}{\epsilon} \partial_\beta \ln Z &= \frac{-1}{\epsilon} \frac{\sum_{s=0}^N (-s\epsilon) e^{-s\epsilon\beta}}{Z} = \langle s \rangle = \\
&= \left( \frac{1 - e^{-\epsilon/\tau}}{1 - e^{-(N+1)\epsilon/\tau}} \right) \left( \frac{-e^{-(N+1)\epsilon/\tau} (-(N+1)\epsilon) (1 - e^{-\epsilon/\tau}) - (-e^{-\epsilon/\tau}) (-\epsilon) (1 - e^{-(N+1)\epsilon/\tau})}{(1 - e^{-\epsilon/\tau})^2} \right) = \\
&= \frac{\epsilon e^{-\epsilon/\tau} (e^{-N\epsilon\beta} (N+1) (1 - e^{-\epsilon\beta}) - (1 - e^{-(N+1)\epsilon/\tau}))}{(1 - e^{-(N+1)\epsilon/\tau}) (1 - e^{-\epsilon/\tau})} \simeq \frac{\epsilon e^{-\epsilon/\tau} (e^{-N\epsilon\beta} (N+1) (1 - e^{-\epsilon\beta}) - 1)}{(1 - e^{-\epsilon\beta})}
\end{aligned}$$

This still does not give the desired approximation. Consider the following:

$$\begin{aligned}
Z &= \frac{1 - e^{-(N+1)\epsilon\beta}}{1 - e^{-\epsilon\beta}} = \frac{e^{\epsilon\beta} - e^{-N\epsilon\beta}}{e^{\epsilon\beta} - 1} \\
\partial_\beta Z &= \frac{\epsilon((e^{\epsilon\beta} + N\beta e^{-N\epsilon\beta})(e^{\epsilon\beta} - 1) - (e^{\epsilon\beta})(e^{\epsilon\beta} - e^{-N\epsilon\beta}))}{(e^{\epsilon\beta} - 1)^2} = \frac{\epsilon(e^{\epsilon\beta}(e^{\epsilon\beta} + Ne^{-N\epsilon\beta} - e^{\epsilon\beta} + e^{-N\epsilon\beta}) - e^{\epsilon\beta} - Ne^{-N\epsilon\beta})}{(e^{\epsilon\beta} - 1)^2} \\
\frac{\partial_\beta Z}{Z} &= \frac{\epsilon(e^{\epsilon\beta}(N+1)e^{-N\epsilon\beta} - e^{\epsilon\beta} - Ne^{-N\epsilon\beta})}{(e^{\epsilon\beta} - 1)(e^{\epsilon\beta} - e^{-N\epsilon\beta})} \simeq \frac{\epsilon(e^{\epsilon\beta}(N+1)e^{-N\epsilon\beta} - e^{+\epsilon\beta} - Ne^{-N\epsilon\beta})}{e^{2\epsilon\beta}} = \\
&= \frac{\epsilon(Ne^{-(N-1)\epsilon\beta} + e^{-(N-1)\epsilon\beta} - e^{\epsilon\beta} - Ne^{-N\epsilon\beta})}{e^{2\epsilon\beta}} = \frac{\epsilon((N+1)e^{-(N-1)\epsilon\beta} - (e^{\epsilon\beta} + Ne^{-N\epsilon\beta}))}{e^{2\epsilon\beta}} = \\
&= \frac{\epsilon \left( \frac{(N+1)e^{\epsilon\beta}}{e^{N\epsilon\beta}} - \frac{n}{e^{N\epsilon\beta}} - e^{\epsilon\beta} \right)}{e^{2\epsilon\beta}} = \frac{\epsilon \left( \frac{(N+1)e^{\epsilon\beta} - N}{e^{N\epsilon\beta}} - e^{\epsilon\beta} \right)}{e^{2\epsilon\beta}} = \frac{\epsilon \left( \frac{Ne^{\epsilon\beta} + e^{\epsilon\beta} - e^{\epsilon\beta} e^{N\epsilon\beta}}{e^{N\epsilon\beta}} \right)}{e^{2\epsilon\beta}} = \epsilon \left( \frac{N - e^{N\epsilon\beta}}{e^{N\epsilon\beta} e^{\epsilon\beta}} \right) \simeq \\
&\simeq \frac{\epsilon(-e^{N\epsilon\beta})}{e^{N\epsilon\beta} e^{\epsilon\beta}} = -\epsilon e^{-\epsilon\beta} \\
&\implies \boxed{\langle s \rangle = e^{-\epsilon/\tau}}
\end{aligned}$$

**Solution 8. Quantum concentration.** Now  $\Psi(x, y, z) = A \sin\left(\frac{n_x \pi x}{L}\right) \sin\left(\frac{n_y \pi y}{L}\right) \sin\left(\frac{n_z \pi z}{L}\right)$ .  $p = \frac{1}{i} \nabla$ ,  $\frac{p^2}{2m} = -\frac{1}{2m} \nabla^2$ .  
Ground orbital:  $n_x = n_y = n_z = 1$ .

$$T = \frac{3}{2m} \left(\frac{\pi}{L}\right)^2 \langle \psi_0 | \psi_0 \rangle = \frac{3}{2m} \left(\frac{\pi}{L}\right)^2$$

where  $\langle \psi_0 | \psi_0 \rangle = 1$ ,  $\psi$  normalized. It was normalized in this way:

$$\int_0^\infty \sin^2\left(\frac{n_x \pi x}{L}\right) dx = \int_0^L \frac{1 - \cos\left(\frac{2n_x \pi x}{L}\right)}{2} dx = \left( \frac{n_x - \sin\left(\frac{2n_x \pi x}{L}\right) \left(\frac{L}{2n_x \pi}\right)}{2} \right) \Bigg|_0^L = \frac{L}{2}$$

$$\langle \psi | \psi \rangle = A^2 \left(\frac{L}{2}\right)^3 = \frac{A^2 L^3}{8} = 1 \text{ or } A^2 = \frac{8}{L^3}$$

Recall that  $n_Q = \left(\frac{m\tau}{2\pi\hbar^2}\right)^{3/2}$ .

Consider the condition that there will be a concentration for which the zero-point quantum kinetic energy is equal to the temperature  $\tau$ :

$$\Rightarrow \frac{3}{2m} \frac{\pi^2}{L^2} = \frac{3}{2m} \pi^2 n^{2/3} = \tau \text{ or } n^{2/3} = \frac{2m\tau}{3\pi^2 \hbar^2} \text{ or } n = \left(\frac{2m\tau}{3\pi^2 \hbar^2}\right)^{3/2}$$

$$\Rightarrow n = \left(\left(\frac{4}{3\pi}\right) \frac{m\tau}{2\pi\hbar^2}\right)^{3/2} = \left(\frac{4}{3\pi}\right)^{3/2} n_Q$$

**Solution 9. Partition function for two systems.**

$$Z(1+2) = \sum_{E_{1+2}} g(E_{1+2}) \exp\left(\frac{-E_{1+2}}{\tau}\right) = \sum_{E_1+E_2=E_0} g(E_1+E_2) \exp\left(\frac{-E_1}{\tau}\right) \exp\left(\frac{-E_2}{\tau}\right) =$$

$$= \sum_{E_1} \sum_{E_2} g(E_1)g(E_2) \exp\left(\frac{-E_1}{\tau}\right) \exp\left(\frac{-E_2}{\tau}\right) = Z(1)Z(2)$$

since systems are independent.

**Solution 10. Elasticity of polymers.**

- (a) Consider  $2s = N_+ - N_-$ ;  $N = N_+ + N_-$ ,  $2s = N_+ - (N - N_+) = 2N_+ - N$ .  $N_+ = \frac{2s+N}{2}$ .  
For  $2s$ , consider  $-2s = N_+ - (N - N_+) = 2N_+ - N$ .  $N_+ = \frac{-2s+N}{2}$ .

$$\Rightarrow g(N, -s) + g(N, s) = \frac{2N!}{\left(\frac{N}{2} + s\right)! \left(\frac{N}{2} - s\right)!}$$

- (b)  $|s| \ll N$

$$\sigma(l) = \ln \left( g\left(N, \frac{l}{2\rho}\right) + g\left(N, \frac{-l}{2\rho}\right) \right) = \ln \left( \frac{2(N!)}{\left(\frac{N}{2} + \frac{l}{2\rho}\right)! \left(\frac{N}{2} + \frac{-l}{2\rho}\right)!} \right) =$$

$$= \ln(2N!) = \left\{ \left(\frac{N}{2} + \frac{l}{2\rho}\right) \ln\left(\frac{N}{2} + \frac{l}{2\rho}\right) - \left(\frac{N}{2} + \frac{l}{2\rho}\right) + \left(\frac{N}{2} - \frac{l}{2\rho}\right) \ln\left(\frac{N}{2} - \frac{l}{2\rho}\right) - \left(\frac{N}{2} - \frac{l}{2\rho}\right) \right\}$$

where we used  $\ln(x + \Delta x) \simeq \ln x + \frac{1}{x} \Delta x$ .

$$\sigma(l) = \ln(2N!) - \left\{ \left(\frac{N}{2} + \frac{l}{2\rho}\right) \left( \ln\left(\frac{N}{2}\right) + \frac{1}{N/2} \frac{l}{2\rho} \right) - \frac{N}{2} + \left(\frac{N}{2} - \frac{l}{2\rho}\right) \left( \ln\left(\frac{N}{2}\right) + \frac{1}{N/2} \left(\frac{-l}{2\rho}\right) \right) \right\} =$$

$$= \ln(2N!) - \left\{ \frac{N}{2} \ln \frac{N}{2} - \frac{N}{2} + \frac{N}{2} \ln \frac{N}{2} - \frac{N}{2} + \frac{l}{2\rho} \ln\left(\frac{N}{2}\right) + \frac{l}{2\rho} + \frac{2}{N} \left(\frac{l}{2\rho}\right)^2 + \frac{-l}{2\rho} - \frac{l}{2\rho} \ln\left(\frac{N}{2}\right) + \frac{l^2}{N2\rho^2} \right\}$$

$$\sigma(l) = \ln \left( \frac{2N!}{\left(\frac{N}{2}\right)! \left(\frac{N}{2}\right)!} \right) - \frac{l^2}{N\rho^2}$$

(c)

$$\frac{\partial \sigma}{\partial l} = \frac{-2l}{N\rho^2} \implies \boxed{f = \frac{2l\tau}{N\rho^2}}$$

**Solution 11. One-dimensional gas.**

$$\epsilon_n = \frac{\hbar^2}{2m} \left(\frac{\pi}{L}\right)^2 n^2 \quad \text{in one dimension}$$

$$Z_1 = \sum_{n=1}^{\infty} \exp\left(\frac{-\hbar^2}{2m} \left(\frac{\pi}{L}\right)^2 n^2 / \tau\right) \implies \int_0^{\infty} dn e^{-\alpha^2 n^2} = \frac{\sqrt{\pi}}{2\alpha}$$

where

$$\alpha^2 = \frac{\hbar^2}{2m\tau} \left(\frac{\pi}{L}\right)^2$$

$$\alpha = \frac{\hbar\pi/L}{\sqrt{2m\tau^{1/2}}}$$

Recall that  $\sigma = \left(\frac{\partial F}{\partial \tau}\right)$ ,  $F = U - \tau\sigma$ , so that

$$\begin{aligned} F &= -\tau \ln Z = -\tau N \ln \frac{\sqrt{\pi}}{2\alpha} = \tau N \ln \left(\frac{2\alpha}{\sqrt{\pi}}\right) = \tau N \ln \left(\frac{2\hbar\pi/L}{\sqrt{\pi}\sqrt{2m\tau^{1/2}}}\right) = \\ &= \tau N \ln \left(\frac{\sqrt{2\pi}\hbar}{\sqrt{mL\tau^{1/2}}}\right) = \frac{1}{2}\tau N \ln \left(\frac{2\pi\hbar^2}{mL^2\tau}\right) \\ \frac{\partial F}{\partial \tau} &= \frac{1}{2}N \ln \left(\frac{2\pi\hbar^2}{mL^2\tau}\right) + \frac{\tau N}{2} \left(\frac{1}{\left(\frac{2\pi\hbar^2}{mL^2\tau}\right)}\right) \left(\frac{-2\pi\hbar^2}{mL^2\tau^2}\right) = \\ &= \frac{1}{2}N \ln \left(\frac{2\pi\hbar^2}{mL^2\tau}\right) + \frac{-\tau N}{2\tau} = \boxed{\frac{N}{2} \left(\ln \left(\frac{2\pi\hbar^2}{mL^2\tau}\right) - 1\right)} \end{aligned}$$

*2015 manuscript.*

## 6 Pressure

cf. pp. 64 of Ch. 3 "Boltzmann Distribution and Helmholtz Free Energy", Kittel and Kroemer (1980) [2].

Consider a system of quantum states of energy  $\epsilon_s$ .Assume  $\epsilon_s$  is a function of volume of the system.Volume is decreased slowly from  $V$  to  $V - \Delta V$  by an applied external force.Let volume change take place sufficiently slowly so that system remains in the same quantum state  $s$  throughout compression.

$$\epsilon_s(V - \Delta V) = \epsilon_s(V) - \frac{d\epsilon_s}{dV} \Delta V + \dots \equiv \text{energy of states after reversible volume change}$$

cf. Eq. (21) of Kittel and Kroemer (1980) [2], pp. 64, Ch. 3.

Now

$$U(V - \Delta V) - U(V) = \Delta U = - \left(\frac{d\epsilon_s}{dV}\right) \Delta V$$

cf. Eq. (22) of Kittel and Kroemer (1980) [2], pp. 64, Ch. 3.

Let  $A$  = area of 1 face of a cube.

Then

$$\Delta V = A(\Delta x + \Delta y + \Delta z) = A\Delta x + A\Delta y + A\Delta z$$

cf. Eq. (23) of Kittel and Kroemer (1980) [2], pp. 66, Ch. 3.

(Total) work done in compression is

$$\Delta U = p_s A(\Delta x + \Delta y + \Delta z) = p_s \Delta V$$

cf. Eq. (24) of Kittel and Kroemer (1980) [2], pp. 66, Ch. 3.

Then

$$p_s = - \left( \frac{\partial \epsilon_s}{\partial V} \right)$$

cf. Eq. (25) of Kittel and Kroemer (1980) [2], pp. 66, Ch. 3.

Take the average of  $p_s = - \left( \frac{\partial \epsilon_s}{\partial V} \right)$  over all states of the ensemble to obtain the average pressure  $p$ :

$$\boxed{p = - \left( \frac{\partial U}{\partial V} \right)_\sigma} \quad (30)$$

cf. Eq. (26) of Kittel and Kroemer (1980) [2], pp. 66, Ch. 3.

Entropy is held constant because the number of states in ensemble is unchanged in reversible compression.

Alternatively, consider that

$$dU = Q - pdV$$

Let  $c = c(t) = (U, V) = (U(t), V(t))$  s.t.  $\sigma$  const.  $\dot{c} = \dot{U} \frac{\partial}{\partial U} + \dot{V} \frac{\partial}{\partial V}$ .

Let  $t = V$ . Then  $\dot{c} = \left( \frac{\partial U}{\partial V} \right)_\sigma + \frac{\partial}{\partial V}$ ,

$$\implies dU(\dot{c}) = \left( \frac{\partial U}{\partial V} \right)_\sigma = -pdV(\dot{c}) = -p \implies \boxed{p = - \left( \frac{\partial U}{\partial V} \right)_\sigma} \quad (31)$$

We can also derive the following thermodynamic identity: for  $U = U(\sigma, V)$ ,  $dU = \tau d\sigma - pdV$ , then for

$$\begin{aligned} \left( \frac{\partial U}{\partial \sigma} \right)_V &= \tau \\ \left( \frac{\partial U}{\partial V} \right)_\sigma &= -p \\ \implies \boxed{- \left( \frac{\partial p}{\partial \sigma} \right)_V &= \left( \frac{\partial \tau}{\partial V} \right)_\sigma} \end{aligned} \quad (32)$$

For *another expression* for pressure  $p$ , consider that entropy  $\sigma$  is a function of internal energy  $U$  and volume  $V$ :

$$\begin{aligned} \sigma &= \sigma(U, V) \\ d\sigma &= \left( \frac{\partial \sigma}{\partial U} \right)_V dU + \left( \frac{\partial \sigma}{\partial V} \right)_U dV \end{aligned}$$

Let  $c = (U, V)$  s.t.  $\sigma$  const., then

$$d\sigma(\cdot c) = 0 = \left( \frac{\partial \sigma}{\partial U} \right)_V \dot{U} + \dot{V} \left( \frac{\partial \sigma}{\partial V} \right)_U$$

Now

$$\frac{1}{\tau} := \left( \frac{\partial \sigma}{\partial U} \right)_V$$

Let  $c = c(t) = (U(t), V(t))$  and  $t = V$

$$\begin{aligned} \frac{1}{\tau} \left( \frac{\partial U}{\partial V} \right)_\sigma + \left( \frac{\partial \sigma}{\partial V} \right)_U &= \frac{-p}{\tau} + \left( \frac{\partial \sigma}{\partial V} \right)_U = 0 \\ p &= \tau \left( \frac{\partial \sigma}{\partial V} \right)_U \end{aligned}$$

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## 7 The First Law of Thermodynamics

$$dU = Q - W \text{ or } Q = dU + W$$

Consider  $W = Q - dU$ .

Consider path in  $M$ ,  $\gamma, \gamma : \mathbb{R} \rightarrow M = (U, V)$

$$\gamma(t) = (U(t), V(t))$$

$$\dot{\gamma} \in \mathfrak{X}(M), \dot{\gamma} = \dot{U} \frac{\partial}{\partial U} + \dot{V} \frac{\partial}{\partial V}$$

$$W(\dot{\gamma}) = p dV(\dot{\gamma}) = p \dot{V} = Q(\dot{\gamma}) - dU(\dot{\gamma}) = Q(\dot{\gamma}) - \dot{U}$$

Suppose  $Q(\dot{\gamma}) = Q(t) dt (\dot{\gamma} \frac{\partial}{\partial t}) = Q(t) \dot{\gamma}$ .

$$\begin{aligned} \int p \dot{V} dt &= \int Q(t) \dot{\gamma} dt - \int \dot{U} dt \\ \implies p \Delta V &= \Delta Q - \Delta U \end{aligned}$$

$p \Delta V$  interpreted as work done by gas.  $\Delta Q$  is heat transferred to gas system.  $-\Delta U$  is the drop in internal energy of gas system as it does work.

### 7.1 Heat capacity

$$Q = Q(\tau, V) = \left( \frac{\partial Q}{\partial \tau} \right)_V d\tau + \left( \frac{\partial Q}{\partial V} \right)_\tau dV$$

So define  $C_V \equiv \left( \frac{\partial Q}{\partial \tau} \right)_V$  or interpret  $C_V$  as energy input by heating at constant volume over ensuing change in temperature.

In this case,  $\left( \frac{\partial U}{\partial \tau} \right)_V$ .

For the case of a monatomic gas,  $U = \frac{d}{2} \tau N$ ,  $\frac{\partial U}{\partial \tau} = \frac{d}{2} N$ .

$$C_V = \frac{d}{2} N$$

$N \equiv$  number of molecules.

Now

$$\begin{aligned} Q &= \Lambda_p dp + C_p d\tau \implies Q \wedge dp = C_p d\tau \wedge dp \\ Q &= \Lambda_V dV + C_V d\tau \implies Q \wedge dp = \Lambda_V dV \wedge dp + C_V d\tau \wedge dp \end{aligned}$$

Now from the thermodynamic identity,  $Q = W + dU = p dV + dU$ ,

$$Q \wedge dp = p dV \wedge dp + dU \wedge dp$$

and from (empirical) ideal gas law,  $pV = N\tau$  (which defines a hypersurface on  $M$ ),

$$dpV + p dV = N d\tau \implies p dV \wedge dp = N d\tau \wedge dp$$

so then

$$Q \wedge dp = N d\tau \wedge dp + dU \wedge dp$$

In the case of the monatomic gas,  $U = \frac{d}{2} \tau N$ , and so  $dU = \frac{d}{2} N d\tau = C_V d\tau$  and so comparing all the equations above, one recovers

$$C_p = N + C_V = \frac{2+d}{2} N$$

EY: 20151019 I'm curious to know how this all generalizes for  $C_V$ ,  $C_P$  heat capacities, regardless of the type of molecule we consider.

## The adiabatic relation for a classical ideal gas

Consider the adiabatic expansion (or contraction!) of a classical ideal gas.

This means that  $Q = 0$ ; there is no heat exchange to or from the gas system.

Recall  $Q = dU + W$ .

If  $Q = 0$ , and supposing  $W = pdV$ , then  $0 = dU + pdV$ .

EY : 20151019 Either by definition, or the thermodynamic identity,  $\tau d\sigma = dU + pdV$ , then  $C_V := \left(\frac{\partial U}{\partial \tau}\right)_V$ . My question is this: for manifold of thermodynamic states  $M$ ,  $M = (U, V)$ , i.e.  $U$  is a global coordinate and  $V$  is a “local” coordinate. One can make a Legendre transformation such that  $M$  is parametrized by  $(\tau, V)$ , where  $\tau$  is the temperature. In general, one should say that  $U = U(\tau, V) \in C^\infty(M)$ , and so  $dU = \frac{\partial U}{\partial \tau} d\tau + \frac{\partial U}{\partial V} dV$ ,  $dU \in \Omega^1(M)$ .

However, for this adiabatic process, we want

$$Q = 0 = dU + W = dU + pdV = C_V d\tau + pdV$$

which implies that  $dU = C_V d\tau$ . What happened to the  $\frac{\partial U}{\partial V} dV$ ? Is it that in this adiabatic process, the internal energy of the gas system goes to either doing work (expansion) or increases due to work being done on it (contraction), and is characterized completely by a drop or increase in its temperature, respectively? And so  $dU = C_V d\tau$ , and  $pdV$  completely describes what’s going on with work done or work done on it?

**Correction (adiabatic question).** On the ideal-gas surface  $(\partial U / \partial V)_\tau = 0$ , so  $dU = C_V d\tau$  on every path. The product  $C_V \tau$  is not that differential. The derivation is §33.1.

Nevertheless, using the (empirical) ideal gas law,  $pV = N\tau$ ,

$$0 = C_V d\tau + pdV = C_V d\tau + \frac{N\tau}{V} dV$$

Consider a path  $\gamma : \mathbb{R} \rightarrow M$  in  $M$ , so that  $\dot{\gamma}(t) = \dot{\tau} \frac{\partial}{\partial \tau} + \dot{V} \frac{\partial}{\partial V} \in \mathfrak{X}(M)$ .

$$\gamma(t) = (\tau(t), V(t))$$

Thus,

$$0 = C_V \dot{\tau} + \frac{N\tau}{V} \dot{V} \text{ so } \frac{\dot{\tau}}{\tau} + \frac{N}{C_V} \frac{\dot{V}}{V}$$

$$\xrightarrow{\int dt} \ln \frac{\tau_f}{\tau_i} + \frac{N}{C_V} \ln \frac{V_f}{V_i} = 0 \text{ or } \ln \tau V^{\frac{N}{C_V}} = \text{const.}$$

Now

$$\frac{N}{C_V} = \frac{C_P - C_V}{C_V} = \gamma - 1$$

which is true, assuming the (empirical) ideal gas law, the thermodynamic identity, and, surely, for the case of a monatomic gas.

Thus

$$\tau_f V_f^{\gamma-1} = \tau_i V_i^{\gamma-1}$$

## Problems

**Problem 4.** *Adiabatic compression.*

A diesel engine doesn’t have a spark plug to ignite and explode the fuel. Instead, the air in the cylinder is compressed so highly that the fuel ignites spontaneously when sprayed into the cylinder.

(a)

$$\frac{\tau_f V_f^{\gamma-1}}{\tau_i V_i^{\gamma-1}} = \frac{P_f V_f^\gamma}{P_i V_i^\gamma} = 1$$

$$\tau_f = \left(\frac{V_i}{V_f}\right)^{\gamma-1} \tau_i$$

Run the Python script `thermo.py` to do the calculations. Here is (some of) the code from `thermo.py` for doing so (one still needs to import the necessary libraries):

```

roomtemp_K = KCconv.subs(T_C,20).rhs # room temperature in Kelvin

Prob0104ans = adia_tV.subs(gamma,1.4).subs(V_f,1).subs(V_i,15).subs(tau_i, roomtemp_K) # answer
to Problem 4 of Chapter 1

Prob0104ans = N( Prob0104ans.lhs ) # 866.016969686253 K
Prob0104ansC = solve( KCconv.subs( T_K, Prob0104ans ), T_C )[0] # 592.866969686253 C
solve( FCconv.subs( T_C, Prob0104ansC ), T_F )[0] # 1099.16054543526 F

```

---

The final temperature is 866.01  $K$  or 592.87  $C$  or 1099.16  $F$

(b) Now from the ideal gas law, which is obeyed at all thermodynamic states

$$\frac{P_f}{P_i} = \frac{V_i}{V_f} \frac{\tau_f}{\tau_i}$$

and so  $\frac{P_f}{P_i} = 44.31$

2015 manuscript.

## 8 Heat capacity at constant pressure

*Manuscript section title: Heat Capacity.*

Let  $\Sigma$  be a (topological) manifold of dim. 2.

Suppose  $U$  is a global coordinate on  $\Sigma$ . Then consider 1 chart  $(U, V)$ .

Consider  $\sigma = \sigma(U, V) \in C^\infty(\Sigma)$ .

Recall the thermodynamic identity

$$\tau d\sigma = dU + p dV$$

Thus

$$d\sigma = \frac{1}{\tau} dU + \frac{p}{\tau} dV \implies \begin{aligned} \frac{p}{\tau} &= \left( \frac{\partial \sigma}{\partial V} \right)_U \\ \frac{1}{\tau} &= \left( \frac{\partial \sigma}{\partial U} \right)_V \end{aligned}$$

From Kittel and Kroemer (1980) [2], pp. 165-166, Chapter 6: Ideal Gas, “Heat Capacity”,

$$Q = Q(\tau, p) = \Lambda_p dp + C_p d\tau = dU + W = dU + p dV$$

Let  $c \in \Sigma$  s.t.  $dp(\dot{c}) = 0$  (constant pressure). And so

$$Q(\dot{c}) = 0 + C_p d\tau(\dot{c}) = \tau d\sigma(\dot{c}) = dU(\dot{c}) + p dV(\dot{c})$$

for  $c = (\tau, 0)$ ,  $\dot{c} = \frac{\partial}{\partial \tau} \in T\Sigma$ .

$$\implies C_p = \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_p = \left( \frac{\partial U}{\partial \tau} \right)_p + p \left( \frac{\partial V}{\partial \tau} \right)_p$$

for heat capacity at constant pressure is larger than  $C_V$  because additional heat must be added to perform the work needed to expand volume of gas against constant pressure.

Now recall for enthalpy

$$H = U + pV$$

$$H = H(\sigma, p)$$

Then

$$dH = dU + p dV + V dp = Q + V dp$$

Thus, for  $c \in \Sigma$ ,  $dp(\dot{c}) = 0$  (constant pressure). Hence

$$dH(\dot{c}) = \left( \frac{\partial H}{\partial \tau} \right)_p = Q(\dot{c}) + 0 = C_p$$

Hence,

$$C_p = \left( \frac{\partial H}{\partial \tau} \right)_p = \left( \frac{\partial Q}{\partial \tau} \right)_p$$

2008 manuscript.

## 9 Thermal Radiation and Planck Distribution

### Problems

#### Solution 1. Number of thermal photons.

We consider a cavity of volume  $V$ , and of edge length  $L$  (so  $V = L^3$ ). So then  $\omega_n = n\pi c/L$ .

Now  $\frac{1}{\exp(\frac{\hbar\omega_n}{\tau}) - 1}$  is the thermal average number of photons in a single mode frequency  $\omega$ . So then

$$\sum \langle s_n \rangle = \sum \frac{1}{\exp(\frac{\hbar\omega_n}{\tau}) - 1}$$

Consider  $(n_x, n_y, n_z)$  on positive octant, and 2 independent polarization  $s$  of em field.

$$\begin{aligned} \sum_n \frac{1}{\exp(\frac{\hbar n \pi c}{L\tau}) - 1} &= \frac{(2)}{8} \int_0^\infty 4\pi n^2 dn \frac{1}{\exp(\frac{\hbar n \pi c}{L\tau}) - 1} = \pi \int_0^\infty \frac{n^2 dn}{\exp(\frac{\hbar n \pi c}{L\tau}) - 1} = \\ &= \pi \left( \frac{L\tau}{\hbar\pi c} \right)^3 \int_0^\infty \left( \frac{x^2 dx}{e^x - 1} \right) = \boxed{N = \frac{V}{\pi^2} \left( \frac{\tau}{\hbar c} \right)^3 (2.404)} \end{aligned}$$

where I used the substitutions

$$\begin{aligned} x &= \frac{\hbar\pi cn}{L\tau} \\ \frac{L\tau x}{\hbar\pi c} &= n \\ (dx) \frac{L\tau}{\hbar\pi c} &= dn \end{aligned}$$

Now  $\sigma(\tau) = (4\pi^2 V/45)(\tau/\hbar c)^3$ , so then

$$\frac{\sigma}{N} = \frac{1}{2.404} \left( \frac{4\pi^4}{45} \right) \simeq \boxed{3.602}$$

Now how was  $\int_0^\infty dx \frac{x^2 dx}{e^x - 1}$  evaluated?

**Solution 2. Surface temperature of a Sun.** Given the **solar constant** of the Earth, the total radiant energy flux density at the Earth from the Sun normal to the incident rays, integrated over all emission wavelengths,

$$\text{solar constant} = 0.136 \text{ J s}^{-1} \text{ cm}^{-2}, \quad (49)$$

(a)

$$4\pi(1.49 \times 10^{11} \text{ m})^2 \cdot 0.136 \text{ J s}^{-1} \text{ cm}^{-2} \left( \frac{10^2 \text{ cm}}{1 \text{ m}} \right)^2 = (4\pi)(1.49)^2 10^{22} \cdot 0.136 \times 10^4 = \boxed{3.8 \times 10^{26} \text{ J} \cdot \text{s}^{-1}}$$

Note that I had used  $1.49 \times 10^{11} \text{ m}$  as the distance of the Earth from the Sun.

(b)  $J_\nu$  = energy flux density or rate of energy emission per unit area.

$$\sigma_B = \frac{\pi^2 k_B^4}{60 \hbar^3 c^2} = 5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}.$$

Note that  $W = \frac{1 \text{ J}}{\text{s}}$ . I will use  $R_\odot = 6.9599 \times 10^{10} \text{ cm}$  as the radius of the Sun.

$$\frac{4 \times 10^{26} \text{ J} \cdot \text{s}^{-1}}{4\pi(6.9599 \times 10^{10} \text{ cm})^2} = J_\nu = \sigma_B T^4$$

$$\frac{4 \times 10^{26} \text{ J} \cdot \text{s}^{-1}}{4\pi(6.9599 \times 10^{10} \text{ cm})^2} \left( \frac{1 \text{ m}^2 \left( \frac{10^2 \text{ cm}}{1 \text{ m}} \right)^2}{5.670 \times 10^{-8} \text{ J/s}} \right) k^4 = T^4$$

$$\boxed{T \simeq 5830 \text{ K}}$$

**Solution 3. Average temperature of the interior of the Sun.**

(a)

$$U = - \int_0^R \frac{G \left( \frac{4}{3} \pi \rho r^3 \right) (4\pi r^2 \rho dr)}{r} = - \frac{16}{3} \pi^2 G \rho^2 \int_0^R r^4 dr = \frac{-16}{15} \pi^2 \rho^2 G R^5; M = \frac{4}{3} \pi R^3 \rho$$

$$U = \frac{-3GM^2}{5R}$$

$$U = - \frac{1}{2} \int_0^R \frac{G \left( \frac{4}{3} \pi \rho r^3 \right) (4\pi r^2 \rho dr)}{r} = \frac{-8}{3} \pi^2 G \rho^2 \int_0^R r^4 dr = \frac{-8}{15} \pi^2 \rho^2 G R^5 =$$

$$= \frac{-8}{15} \pi^2 G R^5 \left( \frac{M}{\frac{4}{3} \pi R^3} \right)^2 = \frac{-3}{10} \frac{GM^2}{R} = 1.14 \times 10^{41} J$$

(b) Using the virial theorem of mechanics, note that

$$\frac{-1}{2} U = \frac{3}{20} \frac{GM^2}{R} = \frac{3}{20} \frac{6.67 \times 10^{-11} kg \cdot \frac{m}{s^2} \cdot \frac{m^2}{kg^2} \cdot 2 \times 10^{33} g \left( \frac{1 kg}{10^3 g} \right)}{7 \times 10^{10} cm \left( \frac{1 m}{10^2 cm} \right)} = 5.72 \times 10^{40} J$$

Now  $\langle s \rangle = \frac{1}{\exp(\hbar\omega/\tau)-1}$ , is the Planck distribution function, giving the thermal average number of photons in a single mode frequency  $\omega$ .

thermal average energy  $\langle \epsilon \rangle = \langle s \rangle \hbar\omega = \frac{\hbar\omega}{\exp(\hbar\omega/\tau)-1}$  for  $\tau \ll \hbar\omega$ ,  $\langle \epsilon \rangle \simeq \tau$

So then  $5.72 \times 10^{40} J = N \langle \epsilon \rangle = N\tau$ .

$$\tau = \frac{5.72 \times 10^{40} J}{1 \times 10^{57} (1.381 \times 10^{-23} J/K)} = \boxed{4.14 \times 10^6 K}$$

**Solution 4. Age of the Sun.**

(a) Consider  $4H \rightarrow_2^4 He$ . Then  $4(1.0078) - 4.0026 = 0.0286 amu$ . Then

$$(0.0286 amu) \left( \frac{1.6726 \times 10^{-27} kg}{1.00727647 u} \right) (3 \times 10^8 m/s)^2 = 4.27 \times 10^{-12} J$$

Given  $M_\odot = 2 \times 10^33 g$ ,

$$(2 \times 10^{30} kg)(0.10) \left( \frac{1}{4 \times (1.0078 amu)} \right) \left( \frac{1.00727647 u}{1.6726 \times 10^{-27} kg} \right) (4.27 \times 10^{-12} J) = 1.28 \times 10^{44} J$$

So  $\boxed{1.28 \times 10^{44} J}$  energy is available.

(b)

$$\frac{1.15 \times 10^{45} J}{4 \times 10^{26} J \cdot s^{-1}} \left( \frac{1 hr}{3600 s} \right) \left( \frac{1 day}{24 hr} \right) \left( \frac{1 yr}{365 days} \right) = 1.02 \times 10^{10} years$$

**Solution 5. Surface temperature of the Earth.**  $J_S = \sigma_b T_\odot^4$  is the radiant power per unit area.

Total emitted radiation energy of the sun is  $J_S 4\pi R_\odot^2$ .

$\frac{4\pi R_\odot^2 J_S}{4\pi R_{ES}^2} = \frac{R_\odot^2}{R_{ES}^2} J_S$  = radiation energy hitting  $1 cm^2$  of Earth's surface in one second

Since the Earth is considered a black-body, the rate of absorption must equal the rate of emission:

$$\frac{R_\odot^2}{R_{ES}^2} \sigma_b T_\odot^4 = \sigma_b T_e^4 \text{ or } T_e^4 = T_\odot^4 \frac{R_\odot^2}{R_{ES}^2} \implies T_e = 5800 K \sqrt{\frac{7 \times 10^{10} cm}{1.5 \times 10^{13} cm}} = 396.2 K = 123 C$$

**Solution 6. Pressure of thermal radiation.**

- (a)  $s_j$  = number of photons in that mode. Suppose modes of  $\omega_j$ ,  $j = 0, 1, 2, \dots$ .  
 $\epsilon_j = s_j \hbar \omega_j$  = total energy in  $j$ th mode,  $s_j$  photons in  $j$ th mode.

$$U = \sum_j \epsilon_j = \sum_j s_j \hbar \omega_j \quad P = -\frac{\partial U}{\partial V} = -\sum_j s_j \hbar \frac{\partial \omega_j}{\partial V}$$

- (b)  $\omega_j = j\pi c/L$ ,  $V = L^3$ . So then  $\omega_j = j\pi c V^{-1/3}$ .

$$\Rightarrow \frac{d\omega_j}{dV} = \frac{-1}{3} j\pi c V^{-4/3} = \frac{-1}{3} \frac{\omega_j}{V}$$

- (c)  $p = \frac{U}{3V}$

- (d) We want the Kinetic pressure at a concentration of  $(1 \text{ mol/cm}^3)$ . Recalling  $P = \frac{Nk_B T}{V}$ ,

$$P = \left( \frac{1 \text{ mol}}{\text{cm}^3} \right) \left( \frac{6.022 \times 10^{23}}{1 \text{ mol}} \right) \left( \frac{10^2 \text{ cm}}{1 \text{ m}} \right)^3 \left( 1.381 \times 10^{-23} \frac{\text{J}}{\text{K}} \right) (2 \times 10^7 \text{ K}) = 1.663 \times 10^{14} \frac{\text{N}}{\text{m}^2}$$

Now for the thermal radiation pressure,

$$p = \frac{U}{3V} = \frac{1}{3} \frac{\pi^2}{15 \hbar^3 c^3} \tau^4 = 4.03 \times 10^{13} \frac{\text{N}}{\text{m}^2}$$

where  $t = 2 \times 10^7 \text{ K}$ .

For the pressures to be equal,

$$\frac{1}{3} \frac{\pi^2}{15 \hbar^3 c^3} k_B^4 T^4 = \frac{Nk_B}{V} T \text{ or } T^3 = \frac{45(\hbar c)^3 Nk_B}{\pi^2 k_B^4 V} \text{ so that } \boxed{T = 3.2 \times 10^7 \text{ K}}$$

### Solution 7. Free energy of a photon gas

- (a)  $Z = \prod_n \frac{1}{1 - e^{-\hbar \omega_n / \tau}}$  Consider  $Z = \sum_{s=0}^{\infty} e^{-s \hbar \omega / \tau} = \frac{1}{1 - e^{-\hbar \omega / \tau}}$  for a single mode.

- (b)  $F = -\tau \ln Z = \tau \sum_n \ln(1 - e^{-\hbar \omega_n / \tau}) \quad \omega_n = \frac{n\pi c}{L}$ .

$$\begin{aligned} F &= \tau \sum_n \ln(1 - e^{-\hbar \omega_n / \tau}) = \tau \int_0^{\infty} \frac{4\pi n^2 dn}{8} (2) \ln(1 - e^{-\hbar n \pi c / \tau L}) = \pi \tau \int_0^{\infty} dn n^2 \ln(1 - e^{-\hbar n \pi c / \tau L}) = \\ &= \pi \tau \left( (n^3 \ln(1 - e^{-\hbar n \pi c / \tau L})) \Big|_0^{\infty} - \int_0^{\infty} dn \frac{n^3 e^{-\hbar n \pi c / \tau L}}{1 - e^{-\hbar n \pi c / \tau L}} \left( \frac{-\hbar \pi c}{\tau L} \right) \right) = -\frac{\hbar \pi^2 c}{L} \int_0^{\infty} dn \frac{n^3}{e^{\hbar n \pi / \tau L} - 1} = \\ &= -\left( \frac{\tau L}{\hbar \pi c} \right)^3 \int_0^{\infty} \frac{x^3}{e^x - 1} = -\frac{(\tau L)^3}{(\hbar^2 c^3)} \frac{\pi^2}{45} \end{aligned}$$

where I used  $x = \frac{\hbar n \pi c}{\tau L}$ .

**Solution 8. Heat shields.** For  $J_u = \sigma_B(T_u^4 - T_l^4)$ , the thermal flux without the heat shield, in the middle region. Plane  $m$  absorbs  $\sigma_B T_u^4 + \sigma_B T_l^4$  and emits  $J_m = \sigma_B(T_u^4 + T_l^4) = \sigma T_m^4$ .

$T_m = (T_u^4 + T_l^4)^{1/4}$ . The key point is that, by symmetry, plane  $m$  emits  $\frac{J_m}{2}$  flux on each side.

$$J_{net} = \sigma_B T_u^4 - \left( \sigma_B \left( \frac{T_u^4 + T_l^4}{2} \right) \right) = \sigma_B \left( \frac{T_u^4 - T_l^4}{2} \right) = \frac{J_u}{2}$$

$J_{net}$  is the same for the other side of the heat shield:

$$J_{net} = -\sigma_B T_l^4 + \sigma_B \left( \frac{T_u^4 + T_l^4}{2} \right) = \sigma_B \left( \frac{T_u^4 - T_l^4}{2} \right)$$

**Solution 9. Photon gas in one dimension.**  $E = E_0 \sin(kx) \cos(\omega t)$  is the form of a solution with  $kL = n\pi$  or  $k = \frac{n\pi}{L}$ , since

$$v^2 E_x x = E_t t \rightarrow v^2 k^2 = \omega^2 \text{ or } \frac{\omega}{k} = v, \omega_n = v \frac{n\pi}{L}$$

$$Z_j = \sum_{s=0}^{\infty} e^{-s\hbar\omega/\tau} = \frac{1}{1 - e^{-\hbar\omega/\tau}}$$

where  $Z_j$  is the partition function for a particular mode frequency  $\omega$ .

$$\begin{aligned} \langle s \rangle &= \frac{\sum_{s=0}^{\infty} s e^{-\frac{s\hbar\omega_n}{\tau}}}{Z} = Z^{-1} \frac{d}{d(-\hbar\omega_n/\tau)} (1 - \exp\left(\frac{-\hbar\omega_n}{\tau}\right))^{-1} = Z^{-1} (1 - \exp\left(\frac{-\hbar\omega}{\tau}\right))^{-2} \exp\left(\frac{-\hbar\omega_n}{\tau}\right) = \\ &= \frac{\exp(-\hbar\omega_n/\tau)}{1 - \exp(-\hbar\omega_n/\tau)} \end{aligned}$$

So

$$\langle s \rangle \hbar\omega_n = \langle \epsilon_n \rangle = \frac{\hbar\omega_n}{\exp\left(\frac{\hbar\omega_n}{\tau}\right) - 1} \quad U = \sum_n \langle \epsilon_n \rangle = \sum_n \frac{\hbar v n \pi / L}{\exp\left(\frac{\hbar v n \pi}{L\tau}\right) - 1}$$

$$\frac{\partial U}{\partial \tau} = \sum_n \frac{-\hbar v n \pi / L}{\left(\exp\left(\frac{\hbar v n \pi}{L\tau}\right) - 1\right)^2} \left( \exp\left(\frac{\hbar v n \pi}{L\tau}\right) \left( \frac{-\hbar v n \pi}{L\tau^2} \right) \right) = \sum_n \frac{\left(\frac{\hbar v \pi}{L\tau}\right)^2 n^2 \exp\left(\frac{\hbar v n \pi}{L\tau}\right)}{\left(\exp\left(\frac{\hbar v n \pi}{L\tau}\right) - 1\right)^2} = \sum_n \frac{\left(\frac{\hbar v \pi}{L\tau}\right)^2 n^2 \exp\left(\frac{\hbar v \pi n}{L\tau}\right)}{\left(\exp\left(\frac{\hbar v \pi}{L\tau}\right) - 1\right)^2}$$

Now  $\sum_n \rightarrow \int_0^\infty dn$  for one-dimensional photon. Let  $\alpha = \frac{\hbar v \pi}{L\tau}$ .

Letting  $x = \alpha n$ ,

$$\frac{\partial U}{\partial \tau} = \left(\frac{nv\pi}{L\tau}\right)^2 \int dn \frac{n^2 \exp(\alpha n)}{(e^{\alpha n} - 1)^2} = \frac{1}{\alpha} \int dx \frac{x^2 e^x}{(e^x - 1)^2} = \frac{1}{\alpha} \left\{ \frac{-x^2}{(e^x - 1)} + \int_0^\infty \frac{x dx}{e^x - 1} \right\}$$

Coefficient of  $k$  term of  $f(k)$  is  $j = \int_0^\infty \frac{x}{e^x - 1} dx$ . Now  $f(k) = \int_0^\infty \frac{\sin(kx)}{e^x - 1} dx$ ,  $\sin(kx) = kx - \frac{(kx)^3}{3!} + \dots$ , so that

$$\frac{\pi^2}{6} = \int_0^\infty \frac{x}{e^x - 1} dx$$

$$\frac{\partial U}{\partial \tau} = \frac{L\tau}{\hbar v \pi} \left( \frac{\pi^2}{6} \right) = \boxed{\frac{L\tau\pi}{6\hbar v} = C_V}$$

**Solution 10. Heat capacity of intergalactic space.**

Given the density 1 atom  $m^{-3}$ , considering thermal radiation at  $2.9 K$ , then  $k_B T = (1.381 \times 10^{-23} J/K)(2.9 K)$ ,  $\hbar c = (1.05457 \times 10^{-34} J \cdot s)(3 \times 10^8 \frac{m}{s})$ .

Recall for radiation, that the energy per unit volume:  $\frac{U}{V} = \frac{\pi^2}{15\hbar^3 c^3} \tau^4$  so that  $\frac{\partial U}{\partial \tau} = \frac{4\pi^2}{15\hbar^3 c^3} \tau^3 V$ .

Assume hydrogen atoms modeled as ideal gas:  $U = \frac{3}{2} N \tau$ ,  $\frac{dU}{d\tau} = \frac{3}{2} N$ .

$$\frac{C_{V_{matter}}}{C_{V_{radiation}}} = \frac{\frac{3}{2} N}{\frac{4\pi^2}{15\hbar^3 c^3} \tau^3 V} = \frac{45(\hbar c)^3 (N/V)}{8\pi^2 (k_B T)^3} = 2.8 \times 10^{-10}$$

**Solution 11. Heat capacity of solids in high temperature limit.**  $\frac{\hbar\omega_n}{\tau} = \frac{\pi\hbar v n}{L\tau}$ ;  $\omega_n = \frac{\pi v n}{L}$ . For  $\tau \ll \hbar\omega_n$ ,  $0 \leq n \leq n_D$ .

$$\exp\left(\frac{\hbar\omega_n}{\tau}\right) - 1 \simeq 1 + \frac{\hbar\omega_n}{\tau} + \left(\frac{\hbar\omega_n}{\tau}\right)^2 \left(\frac{1}{2}\right) + \frac{1}{6} \left(\frac{\hbar\omega_n}{\tau}\right)^3 + \dots - 1$$

By doing long division

$$\frac{\hbar\omega_n}{\exp\left(\frac{\hbar\omega_n}{\tau}\right) - 1} \simeq \frac{\hbar\omega_n}{\frac{\hbar\omega_n}{\tau} \left(1 + \frac{\hbar\omega_n}{2\tau} + \frac{(\hbar\omega_n)^2}{6\tau^2}\right)} = \frac{\tau}{1 + \frac{\hbar\omega_n}{2\tau} + \frac{(\hbar\omega_n)^2}{6\tau^2}} = \tau + -\frac{\hbar\omega_n}{2} + \frac{(\hbar\omega_n)^2}{12\tau} + \dots$$

For  $n_p = (6N/\pi)^{1/3}$

$$U = \frac{3\pi}{2} \int_0^{n_D} dnn^2 \left( \tau - \frac{\hbar\omega_n}{2} + \frac{(\hbar\omega_n)^2}{12\tau} \right) = \frac{3\pi}{2} \int_0^{n_D} dnn^2 \left( \tau - \frac{\hbar}{2} \frac{\pi v n}{L} + \frac{\hbar^2 \pi^2 v^2 n^2}{12\tau L^2} \right) =$$

$$= \frac{3\pi}{2} \left\{ \frac{1}{3} n_D^3 \tau - \frac{\hbar\pi v}{2L} \frac{n_D^4}{4} + \frac{\hbar^2 \pi^2 v^2}{12\tau L^2} \frac{1}{5} n_D^5 \right\} = \frac{\pi}{2} \frac{6N}{\pi} \tau - \frac{3\pi^2 \hbar v}{16L} \left( \frac{6N}{\pi} \right)^{4/3} + \frac{3\hbar^2 \pi^3 v^2 (6N/\pi)^{5/3}}{120\tau L^2}$$

So

$$U = 3N\tau - \frac{3\pi^2 \hbar v}{16L} \left( \frac{6N}{\pi} \right)^{4/3} + \frac{3\hbar^2 \pi^3 v^2 (6N/\pi)^{5/3}}{120\tau L^2}$$

Now  $T = \theta$ ,  $\theta = \left( \frac{\hbar v}{k_B} \right) \frac{(6\pi^2 N)^{1/3}}{L}$

So then

$$U = 3N \frac{\hbar v (6\pi^2 N)^{1/3}}{L} - \frac{3\pi^2 \hbar v}{16L} \frac{6^{1/3} N^{4/3} 6}{\pi^{4/3}} = \frac{15}{8} \frac{(6\pi^2 N)^{1/3} \hbar v}{L} = N k_B \theta \left( \frac{15}{8} \right)$$

$$\frac{U}{\theta} = N k_B \left( \frac{15}{8} \right) = \left( \frac{15}{8} \right) (1.381 \times 10^{-23} \text{ J/K}) \left( \frac{6.022 \times 10^{23} \text{ particles}}{1 \text{ mol}} \right) = 15.59$$

which is very close to experimental values.

**Solution 12. Heat capacity of photons and phonons.** For a photon:  $U = \frac{\pi^2 V}{15\hbar^3 c^3} \tau^4$   $\partial_\tau U = \frac{4\pi^2 V}{15\hbar^3 c^3} \tau^3$ .

phonon:  $U(\tau) = \frac{3\pi^4 N \tau^4}{5(k_B \theta)^3}$ .  $\frac{\partial U}{\partial \tau} = \frac{12\pi^4 N \tau^3}{5(k_B \theta)^3}$

So then

$$C_V = \frac{12\pi^4 10^{22}}{5} \left( \frac{1}{100} \right)^3 = 2.3 \times 10^{18}$$

for a phonon.

For a photon,

$$\frac{4\pi^2}{15} \frac{(1.381 \times 10^{-23} \text{ J/K})^3}{(1.05457 \times 10^{-34} \text{ J} \cdot \text{s} (3 \times 10^{10} \text{ cm/s}))^3} \tau^3 = 220 / K^3 \tau^3$$

Temperature at which to photon contribution equals to phonon contribution:

$$(220/K^3) \tau^3 = 2.3 \times 10^{18} \implies \boxed{\tau = 2.2 \times 10^5 \text{ K}}$$

**Solution 13. Energy fluctuations in a solid at low temperatures.**

$$\tau^2 \left( \frac{\partial U}{\partial \tau} \right)_V = \langle (\epsilon - \langle \epsilon \rangle)^2 \rangle$$

Recall that

$$\left( \frac{\partial U}{\partial \tau} \right)_V = \frac{12}{5} \frac{\pi^4 N \tau^3}{(k_B \theta)^3} \text{ where } U = \frac{3\pi^4 N \tau^4}{5(k_B \theta)^3}$$

$$\frac{\langle (\epsilon - \langle \epsilon \rangle)^2 \rangle}{\langle \epsilon \rangle^2} = \frac{12\pi^4 N \tau^5}{5(k_B \theta)^3} \left( \frac{1}{\frac{9\pi^8 N^2 \tau^8}{25(k_B \theta)^6}} \right) = \frac{20}{3} \left( \frac{1}{\pi^4} \right) \left( \frac{1}{N} \right) \left( \frac{1}{\tau^3} \right) (k_B \theta)^3 = \frac{0.068}{N} \left( \frac{\theta}{T} \right)^3$$

$$\mathcal{F} = \sqrt{\frac{0.070}{10^{15}} \left( \frac{200}{10^{-2}} \right)^3} = 0.02$$

**Solution 14. Heat capacity of liquid  $^4\text{He}$  at low temperatures.**

- (a) Given  $v = 2.383 \times 10^4 \text{ cm s}^{-1}$  and accounting for only longitudinal waves (only longitudinal polarization), then the Debye temperature is

$$\theta = \left( \frac{\hbar v}{k_B} \right) \left( \frac{18\pi^2 N}{V} \right)^{1/3} = \frac{(1.05457 \times 10^{-34} \text{ J} \cdot \text{s})(2.383 \times 10^4 \text{ cm/s})}{1.381 \times 10^{-23} \text{ J/K}} \left( \frac{18\pi^2 0.145 \text{ g}}{\text{cm}^3} \left( \frac{1.00727647 \text{ u} \left( \frac{1 \text{ He}}{4.0026 \text{ u}} \right)}{1.67262 \times 10^{-24} \text{ g}} \right) \right)^{1/3} =$$

$$= \boxed{28.6 \text{ K}}$$

- (b) Recall the derivation for  $U$  for phonons in a solid. Account for only longitudinal waves (only longitudinal polarization).

$$U = \frac{\pi}{2} \int_0^{n_D} dn \frac{n^2 \hbar \omega_n}{\exp(\hbar \omega_n / \tau) - 1} = \frac{\pi}{2} \int_0^{n_D} n^2 dn \frac{\hbar \frac{\pi n v}{L}}{\exp(x) - 1} = \frac{\hbar \pi^2 v}{2L} \int_0^{n_D} \frac{n^3}{\exp(x) - 1} dn$$

With  $\omega_n = \frac{\pi n}{L} v$ ,  $x = \frac{\hbar \pi n v}{L \tau}$  or  $\left(\frac{L \tau}{\hbar \pi v}\right) x = n$ , then

$$U = \left(\frac{\hbar^2 v}{2L}\right) \left(\frac{L \tau}{\hbar \pi v}\right)^4 \int_0^{n_D} \frac{x^3}{e^x - 1} dx$$

For low temperatures,  $\tau$  small so take  $x_D = \left(\frac{\hbar \pi n_D v}{L \tau}\right) = \left(\frac{\hbar \pi v}{L \tau}\right) \left(\frac{18N}{\pi}\right)^{1/3} = \frac{18^{1/3} \hbar \pi^{2/3} n^{1/3} v}{\tau}$  to go to  $\infty$ .

$$U \rightarrow \left(\frac{\pi^2}{2(\hbar v)^3}\right) \frac{\tau^4}{15} V$$

Recall that  $C_V = \left(\frac{\partial U}{\partial \tau}\right)_V$ . Then  $C_V/V = \frac{2}{15} \left(\frac{\pi^2}{(\hbar v)^3}\right) \tau^3$ . Recall  $\tau_B = k_B T$ , and given  $v = 2.383 \times 10^4 \text{ cm/s}$ , then

$$\frac{k_B}{\hbar v} = \frac{(1.381 \times 10^{-23} \text{ J/K})}{(1.05457266 \times 10^{-34} \text{ J} \cdot \text{s})(2.383 \times 10^4 \text{ cm/s})} = 5.495 \times 10^6 (1/\text{K} \cdot \text{cm})$$

So if we take  $C_V/V$  and divide by the given density  $\rho = 0.145 \text{ g/cm}^3$  to get the heat capacity per gram, (and multiply by  $k_B$ , the Boltzmann constant to get the correct units; Kittel and Kroemer likes using dimensionless formulas ) then

$$(C_V/V)/\rho = (k_B) \frac{2}{15} \pi^2 \left(\frac{k_B}{\hbar v}\right)^3 T^3 \left(\frac{\text{cm}^3}{0.145 \text{ g}}\right) = \boxed{0.0208 \times T^3}$$

#### Solution 15. Angular distribution of radiant energy flux.

- (a) Recall

$$u_\omega = \frac{\hbar}{\pi^2 c^2} \frac{\omega^3}{\exp\left(\frac{\hbar \omega}{\tau}\right) - 1}$$

is the radiation energy per unit volume per unit frequency range.

$cu_\omega$  = energy per unit time, per cross sectional area per unit frequency range.

em waves emitted spherically from pt. Q.

Suppose em wave comes in at a funny angle other than directly inward.

Consider area  $da$  that's from the spherical wave from pt. Q. How much of that goes into solid angle  $d\Omega$ ?

$$\implies cu_\omega \cos \theta dA$$

So  $cu_\omega \cos \theta$  is the energy per unit time, per cross-sectional area, per unit frequency range, that enters into some solid angle  $d\Omega$ .

$$\frac{r^2 d\Omega}{4\pi r^2} = \frac{d\Omega}{4\pi}$$

is the fraction of the spectral density that if arrives in solid angle  $d\Omega$ .

$$\implies \boxed{cu_\omega \cos \theta \frac{d\Omega}{4\pi}}$$

is the spectral density of radiant energy flux that arrives in solid angle  $d\Omega$ .

(b)

$$\int \cos \theta \sin \theta d\theta d\varphi = 2\pi \int \frac{\sin 2\theta}{2} d\theta = -\pi \left( \frac{\cos 2\theta}{2} \right) \Big|_0^{\pi/2} = -\pi \left( \frac{-1-1}{2} \right) = \pi$$

$$\Rightarrow \boxed{\frac{cu_\omega}{4}}$$

**Solution 17. Entropy and occupancy.**

$$Z = \sum_{s=0}^{\infty} e^{-s\hbar\omega/\tau} = \frac{1}{1 - e^{-\hbar\omega/\tau}}$$

$$\partial_\tau Z = \sum_{s=0}^{\infty} e^{-s\hbar\omega/\tau} \left( \frac{s\hbar\omega}{\tau^2} \right) = -(1 - e^{-\hbar\omega/\tau})^{-2} (-e^{-\hbar\omega/\tau}) \left( \frac{\hbar\omega}{\tau^2} \right) = Z \frac{\hbar\omega}{\tau^2} \langle s \rangle$$

Then

$$\langle s \rangle = \frac{e^{-\hbar\omega/\tau}}{1 - e^{-\hbar\omega/\tau}} \quad \langle s+1 \rangle = \frac{1}{1 - e^{-\hbar\omega/\tau}}$$

$$\sigma = \partial_\tau (\tau \ln Z) = \ln Z + \tau \frac{\partial_\tau Z}{Z} = \ln \langle s+1 \rangle + \tau \frac{\hbar\omega}{\tau^2} \langle s \rangle = \ln \langle s+1 \rangle + \frac{\hbar\omega}{\tau} \langle s \rangle$$

Now (nyní)

$$\langle s \rangle \ln \left( \frac{\langle s+1 \rangle}{\langle s \rangle} \right) = Z e^{-\hbar\omega/\tau} \ln e^{\hbar\omega/\tau} = \frac{\hbar\omega}{\tau} \langle s \rangle$$

$$\Rightarrow \boxed{\sigma = \langle s+1 \rangle \ln \langle s+1 \rangle + \langle s \rangle \ln \langle s \rangle}$$

**Solution 18. Isentropic expansion of photon gas.**

(a)  $\tau_i V_i^{1/3} = \tau_f V_f^{1/3}$  or

$$\left( \frac{V_i}{V_f} \right)^{1/3} = \frac{\tau_f}{\tau_i} = \frac{2.9 K}{3000 K} = 10^{-3}$$

$r = r(t) = \alpha t$  so that  $\Delta r = \alpha \Delta t$  or  $\frac{\Delta r}{r} = \frac{\Delta t}{t}$

$$\frac{r_f - r_i}{r_f} = 1 - \frac{r_i}{r_f} = \frac{t_f - t_i}{t_f} = 1 - \frac{t_i}{t_f}$$

Knowing that  $\frac{r_i}{r_f} = 10^{-3}$ , then  $\frac{t_i}{t_f} = 10^{-3}$ .

(b) Now

$$\sigma = \tau V^{1/3}$$

$$\left( \frac{\sigma}{\tau} \right)^3 = V$$

For *constant entropy* expansion.

$$\frac{U}{V} = \frac{\pi^2}{15\hbar^3 c^3} \tau^4 \quad U = \frac{\pi^2}{15(\hbar c)^3} V \left( \frac{\sigma^4}{V^{4/3}} \right) = \frac{\pi^2}{15(\hbar c)^3} \frac{\sigma^4}{V^{1/3}}$$

$$\left( \frac{\partial U}{\partial V} \right)_\sigma = \frac{\pi^2}{15(\hbar c)^3} \sigma^4 \left( \frac{-1}{3} V^{-4/3} \right)$$

$$W = \int p dV = \frac{\pi^2}{15(\hbar c)^3} \sigma^4 (-V^{-1/3}) \Big|_{V_i}^{V_f} = \frac{\pi^2 \sigma^4}{15(\hbar c)^3} \left( \frac{1}{V_i^{1/3}} - \frac{1}{V_f^{1/3}} \right) = \frac{\pi^2 V_i^{4/3} \tau_i^4}{15(\hbar c)^3} \left( \frac{1}{V_i^{1/3}} - \frac{\tau_f}{\tau_i V_i^{1/3}} \right) = \frac{\pi^2 V_i \tau_i^3}{15(\hbar c)^3} (\tau_i - \tau_f)$$

**Exercise 19. Reflective heat shield and Kirchhoff's law.**

For a left plane sheet at  $\tau_u$  temperature, right plane sheet at  $\tau_l$  temperature

$$\begin{array}{ll} +J_u = \sigma_b \tau_u^4 & -J_l = \sigma_b \tau_l^4 \\ \text{reflection } -rJ_u = -(1-a)\sigma_b \tau_u^4 & \text{reflection } rJ_l = (1-a)\sigma_b \tau_l^4 \\ \text{absorb (left) } aJ_u = a\sigma_b \tau_u^4 & \text{absorb (left) } aJ_l = a\sigma_b \tau_l^4 \end{array}$$

total absorption:  $a(J_u + J_l) = a\sigma_b(\tau_u^4 + \tau_l^4)$

total emission:  $a\sigma_b(\tau_u^4 + \tau_l^4)$ . By symmetry,  $\frac{a(J_u + J_l)}{2}$  emitted to the left, and to right.

$e$  emissivity, where emissivity is defined so radiation flux emitted by the object is  $e$  times the flux emitted by a blackbody at the same temperature.

By Kirchhoff law, for equilibrium,  $a = e$ ; object must emit at same rate as it absorbs.

$$\begin{aligned} J_{net} &= \frac{-a(J_u + J_l)}{2} - rJ_u + J_u = \frac{-(1-r)(J_u + J_l)}{2} + (1-r)J_u = \frac{(1-r)J_u}{2} - \frac{(1-r)J_l}{2} = (1-r)\frac{(J_u - J_l)}{2} = \\ &= (1-r)\left(\frac{\sigma_b(\tau_u^4 - \tau_l^4)}{2}\right) = \frac{a(J_u + J_l)}{2} + rJ_l - J_l = \boxed{\frac{(1-r)(J_u - J_l)}{2}} \end{aligned}$$

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## 10 Chemical Potential and Gibbs Distribution

### Chemical Potential and Entropy

Let  $\sigma = \sigma(U, V, N)$  (so now  $\Sigma$  has 3 coordinates  $U, V, N$ ). Then

$$d\sigma = \left(\frac{\partial \sigma}{\partial U}\right)_{V,N} dU + \left(\frac{\partial \sigma}{\partial V}\right)_{U,N} dV + \left(\frac{\partial \sigma}{\partial N}\right)_{U,V} dN$$

Consider curve  $c : \mathbb{R} \rightarrow \Sigma$  s.t.  $\dot{c} = \dot{U} \frac{\partial}{\partial U} + \dot{N} \frac{\partial}{\partial N} \in \mathfrak{X}(\Sigma)$  (i.e.  $dV(\dot{c}) = 0$ )  
 $c(t) \in \Sigma$

Let  $\tau = \tau(U, V, N) \in C^\infty(\Sigma)$

$$d\tau \in \Omega^1(\Sigma)$$

Suppose  $d\tau(\dot{c}) = 0$  (i.e.  $c$  is a null curve of  $d\tau$  or  $c$  is an isothermal curve, or i.e. overall temperature change  $d\tau$  is 0)

$$d\sigma(\dot{c}) = \left(\frac{\partial \sigma}{\partial U}\right) dU(\dot{c}) + \left(\frac{\partial \sigma}{\partial N}\right) dN(\dot{c}) \text{ or } \frac{d\sigma(\dot{c})}{dN(\dot{c})} = \left(\frac{\partial \sigma}{\partial U}\right) \frac{dU(\dot{c})}{dN(\dot{c})} + \left(\frac{\partial \sigma}{\partial N}\right)$$

Now recall that originally, considering a system with only thermal contact to a huge heat reservoir/bath (it can be the universe), then only  $N, U$  mattered in describing this system, and so  $\sigma = \sigma(N, U)$ , so that  $\frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial U}\right)_N$ .

Now

$$\frac{dU(\dot{c})}{dN(\dot{c})} = \frac{\dot{c}(U)}{\dot{c}(N)} = \frac{\dot{U}}{\dot{N}} = \left(\frac{\partial U}{\partial N}\right)_{\tau,V}$$

Similarly,

$$\frac{d\sigma(\dot{c})}{dN(\dot{c})} = \frac{\dot{c}(\sigma)}{\dot{N}} = \left(\frac{\partial \sigma}{\partial N}\right)_{\tau,V}$$

So

$$\left(\frac{\partial \sigma}{\partial N}\right)_{\tau,V} = \frac{1}{\tau} \left(\frac{\partial U}{\partial N}\right)_{\tau,V} + \left(\frac{\partial \sigma}{\partial N}\right)_{U,V}$$

Consider  $F \equiv U - \tau\sigma$ . Now consider  $dF = dU - \sigma d\tau - \tau d\sigma$ . Apply  $dF$  onto  $\dot{c}$ , our vector field generated by our particular isothermal, and constant volume curve in  $\Sigma$ :

$$dF(\dot{c}) = dU(\dot{c}) - \sigma d\tau(\dot{c}) - \tau d\sigma(\dot{c}) = \dot{U} - 0 - \tau d\sigma(\dot{c})$$

The right hand side is

$$dF(\dot{c}) = \dot{U} \frac{\partial F}{\partial U} + \dot{N} \left( \frac{\partial F}{\partial N} \right)_{\tau, V} = \dot{U}(1) + \dot{N} \left( \frac{\partial F}{\partial N} \right)_{\tau, V}$$

Left hand side is

$$\dot{U} - \tau \left( \left( \frac{\partial \sigma}{\partial N} \right)_{\tau, V} \dot{N} - \frac{1}{\tau} \left( \frac{\partial U}{\partial N} \right)_{\tau, V} \dot{N} + \left( \frac{\partial \sigma}{\partial U} \right)_{V, N} \dot{U} \right)$$

Subtract  $\dot{U}$  from both sides.  $\dot{U}$  was arbitrary (and so is  $\dot{N}$ ), and so

$$\implies \mu \equiv \left( \frac{\partial F}{\partial N} \right)_{\tau, V} \equiv \left( \frac{\partial U}{\partial N} \right)_{\tau, V} - \tau \left( \frac{\partial \sigma}{\partial N} \right)_{\tau, V} \text{ or } \mu = -\tau \left( \frac{\partial \sigma}{\partial N} \right)_{U, V}$$

### Thermodynamic identity

Generalize thermodynamic identity for  $\sigma = \sigma(U, V, N)$

$$\begin{aligned} d\sigma &= \left( \frac{\partial \sigma}{\partial U} \right)_{V, N} dU + \left( \frac{\partial \sigma}{\partial V} \right)_{U, N} dV + \left( \frac{\partial \sigma}{\partial N} \right)_{U, V} dN = \\ &= \frac{1}{\tau} dU + \frac{p}{\tau} dV - \frac{\mu}{\tau} dN \text{ or } dU = \tau d\sigma - p dV + \mu dN \end{aligned}$$

### Solution 1. Centrifuge.

$$T = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \omega^2.$$

Consider  $\mu_{ext} = -\frac{1}{2} M r^2 \omega^2$  (negative so for bigger  $r$  away from  $r = 0$  axis, lower chemical potential  $\mu$ , so to show “centrifugal force” outwards.

$$\mu_{tot} = \tau \ln(n/n_Q) - \frac{1}{2} M r^2 \omega^2.$$

$\mu_{tot}(r) = \mu_{tot}(0)$  for diffusion equilibrium.

$$\tau \ln \left( \frac{n}{n_Q} \right) - \frac{1}{2} M r^2 \omega^2 = \tau \ln \left( \frac{n(0)}{n_Q} \right) \implies \tau \ln \left( \frac{n(r)}{n(0)} \right) = \frac{1}{2} M r^2 \omega^2$$

$$\frac{n(r)}{n(0)} = \exp \left( \frac{M r^2 \omega^2}{2\tau} \right) \text{ or } \boxed{n(r) = n(0) \exp \left( \frac{M r^2 \omega^2}{2\tau} \right)}$$

### Solution 2. Molecules in the Earth’s atmosphere.

Recall that for ideal gas,  $F = -\tau[N \ln Z_1 - \ln N!]$ ;  $Z_1 = n_Q V = \left( \frac{M\tau}{2\pi\hbar^2} \right)^{3/2} V$ .

$$\mu = \left( \frac{\partial F}{\partial N} \right)_{\tau, V} = -\tau [\log Z_1 - \frac{d}{dN} \ln N!] = \tau \ln \left( \frac{n}{n_Q} \right)$$

where  $n_Q = \left( \frac{M\tau}{2\pi\hbar^2} \right)^{3/2}$ .

$$\text{Now } \mu_{\int} = \tau \log \left( \frac{n}{n_Q} \right)$$

$$\mu_{ext} = \frac{-GM_e}{r} = -\frac{gR_m^2}{r} \text{ since } g = \frac{GM_e}{R_e^2}.$$

$$\mu_{tot} = \tau \ln(n/n_Q) + Mgh$$

In equilibrium, this must be independent of  $r$ :  $\mu_{tot}(r) = \mu_{tot}(R)$ .

$$\tau \ln(n(r)/n_Q) - \frac{gR^2 M}{r} = \tau \ln(n(R)/n_Q) - \frac{MgR^2}{R}$$

$$\begin{aligned}\tau \ln \left( \frac{n(r)}{n(R)} \right) &= Mg \left( \frac{R^2}{r} - R \right) \text{ or } \exp \left( \frac{Mg}{\tau} \left( \frac{R^2}{r} - R \right) \right) = \frac{n(r)}{n(R)} \\ \Rightarrow n(r) &= n(R) \exp \left( \frac{Mg}{\tau} \left( \frac{R^2}{r} - \frac{1}{R} \right) \right)\end{aligned}$$

so that

$$N = 4\pi n(R) \exp \frac{-MgR}{\tau} \int_R^\infty r^2 dr \exp \left( \frac{MgR^2}{\tau r} \right)$$

**Solution 3. Potential energy of gas in gravitational field.**

I looked at the solution from Ed Groth [10] for Problem 3, Homework No. 4, Physics 301 for a hint, which prompted me to recall that diffusive equilibrium between 2 systems means their chemical potential  $\mu_i$ 's are equal.  
1

Problem 3 **Potential energy of gas in gravitational field** of Chapter 5: Chemical Potential and Gibbs Distribution from Kittel and Kroemer (1980) [2]: "Consider a column of atoms each of mass  $M$  at temperature  $\tau$  in a uniform gravitational field  $g$ ."

Consider an infinitesimal layer at height  $h$  and at height  $h + dh$ . At each height  $h$ , the layers of atoms are in *diffusive* and *thermal equilibrium*, and so

$$\tau(h + dh) = \tau(h) = \tau$$

(thermal equilibrium) and

$$\mu(h + dh) = \mu(h)$$

(diffusive equilibrium).

Now for chemical potential  $\mu = \mu(\tau, V, N)$ ,

$$\mu = \mu_{\text{int}} + \mu_{\text{ext}} = \tau \ln \left( n \left( \frac{2\pi\hbar^2}{M\tau} \right)^{3/2} \right) + Mgh$$

Thus, to first order,

$$\tau \frac{1}{n(h)} \frac{dn(h)}{dh} = -Mg \text{ or } n(h) = n(0) \exp \left( \frac{-Mgh}{\tau} \right)$$

The thermal average potential energy per atom is

$$\langle U \rangle = \frac{\int_0^\infty dh Mgh n(h)}{\int_0^\infty dh n(h)} = Mg \frac{\left[ \frac{h \exp \left( \frac{-Mgh}{\tau} \right)}{-Mg/\tau} + \frac{\exp \left( \frac{-Mgh}{\tau} \right)}{(-Mg/\tau)^2} \right] \Big|_0^\infty}{\left( \frac{\exp \left( \frac{-Mgh}{\tau} \right)}{-Mg/\tau} \right) \Big|_0^\infty} = Mg \frac{\tau}{Mg} = \boxed{\tau = \langle U \rangle}$$

From Ch.3, Eq. (64) of Kittel and Kroemer (1980) [2],

$$U = \frac{\sum_{\mathbf{n}} \epsilon_{\mathbf{n}} \exp[-\epsilon_{\mathbf{n}}/\tau]}{Z_1} = \tau^2 \frac{\partial \ln Z_1}{\partial \tau} = \frac{3\tau}{2}$$

So the total heat capacity per atom  $C$  is

$$C = \frac{\partial E}{\partial \tau} = \frac{\partial(\langle U \rangle + U)}{\partial \tau} = \frac{\partial}{\partial \tau} \left( \tau + \frac{3\tau}{2} \right) = \boxed{\frac{5}{2} = C}$$

**Solution 4.**  $\mu = \mu_{\text{int}} + \mu_{\text{ext}} = \tau \ln \left( \frac{n}{n_Q} \right) + \mu_{\text{ext}}$

$$\mu_{\text{ext}} = \tau \ln \left( \frac{10^4 n}{n_Q} \right) - \tau \ln \left( \frac{n}{n_Q} \right) = \tau \ln 10^4 = 300 \text{ K} * 0.0000861733 * eV/K \ln 10^4 = 0.24 \text{ eV}$$

Note that I checked this solution afterwards with the one by John D. Naud, via Ed Groth's materials [10], for Problem 4, Homework No. 4, Physics 301 <sup>2</sup>.

<sup>1</sup>John D. Naud via Ed Groth, Problem 3, Homework No. 4, Physics 301, Fall 1999, <http://grothserver.princeton.edu/~groth/phys301f99/hw04sol.pdf>

<sup>2</sup>John D. Naud via Ed Groth, Problem 4, Homework No. 4, Physics 301, Fall 1999, <http://grothserver.princeton.edu/~groth/phys301f99/hw04sol.pdf>

**Solution 5. Magnetic concentration.**

This problem pertains to Figure 5.6. Recall the example on pp. 127 of Kittel and Kroemer (1980) [2],

**Example: Chemical potential of mobile magnetic particles in a magnetic field.** “Consider system of  $N$  identical particles with magnetic moment  $m$ .”

Now

$$\mu_{\text{tot}}(\uparrow) = \tau \ln \left( \frac{n_{\uparrow}}{n_Q} \right) - mB = \tau \ln n_{\uparrow} + \frac{3\tau}{2} \ln \left( \frac{2\pi\hbar^2}{M\tau} \right) - mB$$

$$\mu_{\text{tot}}(\uparrow)(B=0) = \tau \ln (n_{\uparrow}(B=0)) + \frac{3\tau}{2} \ln \left( \frac{2\pi\hbar^2}{M\tau} \right)$$

$$\mu_{\text{tot}}(\uparrow)(B=20 \text{ kgauss}) = \tau \ln (n_{\uparrow}(B=20 \text{ kgauss})) + \frac{3\tau}{2} \ln \left( \frac{2\pi\hbar^2}{M\tau} \right) - m(20 \text{ kgauss})$$

When  $n_{\uparrow}(B=0) = 2 \times 10^7 \text{ cm}^{-3}$ ,

$$n_{\uparrow}(B=20 \text{ kgauss}) = 2 \times 10^9 \text{ cm}^{-3}$$

$$\mu_{\text{tot}}(\uparrow)(B=0) = \mu_{\text{tot}}(\uparrow)(B=20 \text{ kgauss})$$

so

$$\tau \ln \left( \frac{2 \times 10^9 \text{ cm}^{-3}}{2 \times 10^7 \text{ cm}^{-3}} \right) = m(20 \text{ kgauss})$$

Given  $\tau = 300 \text{ K}$ ,

$$m = \frac{300 \text{ K} \cdot k_B \ln(10^2)}{20 \text{ kgauss}} = 9.54 \times 10^{-18} \text{ erg/gauss} = 10^3 \text{ Bohr magnetons}$$

where we take the Bohr magneton  $\mu_B \equiv e\hbar/2mc$  to be  $9.274 \times 10^{-21} \text{ erg/gauss}$ .

**Solution 6. Gibbs sum for a two level system.** Recall that

$$\mathcal{Z}(\mu, \tau) = \sum_{N=0}^{\infty} \sum_{s(N)} \exp[(N\mu - \epsilon_{s(N)})/\tau] = \sum_{ASN} \exp[(N\mu - \epsilon_{s(N)})/\tau]; \quad \lambda = \exp\left(\frac{\mu}{\tau}\right)$$

(a)  $\mathcal{Z} = 1 + \lambda + \lambda \exp\left(\frac{-\epsilon}{\tau}\right)$ , 1 for  $N=0$ ; for  $N=1$ ,  $\lambda + \lambda \exp\left(\frac{-\epsilon}{\tau}\right)$ .  $\lambda$  for  $\epsilon=0$ .

(b)  $\langle N \rangle = \frac{0(1) + (1)(\lambda + \lambda \exp\left(\frac{-\epsilon}{\tau}\right))}{\mathcal{Z}} = \frac{\lambda(1 + \exp\left(\frac{-\epsilon}{\tau}\right))}{\mathcal{Z}}$

(c)  $\langle N(\epsilon) \rangle = \frac{0(1) + (0)(\lambda) + \lambda \exp\left(\frac{-\epsilon}{\tau}\right)}{\mathcal{Z}} = \frac{\lambda \exp\left(\frac{-\epsilon}{\tau}\right)}{\mathcal{Z}}$

(d)  $\langle \epsilon \rangle = \epsilon \langle N(\epsilon) \rangle = \frac{\lambda \epsilon \exp\left(\frac{-\epsilon}{\tau}\right)}{\mathcal{Z}}$

(e)

$$\mathcal{Z} = 1 + \lambda + \lambda \exp(-\epsilon/\tau) + \lambda^2 \exp(-\epsilon/\tau) = (1 + \lambda)(1 + \lambda \exp(-\epsilon/\tau))$$

where we considered the possibility the orbital at 0 and at  $\epsilon$  are each occupied by one particle at the same time.

So that, for total energy being  $\epsilon$ ,  $\exp[(2\mu - \epsilon)/\tau] = (\exp(\frac{\mu}{\tau}))^2 e^{-\epsilon/\tau} = \lambda^2 e^{-\epsilon/\tau}$ .

**Solution 7. States of positive and negative ionization.**

$$\mathcal{Z} = e^{\frac{\delta}{2\tau}} + \lambda e^{\frac{\Delta}{2\tau}} + \lambda e^{\frac{-\Delta}{2\tau}} + \lambda^2 e^{\frac{-\delta}{2\tau}}$$

$$\partial_{\lambda} \mathcal{Z} = 0 + e^{\Delta/2\tau} + e^{\frac{-\Delta}{2\tau}} + 2\lambda e^{\frac{-\delta}{2\tau}}$$

$$\lambda \partial_{\lambda} \ln \mathcal{Z} = \frac{\lambda(e^{\Delta/2\tau} + e^{-\Delta/2\tau} + 2\lambda e^{-\delta/2\tau})}{e^{\frac{\delta}{2\tau}} + \lambda e^{\frac{\Delta}{2\tau}} + \lambda e^{\frac{-\Delta}{2\tau}} + \lambda^2 e^{\frac{-\delta}{2\tau}}} = 1$$

$$\lambda e^{\Delta/2\tau} + \lambda e^{-\Delta/2\tau} + 2\lambda^2 e^{-\delta/2\tau} = e^{\delta/2\tau} = \lambda e^{\Delta/2\tau} + \lambda e^{-\Delta/2\tau} + \lambda^2 e^{-\delta/2\tau} \implies \lambda^2 e^{-\delta/2\tau} = e^{\delta/2\tau} \\ \lambda^2 = e^{\delta/\tau} \text{ or } 2 \ln \lambda = \delta/\tau$$

$$\boxed{2\tau \ln \lambda = \delta}$$

**Solution 8. Carbon monoxide poisoning.**

(a)  $\mathcal{Z} = 1 + \lambda(O_2)e^{-\epsilon_A/\tau}$

$$P(O_2) = \frac{\lambda(O_2)e^{-\epsilon_A/\tau}}{1 + \lambda(O_2)e^{-\epsilon_A/\tau}} = 0.9 \text{ or } 0.9 = 0.1\lambda(O_2) \exp\left(\frac{-\epsilon_A}{\tau}\right)$$

$$\Rightarrow \ln\left(\frac{9}{\lambda(O_2)}\right) = \frac{-\epsilon_A}{\tau}; \text{ or } \tau \ln\left(\frac{\lambda(O_2)}{9}\right) = \epsilon_A$$

$$\boxed{\epsilon_A = (k_B T) \ln\left(\frac{\lambda(O_2)}{9}\right)} = (8.617 \times 10^{-5} \frac{eV}{k})(273 + 39) \ln\left(\frac{10^{-5}}{9}\right) = -0.3686 eV$$

(b)  $\mathcal{Z} = 1 + \lambda(O_2)e^{-\epsilon_A/\tau} + \lambda(CO)e^{-\epsilon_B/\tau}$

$$P(O_2) = \frac{\lambda(O_2)e^{-\epsilon_A/\tau}}{1 + \lambda(O_2)e^{-\epsilon_A/\tau} + \lambda(CO)e^{-\epsilon_B/\tau}} = 0.1 \Rightarrow 0.1 + 0.1\lambda(CO)e^{-\epsilon_B/\tau} = 0.9\lambda(O_2)e^{-\epsilon_A/\tau}$$

$$\ln\left(\frac{9\lambda(O_2)e^{-\epsilon_A/\tau} - 1}{\lambda(CO)}\right) = \frac{-\epsilon_B}{\tau} \text{ or } \boxed{\epsilon_B = \tau \ln\left(\frac{\lambda(CO)}{9\lambda(O_2)e^{-\epsilon_A/\tau} - 1}\right)} = -0.5511 eV$$

**Solution 9. Absorption of  $O_2$  in a magnetic field.** Recall that  $2j + 1 =$  total number of spin states.  
 $2(1) + 1 = 3$ .  
 Now  $U = -\mathbf{m} \cdot \mathbf{B}$ .

$$\mathcal{Z} = 1 + \lambda(O_2)e^{-\frac{\epsilon_A + \mu_B B}{\tau}} + \lambda(O_2)e^{-\frac{\epsilon_A}{\tau}} + \lambda(O_2)e^{-\frac{(\epsilon_A + \mu_B B)}{\tau}} =$$

$$\mathcal{Z} = 1 + \lambda(O_2)e^{-\epsilon_A/\tau}(1 + 2 \cosh\left(\frac{\mu_B B}{\tau}\right))$$

$$0.91 = \frac{\lambda(O_2)e^{-\epsilon_A/\tau}(1 + 2 \cosh\left(\frac{\mu_B B}{\tau}\right))}{1 + \lambda(O_2)e^{-\epsilon_A/\tau}(1 + 2 \cosh\left(\frac{\mu_B B}{\tau}\right))} \text{ or } 0.91 = 0.09\lambda(O_2)e^{-\epsilon_A/\tau}(1 + 2 \cosh\left(\frac{\mu_B B}{\tau}\right))$$

$$\frac{1}{2} \left( \frac{91}{9} \frac{1}{\lambda(O_2)} e^{\epsilon_A/\tau} - 1 \right) = \cosh\left(\frac{\mu_B B}{\tau}\right) \text{ or } B = \frac{\tau}{\mu_B} \operatorname{arccosh}\left(\frac{1}{2} \left( \frac{91}{9} \frac{1}{\lambda(O_2)} e^{\epsilon_A/\tau} - 1 \right)\right)$$

The Gibbs sum in the limit of zero magnetic field will differ from that of Problem 8 because there the spin multiplicity of the bound state was neglected.

$$\mathcal{Z} = 1 + 3\lambda(O_2)e^{-\epsilon_A/\tau}$$

$$P(O_2) = \frac{3\lambda(O_2)e^{-\epsilon_A/\tau}}{1 + 3\lambda(O_2)e^{-\epsilon_A/\tau}} = 0.9 \text{ or } 0.9 = 0.3\lambda(O_2)e^{-\epsilon_A/\tau}$$

$$\ln \frac{3}{\lambda(O_2)} = -\epsilon_A/\tau \text{ or } \epsilon_A = \tau \ln\left(\frac{\lambda(O_2)}{3}\right) = -0.6227$$

for  $T = 300 K$ , so that  $\tau = 0.049375 eV$ .

$$\Rightarrow B = \frac{\tau}{\mu_B} 0.59927 = \frac{0.049375 \cdot 0.59927}{5.7884 \times 10^{-11} \times 10^6 eV/T} = 511.1 T$$

**Solution 10. Concentration fluctuations.**

(a) Recall that  $\mathcal{Z} = \sum_{N=0}^{\infty} \sum_{s(N)} \exp[(N\mu - \epsilon_{s(N)})/\tau]$ .

$$\frac{\partial^2 \mathcal{Z}}{\partial \mu^2} = \sum_{ASN} \left(\frac{N}{\tau}\right)^2 \exp[(N\mu - \epsilon_{s(N)})/\tau] \quad \langle N^2 \rangle = \frac{\tau^2}{\mathcal{Z}} \frac{\partial^2 \mathcal{Z}}{\partial \mu^2}$$

(b)

$$\begin{aligned}\frac{\partial}{\partial \mu} \langle N \rangle &= \frac{\partial}{\partial \mu} \left( \frac{\tau}{\mathcal{Z}} \left( \frac{\partial \mathcal{Z}}{\partial \mu} \right)_{\tau, V} \right) = \tau \left( \frac{-\left( \frac{\partial \mathcal{Z}}{\partial \mu} \right)_{\tau, V}^2}{\mathcal{Z}^2} + \frac{\left( \frac{\partial^2 \mathcal{Z}}{\partial \mu^2} \right)_{\tau, V}}{\mathcal{Z}} \right) \\ &\Rightarrow \tau \frac{\partial}{\partial \mu} \langle N \rangle = \langle N^2 \rangle - \langle N \rangle^2 = \langle (\Delta N)^2 \rangle\end{aligned}$$

**Solution 11. Equivalent definition of chemical potential.**

Recall that

$$d\sigma = \left( \frac{\partial \sigma}{\partial U} \right)_{V, N} dU + \left( \frac{\partial \sigma}{\partial V} \right)_{U, N} dV + \left( \frac{\partial \sigma}{\partial N} \right)_{U, V} dN \quad (31)$$

$$\mu = -\tau \left( \frac{\partial \sigma}{\partial N} \right)_{U, V} \quad (35)$$

Consider when  $d\sigma = dV = 0$ .

$$\begin{aligned}\Rightarrow \frac{-\mu}{\tau} dN + \left( \frac{\partial \sigma}{\partial U} \right)_{V, N} dU &= 0 \\ \left( \frac{\partial \sigma}{\partial U} \right)_{V, N} \frac{dU}{dN} &= \frac{\mu}{\tau}\end{aligned}$$

Note that  $\frac{dU}{dN} = \left( \frac{\partial U}{\partial N} \right)_{\sigma, V}$ .

Using the definition,  $\left( \frac{\partial \sigma}{\partial U} \right)_{N, V} \equiv \frac{1}{\tau}$ , so then

$$\boxed{\mu = \left( \frac{\partial U}{\partial N} \right)_{\sigma, V}}$$

Now  $F = U - \tau\sigma$ , by definition. Consider the thermodynamic identity,  $dU = \tau d\sigma - pdV + \mu dN$ .

$$\begin{aligned}dF &= dU - \sigma d\tau - \tau d\sigma = \tau d\sigma - pdV + \mu dN - \sigma d\tau - \tau d\sigma = \\ &= -pdV + \mu dN - \sigma d\tau \\ \Rightarrow \left( \frac{\partial F}{\partial N} \right)_{\tau, V} &= \mu(\tau, V, N)\end{aligned}$$

Likewise,  $\mu = \left( \frac{\partial U}{\partial N} \right)_{\sigma, V}$  is equivalent to  $\mu(\tau, V, N) = \left( \frac{\partial F}{\partial N} \right)_{\tau, V}$  through the thermodynamic identity as well.

$$dU = \tau d\sigma - pdV + \mu dN$$

$\sigma, V$  constant.  $d\sigma, dV = 0$ .

$$\Rightarrow \left( \frac{\partial U}{\partial N} \right)_{\sigma, V} = \mu$$

**Solution 12. Ascent of sap in trees.** Given: relative humidity  $r = 0.9$ .  $T = 25^\circ C$

$n_0$  = concentration in saturated air that stands immediately above pool of water of water vapor in air.  
 $rn_0$  = actual concentration of water vapor in air at uppermost leaves is  $rn_0$ .

At pool,  $\mu_{H_2O} = \mu_{\text{vapor}}$  for diffusion equilibrium.

Same condition at uppermost leaves, otherwise there's evaporation:

$$\mu_{\text{sap}} = \mu_{\text{vapor}}(h).$$

$\mu_{\text{sap}} = \mu_{H_2O}$  (no flow going on in water). Thus,  $\mu_{\text{vapor}}(h) = \mu_{\text{vapor}}(0)$  and treat water vapor as an ideal gas.

$$\begin{aligned}\tau \ln \left( \frac{n(h)}{n_Q} \right) + mgh &= \tau \ln \left( \frac{n(0)}{n_Q} \right) \\ \tau \ln \left( \frac{rn_0}{n_0} \right) &= -mgh \Rightarrow \tau \ln \left( \frac{1}{r} \right) = mgh \text{ or } h = \frac{\tau}{mg} \ln \left( \frac{1}{r} \right)\end{aligned}$$

$$h = \frac{(1.381 \times 10^{-23} \frac{1}{K})(298 K)(kg \cdot \frac{m^2}{s^2}) \ln(\frac{10}{9})}{(18 amu) \left( \frac{1.67 \times 10^{-27} kg}{1 amu} \right) (9.8 m/s^2)} = 147.1 m$$

**Solution 13. Isentropic expansion.**

(a)

(b) Recall that  $F = -\tau[N \ln Z_1 - \ln N!]$  where  $Z_1 = n_Q V = \left( \frac{M\tau}{2\pi\hbar^2} \right)^{3/2} V$  and  $\sigma = -\left( \frac{\partial F}{\partial \tau} \right)_{V,W}$ .

$$\begin{aligned} \frac{\partial F}{\partial \tau} &= -[N \ln Z_1 - \ln N!] - \tau \left[ N \frac{1}{Z_1} \left( \frac{M}{2\pi\hbar^2} \right)^{3/2} \frac{3}{2} \tau^{1/2} V \right] = \\ &= -[N \ln Z_1 - \ln N!] - [N \frac{1}{Z_1} n_Q \frac{3}{2} V] = -[N \ln Z_1 - \ln N!] - \frac{3}{2} N \end{aligned}$$

With  $N$  constant and  $Z_1 = \left( \frac{M}{2\pi\hbar^2} \right)^{3/2} \tau^{3/2} V$ , then for an isentropic expansion,  $\tau V^{2/3}$  must remain constant.

**Solution 14. Multiple binding of  $O_2$ .**

(a) Be wary of the multiplicity, how you count, each of the energy states.

$$\mathcal{Z} = 1 + 4\lambda e^{-\epsilon/\tau} + \binom{4}{2} \lambda^2 e^{-2\epsilon/\tau} + \binom{4}{3} \lambda^3 e^{-3\epsilon/\tau} + \lambda^4 e^{-4\epsilon/\tau} = (1 + \lambda e^{-\epsilon/\tau})^4$$

$$P(\epsilon) = \frac{4\lambda e^{-\epsilon/\tau}}{(1 + \lambda e^{-\epsilon/\tau})^4}$$

(b)

$$P(4\epsilon) = \frac{\lambda^4 e^{-\epsilon/\tau}}{(1 + \lambda e^{-\epsilon/\tau})^4} = \frac{e^{-\epsilon/\tau}}{\left( \frac{1}{\lambda + e^{-\epsilon/\tau}} \right)^4}$$

2015 manuscript.

## 11 Gibbs distribution

*Manuscript section title: Reactions. Equilibrium in reactions is §22.*

### 11.1 Gibbs (review)

cf. Chapter 5: Chemical Potential and Gibbs Distribution of Kittel and Kroemer (1980) [2]

Recall that

$$F = U - \tau\sigma$$

$$dF = dU - \tau d\sigma - \sigma d\tau = -pdV - \sigma d\tau \text{ or } dF = +\mu dN - pdV - \sigma d\tau$$

so  $F = F(\tau, V)$ .

Total entropy of system + reservoir  $\sigma = \sigma_{\mathcal{R}} + \sigma_{\mathcal{S}}$

$$\sigma = \sigma_{\mathcal{R}} + \sigma_{\mathcal{S}} = \sigma_{\mathcal{R}}(U - U_{\mathcal{S}}) + \sigma_{\mathcal{S}}(U_{\mathcal{S}}) \simeq \sigma_{\mathcal{R}}(U) - U_{\mathcal{S}} \left( \frac{\partial \sigma_{\mathcal{S}}}{\partial U_{\mathcal{R}}} \right)_{V,N} + \sigma_{\mathcal{S}}(U_{\mathcal{S}})$$

Now

$$\left( \frac{\partial \sigma_{\mathcal{R}}}{\partial U_{\mathcal{R}}} \right)_{V,N} \equiv \frac{1}{\tau}$$

and so

$$\sigma = \sigma_{\mathcal{R}}(U) - \frac{1}{\tau} U_{\mathcal{S}} + \sigma_{\mathcal{S}} = \sigma_{\mathcal{R}}(U) - F_{\mathcal{S}}/\tau$$

$F_S$  must be a minimum with respect to  $U_S$  when system in its most probable state.

EY : 20151218 Consider  $\sigma = \frac{U}{\tau} - \frac{F}{\tau}$ , so

$$d\sigma = \frac{dU}{\tau} - \frac{U d\tau}{\tau^2} - \frac{dF}{\tau} + \frac{F}{\tau^2} d\tau$$

Consider curve  $\gamma : \mathbb{R} \rightarrow \Sigma$ , s.t.  $d\tau(\dot{\gamma}) = 0$ , i.e.  $\gamma$  represents a thermodynamic process at constant temperature. System in thermal equilibrium with reservoir.

Now for every thermodynamic process, entropy must increase, and so

$$d\sigma(\dot{\gamma}) = \frac{dU}{\tau}(\dot{\gamma}) - \frac{dF(\dot{\gamma})}{\tau} \geq 0$$

Thus, if  $\int dU(\dot{\gamma}) < 0$ ,  $\int dF(\dot{\gamma}) < 0$ , and so  $F$  is a minimum.

If  $d\sigma(\dot{\gamma}) = 0$ ,  $dU(\dot{\gamma}) = dF(\dot{\gamma})$ .  $U$  is minimized, so  $F$  is minimized.

Consider  $\mathcal{S}_1, \mathcal{S}_2$  in thermal and diffusive contact,  $\mathcal{S}_1, \mathcal{S}_2$  in thermal contact with large reservoir, with  $V_1, V_2$  held constant,

$$F = F_1 + F_2$$

$$N = N_1 + N_2 \text{ constant}$$

$$dF = \left( \frac{\partial F_1}{\partial N_1} \right)_{\tau, V_1} dN_1 + \left( \frac{\partial F_2}{\partial N_2} \right)_{\tau, V_2} dN_2 = \left[ \left( \frac{\partial F_1}{\partial N_1} \right)_{\tau, V_1} - \left( \frac{\partial F_2}{\partial N_2} \right)_{\tau, V_2} \right] dN_1 = 0$$

and so  $\mu_1 = \mu_2$  for diffusive equilibrium.

Note that

$$\mu(\tau, V, N) := \left( \frac{\partial F}{\partial N} \right)_{\tau, V} \quad (33)$$

**Example: Chemical potential of ideal gas.** If  $\mu_1 > \mu_2$ , then  $dF < 0$ , when  $dN_1 < 0$ , *particles flow from high chemical potential to low chemical potential.*

Recall

$$F = -\tau [N \ln Z_1 - \ln N!]$$

$$Z_1 = n_Q V = \left( \frac{M\tau}{2\pi\hbar^2} \right)^{3/2} V$$

Then

$$\mu = -\tau (\ln Z_1 - \ln N) = \tau \ln (N/Z_1)$$

$$\mu = \tau \ln (n/n_Q)$$

$$\mu = \tau \ln \left( \frac{p}{\tau n_Q} \right)$$

Note that the *total chemical potential*  $\mu$  is

$$\mu = \mu_{\text{tot}} = \mu_{\text{ext}} + \mu_{\text{int}}$$

Examples of  $\mu$ :

$$\mu = \tau \ln \left( \frac{n}{n_Q} \right) + q\Delta V$$

**Example: Variation of barometric pressure with altitude.**

$$\mu = \tau \ln \left( \frac{n}{n_Q} \right) + Mgh$$

### 11.1.1 Chemical Potential and Entropy

cf. pp. 131 of Kittel and Kroemer (1980) [2], Ch.5.

$$\sigma = \sigma(U, V, N)$$

$$\tau d\sigma = dU + p dV - \mu dN$$

$$-\left( \frac{\partial \sigma}{\partial N} \right)_{U, V} = \frac{\mu}{\tau}$$

## 11.2 Gibbs Factor and Gibbs Sum

cf. pp. 134 of Kittel and Kroemer (1980) [2], Ch.5. Consider system  $S$ , with  $(N, \epsilon_S)$ , number of particles and internal energy, respectively, and reservoir  $\mathcal{R}$ , with  $(N_0 - N, U_0 - \epsilon_S)$ , in thermal and diffusive contact.

System probability  $P(N, \epsilon_S) \propto g(N_0 - N, U_0 - \epsilon_S)$

$$\frac{P(N_1, \epsilon_1)}{P(N_2, \epsilon_2)} = \frac{g(N_0 - N_1, U_0 - \epsilon_1)}{g(N_0 - N_2, U_0 - \epsilon_2)}$$

Now by definition of entropy,  $g(N_0, U_0) \equiv \exp[\sigma(N_0, U_0)]$ ,

$$\frac{P(N_1, \epsilon_1)}{P(N_2, \epsilon_2)} = \exp(\Delta\sigma) = \exp(\sigma(N_0 - N_1, U_0 - \epsilon_1) - \sigma(N_0 - N_2, U_0 - \epsilon_2)) = \exp\left[(N_1 - N_2)\frac{\mu}{\tau} - \frac{(\epsilon_1 - \epsilon_2)}{\tau}\right]$$

so

$$\exp\left[\frac{(N\mu - \epsilon)}{\tau}\right]$$

is the Gibbs factor.

Gibbs sum or grand sum or grand partition function  $\mathcal{Z}$  is

$$\mathcal{Z}(\mu, \tau) = \sum_{N=0}^{\infty} \sum_{s(N)} \exp[(N\mu - \epsilon_{s(N)})/\tau] = \sum_{ASN} \exp\left[\frac{(N\mu - \epsilon_{s(N)})}{\tau}\right]$$

where  $ASN$  = overall states of the system for all number of particles.

**Problem 3 Potential energy of gas in gravitational field** of Chapter 5: Chemical Potential and Gibbs Distribution from Kittel and Kroemer (1980) [2]: “Consider a column of atoms each of mass  $M$  at temperature  $\tau$  in a uniform gravitational field  $g$ .”

Consider an infinitesimal layer at height  $h$  and at height  $h + dh$ . At each height  $h$ , the layers of atoms are in *diffusive* and *thermal equilibrium*, and so

$$\tau(h + dh) = \tau(h) = \tau$$

(thermal equilibrium) and

$$\mu(h + dh) = \mu(h)$$

(diffusive equilibrium).

Now for chemical potential  $\mu = \mu(\tau, V, N)$ ,

$$\mu = \mu_{\text{int}} + \mu_{\text{ext}} = \tau \ln \left( n \left( \frac{2\pi\hbar^2}{M\tau} \right)^{3/2} \right) + Mgh$$

Thus, to first order,

$$\tau \frac{1}{n(h)} \frac{dn(h)}{dh} = -Mg \text{ or } n(h) = n(0) \exp\left(\frac{-Mgh}{\tau}\right)$$

The thermal average potential energy per atom is

$$\langle U \rangle = \frac{\int_0^\infty dh Mgh n(h)}{\int_0^\infty dh n(h)} = Mg \frac{\left[ \frac{h \exp\left(\frac{-Mgh}{\tau}\right)}{-Mg/\tau} + \frac{\exp\left(\frac{-Mgh}{\tau}\right)}{(-Mg/\tau)^2} \right] \Big|_0^\infty}{\left( \frac{\exp\left(\frac{-Mgh}{\tau}\right)}{-Mg/\tau} \right) \Big|_0^\infty} = Mg \frac{\tau}{Mg} = \boxed{\tau = \langle U \rangle}$$

From Ch.3, Eq. (64) of Kittel and Kroemer (1980) [2],

$$U = \frac{\sum_{\mathbf{n}} \epsilon_{\mathbf{n}} \exp[-\epsilon_{\mathbf{n}}/\tau]}{Z_1} = \tau^2 \frac{\partial \ln Z_1}{\partial \tau} = \frac{3\tau}{2}$$

So the total heat capacity per atom  $C$  is

$$C = \frac{\partial E}{\partial \tau} = \frac{\partial(\langle U \rangle + U)}{\partial \tau} = \frac{\partial}{\partial \tau} \left( \tau + \frac{3\tau}{2} \right) = \boxed{\frac{5}{2} = C}$$

2008 manuscript.

## 12 Ideal Gas

### Classical Limit

Recall, Ch. 6, Sec. “Classical Limit” of Kittel and Kroemer [2], **an ideal gas is defined as a system of free noninteracting particles in the classical regime.**

$f(\epsilon) \equiv$  average occupancy of an orbital at energy  $\epsilon$

$\epsilon \equiv$  energy of orbital occupied by 1 particle; not energy of system of  $N$  particles

Fermi-Dirac and Bose-Einstein distribution  $f(\epsilon) = \frac{1}{\exp[(\epsilon - \mu)/\tau] \pm 1}$

In order for  $f(\epsilon) \ll 1 \forall$  orbitals,  $\exp[(\epsilon - \mu)/\tau] \gg 1 \forall \epsilon$ .

$$\implies f(\epsilon) \simeq \exp[(\mu - \epsilon)/\tau] = \lambda \exp(-\epsilon/\tau) \quad \lambda \equiv \exp\left(\frac{\mu}{\tau}\right)$$

$f(\epsilon)$ , average occupancy of orbital of energy  $\epsilon$ , is classical distribution function.

In Ch. 3 of Kittel and Kroemer [2],  $Z_1 = n_Q V$ ,  $n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}$ .

$\lambda = e^{\mu/\tau}$

$$N = \lambda Z_1 = \lambda n_Q V \quad \lambda = \frac{N}{n_Q V} = \frac{n}{n_Q}$$

$$\implies \lambda = \exp\left(\frac{\mu}{\tau}\right) = \frac{n}{n_Q}$$

### Heat Capacity

Now

$$Q = Q(\tau, p) = \Lambda_p dp + C_p d\tau = dU + W = dU + pdV$$

Let  $c \in \Sigma$  s.t.  $dp(\dot{c}) = 0$  (constant pressure). And so

$$Q(\dot{c}) = 0 + C_p d\tau(\dot{c}) = \tau d\sigma(\dot{c}) = dU(\dot{c}) + pdV(\dot{c})$$

for  $c = (\tau, 0)$ ,  $\dot{c} = \frac{\partial}{\partial \tau} \in T\Sigma$ .

$$\implies C_p = \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_p = \left( \frac{\partial U}{\partial \tau} \right)_p + p \left( \frac{\partial V}{\partial \tau} \right)_p$$

for heat capacity at constant pressure is larger than  $C_V$  because additional heat must be added to perform the work needed to expand volume of gas against constant pressure.

Now recall for enthalpy

$$H = U + pV$$

$$H = H(\sigma, p)$$

Then

$$dH = dU + pdV + Vdp = Q + Vdp$$

Thus, for  $c \in \Sigma$ ,  $dp(\dot{c}) = 0$  (constant pressure). Hence

$$dH(\dot{c}) = \left( \frac{\partial H}{\partial \tau} \right)_p = Q(\dot{c}) + 0 = C_p$$

Hence,

$$C_p = \left( \frac{\partial H}{\partial \tau} \right)_p = \left( \frac{\partial Q}{\partial \tau} \right)_p$$

Now

$$\int_c dH = H_f - H_i = \int_c Q = \int_c C_p d\tau = C_p(\tau_f - \tau_i)$$

(if  $C_p$  constant over desired temperature range)

## Reversible Isothermal Expansion

$Q = 0$ , insulated gas, no heat flow to or from the gas (“adiabatic”)  
 $\sigma$  constant in system isolated from reservoir, if expansion reversible (slowly)

**What is the pressure after expansion?**

Remember

$$\frac{C_p}{C_V} = \frac{\frac{5}{2}N}{\frac{3}{2}N} = \frac{5}{3} = \gamma$$

So let

$$\frac{1}{\gamma - 1} = \frac{3}{2}$$

$$\sigma(\tau, V) = N(\ln \tau^{\frac{1}{\gamma-1}} + \ln V + \text{constant}) \quad (61)$$

$$\ln \tau^{\frac{1}{\gamma-1}} V = \text{constant or } \tau^{\frac{1}{\gamma-1}} V \text{ constant} \quad (62)$$

$$\implies \tau_1^{\frac{1}{\gamma-1}} V_1 = \tau_2^{\frac{1}{\gamma-1}} V_2 \quad (63)$$

for ideal monoatomic gas.

Use  $pV = N\tau$

$$\implies \frac{\tau_1^{\frac{\gamma}{\gamma-1}}}{p_1} = \frac{\tau_2^{\frac{\gamma}{\gamma-1}}}{p_2} \quad (64)$$

or  $p_1^{\frac{1}{\gamma-1}} V_1^{\frac{\gamma}{\gamma-1}} = p_2^{\frac{1}{\gamma-1}} V_2^{\frac{\gamma}{\gamma-1}}$  or  $p_1 V_1^\gamma = p_2 V_2^\gamma$

I will recap Problem 10 and my solution.

*Isentropic relations of ideal gas.*

(a)  $\gamma = \frac{C_p}{C_V}$ . For isentropic process,  $pV^\gamma = p_i V_i^\gamma$ .

Then, essentially and equivalently, take the exterior derivative:

$$V^\gamma dp + \gamma p V^{\gamma-1} dV = 0 \implies \frac{dp}{p} + \frac{\gamma dV}{V} = 0$$

For

$$\tau V^{\gamma-1} = \tau_i V_i^{\gamma-1}$$

then taking  $d$ :

$$d\tau V^{\gamma-1} + \tau(\gamma-1)V^{\gamma-2}dV = 0 \implies \frac{d\tau}{\tau} + \frac{\gamma-1}{V}dV = 0$$

For

$$\tau^{\frac{\gamma}{1-\gamma}} p = \tau_i^{\frac{\gamma}{1-\gamma}} p_i$$

then taking  $d$ :

$$dp \tau^{\frac{\gamma}{1-\gamma}} + \frac{\gamma}{1-\gamma} \tau^{\frac{\gamma}{1-\gamma}} d\tau p = 0 \implies \frac{dp}{p} + \frac{\gamma}{1-\gamma} \frac{d\tau}{\tau} = 0$$

(b) isentropic bulk moduli

$$B_\sigma = -V \left( \frac{\partial p}{\partial V} \right)_\sigma = \gamma \frac{p_i V_i^\gamma}{V^\gamma} = \gamma p$$

since  $p = \frac{p_i V_i^\gamma}{V^\gamma}$

$$\boxed{B_\sigma = \gamma p}$$

EY : 20150606 I think when one consider small, linear longitudinal perturbations of the gas system, with pressure being the external restoring force, then sound waves propagate (correct me if I'm wrong) and this is the way to derive  $B_\sigma$ .

isothermal bulk moduli  $B_\tau = -V \left( \frac{\partial p}{\partial V} \right)_\tau = \frac{n\tau}{V} = p$

velocity of sound in gas is  $c = \left( \frac{B_\sigma}{\rho} \right)^{1/2} = \left( \frac{\gamma p}{\rho} \right)^{1/2}$

for ideal gas of molecules of mass  $M$ ,  $pV = N\tau$

Manifold interpretation; I'm using Chapter 5 Applications in Physics, Section A Thermodynamics of [3]. Consider a 2-dimensional manifold of equilibrium states  $\mathcal{M}$ , s.t.  $\dim \mathcal{M} = 2$ . Local coordinates are  $(U, V)$  with  $U$  being a global coordinate.

Now  $U = C_V \tau$  so use  $\tau$  as a global coordinate, i.e. the local coordinates of  $\mathcal{M}$  can be  $(\tau, V)$ .

Consider a curve in  $\mathcal{M}$  parametrized by  $\lambda$ :  $(\tau(\lambda), V(\lambda))$ . The corresponding tangent vector  $X = \dot{\tau} \frac{\partial}{\partial \tau} + \dot{V} \frac{\partial}{\partial V} \in$

$\mathfrak{X}(\mathcal{M})$  has components in coordinate vector basis  $\dot{\tau} = \dot{\tau}(\lambda)$

$$\dot{V} = \dot{V}(\lambda)$$

Recall

$$\frac{d\tau}{\tau} + \frac{\gamma - 1}{V} dV = 0$$

Consider applying this 1-form onto  $X$ :

$$\left( \frac{d\tau}{\tau} - \frac{\gamma - 1}{V} dV \right) (X) = \frac{\dot{\tau}}{\tau} + \frac{\gamma - 1}{V} \dot{V} = 0 \implies \frac{d}{d\lambda} (\tau V^{\gamma-1}) = 0$$

Then the equalities of the endpoints of this curve  $(\tau(\lambda), V(\lambda))$  are the equalities above. The interpretation is that the isentropic process draws out a curve in  $\mathcal{M}$  and can be written as a curve or as a tangent vector field (specifically a section of  $T\mathcal{M}$ ).

$$\tau_1 V_1^{\gamma-1} = \tau_2 V_2^{\gamma-1} \quad (66)$$

$$\tau_1^{\frac{\gamma}{1-\gamma}} P_1 = \tau_2^{\frac{\gamma}{1-\gamma}} P_2 \quad (67)$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma \quad (68)$$

for, indeed,

$$p_1 V_1^\gamma = p_2 V_2^\gamma \xrightarrow{pV=N\tau} p_1 \left( \frac{N\tau_1}{p_1} \right)^\gamma = p_2 \left( \frac{N\tau_2}{p_2} \right)^\gamma \text{ so}$$

$$p_1^{1-\gamma} \tau_1^\gamma = p_2^{1-\gamma} \tau_2^\gamma \text{ or } p_1 \tau_1^{\frac{\gamma}{1-\gamma}} = p_2 \tau_2^{\frac{\gamma}{1-\gamma}}$$

$$\implies p_1^{\frac{1-\gamma}{\gamma}} \tau_1 = p_2^{\frac{1-\gamma}{\gamma}} \tau_2 \text{ or } \left( \frac{p_1}{p_2} \right)^{\frac{1-\gamma}{\gamma}} = \frac{\tau_2}{\tau_1} = \left( \frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}}$$

$$T_1 = 300 \text{ K}, \quad V_1/V_2 = \frac{1}{2},$$

$$T_2 = \left( \frac{1}{2} \right)^{2/3} (300 \text{ K}) = 189 \text{ K} \quad (69)$$

The gas is cooled in expansion process by

$$T_1 - T_2 = 300 \text{ K} - 189 \text{ K} = 111 \text{ K} \quad (70)$$

Expansion at constant entropy is important e.g. methods of re Fridgeration.

**What is the change in energy in the expansion?**

$$U_2 - U_1 = C_V (\tau_2 - \tau_1) = \frac{1}{\gamma - 1} N (\tau_2 - \tau_1)$$

## Problems

**Solution 1. Derivative of Fermi-Dirac function.** Recall  $f = \frac{1}{\exp[(\epsilon-\mu)/\tau]+1} = (e^{(\epsilon-\mu)/\tau} + 1)^{-1}$

$$\partial_\epsilon f = -1(e^{(\epsilon-\mu)/\tau} + 1)^{-2}(e^{(\epsilon-\mu)/\tau}) \left( \frac{1}{\tau} \right)$$

$$\partial_\epsilon f(\epsilon = \mu) = \frac{-1}{4\tau}$$

**Solution 2. Symmetry of filled and vacant orbitals.**  $\epsilon = \mu + \delta$

$$f(\epsilon) = f(\mu + \delta) = \frac{1}{e^{\delta/\tau} + 1} = \frac{e^{-\delta/\tau}}{1 + e^{-\delta/\tau}} = 1 + \frac{-1}{e^{-\delta/\tau} + 1} = 1 - f(\mu - \delta)$$

**Solution 3. Distribution function for double occupancy statistics.**

(a)  $\eta = 1 + \lambda e^{-\epsilon/\tau} + \lambda^2 e^{-2\epsilon/\tau}$  where  $\lambda = e^{\mu/\tau}$ .

$$\langle N \rangle = \lambda \partial_\lambda \ln \zeta = \frac{\lambda}{\zeta} (e^{-\epsilon/\tau} + 2\lambda e^{-2\epsilon/\tau}) = \frac{\lambda e^{-\epsilon/\tau} + 2\lambda^2 e^{-2\epsilon/\tau}}{\zeta}$$

(b)  $\zeta = 1 + 2\lambda e^{-\epsilon/\tau} + \lambda^2 e^{-2\epsilon/\tau}$

$$\langle N \rangle = \lambda \partial_\lambda \ln \lambda = \frac{2\lambda e^{-\epsilon/\tau} + 2\lambda^2 e^{-2\epsilon/\tau}}{\zeta} = \frac{2\lambda(e^{-\epsilon/\tau})(1 + \lambda e^{-\epsilon/\tau})}{\zeta}$$

**Solution 4. Energy of gas of extreme relativistic particles.** For  $\epsilon_p \simeq p$ ,  $\sum_s e^{-p/\tau} = Z$ .

With the factor 2 for the 2 possible polarizations,

$$\begin{aligned} Z &= (2) \frac{4\pi}{8} \int_0^\infty p^2 dp e^{-p/\tau} = \pi \{ p^2 e^{-p/\tau}(-\tau) \Big|_0^\infty - \int_0^\infty 2p e^{-p/\tau}(-\tau) \} = 2\tau\pi \int_0^\infty p e^{-p/\tau} dp = \\ &= 2\tau\pi \{ p e^{-p/\tau}(-\tau) \Big|_0^\infty - \int_0^\infty e^{-p/\tau}(-\tau) dp \} = 2\tau^3\pi \end{aligned}$$

$$U = \tau^2 \frac{\partial \ln Z}{\partial \tau} = \tau^2 \partial_\tau \{ \ln(2\pi\tau^3) \} = \tau^2 \partial_\tau (3 \ln \tau) = \boxed{U = 3\tau}$$

**Solution 5. Integration of the thermodynamic identity for an ideal gas.** For constant  $N$ , recall

$$d\sigma = \frac{dU}{\tau} + \frac{pdV}{\tau} = \frac{1}{\tau} \left( \frac{\partial U}{\partial \tau} \right)_V d\tau + \frac{1}{\tau} \left( \frac{\partial U}{\partial V} \right)_\tau dV + \frac{pdV}{\tau}$$

$C_V = \left( \frac{\partial U}{\partial \tau} \right)_V$ , and for an ideal gas  $pV = N\tau$ .

$$\int d\sigma = \sigma = C_V \ln \tau + N \ln V + \int \frac{1}{\tau} \left( \frac{\partial U}{\partial \tau} \right)_\tau dV + \sigma_1$$

Now  $U = \frac{3}{2}\tau$  for an ideal gas, so  $\left( \frac{\partial U}{\partial V} \right)_\tau = 0$ .

$$\implies \sigma = C_V \ln \tau + N \ln V + \sigma_1$$

$\sigma_1$  independent constant of  $\tau$  and  $V$ .

**Solution 6. Entropy of mixing.**

**Correction (empty solution).** The 2008 manuscript stops at this heading. The mixing entropy, the identical-gas cancellation, and the ideal-mixture chemical potential are derived in §14.

**Solution 7. Relation of pressure and energy density.**

(a) Recall that  $U = U(\sigma, V, N)$ .  $p = -\left(\frac{\partial U}{\partial V}\right)_N$ , and

$$U = \frac{\sum_s \epsilon_s e^{-\epsilon_s/\tau}}{Z}$$

$$\frac{\partial U}{\partial V} = \sum_s \left( \left( \frac{\partial \epsilon_s}{\partial V} \right)_N e^{-\epsilon_s/\tau} + \epsilon_s \left( \frac{-1}{\tau} \left( \frac{\partial \epsilon_s}{\partial V} \right)_N e^{-\epsilon_s/\tau} \right) / Z - \sum_s \left( -\frac{\partial \epsilon_s}{\partial V} \right) \left( \frac{1}{\tau} \right) e^{-\epsilon_s/\tau} \sum_{s_2} \epsilon_{s_2} e^{-\epsilon_{s_2}/\tau} / Z^2 \right)$$

Now  $p = -\left(\frac{\partial U}{\partial V}\right)_N$ .

So if the system is in state  $s$ ; then  $p_s = -\left(\frac{\partial \epsilon_s}{\partial V}\right)_N$ .

$$\Rightarrow \boxed{\langle p \rangle = p = \frac{\sum_s -\left(\frac{\partial \epsilon_s}{\partial V}\right)_N e^{-\epsilon_s/\tau}}{Z}}$$

(b) Now  $\epsilon_s = \frac{\pi}{L^2} \frac{n^2 \hbar^2}{2M} = \frac{\pi}{V^{2/3}} \frac{n^2 \hbar^2}{2M}$

$$\left( \frac{\partial \epsilon_s}{\partial V} \right)_N = -\frac{2}{3} V^{-5/3} \frac{\pi n^2 \hbar^2}{2M} = \boxed{-\frac{2}{3V} \epsilon_s}$$

(c)

$$p = \frac{\sum_s \frac{-2\epsilon_s}{3V} e^{-\epsilon_s/\tau}}{Z} = \boxed{\frac{2}{3} \left( \frac{1}{V} \right) U}$$

#### Solution 8. Time for a large fluctuation.

(a) Recall  $\sigma = N[\ln\left(\frac{n_Q}{n}\right) + \frac{5}{2}]$ .  $n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}$ .

$$\sigma_f = N[\ln\left(\frac{n_{QA} 2V}{N}\right) + \frac{5}{2}]$$

$$\sigma_i = N[\ln\left(\frac{n_{QA} V}{N}\right) + \frac{5}{2}]$$

Now

$$g \sim e^\sigma \sim \exp\left(\ln\left(\frac{n_Q}{n}\right)^N + \frac{5N}{2}\right) = e^{\frac{5N}{2}} \left(\frac{n_Q}{n}\right)^N$$

$$n_Q = 4 \times \frac{(938 \text{ MeV}/c^2)((0.8617 \times 10^{-4} \text{ eV})/K)(300 \text{ K})}{2\pi(6.582122 \times 10^{-22} \text{ MeV} \cdot \text{s})^2} \left(\frac{1 \text{ MeV}}{10^6 \text{ eV}}\right) \left(\frac{1 \text{ c}}{3 \times 10^8 \text{ m/s}}\right)^2 = 7.88 \times 10^{30}/m^3$$

Now  $PV = N\tau$ . Then

$$\frac{P}{\tau} = \frac{N}{V} = \frac{(1 \text{ atm})\left(\frac{1.013 \times 10^5 \text{ N/m}^2}{1 \text{ atm}}\right)\left(\frac{\text{kg} \cdot \text{m/s}^2}{1 \text{ N}}\right)}{1.381 \times 10^{-23} \text{ J/K}(300 \text{ K})} = 2.445 \times 10^{25}/m^3$$

Now  $1 \text{ L} = 10^{-3} m^3$  and so for  $0.1 \text{ L}$ ,

$$(2.445 \times 10^{25} \frac{1}{m^3})(10^{-4} m^3) = 2.445 \times 10^{21}$$

With

$$\begin{aligned} \left(\frac{7.88 \times 10^{30}/m^3}{2.445 \times 10^{25}/m^3}\right) &= 3.22 \times 10^5 \\ \Rightarrow g \sim e^{\frac{5}{2}(2.445 \times 10^{21})} (3.22 \times 10^5)^{2.445 \times 10^{21}} \end{aligned}$$

(b)

(c)

**Solution 9. Gas of atoms with internal degree of freedom.**

For an ideal monatomic gas, assume noninteracting.

(a)  $\lambda_{int} = \lambda$ .  $\lambda = \exp(\mu/\tau)$  ideal gas.  $\mu = \tau \ln(n/n_Q)$ ;  $n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}$ .

$$\boxed{\lambda_{ext} = \exp(-\Delta/\tau) \text{ or } 1}$$

$Z_1$  is the usual canonical partition function,  $Z_1 = n_Q V$  where  $n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2}$

$$Z_1 = \sum_s (\lambda \exp(-\epsilon_s/\tau)(1) + \lambda \exp\left(\frac{-\epsilon_s}{\tau}\right) \exp\left(\frac{-\Delta}{\tau}\right)) Z_1 = \lambda(1 + e^{-\Delta/\tau}) Z_1$$

$$(Z_1)^N = (\lambda(1 + e^{-\Delta/\tau}) Z_1)^N$$

(b)

(c)

**Solution 10. Isentropic relations of ideal gas.**

(a) Isentropic process, so  $pV^\gamma = p_i V_i^\gamma$ .

$$V^\gamma dp + \gamma p V^{\gamma-1} dV = 0 \implies \boxed{\frac{dp}{p} + \frac{\gamma}{V} dV = 0}$$

Dealing with an ideal gas,  $pV = N\tau$  still applies.

$$\begin{aligned} \tau V^{\gamma-1} &= \tau_i V_i^{\gamma-1} \\ d\tau V^{\gamma-1} + \tau(\gamma-1) V^{\gamma-2} dV &= 0 \\ \implies \boxed{\frac{d\tau}{\tau} + \frac{(\gamma-1)}{V} dV} &= 0 \end{aligned}$$

Using  $pV = N\tau$  again, note that  $p^{1-\gamma} \tau^\gamma = \text{constant}$ .

$$(1-\gamma)p^{-\gamma} dp \tau^\gamma + p^{1-\gamma} \gamma \tau^{\gamma-1} d\tau = 0 \implies \boxed{\frac{dp}{p} + \frac{\gamma}{1-\gamma} \frac{d\tau}{\tau} = 0}$$

(b) Using  $p = \frac{p_i V_i^\gamma}{V^\gamma}$ ,

$$\begin{aligned} \frac{\partial p}{\partial V} &= \frac{-\gamma p_i V_i^\gamma}{V^{\gamma+1}} \\ \implies -V \frac{\partial p}{\partial V} &= \frac{\gamma p_i V_i^\gamma}{V^\gamma} = \gamma p \end{aligned}$$

So that  $B_\sigma = -V(\partial p/\partial V)_\sigma = \gamma p$ , the isentropic bulk moduli.

$$B_\tau = -V \left( \frac{\partial P}{\partial V} \right)_\tau = \frac{n\tau}{V} = p$$

since

$$\begin{aligned} pV &= n\tau \\ p &= \frac{n\tau}{V} \\ \frac{\partial p}{\partial V} &= \frac{-n\tau}{V^2} \end{aligned}$$

### 13 Ideal gas (summary)

The gist of concepts involving the ideal gas, from Chapters 3,5,6 of Kittel and Kroemer [2] are the following. Consider the partition function for 1 atom in a box,  $Z_1$

$$Z_1 = \frac{n_Q}{n} = n_Q V \text{ where } n_Q := \left( \frac{M\tau}{2\pi\hbar^2} \right)^{d/2}$$

where  $d$  is the dimension of space, usually  $d = 3$ .

From Kittel and Kroemer (1980) [2], Example :  $N$  atoms in a box, Chapter 3: “Boltzmann Distribution and Helmholtz Free Energy”,

state of energy  $\epsilon_\alpha(1) + \epsilon_\beta(2) + \dots + \epsilon_\xi(N)$ ,  $\alpha, \beta, \dots, \xi$  denote orbital indices of atoms in successive boxes. each entry occurs  $N!$  times in  $Z_1^N$  (EY: 20151022)  $N!$  ways to fill  $\alpha, \beta, \dots, \xi$  orbitals with  $N$  distinguishable particles. Thus,

$$Z_N = \frac{1}{N!} Z_1^N = \frac{1}{N!} (n_Q V)^N$$

Then for  $N$  *indistinguishable particles*, the partition function of the system  $Z_N$  is

$$Z_N = \frac{1}{N!} Z_1^N = \frac{1}{N!} (n_Q V)^N \quad (43)$$

Then the Helmholtz free energy  $F$  is

$$F = -\tau \ln Z_N = -\tau [N \ln (n_Q V) - (N \ln N - N)] = \tau N \left[ \ln \left( \frac{n}{n_Q} \right) - 1 \right] \quad (44)$$

and so

$$\begin{aligned} \frac{\partial F}{\partial \tau} &= -N \ln \left( \frac{n_Q V}{N} \right) - N - \tau N \frac{d}{2} \frac{1}{\tau} = -N \left[ \ln \left( \frac{n_Q}{n} \right) + \left( \frac{d}{2} + 1 \right) \right] \\ \sigma &= \frac{-\partial F}{\partial \tau} = N \left[ \ln \left( \frac{n_Q}{n} \right) + \left( \frac{d}{2} + 1 \right) \right] \end{aligned} \quad (45)$$

Also, from using

$$\begin{aligned} F &:= U - \tau \sigma \\ dF &= dU - \tau d\sigma - \sigma d\tau = -pdV - \sigma d\tau \end{aligned}$$

then

$$p = \frac{\tau N}{V} \text{ or } pV = \tau N$$

Keep in mind that

$$\mu(\tau, V, N) \equiv \left( \frac{\partial F}{\partial N} \right)_{\tau, V}$$

I shall attempt to generalize the Sackur-Tetrode equation to consider vibrational and rotational energy modes, and so we should review its derivation.

Consider the example of a molecule with translational *and* rotational degrees of freedom. Note that  $\mathbf{n} = (n_x, n_y, n_z)$  are good quantum numbers discretizing the translational degrees of freedom, and  $j \in \mathbb{N} = \{0, 1, \dots\}$  is a good quantum number discretizing the rotational degrees of freedom. Then the energy (eigenvalue) for quantum state (described by particular values of )  $\mathbf{n}, j$  is

$$\epsilon_{\mathbf{n}, j} = \frac{\hbar^2}{2M} \left( \frac{\pi}{L} \right)^2 \mathbf{n}^2 + j(j+1)\epsilon_0 \equiv \frac{\hbar^2}{2M} \left( \frac{\pi}{L} \right)^2 (n_x^2 + n_y^2 + n_z^2) + j(j+1)\epsilon_0$$

keeping in mind the multiplicity of the energy state from the multiplicity of the rotational energy,  $g(j) = 2j+1$ .

Then

$$\exp \left( -\frac{1}{\tau} \epsilon_{\mathbf{n}, j} \right) = \exp \left( -\frac{\hbar^2}{2M\tau} \left( \frac{\pi}{L} \right)^2 \mathbf{n}^2 \right) e^{j(j+1)\epsilon_0/\tau}$$

Then the partition function for 1 molecule in a box is

$$Z_1 = \sum_{\mathbf{n}, j} g(j) \exp \left( \frac{-\hbar^2}{2\pi\tau} \left( \frac{\pi}{L} \right)^2 \mathbf{n}^2 \right) e^{j(j+1)\epsilon_0} = \sum_{\mathbf{n}} \exp \left( \frac{-\hbar^2}{2\pi\tau} \left( \frac{\pi}{L} \right)^2 \mathbf{n}^2 \right) \sum_j g(j) e^{j(j+1)\epsilon_0} = n_Q V \cdot \sum_j g(j) e^{j(j+1)\epsilon_0} = n_Q V \frac{\tau}{\epsilon_0}$$

with  $n_Q := \left( \frac{M\tau}{2\pi\hbar^2} \right)^2$ .

**Correction (rotational factor).** The Boltzmann factor of  $j(j+1)\epsilon_0$  carries a minus sign and a factor  $1/\tau$ . In three dimensions the power on  $M\tau/(2\pi\hbar^2)$  is 3/2. The integral that produces  $\tau/\epsilon_0$  is §33.7.

From the indistinguishability of quantum particles, for  $N$  of these particles,

$$Z_N = \frac{(n_Q V \frac{\tau}{\epsilon_0})^N}{N!}$$

and noting that

$$\ln Z_N = N \left[ \ln(n_Q V) + \ln \left( \frac{\tau}{\epsilon_0} \right) \right] - \ln N! \simeq N \ln \left( \frac{n_Q}{n} \right) + N \ln \left( \frac{\tau}{\epsilon_0} \right) + N$$

then

$$F := -\tau \ln Z_N = -\tau N \left[ \ln \left( \frac{n_Q}{n} \right) + 1 \right] + -\tau N \ln \left( \frac{\tau}{\epsilon_0} \right) \equiv -\tau N \left[ \ln \left( \frac{n_Q}{n} \right) + 1 \right] + F(\text{int})$$

$$\sigma := - \left( \frac{\partial F}{\partial \tau} \right) = N \left[ \ln \left( \frac{n_Q}{n} \right) + \left( \frac{d}{2} + 1 \right) \right] + N \left( \ln \left( \frac{\tau}{\epsilon_0} \right) + 1 \right)$$

Examining the steps above, it's clear how to generalize to when the form of the energy eigenvalues in the vibrational and rotational degrees of freedom (internal degrees of freedom), described by quantum numbers  $\mathbf{k}$ , is unknown:

$$\epsilon_{\mathbf{n}, \mathbf{k}} = \frac{\hbar^2}{2M} \left( \frac{\pi}{L} \right)^2 \mathbf{n}^2 + \epsilon'_{\text{int}}(\mathbf{k})$$

$g(\mathbf{k}) \in \mathbb{Z}^+$  is the multiplicity of the internal degrees of freedom

$$Z_1 = n_Q V \cdot Z_{1, \text{int}} \text{ with } Z_{1, \text{int}} = \sum_{\mathbf{k}} g(\mathbf{k}) e^{-\frac{1}{\tau} \epsilon'_{\text{int}}(\mathbf{k})}$$

$$Z_N = \frac{(n_Q V)^N}{N!} (Z_{1, \text{int}})^N \implies \ln Z_N = N \left( \ln \left( \frac{n_Q}{n} \right) + 1 \right) + N \ln Z_{1, \text{int}}$$

$$F = -\tau N \left[ \ln \left( \frac{n_Q}{n} \right) + 1 \right] + F(\text{int})(\tau)$$

$$\sigma = N \left[ \ln \left( \frac{n_Q}{n} \right) + \left( \frac{d}{2} + 1 \right) \right] + \sigma_{\text{int}}$$

with  $F(\text{int})(\tau)$  emphasizing that we could surmise or conjecture that the Helmholtz free energy contribution due to internal degrees of freedom can be a function of temperature  $\tau$  (cf. Eq. (42.4) of Landau and Lifshitz (1980) [7]).

## 14 Binary mixtures

The 2008 chapter heading was empty. The entropy-of-mixing solution under the ideal-gas heading is also empty. The 2015 manuscript derives the distinct-gas increase and the identical-gas cancellation, and its general two-volume formula repeats  $V_A$  in the second logarithm. The derivation here is the one to use. It uses the Sackur-Tetrode entropy already obtained in that manuscript,

$$\sigma = N \left[ \ln \left( \frac{n_Q}{n} \right) + 1 + \frac{d}{2} \right], \quad n = \frac{N}{V}, \quad n_Q = \left( \frac{M\tau}{2\pi\hbar^2} \right)^{d/2}, \quad (46)$$

for a classical monatomic ideal gas of one species in  $d$  dimensions. The usual case is  $d = 3$ , where  $1 + d/2 = 5/2$ . The hypotheses are: the gas is classical,  $n \ll n_Q$ ; the particles of one species are indistinguishable, which has already put the  $1/N!$  into (46); different species are distinguishable from each other; and there is no interaction energy, so entropies of different species in the same volume add.

## 14.1 Two different species

Put species  $A$  in volume  $V_A$  and species  $B$  in volume  $V_B$ , at the same  $\tau$ , and then let each occupy the total volume

$$V = V_A + V_B \quad (47)$$

without changing  $N_A$ ,  $N_B$ , or  $\tau$ . A partition removed at fixed total volume is this process. Because  $n_Q$  depends on  $\tau$  and on the mass, and  $\tau$  is fixed,

$$\begin{aligned} \sigma_A &= N_A \left[ \ln \left( \frac{n_{QA} V_A}{N_A} \right) + 1 + \frac{d}{2} \right], \\ \sigma'_A &= N_A \left[ \ln \left( \frac{n_{QA} V}{N_A} \right) + 1 + \frac{d}{2} \right]. \end{aligned} \quad (48)$$

Subtract:

$$\sigma'_A - \sigma_A = N_A \ln \left( \frac{V}{V_A} \right). \quad (49)$$

The same subtraction for  $B$ , with  $V_B$  in the denominator, gives the mixing entropy

$$\Delta\sigma = N_A \ln \left( \frac{V}{V_A} \right) + N_B \ln \left( \frac{V}{V_B} \right). \quad (50)$$

The 2015 general formula writes  $V_A$  in both denominators. The second term of (50) is the  $B$  contribution (49) with the labels exchanged.

If  $N_A = N_B = N$  and  $V_A = V_B$ , then  $V/V_A = 2$  and

$$\Delta\sigma = 2N \ln 2. \quad (51)$$

That is the result in the 2015 manuscript for equal amounts of two different gases.

If the initial pressures and temperature agree, the ideal-gas law  $pV_i = N_i\tau$  gives  $V_i/V = N_i/N$  with  $N = N_A + N_B$ . Then

$$\Delta\sigma = -N_A \ln x_A - N_B \ln x_B, \quad x_i = \frac{N_i}{N}. \quad (52)$$

Each  $-x \ln x$  is nonnegative for  $x \in (0, 1]$ , so  $\Delta\sigma \geq 0$ .

## 14.2 The same species

Now let both sides be the same species, the same  $n_Q$ , with  $N$  particles in volume  $V$  on each side. The initial entropy is the sum of two separate applications of (46):

$$\sigma_i = 2N \left[ \ln \left( \frac{n_Q V}{N} \right) + 1 + \frac{d}{2} \right]. \quad (53)$$

After the partition is removed there is one system of  $2N$  indistinguishable particles in volume  $2V$ . One application of (46) gives

$$\sigma_f = 2N \left[ \ln \left( \frac{n_Q (2V)}{2N} \right) + 1 + \frac{d}{2} \right] = \sigma_i. \quad (54)$$

The ratio  $(2V)/(2N)$  equals  $V/N$ , so

$$\Delta\sigma = 0. \quad (55)$$

Counting the final state as two distinguishable species, each  $N$  in volume  $2V$ , would instead add  $2N \ln 2$ . The single factorial  $(2N)!$  removes that amount. With (8) truncated at  $n \ln n - n$ ,

$$\begin{aligned} \ln((2N)!) - 2 \ln(N!) &= (2N \ln(2N) - 2N) - 2(N \ln N - N) \\ &= 2N \ln 2 + 2N \ln N - 2N \ln N \\ &= 2N \ln 2. \end{aligned} \quad (56)$$

That is the Gibbs cancellation: the extra  $2N \ln 2$  from doubling the volume available to each group is the same  $2N \ln 2$  that appears when two groups of  $N$  identical particles are entered as one group of  $2N$ .

### 14.3 Chemical potential in an ideal mixture

For one species, the Helmholtz free energy consistent with (46) and with  $F = U - \tau\sigma$ ,  $U = (d/2)N\tau$ , is the expression in the 2015 ideal-gas summary,

$$F = N\tau \left[ \ln \left( \frac{n}{n_Q} \right) - 1 \right]. \quad (57)$$

Differentiate at fixed  $\tau$  and  $V$ . The  $N$  inside  $n = N/V$  contributes one power of  $N$ :

$$\begin{aligned} \left( \frac{\partial F}{\partial N} \right)_{\tau, V} &= \tau \left[ \ln \left( \frac{n}{n_Q} \right) - 1 \right] + N\tau \cdot \frac{1}{N} \\ &= \tau \ln \left( \frac{n}{n_Q} \right). \end{aligned} \quad (58)$$

So  $\mu = \tau \ln(n/n_Q)$ .

In a mixture of ideal gases the species share  $V$  and  $\tau$ , and the free energy is the sum of (57) over species because the interaction energy was assumed to vanish. For species  $j$ ,

$$\mu_j = \tau \ln \left( \frac{n_j}{n_{Qj}} \right) = \tau \ln x_j + \tau \ln \left( \frac{n}{n_{Qj}} \right), \quad (59)$$

where  $n_j = N_j/V$ ,  $n = \sum_i n_i$ , and  $x_j = n_j/n$ . The mixture pressure is  $p = n\tau$ . A pure gas of species  $j$  at the same  $\tau$  and at that same total pressure  $p$  would have density  $n$ , hence chemical potential  $\tau \ln(n/n_{Qj})$ . Call that pure-gas value  $\mu_j^0(\tau, p)$ . Then

$$\mu_j(\tau, p, x_j) = \mu_j^0(\tau, p) + \tau \ln x_j. \quad (60)$$

The partial pressure follows from the same ideal-gas law applied to  $n_j$ :

$$p_j = n_j\tau = x_j p, \quad \sum_j p_j = p. \quad (61)$$

Equations (50), (55), (60), and (61) are the content that the empty chapter and the empty mixing solution did not contain. *2015 manuscript.*

## 15 Mixtures of ideal gases

### 15.1 Gibbs paradox

Kittel and Kroemer (1980) [2] Problem 6 **Entropy of mixing** of Chapter 6: Ideal Gas,

Prof. Steven Anlage, Introduction to Thermodynamics and Statistical Mechanics, Spring 2011, University of Maryland Physics 404, Homework 8 Solutions, Question 9,<sup>3</sup>

Consider adding together the entropies of ideal gas A with initial volume  $V$ , and molecular mass  $M_A$  (so it has a  $n_{Q_A}$  quantum concentration), and of ideal gas B with initial volume  $V$ , molecular mass  $M_B$ ,  $n_{Q_B}$  quantum concentration:

$$\sigma_{\text{initial}} = \sigma_A + \sigma_B = N_A \ln \left( \frac{n_{Q_A} V}{N_A} \right) + N_A \left( 1 + \frac{d}{2} \right) + N_B \ln \left( \frac{n_{Q_B} V}{N_B} \right) + N_B \left( 1 + \frac{d}{2} \right)$$

After diffusive equilibrium is reached means that each of the gases now occupy a volume  $2V$ , gas A is free to roam in volume  $2V$ , as is gas B. Thus, the entropy increases:

$$\sigma_{\text{final}} = N_A \ln \left( \frac{n_{Q_A} 2V}{N_A} \right) + N_A \left( 1 + \frac{d}{2} \right) + N_B \ln \left( \frac{n_{Q_B} 2V}{N_B} \right) + N_B \left( 1 + \frac{d}{2} \right)$$

and so

$$\Delta\sigma = (N_A + N_B) \ln 2 = 2N \ln 2 \text{ (if } N_A = N_B = N)$$

<sup>3</sup>[http://www.physics.umd.edu/courses/Phys404/Anlage\\_Spring11/hw8soln5.pdf](http://www.physics.umd.edu/courses/Phys404/Anlage_Spring11/hw8soln5.pdf)

In general,

$$\Delta\sigma = N_A \ln \left( \frac{V_A + V_B}{V_A} \right) + N_B \ln \left( \frac{V_A + V_B}{V_B} \right)$$

**Correction (mixing volumes).** The second logarithm is  $\ln((V_A + V_B)/V_B)$ . The derivation is §14, equation (50).

For identical atoms ( $A \equiv B$ ),

$$\sigma_A = N \ln \left( \frac{n_Q V}{N} \right) + N \left( 1 + \frac{d}{2} \right) = \sigma_B$$

and so

$$\sigma_{\text{initial}} = \sigma_A + \sigma_B = 2N \ln \left( \frac{n_Q V}{N} \right) + 2N \left( 1 + \frac{d}{2} \right)$$

For the final entropy after mixing, treat the system as a (whole) system of  $2N$  indistinguishable particles occupying a volume of  $2V$ :

$$\sigma_{\text{final}} = 2N \ln \left( \frac{n_Q(2V)}{2N} \right) + 2N \left( 1 + \frac{d}{2} \right)$$

and so  $\Delta\sigma = 0$

From Sec. 93. Mixtures of ideal gases of Landau and Lifshitz (1980) [7], additivity of thermodynamic quantities (such as energy and entropy) holds only if interaction between various parts of body is negligible.

But for mixture of several substances, e.g. mixture of liquids, thermodynamic quantities are not equal to sums of thermodynamic quantities for individual components of the mixture.

Mixtures of ideal gases are an exception, since interaction between molecules is by definition negligible.

But for partial pressure of  $i$ th gas,  $p_i = \frac{N_i \tau}{V} = \frac{N_i p}{N}$ ,  $N$  = total number of molecules in mixture,  $N_i$  number of molecules of  $i$ th gas.

So

$$p_j = \frac{N_j \tau}{V} = \frac{N_j}{N} p = X_j p$$

where  $X_j$  is mole fraction or particle fraction.

2008 manuscript.

## 16 Fermi and Bose Gases

### Problems

**Problem 1. Density of orbitals in one and two dimensions.**

(a) Show that the density of orbitals of a free electron in one dimension is

$$\mathcal{D}_1(\epsilon) = (L/\pi)(2m/\hbar^2 \epsilon)^{1/2}, \quad (62)$$

where  $L$  is the length of the line.

(b) Show that in two dimensions, for a square of area  $A$ ,

$$\mathcal{D}_2(\epsilon) = Am/\pi \hbar^2 \quad (63)$$

independent of  $\epsilon$ .

**Solution 1.** Recall, for the free electron:  $H = \frac{p^2}{2m} = \frac{\hbar^2}{2m} \nabla^2$

$\implies \epsilon_s = \frac{\hbar^2}{2m} \frac{n_x^2 \pi^2}{L^2}$  for 1 dim.,  $\epsilon_s = \frac{\hbar^2}{2m} \frac{(n_x^2 + n_y^2) \pi^2}{L^2}$  for 2-dim.

If  $\epsilon_F$  = Fermi energy, energy of the highest filled orbital,

1-dim:  $N = 2n_F$ . 2-dim.:  $N = (2) \left( \frac{1}{4} \right) (\pi n_F^2) = \frac{\pi n_F^2}{2}$

2 factor for 2 possible spin states.

$$\begin{aligned} \text{1-dim: } \epsilon_F &= \frac{\hbar^2}{2m} \left(\frac{N}{2}\right)^2 \frac{\pi^2}{L^2} = \frac{1}{2m} \left(\frac{\hbar\pi N}{2}\right)^2 \left(\frac{1}{V^2}\right) = \frac{1}{2m} \left(\frac{\hbar m}{2}\right)^2 n^2 \quad N = \sqrt{2m\epsilon_F} \left(\frac{2V}{\hbar\pi}\right)^2 \\ \text{2-dim: } \epsilon_F &= \frac{\hbar^2}{2m} \left(\frac{2N}{\pi}\right) \left(\frac{\pi}{L}\right)^2 = \frac{\hbar^2}{m} \frac{N}{\pi} \frac{\pi^2}{V} = \frac{\hbar^2}{m} \frac{N\pi}{V} = \frac{\hbar^2}{m} n \quad N = \frac{mV\epsilon_F}{\pi\hbar^2} \end{aligned}$$

$$\text{1-dim.: } \mathcal{D}(\epsilon) = \frac{dN}{d\epsilon} = \sqrt{2m} \left(\frac{2V}{\hbar\pi}\right)^2 \frac{1}{2} (\epsilon_F)^{-1/2} = \frac{N}{2\epsilon} = \left(\frac{L}{\pi}\right) \left(\frac{2m^2}{\hbar}\epsilon\right)^{1/2}$$

$$\text{2-dim.: } N\epsilon = Am/\pi\hbar^2$$

**Problem 2. Energy of relativistic Fermi gas.** For electrons with an energy  $\epsilon \gg mc^2$ , where  $m$  is the rest mass of the electron, the energy is given by  $\epsilon \simeq pc$ , where  $p$  is the momentum. For electrons in a cube of volume  $V = L^3$  the momentum is of the form  $(\pi\hbar/L)$ , multiplied by  $(n_x^2 + n_y^2 + n_z^2)^{1/2}$ , exactly as for the nonrelativistic limit.

(a) Show that in this extreme relativistic limit the Fermi energy of a gas of  $N$  electrons is given by

$$\epsilon_F = \hbar\pi c(3n/\pi)^{1/3}, \quad (64)$$

where  $n = N/V$ .

(b) Show that the total energy of the ground state of the gas is

$$U_0 = \frac{3}{4}N\epsilon_F. \quad (65)$$

The general problem is treated by F. Jüttner, Zeitschrift für Physik **47**, 542 (1928).

**Solution 2.**

$$(a) \quad \epsilon \simeq pc = \frac{\hbar n\pi}{L}c, \quad \epsilon_F = \frac{\hbar n_F\pi}{L}c.$$

$$\text{Recall, for 3-dim.: } N = (2) \left(\frac{1}{8}\right) \left(\frac{4}{3}\pi n_F^3\right) = \frac{\pi}{3}n_F^3. \quad n_F = \left(\frac{3N}{\pi}\right)^{1/3}.$$

$$\epsilon_F = \frac{\pi\hbar}{L} \left(\frac{3N}{\pi}\right)^{1/3} = \hbar\pi c \left(\frac{3n}{\pi}\right)^{1/3}$$

(b)

$$\begin{aligned} U_0 &= 2 \sum_{n \leq n_F} \epsilon_n = 2 * \frac{1}{8} * 4\pi \int_0^{n_F} dn n^2 \frac{\hbar n\pi c}{L} = \frac{\hbar\pi^2 c}{L} \int_0^{n_F} dn n^3 = \frac{\hbar\pi^2 c}{4L} n_F^4 = \\ &= \frac{\hbar\pi^2 c}{4L} \left(\frac{\epsilon_F L}{\hbar\pi c}\right)^4 = \frac{\hbar\pi^2 c}{4L} \left(\frac{3N}{\pi}\right) \left(\frac{\epsilon_F L}{\hbar\pi c}\right) = \boxed{\frac{3}{4}N\epsilon_F} \end{aligned}$$

**Problem 3. Pressure and entropy of degenerate Fermi gas.**

(a) Show that a Fermi electron gas in the ground state exerts a pressure

$$p = \frac{(3\pi^2)^{2/3}}{5} \frac{\hbar^2}{m} \left(\frac{N}{V}\right)^{5/3} \quad (66)$$

In a uniform decrease of the volume of a cube every orbital has its energy raised: The energy of an orbital is proportional to  $1/L^2$  or to  $1/V^{2/3}$ .

(b) Find an expression for the entropy of a Fermi electron gas in the region  $\tau \ll \epsilon_F$ . Notice that  $\sigma \rightarrow 0$  as  $\tau \rightarrow 0$ .

**Solution 3.**

$$(a) \quad \text{Recall } U_0 = \frac{3}{5}N \frac{\hbar^2}{2m} (3\pi^2 N)^{2/3} V^{-2/3}$$

$$\frac{\partial U}{\partial V} = \frac{3}{5}N \frac{\hbar^2}{2m} (3\pi^2 N)^{2/3} \left(\frac{-2}{3}\right) V^{-5/3} = \frac{-1}{5} \frac{\hbar^2}{m} (3\pi^2)^{2/3} \left(\frac{N}{V}\right)^{5/3}$$

So then

$$\boxed{p = \frac{-\partial U_0}{\partial V} = \frac{1}{5} \frac{\hbar^2}{m} (3\pi^2)^{2/3} \left(\frac{N}{V}\right)^{5/3}}$$

- (b) Recall that  $\epsilon_F \equiv \tau_F = \frac{\hbar^2}{2m} \left( \frac{3\pi^2 N}{V} \right)^{2/3}$  and that the heat capacity of an electron gas is  $C_{el} = \frac{1}{2} \pi^2 N \frac{\tau}{\tau_F} = \frac{\partial U}{\partial \tau}$ , which *helps directly* with finding the entropy.

$$\sigma(\tau) - \sigma(\tau_0) = \int_{\tau_0}^{\tau} \frac{1}{\tau} dU = \int_{\tau_0}^{\tau} \frac{1}{\tau} \frac{1}{2} \pi^2 N \frac{\tau}{\tau_F} d\tau = \frac{\pi^2 N}{2\tau_F} \tau$$

Let  $\sigma(\tau_0 = 0) = 0$ ,

$$\sigma(\tau) = \frac{\pi^2 N}{2\tau_F} \tau$$

**Problem 4. Chemical potential versus temperature.** Explain graphically why the initial curvature of  $\mu$  versus  $\tau$  is upward for a fermion gas in one dimension and downward in three dimensions (Figure 7.7). *Hint:* The  $\mathcal{D}_1(\epsilon)$  and  $\mathcal{D}_3(\epsilon)$  curves are different, where  $\mathcal{D}_1$  is given in Problem 1. It will be found useful to set up the integral for  $N$ , the number of particles, and to consider from the graphs the behavior of the integrand between zero temperature and a finite temperature.

**Solution 4.** Recall,  $N = \int_0^{\epsilon_F} d\epsilon \mathcal{D}(\epsilon)$ .

1-dim:

$$\begin{aligned} N &= \int_0^{\epsilon_F} d\epsilon \mathcal{D}_1(\epsilon) = \int_0^{\epsilon_F} d\epsilon \left( \frac{L}{\pi} \right) \left( \frac{2m}{\hbar^2 \epsilon} \right)^{1/2} = \left( \frac{L}{\pi} \right) \frac{(2m)^{1/2}}{\hbar} \int_0^{\epsilon_F} d\epsilon \epsilon^{-1/2} = \\ &= \left( \frac{L}{\pi} \right) \frac{(2m)^{1/2}}{\hbar} (2\epsilon^{1/2}) \Big|_0^{\epsilon_F} = \left( \frac{L}{\pi} \right) \frac{(2m)^{1/2}}{\hbar} 2\epsilon_F^{1/2} \end{aligned}$$

3-dim:

$$N = \int_0^{\epsilon_F} d\epsilon \frac{V}{2\pi^2} \left( \frac{2m}{\hbar^2} \right)^{3/2} \epsilon^{1/2} = \frac{V}{2\pi^2} \left( \frac{2m}{\hbar^2} \right)^{3/2} \left( \frac{2}{3} \epsilon^{3/2} \right) \Big|_0^{\epsilon_F} = \frac{V}{3\pi^2} \left( \frac{2m}{\hbar^2} \right)^{3/2} \epsilon_F^{3/2}$$

Note the difference in the concavity of the  $N(\epsilon)$  curves.

**Solution 5.**

- (a) For  ${}^3\text{He}$ , given  $I = 1/2$ , density of liquid  $0.081 \text{ g cm}^{-3}$ , we want to find  $v_F$ ,  $\epsilon_F$ ,  $\tau_F$ .

$$\begin{aligned} \epsilon_F &= \left( \frac{\hbar^2}{2m} \right) (3\pi^2 n)^{2/3} = \\ &= \frac{(6.582 \times 10^{-22} \text{ MeV} \cdot \text{s})^2}{2 \cdot 3 \cdot 938 \text{ MeV}/c^2 \left( \frac{1 \text{ c}}{3 \times 10^{10} \text{ cm/s}} \right)^2} \left( \left( 3\pi^2 \frac{0.081 \text{ g}}{\text{cm}^3} \right) \left( \frac{1 \text{ kg}}{10^3 \text{ g}} \right) \left( \frac{1 \text{ u}}{1.67 \times 10^{-27} \text{ kg}} \right) \left( \frac{1 {}^3\text{He}}{3 \text{ u}} \right) \right)^{2/3} = \\ &= 4.24 \times 10^{-10} \text{ MeV} = 4.24 \times 10^{-4} \text{ eV} \end{aligned}$$

Now suppose we have a nonrelativistic gas. Then  $\frac{1}{2} m v_F^2$  or  $v_F^2 = \frac{2\epsilon_F}{m}$ .

$$v_F = 1.675 \times 10^4 \frac{\text{cm}}{\text{sec}}$$

$$T_F = \frac{4.24 \times 10^{-4} \text{ eV}}{0.8619 \times 10^{-4} \text{ eV/K}} = 4.92 \text{ K}$$

- (b)

$$C_{el} = \frac{\pi^2}{3} \mathcal{D}(\epsilon_F) \tau = \frac{\pi^2}{3} \frac{3N}{2\tau_F} \tau = \frac{\pi^2}{2} N \frac{\tau}{\tau_F} = 1.003 k_B T N$$

**Solution 6. Mass-radius relationship for white dwarfs.**

- (a)

$$U = \int \rho(r) \phi(r) r^2 4\pi dr = -4\pi \int \rho \frac{G \frac{4}{3} \pi r^3 \rho}{r} r^2 dr = 4\pi \frac{M}{\frac{4}{3} \pi R^3} G \frac{4}{3} \pi \frac{1}{5} R^5 \frac{M}{\frac{4}{3} \pi R^3} = \boxed{\frac{-3GM^2}{5R}}$$

(b)

$$\epsilon_F = \left( \frac{\hbar^2}{2m} \right) (3\pi^2 n)^{2/3}$$

is the Fermi energy.

With  $V = \frac{4}{3}\pi R^3$ ,

$$\begin{aligned} T_{tot} &= N \frac{1}{2} m v^2 = N \epsilon_F = N \frac{\hbar^2}{2m} \left( \frac{3\pi^2 N}{V} \right)^{2/3} = \frac{(3\pi^2)^{2/3}}{2} \left( \frac{\hbar^2}{m} \frac{N^{5/3}}{V^{2/3}} \right) = \frac{(3\pi^2)^{2/3}}{2} \frac{\hbar^2}{m} \frac{N^{5/3}}{\left( \frac{4}{3}\pi \right)^{2/3} R^2} = \\ &= \frac{\left( \frac{9\pi}{4} \right)^{2/3}}{2} \frac{\hbar^2 N^{5/3}}{m R^2} \simeq \hbar^2 \frac{(M/M_H)^{5/3}}{m R^2} \end{aligned}$$

since  $N = \frac{M}{M_H}$  since  $M_H \ll m$ .

(c)

$$\frac{\hbar^2 M^{5/3}}{m M_H^{5/3} R^2} = \frac{G M^2}{R} \implies M^{1/3} R \simeq \frac{\hbar^2 / G}{m M_H^{5/3}}$$

$$\frac{(6.582 \times 10^{-22} \text{ MeV} \cdot \text{s})(1.05457 \times 10^{-34} \text{ J} \cdot \text{s})}{0.511 \text{ MeV}/c^2 ((1.67 \times 10^{-27} \text{ kg}) \left( \frac{10^3 \text{ g}}{1 \text{ kg}} \right)^{5/3})} \left( \frac{10^3 \text{ g}}{1 \text{ kg}} \right)^2 / (6.67 \times 10^{-11} \frac{\text{m}^3/\text{s}^2}{\text{kg}}) \left( \frac{(3 \times 10^8 \text{ m/s})^2}{1 \text{ c}^2} \right)$$

Dealing with units and dimensions,

$$J = \text{kg} \cdot \frac{\text{m}^2}{\text{s}^2}; \quad \frac{N \cdot \text{m}^2}{\text{kg}^2} = \text{kg} \cdot \frac{\text{m}^3/\text{s}^2}{\text{kg}^2} = \frac{\text{m}^3/\text{s}^2}{\text{kg}}$$

$$\frac{(6.582 \times 10^{-22} \text{ MeV} \cdot \text{s})(1.05457 \times 10^{-34} \text{ kg} \cdot \frac{\text{m}^2}{\text{s}}) \left( \frac{10^2 \text{ cm}}{1 \text{ m}} \right)^2 \left( \frac{10^3 \text{ g}}{1 \text{ kg}} \right)^2}{(0.511 \text{ MeV}/c^2)(1.67 \times 10^{-24} \text{ g})^{5/3} (6.67 \times 10^{-11} \frac{\text{m}^3/\text{s}^2}{\text{kg}} \times \left( \frac{10^2 \text{ cm}}{1 \text{ m}} \right)^3)} \left( \frac{(3 \times 10^{10} \text{ cm/s})^2}{1 \text{ c}^2} \right) 10^{20} \text{ g}^{1/3} \text{ cm}$$

(d)

$$\begin{aligned} \rho &= \frac{M}{\frac{4}{3}\pi R^3} = \frac{M}{\frac{4}{3}\pi \left( \frac{10^{20} \text{ g}^{1/3} \text{ cm}}{M^{1/3}} \right)^3} = \frac{M^2}{\frac{4}{3}\pi 10^{60} \text{ g cm}^3} = \frac{(2 \times 10^{33} \text{ g})^2}{\frac{4}{3}\pi 10^{60} \text{ g cm}^3} = \frac{3 \times 10^{66} \text{ g}^2}{\pi 10^{60} \text{ g cm}^3} \\ &= 10^6 \frac{\text{g}}{\text{cm}^3} \end{aligned}$$

(e)

$$M^{1/3} R \simeq \frac{\hbar^2 / G}{m M_H^{5/3}} 10^{17} \text{ g}^{1/3} \text{ cm} \implies R = \frac{10^{17} \text{ g}^{1/3} \text{ cm}}{(2 \times 10^{33} \text{ g})^{1/3}} = 7.937 \text{ km}$$

**Solution 7. Photon condensation.**  $N_e = 2.404 V \tau^3 / \pi^2 \hbar^3 c^3$ .

The condition is that  $N = N_e$ :

$$N = N_e = \frac{2.404 V \tau^3}{\pi^2 \hbar^3 c^3}$$

$$\boxed{\tau = \left( \frac{\pi^2 \hbar^3 c^3 N}{2.404 V} \right)^{1/3}}$$

With a concentration of  $10^{20} \text{ cm}^{-3}$ ,  $T = 1.7 \times 10^6 \text{ K}$  (the critical temperature in  $K$  below which  $N_e < N$ ).

**Solution 8. Energy, heat capacity, and entropy of degenerate boson gas.**

Recall that the distribution function for bosons is

$$f(\epsilon, \tau) = \frac{1}{\exp[(\epsilon - \mu)/\tau] - 1}$$

Consider  $N$  noninteracting bosons of spin zero.

$\epsilon = 0$  for ground state. Thus,  $f(0, \tau) = \frac{1}{\exp(\frac{-\mu}{\tau}) - 1}$ . Recall that

$$N_\epsilon(\tau) = \int_0^\infty d\epsilon \mathcal{D}(\epsilon) f(\epsilon, \tau) = \int_0^\infty d\epsilon \frac{V}{4\pi^2} \left( \frac{2M}{\hbar^2} \right)^{3/2} \epsilon^{1/2} \frac{1}{\lambda^{-1} \exp(\frac{\epsilon}{\tau}) - 1}$$

Recall,

$$\begin{aligned} U &= 0 \left( \frac{1}{\exp(-\mu/\tau) - 1} \right) + \int_0^\infty \frac{\epsilon}{e^{(\epsilon-\mu)/\tau} - 1} \mathcal{D}(\epsilon) d\epsilon = \int_0^\infty \frac{\epsilon}{e^{(\epsilon-\mu)/\tau} - 1} \frac{V}{4\pi^2} \left( \frac{2M}{\hbar^2} \right)^{3/2} \epsilon^{1/2} d\epsilon = \\ &= \frac{V}{4\pi^2} \left( \frac{2M}{\hbar^2} \right)^{3/2} \int_0^\infty \frac{\epsilon^{3/2}}{e^{(\epsilon-\mu)/\tau} - 1} \end{aligned}$$

$\tau < \tau_\epsilon$ , so  $\lambda \approx 1$ , for  $N_0$  to be sufficiently large.

$$\implies U = \frac{V}{4\pi^2} \left( \frac{2M}{\hbar^2} \right)^{3/2} \int_0^\infty \frac{\epsilon^{3/2}}{e^{\epsilon/\tau} - 1}$$

Using  $\begin{matrix} x = \epsilon/\tau \\ dx = d\epsilon/\tau \end{matrix}$

$$\implies U = \frac{V}{4\pi^2} \left( \frac{2M}{\hbar^2} \right)^{3/2} \tau^{5/2} \int_0^\infty \frac{x^{3/2} dx}{e^x - 1} = B_0 \tau^{5/2} C_0$$

It's true that  $\partial_\tau(e^{\epsilon/\tau}) = e^{\epsilon/\tau} \left( \frac{-\epsilon}{\tau^2} \right)$

$$C_V = \left( \frac{5}{2} \right) \frac{V}{4\pi^2} \left( \frac{2M}{\hbar^2} \right)^{3/2} \tau^{3/2} \int_0^\infty \frac{x^{3/2} dx}{e^x - 1}$$

Now  $\frac{1}{\tau} = \left( \frac{\partial \sigma}{\partial U} \right)_V$ .

$$\begin{aligned} \sigma(U) - \sigma(U_0) &= \int \frac{1}{\tau} dU = \int \frac{V B_0 \frac{5}{2} \tau^{3/2} C_0 d\tau}{\tau} = V B_0 \frac{5}{2} C_0 \int \tau^{1/2} d\tau = \\ &\xrightarrow{dU = V B_0 \frac{5}{2} \tau^{3/2} C_0 d\tau} V B_0 \frac{5}{2} C_0 \frac{2}{3} \tau^{3/2} \Big|_0^\tau \\ \sigma(U) &= V B_0 \frac{5}{3} C_0 \tau^{3/2} = V B_0 \frac{5}{3} C_0 \frac{U^{3/5}}{(V B_0 C_0)^{3/5}} = (V B_0 C_0)^{2/5} \frac{5}{3} U^{3/5} = \\ &= \left( \frac{V}{4\pi^2} \left( \frac{2M}{\hbar^2} \right)^{3/2} \int_0^\infty \frac{x^{3/2} dx}{e^x - 1} \right)^{2/5} \frac{5}{3} U^{3/5} \end{aligned}$$

where we had used  $U = V B_0 \tau^{5/2} C_0$  or  $\left( \frac{U}{V B_0 C_0} \right)^{3/5} = \tau^{3/2}$ .

**Solution 9. Boson gas in one dimension.**

In one-dim.,

$$\begin{aligned} \epsilon_s &= \frac{\hbar^2}{2m} \frac{\pi^2 n^2}{L^2} = \frac{\hbar^2}{2m} \frac{\pi^2 n^2}{V^2} \text{ or } \frac{2mV^2}{\hbar^2 \pi^2} \epsilon_s = n^2 \\ \implies n &= \left( \frac{2mV^2}{\hbar^2 \pi^2} \right)^{1/2} \epsilon^{1/2} \implies \frac{dn}{d\epsilon} = \left( \frac{2mV^2}{\hbar^2 \pi^2} \right)^{1/2} \frac{1}{2\epsilon^{1/2}} \end{aligned}$$

Note the difference with 3-dim.: For spinless bosons,

$$\begin{aligned} \mathcal{D}(n) dn &= \frac{4\pi n^2 dn}{8} = \frac{\pi}{2} \left( \frac{2mL^2}{\hbar^2 \pi^2} \epsilon_s \right) \left( \frac{2mL^2}{\hbar^2 \pi^2} \right)^{1/2} \frac{1}{2\epsilon^{1/2}} d\epsilon = \\ &= \frac{\pi}{4} \left( \frac{2mL^2}{\hbar^2 \pi^2} \right)^{3/2} \epsilon^{1/2} d\epsilon = \mathcal{D}(\epsilon) d\epsilon \end{aligned}$$

$$\begin{aligned}
N_\epsilon(\tau) &= \int_0^\infty d\epsilon \mathcal{D}(\epsilon) f(\epsilon, \tau) = \int_0^\infty d\epsilon \left( \frac{2mV^2}{\hbar^2 \pi^2} \right)^{1/2} \frac{1}{2\epsilon^{1/2}} \frac{1}{\lambda^{-1}} \exp\left(\frac{\epsilon}{\tau}\right) - 1 = \\
&= \frac{1}{2} \left( \frac{2mV^2}{\hbar^2 \pi^2} \right)^{1/2} \int_0^\infty \frac{d\epsilon}{\epsilon^{1/2}} (\lambda^{-1} \exp\left(\frac{\epsilon}{\tau}\right) - 1)
\end{aligned}$$

For  $\epsilon \rightarrow 0$ ,  $N_\epsilon(\tau) \rightarrow \infty$  which is not characteristic of a Boson.

**Solution 10. Relativistic white dwarf stars.**  $\epsilon \simeq pc$   $\lambda = 2\pi\hbar/p$ .

Virial theorem:

$$\begin{aligned}
2\langle T \rangle &= -\langle U \rangle & 2\langle T \rangle &= k\langle U \rangle \\
2\left(\frac{3}{4}N\epsilon_F\right) &= \frac{3}{2}N^{4/3}\hbar\pi c \left(\frac{3}{\pi}\right)^{1/3} \frac{1}{L} = -\left(-\frac{3GM^2}{5R}\right) = \frac{3GM^2}{5R} \\
\epsilon_F &= \hbar\pi c \left(\frac{3n}{\pi}\right)^{1/3}
\end{aligned}$$

Approximating the sphere as a box,

$$\begin{aligned}
L^3 &= \frac{4}{3}\pi R^3 & \left(\frac{3}{4\pi}\right)^{1/3} L &= R \\
L &= \left(\frac{4}{3}\right)^{1/3} \pi^{1/3} R
\end{aligned}$$

$$N^{4/3} = \left(\frac{GM^2}{5R}\right) \left(\frac{2L}{\hbar\pi^{2/3}c3^{1/3}}\right) \Rightarrow N = \left(\left(\frac{GM^2}{5R}\right) \left(\frac{2L}{\hbar\pi^{2/3}c3^{1/3}}\right)\right)^{3/4} = \left(\left(\frac{2GM^2}{5}\right) \frac{4^{1/3}}{\hbar 3^{2/3}\pi^{1/3}c}\right)^{3/4}$$

Now  $M = Nm_H$ , where  $m_H$  is the mass of hydrogen, so

$$\begin{aligned}
1 &= \left(\left(\frac{2Gm_H^2}{5}\right) \frac{4^{1/3}}{\hbar 3^{2/3}\pi^{1/3}c}\right)^{3/4} N^{1/2} \\
\Rightarrow N &= \left(\frac{5\hbar(3^{2/3})\pi^{1/3}c}{4^{1/3}(2Gm_H^2)}\right)^{3/2} = \left(\frac{(1.05457 \times 10^{-34} \text{ J} \cdot \text{s})(3 \times 10^8 \text{ m/s})}{(6.67 \times 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2})(1.67 \times 10^{-27} \text{ kg})^2}\right)^{3/2} \cdot \left(\frac{5(3^{2/3})\pi^{1/3}}{4^{1/3}2}\right)^{3/2} = 2.2 \times 10^{58}
\end{aligned}$$

2008 manuscript.

## 17 Heat and Work

### Energy and Entropy transfer: Definition of Heat and Work

Consider  $\Sigma$ , a manifold consisting of points representing thermodynamic states of a single system. For instance, for global coordinates  $(U, V)$ ,

$(U, V) \in \Sigma$ .

Consider  $W, Q \in \Omega^1(\Sigma)$ , 1-forms on  $\Sigma$ .

Now define  $Q$  as

$$Q \equiv \tau d\sigma$$

with  $\sigma = \sigma(U, V) \in C^\infty(\Sigma)$

$\tau = \tau(U, V) \in C^\infty(\Sigma)$

Recall energy conservation in this form:

$$dU = W + Q$$

Consider pure heat and, so, no work. Now  $Q = \tau d\sigma$

## Heat Engines: Conversion of Heat into Work

Consider curve  $c : \mathbb{R} \rightarrow \Sigma$  s.t.  $c$  generates vector field  $\dot{c} = \frac{\partial}{\partial \sigma}$  ( $U$  is suited for this).

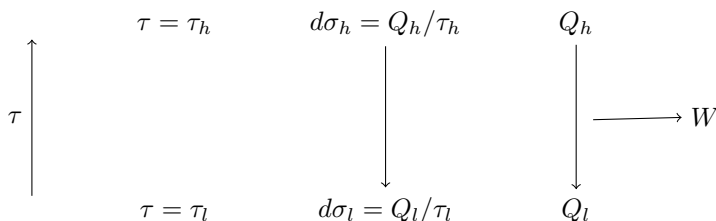
$$c(t) \in \Sigma$$

Act on this vector field  $\frac{\partial}{\partial \sigma} \in \mathfrak{X}(\Sigma)$  with  $Q$ , i.e.

$$Q\left(\frac{\partial}{\partial \sigma}\right) = \tau$$

This is what's meant when it's said "reversible heat transfer accompanying 1 unit of entropy is given by temperature  $\tau$ " [2].

Consider Figure 8.1 on page 229 of Kittel and Kroemer [2]. Roughly it looks like this:



But what's really going on?

Consider  $Q_h = dU$ , the initial heat input at high temperature (I'll show that later)  $\tau_h$ . Consider a curve  $c_0 \in \Sigma$  s.t.  $\dot{c}_0 = \dot{\sigma}_0 \frac{\partial}{\partial \sigma}$ . Then

$$Q_h(\dot{c}_0) = dU(\dot{c}_0) = \left(\frac{\partial U}{\partial \sigma}\right)_V = \tau \dot{\sigma}_0 \equiv \tau_h \dot{\sigma}_0$$

We can integrate the 1-form  $dU \in \Omega^1(\Sigma)$  for 2 reasons: mathematically, it is an exact form. Physically, we are considering a reversible process, passing through thermodynamic states of the system, starting with the system being in energy  $U_0$  and ending up with energy  $U_1$ , and all the energy states ( $\in \mathbb{R}$ ) in between.

$$\implies U_1 - U_0 = \int_0^1 \tau_h d\sigma$$

If this is conducted all at temperature  $\tau_h$  during the whole process, then  $U_1 - U_0 = \tau_h \int_0^1 d\sigma = \tau_h(\sigma_1 - \sigma_0)$ . It's in this case that  $d\sigma$  is an exact form and can be integrated over that curve  $c_0$ .

## Legendre transforms revisited

Let's recall 2 of our favorite thermodynamic potentials,  $U$ , and Helmholtz free energy  $F$ . They are related by Legendre transformations that transform 1 coordinate into its conjugate coordinate, somewhat like how the Legendre transform transforms that Lagrangian in canonical coordinates into a Hamiltonian written with the conjugate momentum. However, I do want to point out that, for Lagrangians and Hamiltonians, the Legendre transformation is a fiber derivative between tangent bundle to the cotangent bundle on the manifold. In our current case, we want a mundane Legendre transformation between convex function to another convex function, a coordinate transformation by a  $C^\infty$  function, not a morphism between vector spaces.

Recall  $F$ . It's defined as such:

$$F \equiv U - \tau \sigma, \text{ so}$$

$$dF = dU - \tau d\sigma - \sigma d\tau = -\sigma d\tau - p dV$$

Consider curve  $c : \mathbb{R} \rightarrow \Sigma$

$$c(t) \in \Sigma$$

Consider 2 curves that generate vector fields:

$$\dot{c} = \dot{\tau} \frac{\partial}{\partial \tau} \text{ or } \dot{c} = \dot{V} \frac{\partial}{\partial V}$$

Now, in general, mathematically,

$$dF = + \left( \frac{\partial F}{\partial \tau} \right)_V d\tau + \left( \frac{\partial F}{\partial V} \right)_\tau dV$$

Thus,

$$\begin{aligned} \left( \frac{\partial F}{\partial \tau} \right)_V &= -\sigma \\ \left( \frac{\partial F}{\partial V} \right)_\tau &= -p \end{aligned}$$

Maxwell relations are easily derived:

$$\frac{\partial^2 F}{\partial V \partial \tau} = \frac{\partial^2 F}{\partial \tau \partial V} \text{ so } \left( \frac{\partial \sigma}{\partial V} \right)_\tau = \left( \frac{\partial p}{\partial \tau} \right)_V$$

So-called natural coordinates for  $F$  are  $\tau, V$ . So  $\Sigma \ni (\tau, V)$  (i.e. after a Legendre transformation, the coordinates become  $(\tau, V)$  for each thermodynamic state.

Recall  $U$  as a thermodynamic potential. Using energy conservation and how  $Q$  is defined,

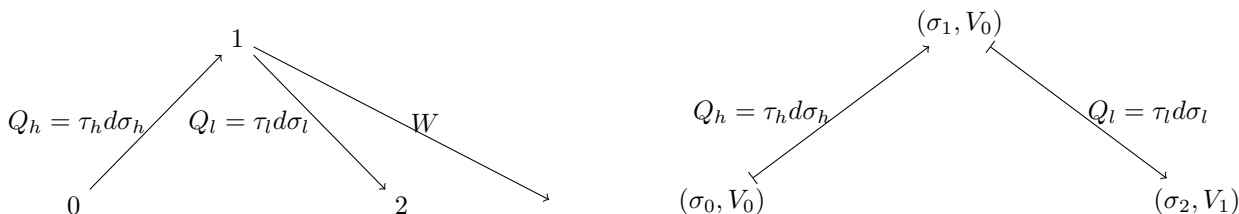
$$dU = Q + W = \tau d\sigma + p dV$$

**Correction (work sign).** The identity used in the rest of these notes is  $dU = \tau d\sigma - p dV$ . The plus sign on  $p dV$  in the display above disagrees with that identity and with the Carnot integral later in this section. See §33.3.

Natural coordinates are  $\sigma, V$  for  $U$ . So  $\Sigma \ni (\sigma, V)$ .

**heat engine**

ideal heat engine:

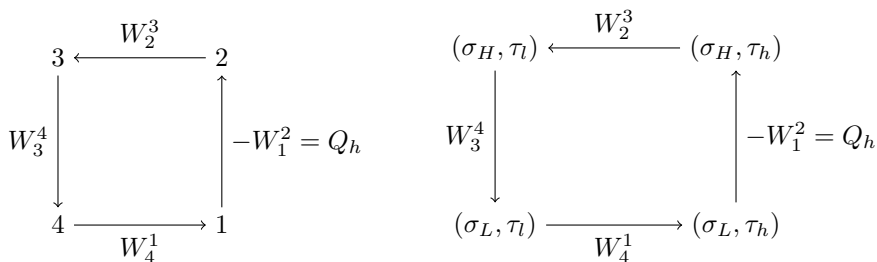


$$\begin{aligned} W + Q_l &= Q_h \text{ or } W = Q_h - Q_l \\ W &= Q_h - Q_l \leq \frac{\tau_h - \tau_l}{\tau_h} Q_h = \eta_C Q_h \end{aligned}$$

$$\sigma_l = \sigma_h \text{ so } \frac{Q_h}{\tau_h} = \frac{Q_l}{\tau_l}$$

**Carnot efficiency**  $\eta_C \equiv \frac{\tau_h - \tau_l}{\tau_h}$  is the ratio of the work generated to the heat added, in the reversible process.

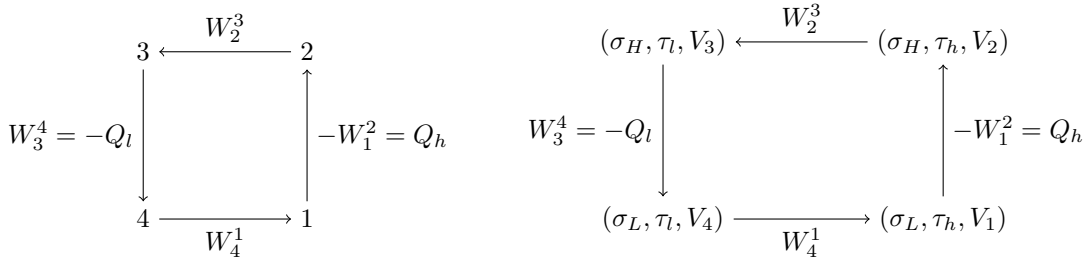
**Carnot cycle**



The total work is as such:  $\oint dU = 0$  for 2 reasons: mathematically, the integration of an exact 1-form around a closed curve is 0, and physically, we return the system back to its original state, as this is a reversible process.

$$\oint dU = 0 = \oint \tau d\sigma - \oint p dV \implies -\oint W = \oint \tau d\sigma = [\tau_h(\sigma_H - \sigma_L) + 0 + \tau_l(\sigma_L - \sigma_H) + 0] = (\tau_h - \tau_l)(\sigma_H - \sigma_L)$$

**Example: Carnot cycle for an ideal gas**



with

isothermal expansion  $Q_h = -W_1^2 = \int_1^2 p dV = N\tau_h \ln\left(\frac{V_2}{V_1}\right)$

adiabatic expansion  $W_2^3 = -\int_2^3 dU = U(\tau_h) - U(\tau_l) = C_V(\tau_h - \tau_l)$

$$\tau_l V_3^{\gamma-1} = \tau_h V_2^{\gamma-1} \text{ or } \frac{V_3}{V_2} = \left(\frac{\tau_h}{\tau_l}\right)^{\frac{1}{\gamma-1}}$$

isothermal compression  $-Q_l = W_3^4 = \int_3^4 -p dV = N\tau_l \ln\frac{V_3}{V_4}$

$$\frac{V_4}{V_1} = \left(\frac{\tau_h}{\tau_l}\right)^{\frac{1}{\gamma-1}}$$

adiabatic compression  $W_4^1 = C_V(\tau_h - \tau_l)$

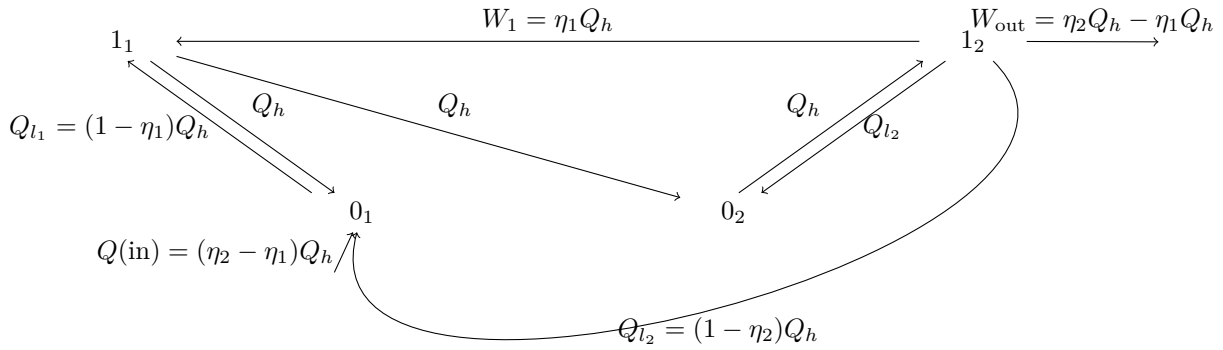
EY : 20150911 I don't have a good reason why  $C_V$  which is defined for constant  $V$ , that  $C_V \equiv \left(\frac{\partial U}{\partial \tau}\right)_V$ , can be used in the isentropic (i.e. adiabatic) expansion from  $2 \rightarrow 3$ .

**Correction (heat capacity on an adiabatic leg).** The subscript on  $C_V = (\partial U / \partial \tau)_V$  names the derivative. On the ideal-gas surface  $U = U(\tau)$ , so  $dU = C_V d\tau$  along the adiabatic leg as well. The argument is §33.1.

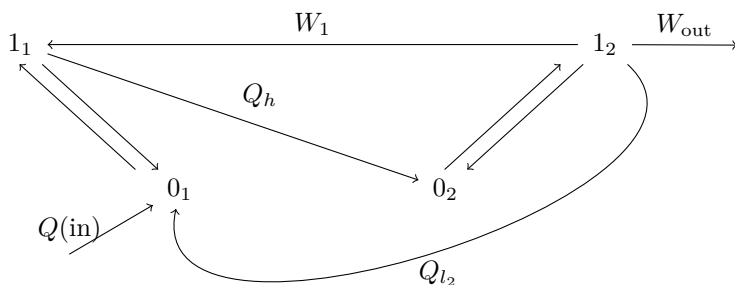
The total work done is

$$W = N(\tau_h - \tau_l) \ln\left(\frac{V_2}{V_1}\right)$$

**Energy Conversion and the Second Law of Thermodynamics**



$Q_{l_2}$  waste heat



So with  $Q(\text{in})$  heat in,  $W_{\text{out}}$  net work can be done. But that's a decrease in overall entropy. This violates the law of increasing entropy.

Define  $H = U + pV$ .  $H \in C^\infty(\Sigma)$ , where  $\Sigma$  is the manifold of equilibrium (and non-equilibrium) states of the system.

### Path Dependence of Heat and Work

Mathematically,  $Q$  and  $W$  are not necessarily *exact* 1-forms. So they are path-dependent.

EY : 20150911 That  $Q, W$  are not necessarily exact 1-forms would imply that  $\Sigma$  has some nontrivial, interesting *topological* features.

**Correction (topology).** Path dependence of the reversible heat is  $dQ = d\tau \wedge d\sigma \neq 0$ . That failure is local on an open set of  $\mathbb{R}^2$ . See §33.2.

## Heat and Work at Constant Temperature or Constant Pressure

### isothermal work

$$dU = W + Q = W + \tau d\sigma$$

$$F = U - \tau\sigma$$

$$dF = dU - \tau d\sigma - \sigma d\tau = W - \sigma d\tau$$

If  $d\tau = 0$ , on an isothermal curve,

$dF = W$ ,  $W$  becomes an exact 1-form, with potential function  $F$ , the Helmholtz free energy.

### isobaric heat and work

e.g. boiling of liquid. When liquid boils under atmospheric pressure, vapor pressure displacing atmospheric pressure does work against atmospheric pressure. isobaric process.

Consider this change of volume:

$dx = \frac{dV}{A}$ . Now

$p_{\text{eq}}$  = vapor pressure.

$$F = p_{\text{eq}}A = p_{\text{atm}}A \text{ (force equilibrium)}$$

$$W = -p_{\text{atm}}A dx = -p_{\text{atm}}dV$$

$\begin{array}{c} (\sigma_1, V_1) \\ \uparrow \\ \downarrow \\ (\sigma_0, V_0) \end{array}$

$W = -p_{\text{atm}}dV \equiv -pdV = -d(pV)$  is part of total work done on system.

If  $-d(pV) > 0$ , work provided by environment and is “free”.  
 If  $-d(pV) < 0$ , work delivered to environment and not extractable from system for other purposes.

$$W + d(pV) = dU - Q + d(pV) = dH - Q$$

Recall that for enthalpy  $H = U + pV$ ,

$$dH = dU + Vdp + pdV = dU - W + Vdp = \tau d\sigma + Vdp$$

$\sigma, p$  are natural coordinates of  $H$ .

$$dH - Q = W + d(pV)$$

An isobaric curve s.t.  $dp = 0$ ,

$$dH = Q + W + d(pV)$$

so

$$Q + W \text{ is an exact 1-form of } H - pV \implies d(H - pV) = W + Q.$$

2 classes of constant pressure processes:

(a)

$$W + d(pV) = 0$$

$$dH = Q$$

e.g. liquid evaporation from open vessel, because no effective work is done.

heat of evaporation is enthalpy difference between vapor phase and liquid phase

(b) constant temperature and constant pressure.

$$G = F + pV = U - \tau\sigma + pV$$

$$dG = dF + Vdp + pdV = dU - \tau d\sigma - \sigma d\tau + Vdp + pdV = Vdp - \sigma d\tau$$

$$dG = W - \sigma d\tau + d(pV) = W + d(pV) - \sigma d\tau$$

with natural variables are  $p, \tau$

at constant temperature,  $W + d(pV)$  is exact 1-form,  $dG$

## Problems

### Solution 1. Heat pump.

(a) For a heat pump,

$$\text{input: } \sigma_h = \frac{Q_h}{\tau_h}$$

$$\text{output: } \sigma_l = \frac{Q_l}{\tau_l}$$

Reversible condition:  $\sigma_h = \sigma_l = \frac{Q_h}{\tau_h} = \frac{Q_l}{\tau_l}$  so that  $Q_h = \frac{\tau_h}{\tau_l} Q_l$ .

$Q_h - Q_l = Q_h - \frac{\tau_l}{\tau_h} Q_h = \frac{\tau_h - \tau_l}{\tau_h} Q_h$  net heat inputted to pump heat.

Thus,

$$\frac{W}{Q_h} = \eta_c = \frac{\tau_h - \tau_l}{\tau_h}$$

If heat pump is not reversible,  $\sigma_h > \sigma_l$ , so that  $\frac{Q_h}{\tau_h} > \frac{Q_l}{\tau_l}$  or  $\frac{\tau_l}{\tau_h} Q_h > Q_l$ ,

$$\frac{W}{Q_h} = \frac{Q_h - Q_l}{Q_h} < \frac{Q_h - \frac{\tau_l}{\tau_h} Q_h}{Q_h} = \eta_{c, ideal}$$

(b)  $Q_h$  = electricity consumed by reversible heat pump.

Carnot engine:  $W = (\tau_{hh} - \tau_l)(\sigma_{hh} - \sigma_l)$ , with  $\sigma_{hh} = \frac{Q_{hh}}{\tau_{hh}}$ , and  $\sigma_l = \frac{Q_l}{\tau_l}$

Condition that electricity consumed by reversible heat pump:

$$W = (\tau_{hh} - \tau_l) \left( \frac{Q_{hh}}{\tau_{hh}} - \frac{Q_l}{\tau_l} \right) = Q_h$$

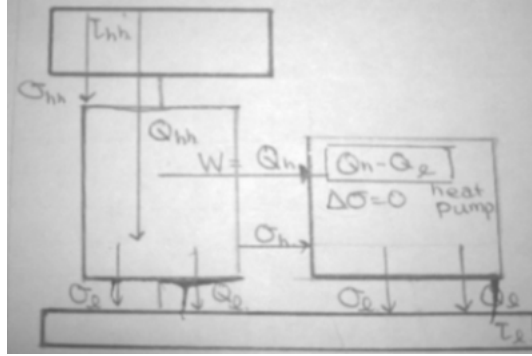
Note we let  $\sigma_l = \frac{Q_l}{\tau_l}$  since both heat pump and Carnot engine are reversible.

$$\Rightarrow \left( \frac{Q_{hh}}{\tau_{hh}} - \frac{Q_h}{\tau_h} \right) = \frac{Q_h}{\tau_{hh} - \tau_l} \Rightarrow \frac{Q_{hh}}{\tau_{hh}} = Q_h \left( \frac{1}{\tau_{hh} - \tau_l} + \frac{1}{\tau_h} \right) \Rightarrow \boxed{\frac{Q_{hh}}{Q_h} = \frac{\tau_{hh}(\tau_h + \tau_{hh} - \tau_l)}{\tau_h(\tau_{hh} - \tau_l)}}$$

For  $T_{hh} = 600\text{ K}$ ,  $T_h = 300\text{ K}$ ,  $T_l = 270\text{ K}$ ,

$$\frac{Q_{hh}}{Q_h} = \frac{600(300 + 600 - 270)}{300(600 - 270)} = 3.82$$

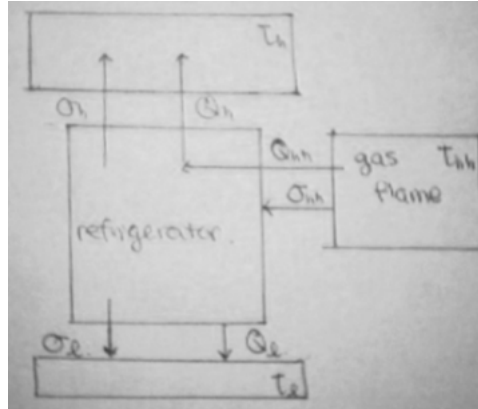
(c) See Figure (17).



Problem 8.1(c)

**Solution 2. Absorption refrigerator.**

(a) See Figure (17).



Problem 8.2(a)

(b) Given  $\tau_{hh} > \tau_h$ ,

by energy conservation:  $Q_{hh} + Q_l - Q_h = 0$

reversible refrigerator:  $\sigma_{hh} + \sigma_l - \sigma_h = 0$ ,

$$\Rightarrow \frac{Q_{hh}}{\tau_{hh}} + \frac{Q_l}{\tau_l} - \frac{Q_h}{\tau_h} = 0$$

$$\frac{Q_{hh}}{\tau_{hh}} + \frac{Q_l}{\tau_l} = \frac{Q_{hh} + Q_l}{\tau_h} \text{ or } Q_{hh} \left( \frac{1}{\tau_{hh}} - \frac{1}{\tau_h} \right) = Q_l \left( \frac{1}{\tau_h} - \frac{1}{\tau_l} \right)$$

$$\frac{Q_l}{Q_{hh}} = \left( \frac{1}{\tau_{hh}} - \frac{1}{\tau_h} \right) / \left( \frac{1}{\tau_h} - \frac{1}{\tau_l} \right) = \left( \frac{\tau_h - \tau_{hh}}{\tau_h \tau_{hh}} \right) / \left( \frac{\tau_l - \tau_h}{\tau_h \tau_l} \right) = \boxed{\left( \frac{\tau_{hh} - \tau_h}{\tau_h - \tau_l} \right) \left( \frac{\tau_l}{\tau_{hh}} \right) = \frac{Q_l}{Q_{hh}}}$$

Note that  $Q_l - Q_h = Q_l - (Q_{hh} + Q_l) = -Q_{hh}$ ; we've removed  $Q_{hh}$  heat from refrigerator's inside.

**Solution 3. Photon Carnot engine.** Recall, photons are relativistic:  $\epsilon = pc$ . Recall  $p = \frac{\hbar}{i} \nabla$ .  $\implies \epsilon_s = pc = \hbar k_s c = \left( \frac{n_s \pi}{L} \right) \hbar c$ .

Recalling that there are 2 polarization states for a photon in 3-dim. space,

$$\begin{aligned} U &= (2) \left( \frac{1}{8} \right) (4\pi) \int_0^\infty n^2 dn \left( \frac{n\pi}{L} \hbar c \right) e^{-\frac{n\pi}{L} \hbar c / \tau} = \frac{\pi^2 \hbar c}{L} \int_0^\infty n^3 dn \exp\left( \frac{-\pi \hbar c}{L\tau} n \right) = \\ &= \left( \frac{\pi^2 \hbar c}{L} \right) \left\{ n^3 \exp\left( \frac{-\pi \hbar c}{L\tau} n \right) \left( \frac{L\tau}{-\pi \hbar c} \right) \Big|_0^\infty - \int_0^\infty 3n^2 \exp\left( \frac{-\pi \hbar c}{L\tau} n \right) \left( \frac{L\tau}{-\pi \hbar c} \right) dn \right\} = \\ &= \left( \frac{\pi^2 \hbar c}{L} \right) \left\{ (-1) \left( \frac{-L\tau}{\pi \hbar c} \right) 3 \int_0^\infty n^2 \exp\left( \frac{-\pi \hbar c}{L\tau} n \right) dn \right\} = \\ &= \left( \frac{\pi^2 \hbar c}{L} \right) \left\{ (-1) \left( \frac{-L\tau}{\pi \hbar c} \right) 3 \left\{ n^2 \exp\left( \frac{-\pi \hbar c}{L\tau} n \right) \left( \frac{L\tau}{-\pi \hbar c} \right) \Big|_0^\infty - \int_0^\infty 2n \exp\left( \frac{-\pi \hbar c}{L\tau} n \right) \left( \frac{L\tau}{-\pi \hbar c} \right) dn \right\} \right\} = \\ &= \left( \frac{\pi^2 \hbar c}{L} \right) (-1)^2 \left( \frac{-L\tau}{\pi \hbar c} \right)^2 3(2) \int_0^\infty n \exp\left( \frac{-\pi \hbar c}{L\tau} n \right) \left( \frac{L\tau}{-\pi \hbar c} \right) dn = \\ &= \left( \frac{\pi^2 \hbar c}{L} \right) (-1)^3 \left( \frac{-L\tau}{\pi \hbar c} \right)^3 3(2)(1) \int_0^\infty \exp\left( \frac{-\pi \hbar c}{L\tau} n \right) dn = \left( \frac{\pi^2 \hbar c}{L} \right) (-1)^3 \left( \frac{-L\tau}{\pi \hbar c} \right)^4 3(2)1 \left( \exp\left( \frac{-\pi \hbar c}{L\tau} n \right) \right) \Big|_0^\infty = \\ &= 6 \left( \frac{L}{\pi^2 \hbar c} \right)^3 \tau^4 = \boxed{6 \frac{V}{(\pi^2 \hbar c)^3} \tau^4 = U} \end{aligned}$$

To get the entropy, recall,  $\left( \frac{\partial \sigma}{\partial U} \right)_V = \frac{1}{\tau}$ , and using this is usually the *most direct way to obtain entropy*.

$$\implies d\sigma = \int \frac{dU}{\tau} = \int \frac{6V}{(\pi^2 \hbar^2 c)^3} 4\tau^3 \frac{d\tau}{\tau} = \frac{6V}{(\pi^2 \hbar^2 c)^3} \frac{1}{3} \tau^3 \implies \boxed{\sigma(\tau) = \frac{8V\tau^3}{(\pi^2 \hbar^2 c)^3}}$$

Consider

Isothermal expansion: Helmholtz free energy  $F$  is needed.

$$F = U - \tau \sigma = \frac{6V}{(\pi^2 \hbar^2 c)^3} \tau^4 - \tau \frac{8V\tau^3}{(\pi^2 \hbar^2 c)^3} = -\frac{2V\tau^4}{(\pi^2 \hbar^2 c)^3}$$

Then

$$\begin{aligned} p &= - \left( \frac{\partial F}{\partial V} \right)_{\tau, N} = \frac{2\tau^4}{(\pi^2 \hbar^2 c)^3} & \sigma_{12} &= \frac{8\tau_h^3}{(\pi^2 \hbar^2 c)^3} (V_2 - V_1) \\ W_{12} &= p(V_2 - V_1) = \frac{2\tau_h^4}{(\pi^2 \hbar^2 c)^3} (V_2 - V_1) & \Delta Q_{12} &= \frac{\tau}{\Delta} \sigma = \frac{8\tau_h^4}{(\pi^2 \hbar^2 c)^3} (V_2 - V_1) \end{aligned}$$

Isentropic expansion:  $\implies V_2 \tau_h^3 = V_3 \tau_l^3$  or  $V_3 = V_2 \left( \frac{\tau_h}{\tau_l} \right)^3$ .

So for this isentropic process,  $V_2 \tau_h^3 = V \tau^3$ ,

$$\begin{aligned} U &= \frac{6V}{(\pi^2 \hbar^2 c)^3} \left( \frac{V_2 \tau_h^3}{V} \right)^{4/3} = \frac{6(V_2 \tau_h^3)^{4/3}}{(\pi^2 \hbar^2 c)^3} V^{-1/3} \\ p &= \frac{-\partial U}{\partial V} = -\frac{6(V_2 \tau_h^3)^{4/3}}{(\pi^2 \hbar^2 c)^3} \left( \frac{-1}{3} V^{-4/3} \right) = \frac{2(V_2 \tau_h^3)^{4/3}}{(\pi^2 \hbar^2 c)^3} V^{-4/3} \end{aligned}$$

$$W_{23} = \int p dV = \int \frac{2(V_2 \tau_h^3)^{4/3}}{(\pi^2 \hbar^2 c)^3} V^{-4/3} dV = \frac{2(V_2 \tau_h^3)^{4/3}}{(\pi^2 \hbar^2 c)^3} (-3V^{-1/3}) \Big|_{V_2}^{V_3} = \frac{-6(V_2 \tau_h^3)^{4/3}}{(\pi^2 \hbar^2 c)^3} \left( \frac{1}{V_3^{1/3}} - \frac{1}{V_2^{1/3}} \right) =$$

$$= \frac{6V_2 \tau_h^4}{(\pi^2 \hbar^2 c)^3} \left( 1 - \frac{\tau_l}{\tau_h} \right)$$

$$\text{Isothermal compression: } W_{34} = \frac{2\tau_l^4}{(\pi^2 \hbar^2 c)^3} (V_4 - V_3) = \frac{2\tau_h^3 \tau_l}{(\pi^2 \hbar^2 c)^3} (V_1 - V_2).$$

$$\sigma_{34} = \frac{8\tau_l^3}{(\pi^2 \hbar c)^3} (V_4 - V_3) = \frac{8\tau_h^3}{(\pi^2 \hbar c)^3} (V_1 - V_2).$$

$$\text{Isentropic compression: } V_4 \tau_l^3 = V_1 \tau_h^3 \text{ or } V_4 = V_1 \left( \frac{\tau_h}{\tau_l} \right)^3.$$

$$W_{41} = \frac{6(V_4 \tau_l^3)^{4/3}}{(\pi^2 \hbar^2 c)^3} \left( \frac{1}{V_4^{1/3}} - \frac{1}{V_1^{1/3}} \right) = \frac{-6V_1 \tau_h^4}{(\pi^2 \hbar^2 c)^3} \left( 1 - \frac{\tau_l}{\tau_h} \right)$$

$$\Delta W = \frac{2\tau_h^4}{(\pi^2 \hbar^2 c)^3} (V_2 - V_1) + \frac{6V_2 \tau_h^4}{(\pi^2 \hbar^2 c)^3} \left( 1 - \frac{\tau_l}{\tau_h} \right) + \frac{2\tau_h^3 \tau_l}{(\pi^2 \hbar^2 c)^3} (V_1 - V_2) + \frac{-6V_1 \tau_h^4}{(\pi^2 \hbar^2 c)^3} \left( 1 - \frac{\tau_l}{\tau_h} \right) = \frac{8\tau_h^4 (V_2 - V_1)}{(\pi^2 \hbar^2 c)^3} \left( 1 - \frac{\tau_l}{\tau_h} \right)$$

$$Q_h = \frac{8\tau_h^4 (V_2 - V_1)}{(\pi^2 \hbar c)^3} \implies \boxed{\frac{\Delta W}{Q_h} = 1 - \frac{\tau_l}{\tau_h}}$$

**Solution 4. Heat engine-refrigerator cascade.** Consider the heat engine as a Carnot cycle.

$$W + W_r = (\tau_h - \tau_l) \sigma_h$$

where  $W_r$  = work consumed by refrigerator.

$$\sigma_h = \frac{Q_h}{\tau_h} = \frac{Q_l}{\tau_l} = \sigma_l$$

This must be true for *any* heat engine undergoing Carnot cycle; furthermore, we can say it's the most efficient heat engine possible.

reversible refrigerator:  $Q_L + W_r = Q_H$ , (by  $E$ -conservation.)

$$\sigma_L = \sigma_H = \frac{Q_L}{\tau_L} = \frac{Q_H}{\tau_H}, \text{ (by reversible condition)}$$

Note,  $Q_l$  is energy transfer from heat engine to  $\tau_l$  reservoir.  $Q_L$  is energy transfer from  $\tau_l$  reservoir to refrigerator.  $Q_L \geq Q_l$ , otherwise, no cooling, no thermal energy extracted from  $\tau_l$  reservoir to lower its temperature.  $Q_L = Q_l$  at equilibrium; no further cooling,  $\tau_r$  reached.

Note that  $\tau_l$  is given as the environmental temperature. Assume refrigerator throws out  $Q_H$  heat into the environment.  $\rightarrow \tau_H = \tau_l$ . Since  $Q_L$  heat input into refrigerator from a  $\tau_l$  reservoir now lowered to  $\tau_r$ ,  $\tau_l \rightarrow \tau_r$ .

$$W_r = \frac{\tau_l}{\tau_r} Q_L - Q_L = \left( \frac{\tau_l}{\tau_r} - 1 \right) Q_L$$

since for a reversible refrigerator,  $\sigma_L = \sigma_H = \frac{Q_L}{\tau_L} = \frac{Q_H}{\tau_H}$ .

$$\implies \frac{W}{Q_h} = \left( 1 - \frac{\tau_r}{\tau_h} \right) - \left( \frac{\tau_l}{\tau_r} - 1 \right) \frac{Q_L}{Q_h} = \left( 1 - \frac{\tau_r}{\tau_h} \right) - \left( \frac{\tau_l}{\tau_r} - 1 \right) \left( \frac{\tau_r}{\tau_h} \right) = \boxed{1 - \frac{\tau_l}{\tau_h}}$$

Combinations of reversible systems = reversible system.

**Solution 5. Thermal pollution.** Given  $T_l = 20^\circ C$   
 $T_h = 500^\circ C$ . Consider a Carnot cycle.

$$W = (\tau_h - \tau_l) \sigma_l = (\tau_h - \tau_l) \frac{Q_l}{\tau_l} = \left( \frac{\tau_h}{\tau_l} - 1 \right) Q_l = \left( \frac{500}{20} - 1 \right) 1500 \text{ MW} = \boxed{36000 \text{ MW}}$$

If improvements in hot-steam technology would permit raising  $T_h$  by  $100^\circ C$ ,

$$W = \left( \frac{600}{20} - 1 \right) 1500 \text{ MW} = (29)(1500 \text{ MW}) = 43500 \text{ MW}$$

There was a 17.2% increase in output.

**Solution 6. Room air conditioner.**

(a)

$$\begin{aligned}
W &= (\tau_h - \tau_l) \frac{Q_l}{\tau_l} = \left( \frac{\tau_h}{\tau_l} - 1 \right) Q_l \\
P &= \left( \frac{\tau_h}{\tau_l} - 1 \right) \frac{dQ_l}{dt} = \left( \frac{\tau_h}{\tau_l} - 1 \right) A(\tau_h - \tau_l) \Rightarrow \frac{P}{A} \tau_l = (\tau_h - \tau_l)(\tau_h - \tau_l) = \tau_h^2 - 2\tau_h \tau_l + \tau_l^2 \\
&\Rightarrow \tau_l^2 - 2\tau_h \tau_l - \frac{P}{A} \tau_l + \tau_h^2 = 0 \\
&\boxed{\tau_l = \tau_h + \frac{P}{2A} - \sqrt{(\tau_h + \frac{P}{2A})^2 - \tau_h^2}}
\end{aligned}$$

(b) For  $T_l = 17^\circ C = 290 K$ ,  $T_h = 310 K$ ,

$$A = \frac{P \tau_l}{(\tau_h - \tau_l)^2} = \frac{(2 kW)(290 K)}{(310 - 290)^2} = \frac{580 \times 10^3 W}{400 K} = \boxed{1450 \frac{W}{K}}$$

**Solution 7. Light bulb in a refrigerator**

Carnot refrigerator draws 100 W. For any Carnot cycle,

$$W = (\tau_h - \tau_l) \frac{Q_l}{\tau_l} = \left( \frac{\tau_h}{\tau_l} - 1 \right) Q_l = \left( 1 - \frac{\tau_l}{\tau_h} \right) Q_h$$

Carnot refrigerator expels  $Q_h$  thermal energy to hot  $\tau_h$  environment and *inputs*  $Q_l$  thermal energy from  $\tau_l$  reservoir.

$$Q_l + W = Q_h$$

Work  $W$  must be drawn by Carnot refrigerator to do work. Suppose Carnot cycle part of the refrigerator must input in heat from light bulb to cool down its inside, i.e. consider Carnot refrigerator in *equilibrium with light bulb*, now *inputting in heat from light bulb*  $Q_{ext}$ , and drawing in work to expend out  $Q_h$  thermal energy into the environment.

$$\Rightarrow \boxed{Q_{ext} = Q_l}$$

 $\dot{W} = \dot{Q}_l$  in this case, so

$$\begin{aligned}
\left( \frac{\tau_h}{\tau_l} - 1 \right) Q_l - Q_l &= 0 \text{ or } \left( \frac{\tau_h}{\tau_l} - 2 \right) Q_l = 0 \\
\Rightarrow \boxed{\tau_l = \frac{\tau_h}{2}} &= \frac{300 K}{2} = 150 K
\end{aligned}$$

**Solution 8. Geothermal energy.**Given  $\Delta Q_h = -MCdT_h$ , $T_l$  lower reservoir temperature stays constant.  $\tau_h$  decreasing,  $d\tau_h < 0$ .

$$\begin{aligned}
\Delta W &= (\tau_h - \tau_l) \frac{\Delta Q_h}{\tau_h} = \left( 1 - \frac{\tau_l}{\tau_h} \right) (-MC) \frac{d\tau_h}{k_B} \\
\Rightarrow W &= - \left( \frac{MC}{k_B} \right) (\tau_h - \tau_l \ln \tau_h) \Big|_{\tau_i}^{\tau_f} = - \left( \frac{MC}{k_B} \right) \left( \tau_l \ln \left( \frac{\tau_i}{\tau_f} \right) - (\tau_i - \tau_f) \right)
\end{aligned}$$

For  $M = 10^{17} g$ ,  $C = 1 J/g \cdot K$ ,  $T_l = 20^\circ C = 293 K$ ,  $T_i = 600^\circ C = 873 K$ ,  $T_f = 110^\circ C = 383 K$ 

$$W = 2.486 \times 10^{19} J$$

Note that  $10^{14} kWh = 10^{17} \frac{J}{s} \cdot h \left( \frac{3600 sec}{1 h} \right) = 3.6 \times 10^{20} J$ .**Solution 9. Cooling of nonmetallic solid to  $T = 0$ .** Recall that  $C = aT^3 = \left( \frac{\partial U}{\partial T} \right)_V$ . Then  $dQ_l = aT_l^3 dT_l$ . Now  $d\tau_l < 0$  since  $\tau_l$  decreasing.For the refrigerator:  $Q_l + W = Q_h$ .

$$\begin{aligned}
dW &= (\tau_h - \tau_l) \frac{Q_l}{\tau_l} = - \left( \frac{\tau_h}{\tau_l} - 1 \right) (a\tau_l^3 d\tau_l) \left( \frac{1}{k_B^4} \right) = \frac{-a}{k_B^4} \left( \frac{\tau_h}{\tau_l} - 1 \right) \tau_l^3 d\tau_l = \frac{-a}{k_B^4} (\tau_h \tau_l^2 - \tau_l^3) d\tau_l \\
W &= \frac{-a}{k_B^4} \left( \tau_h \frac{1}{3} \tau_l^3 - \frac{1}{4} \tau_l^4 \right) \Big|_{\tau_h}^0 = \boxed{\frac{aT_h^3}{12k_B} = W}
\end{aligned}$$

## 18 Cryogenics

The 2008 heading was empty. The process written here is cooling by adiabatic demagnetization of the two-state paramagnet of §2. The energy is the expectation of the spin Hamiltonian, not a Legendre transform that moves the field term to the other side of the identity.

### 18.1 Canonical free energy of $N$ independent moments

Each element has energies  $-mB$  and  $+mB$ . The single-element partition function is

$$Z_1 = e^{mB/\tau} + e^{-mB/\tau} = 2 \cosh x, \quad x = \frac{mB}{\tau}. \quad (67)$$

The  $N$  elements are fixed on sites and are distinguishable by those sites, so there is no  $1/N!$ . The Helmholtz free energy is

$$F(\tau, B) = -N\tau \ln Z_1 = -N\tau \ln(2 \cosh x). \quad (68)$$

Hold  $B$  fixed and differentiate in  $\tau$ . Then  $\partial x / \partial \tau = -x/\tau$ , and

$$\begin{aligned} \frac{\partial}{\partial \tau} (\tau \ln(2 \cosh x)) &= \ln(2 \cosh x) + \tau \cdot \tanh x \cdot \left(-\frac{x}{\tau}\right) \\ &= \ln(2 \cosh x) - x \tanh x. \end{aligned} \quad (69)$$

Therefore

$$\sigma = -\left(\frac{\partial F}{\partial \tau}\right)_B = N[\ln(2 \cosh x) - x \tanh x]. \quad (70)$$

At  $B = 0$ , one has  $x = 0$  and  $\sigma = N \ln 2$ , which is  $\ln(2^N)$ . The mean energy per element is  $-\partial \ln Z_1 / \partial \beta$  with  $\beta = 1/\tau$ :

$$U = -NmB \tanh x = -N\tau x \tanh x. \quad (71)$$

The total moment along the field, defined by  $U = -\mathcal{M}B$ , is

$$\mathcal{M} = Nm \tanh x. \quad (72)$$

Check  $F = U - \tau\sigma$ :

$$\begin{aligned} U - \tau\sigma &= -N\tau x \tanh x - N\tau \ln(2 \cosh x) + N\tau x \tanh x \\ &= F. \end{aligned} \quad (73)$$

The  $\tau$  derivative at fixed  $B$  was arranged so that  $(\partial F / \partial \tau)_B = -\sigma$ . The  $B$  derivative at fixed  $\tau$  is

$$\left(\frac{\partial F}{\partial B}\right)_\tau = -N\tau \cdot \tanh x \cdot \frac{m}{\tau} = -\mathcal{M}, \quad (74)$$

because  $\partial x / \partial B = m/\tau$ . Thus

$$dF = -\sigma d\tau - \mathcal{M} dB. \quad (75)$$

From  $U = F + \tau\sigma$ ,

$$\begin{aligned} dU &= dF + \sigma d\tau + \tau d\sigma \\ &= -\mathcal{M} dB + \tau d\sigma. \end{aligned} \quad (76)$$

Equation (76) is the thermodynamic identity for this choice of  $U$ . A convention that uses  $U' = U + \mathcal{M}B$  moves the work term to  $+B d\mathcal{M}$ . That  $U'$  is not the expectation of  $-mB \sum_i \epsilon_i$ .

For  $|x| \ll 1$ ,  $\tanh x = x + O(x^3)$ , so

$$U = -NmB(x + O(x^3)) = -\frac{Nm^2 B^2}{\tau} + O(x^3) N\tau. \quad (77)$$

The leading term is (22), from the microcanonical Gaussian. The two counts agree in the regime where both apply.

## 18.2 Isothermal magnetization

Hold  $\tau = \tau_i$  and increase  $B$  from 0 to  $B_i$ . Then  $x$  goes from 0 to  $x_i = mB_i/\tau_i$ . The entropy change of the spins is

$$\Delta\sigma = N[\ln(\cosh x_i) - x_i \tanh x_i]. \quad (78)$$

The subtraction of  $N \ln 2$  produced  $\ln(\cosh x_i) = \ln(2 \cosh x_i) - \ln 2$ . Define  $f(x) = \ln(\cosh x) - x \tanh x$ . Then  $f(0) = 0$  and

$$f'(x) = \tanh x - \tanh x - x \operatorname{sech}^2 x = -x \operatorname{sech}^2 x. \quad (79)$$

For  $x > 0$ ,  $f'(x) < 0$ , so  $f(x) < 0$  and  $\Delta\sigma < 0$ . The heat entering the spin system along this reversible isotherm is

$$Q = \tau_i \Delta\sigma < 0. \quad (80)$$

The bath receives  $-Q$ .

## 18.3 Adiabatic demagnetization

After the magnetization, isolate the sample and reduce the field. Two cases use (70).

If the spins are the whole system, isolation at fixed entropy means  $\sigma(x)$  is constant. Equation (70) depends on  $\tau$  and  $B$  only through  $x = mB/\tau$ . Constant  $\sigma$  therefore means constant  $x$ , hence

$$\frac{\tau}{B} = \text{constant} \quad (81)$$

along that path. Reducing  $B$  reduces  $\tau$  in the same ratio. Reaching  $B = 0$  at this frozen entropy would require  $\tau \rightarrow 0$ , because  $\sigma < N \ln 2$  whenever  $x \neq 0$ , while  $\sigma(B = 0) = N \ln 2$  at every finite  $\tau$ .

A lattice in the Debye regime has heat capacity  $C_V = A\tau^3$  with  $A$  constant, at temperatures far below the Debye temperature. Then

$$\left( \frac{\partial \sigma_{\text{latt}}}{\partial \tau} \right)_V = \frac{C_V}{\tau} = A\tau^2, \quad \sigma_{\text{latt}} = \frac{A}{3}\tau^3, \quad (82)$$

where the integration constant is zero because  $\sigma_{\text{latt}} \rightarrow 0$  as  $\tau \rightarrow 0$ . Assume the spins and the lattice share one  $\tau$  during the demagnetization. That is an equilibrium assumption. If the spins decouple from the lattice, (81) applies to the spins and the lattice entropy stays where it was. When they do share  $\tau$ , isolation conserves the sum

$$N[\ln(2 \cosh x) - x \tanh x] + \frac{A}{3}\tau^3 = \text{constant}, \quad x = \frac{mB}{\tau}. \quad (83)$$

The constant is fixed by the state at the end of the isothermal step:  $x_i$  and  $\tau_i$ . Lowering  $B$  at fixed  $\tau$  would increase the spin entropy toward  $N \ln 2$ . With the sum fixed, that increase is paid for by a decrease of  $\sigma_{\text{latt}}$ , which by (82) means a lower  $\tau$ . Equation (83) is the equation for the final temperature once a final field and a value of  $A$  are given. No particular  $A$  is assumed here.

The refrigerator cycle is then: magnetize in contact with the bath, isolate, demagnetize. The heat  $-Q$  of (80) left the sample during the contact step. The later drop in  $\tau$  is the useful cooling. Repeating the cycle requires a heat switch, because the sample has to be put back into contact with a bath at the new, lower temperature if another isothermal magnetization is to dump entropy again. The switch is a thermal path that can be opened and closed. The entropy accounting above does not model the switch; it states where the contact is required. *2015 manuscript.*

## 19 Thermodynamic potentials

*In the 2015 manuscript these subsections stood in the section Reactions, after the Gibbs sum.*

### 19.1 Gibbs Free Energy

cf. Section “Gibbs Free Energy” of Ch. 9 Gibbs Free Energy and Chemical Reactions from Kittel and Kroemer (1980) [2]

Recall

$$F = U - \tau\sigma$$

$$dF = dU - \tau d\sigma - \sigma d\tau = -\sigma d\tau - p dV + \mu dN$$

where the *generalized* thermodynamic identity

$$dU = Q + W + \mu dN = \tau d\sigma - p dV + \mu dN$$

was used.

Now

$$\begin{aligned} G &\equiv U - \tau\sigma + pV = F + pV \\ dG &= dU - \tau d\sigma - \sigma d\tau + p dV + V dp = \mu dN - \sigma d\tau + V dp \end{aligned}$$

and so  $G = G(N, \tau, p)$ .

Now

$$\begin{aligned} \sigma &= \frac{1}{\tau} [U + pV - G] \\ \implies d\sigma &= \frac{-1}{\tau^2} d\tau (U + pV - G) + \frac{1}{\tau} (dU + V dp + p dV - dG) \end{aligned}$$

Consider a curve  $\gamma : \mathbb{R} \rightarrow \Sigma$  s.t.  $dU(\dot{\gamma}) = dp(\dot{\gamma}) = dV(\dot{\gamma}) = 0$  corresponding to minimized internal energy, constant pressure, and constant volume. Then

$$d\sigma(\dot{\gamma}) = 0 + 0 + 0 + -dG(\dot{\gamma}) \geq 0$$

and so  $dG(\dot{\gamma}) \leq 0$ , so  $G$  must be a minimum, because  $d\sigma(\dot{\gamma}) \geq 0$  represents the fact that entropy must always increase for any thermodynamic process.  $G$  must move to a minimum at equilibrium, i.e. “For any irreversible change taking place entirely within  $\mathcal{S}$  will increase  $\sigma$  and thus decrease  $G_{\mathcal{S}}$ ” [2] for  $\mathcal{S}$  signifying the system.

$\tau, p$  are **intensive quantities**; they do not change value when 2 identical systems are put together.

$U, \sigma, V, G$  are linear in  $N$ ; their values doubles when 2 identical systems are put together; apart from interface effects

$U, \sigma, V, N, G$  are **extensive quantities**.

$$G = N\varphi(\tau, p)$$

$$\left( \frac{\partial G}{\partial N} \right)_{p, \tau} = \varphi(\tau, p) = \mu \implies G(N, \tau, p) = N\mu(\tau, p)$$

If more than 1 chemical species present,

$$\boxed{\begin{aligned} G &= \sum_j N_j \mu_j \\ \tau d\sigma &= dU + p dV - \sum_j \mu_j dN_j \\ dG &= \sum_j \mu_j dN_j - \sigma d\tau + V dp \end{aligned}} \quad (84)$$

## 19.2 Summary of Thermodynamic Potentials

With Eq. 84, I will summarize the thermodynamic potentials we have, obtained through Legendre transformations:

$$Q = dU - W = dU + p dV$$

For curve or thermodynamic process in  $\Sigma$ ,  $\gamma$ , s.t.  $dp(\dot{\gamma}) = 0$  (thermodynamic process at *constant pressure*),

$$\begin{aligned} Q(\dot{\gamma}) &= dH(\dot{\gamma}) - V dp(\dot{\gamma}) = dH(\dot{\gamma}) - 0 \\ \implies \int_{\gamma} Q(\dot{\gamma}) &= \int dH(\dot{\gamma}) = H_f - H_i \quad \text{for constant } p \end{aligned}$$

with

$$\begin{aligned} H &= U + pV \\ dH &= dU + p dV + V dp = Q + V dp \end{aligned}$$

For the thermodynamic potential

$$\begin{aligned}
dU &= \tau d\sigma - p dV \\
\boxed{U &= U(\sigma, V)} \\
dU &= \tau d\sigma - p dV + \sum_j \mu_j dN_j \quad U = U(\sigma, V, N_j) \\
F &= U - \tau \sigma \\
dF &= dU - \tau d\sigma - \sigma d\tau = -p dV - \sigma d\tau \implies \boxed{F = F(\tau, V)} \\
dF &= Q + W - d(\tau \sigma) \\
G &= F + pV \\
dG &= -\sigma d\tau + V dp \implies \boxed{G = G(\tau, p)} \\
dG &= (Q - p dV) + p dV + V dp - d(\tau \sigma) = Q + V dp - d(\tau \sigma)
\end{aligned} \tag{85}$$

Let  $\Delta H := H_f - H_i$ .

If  $\Delta H < 0$ ,  $\int_\gamma Q(\dot{\gamma}) < 0$ , heat transferred to surroundings: reaction is *exothermic*.

If  $\Delta H > 0$ ,  $\int_\gamma Q(\dot{\gamma}) > 0$ , heat transferred to system: reaction is *endothermic*. Now  $dH = Q + V dp$ . Suppose  $Q = C_p(\tau) d\tau$ . then for constant pressure,  $dp(\dot{\gamma}) = 0$  and so

$$\int_\gamma dH = \Delta H = \int_\gamma C_p(\tau) d\tau$$

For constant volume,  $dV(\dot{\gamma}_V) = 0$ , and so

$$Q(\dot{\gamma}_V) \equiv Q_V = dU(\dot{\gamma}_V)$$

### 19.2.1 Relations between derivatives of thermodynamic quantities

For an adiabatic process,  $Q = 0$ .

For  $Q = Q(\tau, V)$ ,

$$Q = \tau d\sigma = C_V d\tau + \left( \frac{\partial Q}{\partial V} \right)_\tau dV \tag{86}$$

and  $\sigma = \sigma(\tau, V)$ , so

$$\tau d\sigma = \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_V d\tau + \tau \left( \frac{\partial \sigma}{\partial V} \right)_\tau dV \tag{87}$$

So by comparing terms between Eq. 86 and Eq. 87, conclude that

$$\left( \frac{\partial Q}{\partial V} \right)_\tau = \tau \left( \frac{\partial \sigma}{\partial V} \right)_\tau \tag{88}$$

For curve  $c = c(t) = (\tau(t), V(t))$  s.t.  $\sigma = \sigma(\tau, V) = \text{constant}$ ,

$$\begin{aligned}
\dot{c} &= \dot{\tau} \frac{\partial}{\partial \tau} + \dot{V} \frac{\partial}{\partial V} \\
Q(\dot{c}) &= 0 = C_V \dot{\tau} + \dot{V} \left( \frac{\partial Q}{\partial V} \right)_\tau \quad \text{or} \quad \dot{\tau} = \frac{-\dot{V}}{C_V} \left( \frac{\partial Q}{\partial V} \right)_\tau
\end{aligned}$$

Let  $t = V$ . Let  $\dot{\tau} = \left( \frac{\partial \tau}{\partial V} \right)_\sigma$ ,  $\dot{V} = 1$ . So using Eq. 88,

$$\left( \frac{\partial \tau}{\partial V} \right)_\sigma = \frac{-\tau}{C_V} \left( \frac{\partial \sigma}{\partial V} \right)_\tau \tag{89}$$

Likewise, for  $Q = Q(\tau, p)$ ,  $\sigma = \sigma(\tau, p)$ ,

$$Q = Q(\tau, p) = C_p d\tau + \left( \frac{\partial Q}{\partial p} \right)_\tau dp = \tau d\sigma = \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_p d\tau + \tau \left( \frac{\partial \sigma}{\partial p} \right)_\tau dp$$

and repeating all steps for  $p$ , but instead using  $V$  (what I call "label symmetry"),

$$\left(\frac{\partial\tau}{\partial p}\right)_\sigma = \frac{-\tau}{C_p} \left(\frac{\partial\sigma}{\partial p}\right)_\tau \quad (90)$$

Divide Eq. 90 by Eq. 89:

$$\frac{\left(\frac{\partial\tau}{\partial p}\right)_\sigma}{\left(\frac{\partial\tau}{\partial V}\right)_\sigma} = \frac{C_V}{C_p} \frac{\left(\frac{\partial\sigma}{\partial p}\right)_\tau}{\left(\frac{\partial\sigma}{\partial V}\right)_\tau} \quad (91)$$

Since  $F = F(\tau, V)$ ,

$$\begin{aligned} dF &= -pdV - \sigma d\tau \\ \left(\frac{\partial F}{\partial\tau}\right)_V &= -\sigma \\ \left(\frac{\partial F}{\partial V}\right)_\tau &= -p \end{aligned}$$

and it is sufficient that  $\left(\frac{\partial F}{\partial\tau}\right)_V$ ,  $\left(\frac{\partial F}{\partial V}\right)_\tau$  are continuous on a neighborhood of  $(\tau, V)$  and  $\frac{\partial^2 F}{\partial\tau\partial V}$  exists and is continuous on neighborhood of  $(\tau, V)$ , then

$$\left(\frac{\partial^2 F}{\partial\tau\partial V}\right) = \left(\frac{\partial^2 F}{\partial V\partial\tau}\right)$$

(cf. search for Clairaut's Thm., mean value Thm., or Fubini's Thm.)

$$\implies \left(\frac{\partial\sigma}{\partial V}\right)_\tau = \left(\frac{\partial p}{\partial\tau}\right)_V \quad (\text{from Helmholtz's free energy } F) \quad (92)$$

Similarly, with the Gibbs free energy  $G$  given in Eq. 85,

since  $G = G(\tau, p)$ ,

$$dG = -\sigma d\tau + V dp$$

then

$$\begin{aligned} \left(\frac{\partial G}{\partial\tau}\right)_p &= -\sigma \\ \left(\frac{\partial G}{\partial p}\right)_\tau &= V \end{aligned}$$

so

$$-\left(\frac{\partial\sigma}{\partial p}\right)_\tau = \left(\frac{\partial V}{\partial\tau}\right)_p \quad (\text{from Gibbs free energy } G) \quad (93)$$

Use Eqns. 92, 93 in Eq. 91:

$$\frac{\left(\frac{\partial\tau}{\partial p}\right)_\sigma}{\left(\frac{\partial\tau}{\partial V}\right)_\sigma} = \frac{-C_V}{C_p} \frac{\left(\frac{\partial V}{\partial\tau}\right)_p}{\left(\frac{\partial p}{\partial\tau}\right)_V} \quad (94)$$

Consider an *adiabatic* process, such as adiabatic compression. Then  $Q = \tau d\sigma = 0$ . So choose curves in  $\Sigma$  s.t.  $\sigma$  is constant along the curve.

Let  $\sigma(\tau, V)$ ,  $d\sigma = \left(\frac{\partial\sigma}{\partial\tau}\right)_V d\tau + \left(\frac{\partial\sigma}{\partial V}\right)_\tau dV$ .

Let  $c = c(t) = (\tau(t), p(t))$  s.t.  $\sigma$  constant. Then

$$\begin{aligned} d\sigma(\dot{c}) = 0 &= \left(\frac{\partial\sigma}{\partial\tau}\right)_V \dot{\tau} + \left(\frac{\partial\sigma}{\partial V}\right)_\tau \dot{p} \left(\frac{\partial V}{\partial p}\right)_\sigma \implies \\ \left(\frac{\partial V}{\partial p}\right)_\sigma &= \frac{-\dot{\tau} \left(\frac{\partial\sigma}{\partial\tau}\right)_V}{\dot{p} \left(\frac{\partial\sigma}{\partial V}\right)_\tau} \xrightarrow{t=p} \frac{-\left(\frac{\partial\tau}{\partial p}\right)_\sigma \left(\frac{\partial\sigma}{\partial\tau}\right)_V}{\left(\frac{\partial\sigma}{\partial V}\right)_\tau} \end{aligned} \quad (95)$$

Now let  $c = c(t) = (\tau, V) = (\tau(t), V(t))$  s.t.  $\sigma$  constant.

$$d\sigma(\dot{c}) = 0 = \left(\frac{\partial\sigma}{\partial\tau}\right)_V \dot{\tau} + \left(\frac{\partial\sigma}{\partial V}\right)_\tau \dot{V} \text{ or } \frac{\left(\frac{\partial\sigma}{\partial\tau}\right)_V}{\left(\frac{\partial\sigma}{\partial V}\right)_\tau} = \frac{-\dot{V}}{\dot{\tau}} \xrightarrow{t=V} \frac{-1}{\left(\frac{\partial\tau}{\partial V}\right)_\sigma}$$

Then plugging in the previous expression into Eq. 95, we obtain

$$\Rightarrow \left(\frac{\partial V}{\partial p}\right)_\sigma = \frac{\left(\frac{\partial\tau}{\partial p}\right)_\sigma}{\left(\frac{\partial\tau}{\partial V}\right)_\sigma} \quad (96)$$

Using label symmetry for  $p, V \mapsto V, p$ , then

$$\begin{aligned} \left(\frac{\partial p}{\partial V}\right)_\sigma &= \frac{-\left(\frac{\partial\tau}{\partial V}\right)_\sigma \left(\frac{\partial\sigma}{\partial\tau}\right)_p}{\left(\frac{\partial\sigma}{\partial p}\right)_\tau} = \frac{\left(\frac{\partial\tau}{\partial V}\right)_\sigma}{\left(\frac{\partial\tau}{\partial p}\right)_\sigma} \text{ since} \\ \frac{\left(\frac{\partial\sigma}{\partial\tau}\right)_p}{\left(\frac{\partial\sigma}{\partial p}\right)_\tau} &= \frac{-1}{\left(\frac{\partial\tau}{\partial p}\right)_\sigma} \end{aligned}$$

The previous expression and its derivation from first principles (namely that  $d\sigma$  is a 1-form on 2-dim.  $\Sigma$  and thermodynamic quantities are coordinates for  $\Sigma$ ) leads me to strongly exhort the following: stop memorizing and using "hand-wavey" formulae that claim to transform partial derivatives "to different independent variables" such as the "Jacobians" in Landau and Lifshitz (1980) [7], pp. 53, Ch. 11 "Thermodynamic Quantities", Sec. 16 "Relations between the Derivatives of Thermodynamic Quantities", Eq. (IV) and related:

$$\frac{\partial(u, v)}{\partial(x, y)} = \frac{\partial(u, v)}{\partial(t, s)} \cdot \frac{\partial(t, s)}{\partial(x, y)}$$

or in Appendix 4, "Summary of Thermodynamic Relations", pp. 577 of Sabersky, et. al. (1998) [8], which claim that the "rules of differential calculus" (???) allows for some to write, for  $z = z(x, y)$ ,

$$\begin{aligned} dz &= \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy \text{ or} \\ \left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial y}{\partial z}\right)_x &= -1 \end{aligned}$$

For the latter, the "differential",  $dx, dy$  is ill-defined. It is "an outdated concept from the time of Leibniz which was used to define derivatives and integrals before limits came along". For the former, the "Jacobian" is not general enough (at least for me), being only applicable to Euclidean spaces and is another unnecessary formula to memorize.

Instead, consider the coordinate transformation of 2-dim. manifold  $\Sigma$ :

$$(\tau, V) \mapsto (\tau, p)$$

Consider then that in general,  $p = p(\tau, V)$ , so that

$$dp = \left(\frac{\partial p}{\partial\tau}\right)_V d\tau + \left(\frac{\partial p}{\partial V}\right)_\tau dV$$

Consider curve  $c \subset \Sigma$ ,  $c = c(t) = (\tau, p) = (\tau(t), p(t))$ , so that

$$\dot{c} = (\dot{\tau}, \dot{p}) = \dot{\tau} \frac{\partial}{\partial\tau} + \dot{p} \frac{\partial}{\partial p}$$

Then

$$dp(\dot{c}) = \dot{p} = \dot{\tau} \left(\frac{\partial p}{\partial\tau}\right)_V + \dot{p} \left(\frac{\partial p}{\partial V}\right)_\tau \left(\frac{\partial V}{\partial p}\right)_\tau \quad (97)$$

where we used the fact that  $dV \left( \frac{\partial}{\partial \tau} \right) = 0$  ( $\tau$  and  $V$  are still independent coordinates for  $\Sigma$ ) and  $dV \left( \frac{\partial}{\partial p} \right) = \left( \frac{\partial V}{\partial p} \right)_\tau$  since  $\tau$  is held constant for the component of  $\dot{c} \in \mathfrak{X}(\Sigma)$  in  $p$  (i.e.  $\frac{\partial}{\partial p}$ ).

Consider letting parameter  $t = V$  for  $c$ . Then

$$\dot{\tau} = \left( \frac{\partial \tau}{\partial V} \right)_p, \quad \dot{p} = \left( \frac{\partial p}{\partial V} \right)_\tau$$

Consider letting  $t = p$ . Then  $\dot{\tau} = \left( \frac{\partial \tau}{\partial p} \right)_V$  and  $\dot{p} = 1$ .  $V$  is held constant for  $\dot{\tau} = \left( \frac{\partial \tau}{\partial p} \right)_V$  because of how  $p = p(\tau, V)$  for

$$dp = \left( \frac{\partial p}{\partial \tau} \right)_V d\tau + \left( \frac{\partial p}{\partial V} \right)_\tau dV$$

so

$$\Rightarrow 1 = \left( \frac{\partial \tau}{\partial p} \right)_V \left( \frac{\partial p}{\partial \tau} \right)_V + \left( \frac{\partial p}{\partial V} \right)_\tau \left( \frac{\partial V}{\partial p} \right)_\tau \quad (98)$$

Plug in Eq. 98 into Eq. 97:

$$\dot{p} = \dot{\tau} \left( \frac{\partial p}{\partial \tau} \right)_V + \dot{p} \left[ 1 - \left( \frac{\partial \tau}{\partial V} \right)_p \left( \frac{\partial p}{\partial \tau} \right)_V \right] \Rightarrow \dot{p} \left( \frac{\partial \tau}{\partial V} \right)_p \left( \frac{\partial p}{\partial \tau} \right)_V = \dot{\tau} \left( \frac{\partial p}{\partial \tau} \right)_V \text{ or}$$

$$\boxed{\left( \frac{\partial p}{\partial V} \right)_\tau = \frac{\left( \frac{\partial \tau}{\partial V} \right)_p}{\left( \frac{\partial \tau}{\partial p} \right)_V}} \quad (99)$$

Consider the coordinate transformation  $(\tau, p) \mapsto (\tau, V)$ . By "label symmetry" (i.e. substitute  $p$  for  $V$ ,  $V$  for  $p$ , for example),

$$\boxed{\left( \frac{\partial V}{\partial p} \right)_\tau = \frac{\left( \frac{\partial \tau}{\partial p} \right)_V}{\left( \frac{\partial \tau}{\partial V} \right)_p}} \quad (100)$$

In fact, furthermore, by label symmetry, consider  $(\tau, p) \mapsto (\tau, \rho)$ ,  $\rho \equiv \text{density}$ , then

$$\boxed{\left( \frac{\partial p}{\partial \rho} \right)_\tau = \frac{\left( \frac{\partial \tau}{\partial \rho} \right)_p}{\left( \frac{\partial \tau}{\partial p} \right)_\rho}} \quad (101)$$

Compare the previous expressions to the equation before (16.14) and (16.14) of Landau and Lifshitz (1980) [7], pp. 54,

$$\begin{aligned} \left( \frac{\partial V}{\partial P} \right)_S &= \frac{\partial(V, S)}{\partial(P, S)} = \frac{\partial(V, S)/\partial(V, T)}{\partial(P, S)/\partial(P, T)} \cdot \frac{\partial(V, T)}{\partial(P, T)} = \frac{(\partial S/\partial T)_V}{(\partial S/\partial T)_P} \cdot \left( \frac{\partial V}{\partial P} \right)_T \\ \left( \frac{\partial V}{\partial P} \right)_S &= \frac{C_V}{C_P} \left( \frac{\partial V}{\partial P} \right)_T \end{aligned}$$

Similarly with Eq. (A-4.9a) on pp. 577 with Appendix 4 of Sabersky, et. al. (1998) [8]:

$$\left( \frac{\partial p}{\partial \rho} \right)_s = -\frac{(\partial s/\partial \rho)_p}{(\partial s/\partial p)_\rho} = -\frac{(\partial s/\partial T)_p(\partial T/\partial \rho)_p}{(\partial s/\partial p)_\rho(\partial T/\partial p)_\rho} = \frac{(\partial s/\partial T)_p}{(\partial s/\partial T)_\rho} \left( \frac{\partial p}{\partial \rho} \right)_T$$

The expression we'll need uses  $V$ :

$$\begin{aligned} V &= V(\tau, p) \Rightarrow \\ dV &= \left( \frac{\partial V}{\partial \tau} \right)_p d\tau + \left( \frac{\partial V}{\partial p} \right)_\tau dp \end{aligned}$$

Let  $c = (\tau, p)$  be a curve for when  $V$  is constant.

$$dV(\dot{c}) = 0 = \left(\frac{\partial V}{\partial \tau}\right)_p \dot{\tau} + \left(\frac{\partial V}{\partial p}\right)_\tau \dot{p} \xrightarrow{t=\tau} -\left(\frac{\partial V}{\partial \tau}\right)_p = \left(\frac{\partial V}{\partial p}\right)_\tau \left(\frac{\partial p}{\partial \tau}\right)_V \text{ or}$$

$$\left(\frac{\partial V}{\partial p}\right)_\tau = -\frac{\left(\frac{\partial V}{\partial \tau}\right)_p}{\left(\frac{\partial p}{\partial \tau}\right)_V} \quad (102)$$

Using label symmetry for  $V, p \mapsto p, V$ , then also

$$\left(\frac{\partial p}{\partial V}\right)_\tau = -\frac{\left(\frac{\partial p}{\partial \tau}\right)_V}{\left(\frac{\partial V}{\partial \tau}\right)_p}$$

Then using both Eq. 96 and the previous equation in Eq. 94, then we get

$$\boxed{\left(\frac{\partial V}{\partial p}\right)_\sigma = \frac{C_V}{C_p} \left(\frac{\partial V}{\partial p}\right)_\tau} \quad (103)$$

This is the equation for the *adiabatic compressibility*  $\left(\frac{\partial V}{\partial P}\right)_\sigma$  of a body. The inequality  $C_p > C_v$  implies that adiabatic compressibility is always smaller in absolute value than isothermal compressibility. Compare this to Eq. (16.14) of Landau and Lifshitz (1980) and Eq. (A-4.10) of Appendix 4, "Summary of Thermodynamic Relations", pp. 577 of Sabersky, et. al. (1998) [8].

The previous equation can be expressed as

$$\left(\frac{\partial p}{\partial V}\right)_\sigma = \frac{C_p}{C_V} \left(\frac{\partial p}{\partial V}\right)_\tau$$

From the definition of density  $\rho$ ,  $\rho := \frac{MN}{V}$  or  $V = \frac{MN}{\rho}$ , and assuming  $N$  constant, then

$$\frac{\partial}{\partial V} = \frac{\partial \rho}{\partial V} \frac{\partial}{\partial \rho} = \frac{-MN}{V^2} \frac{\partial}{\partial \rho} = \frac{-\rho^2}{MN} \frac{\partial}{\partial \rho}$$

so that

$$\left(\frac{\partial p}{\partial \rho}\right)_\sigma = \frac{C_p}{C_V} \left(\frac{\partial p}{\partial \rho}\right)_\tau$$

2008 manuscript.

## 20 Gibbs Free Energy and Chemical Reactions

### Solution 1. Thermal expansion near absolute zero

(a)

$$\begin{aligned} \left(\frac{\partial G}{\partial \tau}\right)_{N,p} &= -\sigma & \left(\frac{\partial G}{\partial p}\right)_\tau &= V & \Rightarrow & \boxed{\left(\frac{\partial V}{\partial \tau}\right)_p = -\left(\frac{\partial \sigma}{\partial p}\right)_\tau} \\ \left(\frac{\partial^2 G}{\partial p \partial \tau}\right)_\tau &= -\left(\frac{\partial \sigma}{\partial p}\right)_\tau & \left(\frac{\partial^2 G}{\partial \tau \partial p}\right)_p &= \left(\frac{\partial V}{\partial \tau}\right)_p & & \\ \left(\frac{\partial G}{\partial N}\right)_p &= \mu & \left(\frac{\partial G}{\partial p}\right)_\tau &= V & \Rightarrow & \boxed{\left(\frac{\partial V}{\partial N}\right)_p = \left(\frac{\partial \mu}{\partial p}\right)_N} \\ \left(\frac{\partial^2 G}{\partial p \partial N}\right)_N &= \left(\frac{\partial \mu}{\partial p}\right)_N & \left(\frac{\partial^2 G}{\partial N \partial p}\right)_p &= \left(\frac{\partial V}{\partial N}\right)_p & & \\ \left(\frac{\partial^2 G}{\partial \tau \partial N}\right)_N &= \left(\frac{\partial \mu}{\partial \tau}\right)_N & \left(\frac{\partial^2 G}{\partial N \partial \tau}\right)_\tau &= -\left(\frac{\partial \sigma}{\partial N}\right)_\tau & \Rightarrow & \boxed{\left(\frac{\partial \mu}{\partial \tau}\right)_N = -\left(\frac{\partial \sigma}{\partial N}\right)_\tau} \end{aligned}$$

(b)  $\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial \tau}\right)_p = \frac{-1}{V} \left(\frac{\partial \sigma}{\partial p}\right)_\tau = 0$  as  $\tau \rightarrow 0$  since  $\sigma \rightarrow \text{constant}$  as  $\tau \rightarrow 0$  by third law of thermodynamics.

### Solution 2. Thermal ionization of hydrogen.

(a) Given  $e + H^+ \rightleftharpoons H$ , note that  $e + H^+ - H = 0$ . Recall

$$\frac{[e][H^+]}{[H]} = K(\tau) = \prod_j n_{Q_j}^{\nu_j} \exp[-\nu_j F_j(int)/\tau]$$

where  $n_Q = \left(\frac{M\tau}{2\pi\hbar^2}\right)^{3/2} V$ .

For dissociation of  $H$  into  $e^- + H^+$  choose zero of internal energy of each composite particle (here  $H$ ) to coincide with energy of dissociated particles (here  $H^+$ ,  $e^-$ ) at rest; place energy of ground state of composite particle  $H$  at  $-I$ ,  $I$  is energy required in reaction to dissociate composite particle into its constituents and is taken to be positive, i.e. the ionization energy.

$$K(\tau) = (n_{e^-})^1 \exp[-F_{int}(e^-)/\tau] \cdot (n_{H^+})^1 \exp[-F_{int}(H^+)/\tau] (n_H)^{-1} \exp(-(-1)(F_{int}(H)/\tau))$$

Note that  $n_{H^+} \simeq n_H$ . Let  $n_{e^-} = n_Q$ . *Importantly*, note

$$\boxed{F_{int}(e^-) + F_{int}(H^+) - F_{int}(H) = I}$$

$F_{int}(H)$  is at a lower free energy than  $e^-$  and  $H^+$ .

$$\implies K(\tau) = n_Q e^{-I/\tau} \implies \boxed{\frac{[e][H^+]}{[H]} = n_Q e^{-I/\tau}}$$

(b) By charge conservation,  $[e] = [H^+]$ , so that

$$[e] = [H]^{1/2} n_Q^{1/2} \exp(-I/2\tau)$$

Given  $[H] \simeq 10^{23} \text{ cm}^{-3}$ ,  $m_e = 0.511 \text{ MeV}/c^2$ ,  $T = 5000 \text{ K}$ ,  $I = 13.6 \text{ eV}$  ionization energy,

$$[e] = (10^{23} \text{ cm}^{-3})^{1/2} (2.92 \times 10^{10} / \text{cm}^3)^{1/2} \exp(-13.6 \text{ eV} / 2k_B 5000 \text{ K}) = 1.3 \times 10^{15} \text{ cm}^{-3}$$

*Note that  $H(exc)$  and  $H$  are just two different states of atomic hydrogen. Their concentrations must therefore be proportional to the probability of occurrence of these states, and the ratio of probabilities is the ratio of the respective Boltzmann*

$$\frac{[H(exc)]}{[H]} = \frac{p(H(exc))}{p(H)}$$

*If  $\epsilon_{H(exc)}$  is the internal energy of the first excited state and  $\epsilon_H$  is the internal energy of the ground state of atomic hydrogen, we are given that  $\epsilon_{H(exc)} - \epsilon_H = \frac{3}{4}I$ . **We also need to take into account the fact that the first excited electronic state of hydrogen is 4-fold degenerate i.e. one 2s-orbital and three 2p-orbitals.**<sup>4</sup> Therefore,*

$$\frac{[H(exc)]}{[H]} = \frac{p(H(exc))}{p(H)} = \frac{4e^{-\epsilon_{H(exc)}/\tau}}{e^{-\epsilon_H/\tau}} = 4e^{-3I/4\tau}$$

$$[H(exc)] = 4[H]e^{-3I/4\tau} = \boxed{2.092 \times 10^{13} \text{ cm}^{-3}}$$

$$\frac{[e]}{[H(exc)]} = 62$$

### Solution 3. Ionization of donor impurities in semiconductors.

Consider a pentavalent impurity (donor), introduced in place of tetravalent silicon atom, acts like a hydrogen atom in free space.

<sup>4</sup>(from solutions to Homework 8, Ph12c, Caltech, June 6, 2008, by Prabha Mandayam, Heywood Tam)

Then recall, from Problem 3,  $e + H^+ \rightleftharpoons H$ , and so

$$\frac{[e][H^+]}{[H]} = n_Q \exp(-I/\tau)$$

where  $n_Q \equiv \left(\frac{m\tau}{2\pi\hbar^2}\right)^{3/2}$  refers to the electron.

By charge conservation,  $[e] = [H^+]$ .

So

$$[e] = [H]^{1/2} n_Q^{1/2} \exp(-I/2\tau)$$

Now given  $[H] = 10^{17}$  donors per  $\text{cm}^3$ ,  $\tau = 100 \text{ K}$ ,

$$n_Q = \left(\frac{m^*\tau}{2\pi\hbar^2}\right)^{3/2} = \left(\frac{0.3 \cdot 0.511 \text{ MeV}/c^2 \cdot 100 \text{ K} \cdot 0.00008617 \text{ eV/K}}{2\pi(4.136 \times 10^{-15} \text{ eV/s})^2}\right)^{3/2} = 1263. \text{ donors per cm}^3$$

Now  $F = \frac{q_1 q_2}{r^2}$ ,

$$W = \int -F dr = - \int_{r_0}^{\infty} \frac{q_1 q_2}{r^2} dr = + \left(\frac{q_1 q_2}{r}\right) \Big|_{r_0}^{\infty} = \frac{-q_1 q_2}{r_0}$$

and so  $I = \frac{e^2}{\epsilon r_0} = \frac{1}{\epsilon}(13.6 \text{ eV})$  where  $13.6 \text{ eV}$  is the ionization energy of an electron from a (real) hydrogen atom.

So

$$[e] = 2. \times 10^{-18} \text{ donors per cm}^3$$

**Correction (donor density).** The value  $1263 \text{ cm}^{-3}$  for  $n_Q$  does not evaluate  $(m\tau/(2\pi\hbar^2))^{3/2}$ . The hydrogenic binding energy and the quadratic for  $n$  are in §21. At  $m^* = 0.3 m_e$ ,  $\epsilon_r = 11.7$ , and  $T = 100 \text{ K}$ , that calculation gives  $n = 2.965 \times 10^{16} \text{ cm}^{-3}$ .

#### Solution 4. Biopolymer growth.

Recall that  $G(N, p, \tau) = N\mu(p, \tau)$ , since  $G$  was chosen to be an *extensive* quantity (it scales with size). For more than one chemical species  $G = \sum_j N_j \mu_j$ .  
 $dF = 0$  for equilibrium, for constant  $P, \tau$ .

$\mu_j$  = chemical potential of species  $j$ ,  $\mu_j = (\partial G / \partial N_j)_{\tau, p}$ .

Given  $\sum_i \nu_j A_j$ , e.g.  $H_2 + Cl_2 = 2HCl$ ,  
 $dG = (\sum_j \nu_j \mu_j) d\tilde{N}$  where  $dN_j = \nu_j d\tilde{N}$ ,  $dG = 0 \rightarrow \sum_j \nu_j \mu_j = 0$ .

Recall the mass action law derivation: assume constituents act as ideal gases;  $\mu_j = \tau(\ln n_j - \ln c_j)$ ,  $n_j$  concentration of species  $j$ ;  $c_j \equiv n_{Q_j} Z_j(int)$ .

$$\sum_j \nu_j \ln n_j = \sum_j \nu_j \ln c_j \implies \sum_j \ln n_j^{\nu_j} = \sum_j \ln c_j^{\nu_j} = \ln \prod_j n_j^{\nu_j} = \ln K(\tau)$$

$$\prod_j n_j^{\nu_j} = K(\tau) \quad \text{mass action law}$$

(a) By mass action law,  $\frac{[\text{monomer}][N\text{mer}]}{[(N+1)\text{mer}]} = \frac{[1][N]}{[N+1]} = K_N$ .

$$\begin{aligned} \frac{[1][1]}{[2]} &= K_1 & \frac{[1]^2 [1][2]}{[2] [3]} &= \frac{[1]^3}{[3]} = K_1 K_2 \\ \frac{[1]^{j+1} [1][j+1]}{[j+1] [j+2]} &= \prod_{l=1}^j K_l K_{j+1} = \frac{[1]^{j+2}}{[j+2]} = \prod_{l=1}^{j+1} K_l \end{aligned} \implies \boxed{[N+1] = [1]^{N+1} / K_1 K_2 K_3 \dots K_N}$$

(b) Recall that  $K(\tau) = \prod_j n_{Q_j}^{\nu_j} \exp[-\nu_j F_j(int)/\tau]$

$$K_N = \frac{n_Q(N)n_Q(1)}{n_Q(N+1)} \exp\left[\frac{-F_N}{\tau} - \frac{F_1}{\tau}\right] \exp\left[\frac{F_{N+1}}{\tau}\right] = \frac{n_Q(N)n_Q(1)}{n_Q(N+1)} \exp\left[\frac{-(F_N + F_1 - F_{N+1})}{\tau}\right]$$

where  $n_Q(N) = \left(\frac{M_N \tau}{2\pi \hbar^2}\right)^{3/2}$  and  $M_N$  is the mass of  $N$ mer molecules,  $F_N$  is the free energy of one  $N$ mer molecule.

(c) Assume  $N \ll 1$  so  $n_Q(N) \simeq n_Q(N+1)$ . Assume  $[1] = 10^{20} \text{ cm}^{-3}$ . Assume  $\Delta F = F_{N+1} - F_N - F_1 = 0$ , meaning zero free energy change in the basic reaction step. We're given the molecular weight of the monomer to be 200.

We want  $\frac{[N+1]}{[N]}$  at room temperature. Now  $K_N \simeq n_Q(1) = \left(\frac{M_1 \tau}{2\pi \hbar^2}\right)^{3/2}$ .

$$\frac{[1][N]}{[N+1]} = n_Q(1) = \left(\frac{M_1 \tau}{2\pi \hbar^2}\right)^{3/2} \quad \text{or} \quad \frac{[N+1]}{[N]} = \frac{[1]}{n_Q(1)} = \left(\frac{2\pi \hbar^2}{M_1 \tau}\right)^{3/2} 10^{20} \text{ cm}^{-3}$$

$$\text{Note that } \left(\frac{2\pi(6.582 \times 10^{-22} \text{ MeV} \cdot \text{s})^2}{200(938 \text{ MeV}/c^2)(0.8617 \times 10^{-4} \text{ eV}/K)(298 \text{ K})} \left(\frac{3 \times 10^{10} \text{ cm/s}}{1 \text{ c}}\right)^2\right)^{3/2} = 0.3627 \times 10^{-27} \text{ cm}.$$

$$\Rightarrow \boxed{\frac{[N+1]}{[N]} = 3.627 \times 10^{-8}}$$

(d) We want the condition

$$1 < \frac{[N+1]}{[N]} = \frac{[1]}{n_Q(1)} \exp\left(\frac{-\Delta F}{\tau}\right) \quad \text{or} \quad \ln \frac{n_Q(1)}{[1]} < \frac{-\Delta F}{\tau}$$

$$\Rightarrow \Delta F < \tau \ln \frac{[1]}{n_Q(1)} = \boxed{-0.44 \text{ eV}}$$

## 21 Semiconductor statistics

The 2008 heading was empty. The preserved solution of the donor-impurity problem, in the Gibbs chapter, adapts the hydrogen ionization mass action and then evaluates a number. The quantum concentration in that arithmetic is not the concentration defined by the formula next to it. This section writes the mass action, the unit-consistent  $n_Q$ , and the hydrogenic binding energy, and it solves the quadratic that the low-ionization shortcut does not justify at the corrected numbers. The preserved solution is unchanged in the Gibbs chapter; a pointer there sends the numerical claim here.

### 21.1 Classical electrons and holes

Use Kittel and Kroemer's orbital quantum concentration, which does not include a spin factor,

$$n_Q(m, \tau) = \left(\frac{m\tau}{2\pi \hbar^2}\right)^{3/2}. \quad (104)$$

In electron-volt units the same quantity is

$$n_Q = \left(\frac{(mc^2)(k_B T)}{2\pi(\hbar c)^2}\right)^{3/2}, \quad \tau = k_B T. \quad (105)$$

The factors of  $c$  are required because  $mc^2$  is an energy and  $\hbar c$  is an energy times a length. A quotient of  $mc^2$  and  $(\hbar)^2$  with  $\hbar$  in  $\text{eV} \cdot \text{s}$  is not a number per volume.

For a parabolic conduction band edge at energy  $\epsilon_c$ , the classical density of electrons, still without a spin factor 2, is

$$n = n_Q(m_e^*, \tau) \exp((\mu - \epsilon_c)/\tau). \quad (106)$$

For holes in a valence band edge at  $\epsilon_v$ ,

$$p = n_Q(m_h^*, \tau) \exp((\epsilon_v - \mu)/\tau). \quad (107)$$

The product is independent of  $\mu$ :

$$np = n_Q(m_e^*, \tau) n_Q(m_h^*, \tau) \exp(-(\epsilon_c - \epsilon_v)/\tau). \quad (108)$$

In an intrinsic sample the only source of carriers is the promotion across the gap, so charge neutrality says  $n = p$ . Then each density is the square root of (108). A spin factor 2 in both  $n$  and  $p$  multiplies the product by 4 and each density by 2. The preserved hydrogen solution does not put that factor into  $n_Q$ , and the donor arithmetic below follows that choice. The omission is a convention to keep visible, not a claim that spin is absent.

## 21.2 Donors

The preserved solution treats a donor as a hydrogenic center, with the reaction written there as  $e + H^+ \rightleftharpoons H$ , and with the mass action

$$\frac{n n_+}{n_0} = n_Q(m^*, \tau) \exp(-I/\tau). \quad (109)$$

Here  $n$  is the conduction-electron density,  $n_+$  the density of ionized donors, and  $n_0$  the density of neutral donors. Charge neutrality, with no acceptors, is  $n = n_+$ . Number conservation is  $n_0 + n_+ = N_d$ , so  $n_0 = N_d - n$ . Equation (109) becomes

$$\frac{n^2}{N_d - n} = K, \quad K = n_Q(m^*, \tau) e^{-I/\tau}. \quad (110)$$

Rearrange:

$$n^2 + Kn - KN_d = 0. \quad (111)$$

The positive root is

$$n = \frac{-K + \sqrt{K^2 + 4KN_d}}{2}. \quad (112)$$

If  $n \ll N_d$ , then  $N_d - n$  may be replaced by  $N_d$  and  $n \approx \sqrt{KN_d}$ . That replacement has to be checked against the root it produces.

The preserved numerical inputs are  $N_d = 10^{17} \text{ cm}^{-3}$ ,  $T = 100 \text{ K}$ , and  $m^*/m_e = 0.3$ . The constants used to evaluate (105) are

$$\begin{aligned} k_B &= 8.617333262145 \times 10^{-5} \text{ eV/K}, \\ m_e c^2 &= 0.5109989461 \times 10^6 \text{ eV}, \\ \hbar c &= 197.3269718 \text{ eV} \cdot \text{nm}. \end{aligned} \quad (113)$$

They give

$$n_Q(0.3 m_e, k_B \cdot 100 \text{ K}) = 3.9677 \times 10^{17} \text{ cm}^{-3}. \quad (114)$$

The value  $1263 \text{ cm}^{-3}$  in the preserved solution is the result of dividing an energy by  $(\text{eV} \cdot \text{s})^2$  and calling the quotient a concentration. Equation (114) is the evaluation of the displayed formula.

The binding energy in the preserved solution is written  $I = (13.6 \text{ eV})/\epsilon$ , which divides the hydrogen potential by a dielectric constant and holds the Bohr radius fixed. For a hydrogenic donor the radius scales as  $\epsilon_r m_e/m^*$ , and the energy scales as

$$I = (13.6 \text{ eV}) \frac{m^*/m_e}{\epsilon_r^2}. \quad (115)$$

The static dielectric constant of silicon is not in the preserved solution. The usual value  $\epsilon_r = 11.7$  is used here so that (115) has a number, and it is an added assumption:

$$I = (13.6 \text{ eV}) \frac{0.3}{(11.7)^2} = 0.02980 \text{ eV}. \quad (116)$$

Then  $I/\tau = I/(k_B \cdot 100 \text{ K}) = 3.458$ , and

$$K = 1.249 \times 10^{16} \text{ cm}^{-3}. \quad (117)$$

The quadratic (112) gives

$$n = 2.965 \times 10^{16} \text{ cm}^{-3}, \quad \frac{n}{N_d} = 0.296. \quad (118)$$

The shortcut  $\sqrt{KN_d} = 3.53 \times 10^{16} \text{ cm}^{-3}$  is about twenty percent higher, and it is not consistent with  $n \ll N_d$ . The reported density is (118). Changing  $m^*/m_e$  or  $\epsilon_r$  inside their experimental range moves  $n$  by more than this arithmetic does. The preserved value  $2 \times 10^{-18} \text{ cm}^{-3}$  comes from  $n_Q \sim 10^3 \text{ cm}^{-3}$  together with a binding energy of order one electron-volt. Both inputs differ from (114) and (116).

A spin degeneracy  $g = 2$  of the neutral donor, or a factor 2 in the conduction-band density, replaces  $K$  by  $K/2$  or by  $2K$ . Either change is outside the mass action as written in the preserved solution, so it is not folded into (118). *2015 manuscript.*

## 22 Equilibrium in Reactions

cf. Section Equilibrium in Reactions for Chapter 9: Gibbs Free Energy and Chemical Reactions of Kittel and Kroemer (1980) [2]

$$\nu_1 A_1 + \nu_2 A_2 + \cdots + \nu_l A_l = 0 \text{ or } \sum_j \nu_j A_j = 0 \quad (119)$$

where  $\nu_j \in \mathbb{Z}$ ,  $A_j$  chemical species, where we consider the finite set of chemical elements, and nonnegative integers  $\mathbb{Z}^+$  for each element, signifying how many of the element in the molecule.

Now clearly,

$$dN_j = \nu_j d\hat{N}$$

where  $d\hat{N} \equiv$  how many times reaction takes place.

Now

$$dG = \sum_j \mu_j dN_j - \sigma d\tau + V dp$$

and so for constant temperature, constant pressure,

$$dG = \sum_j \mu_j dN_j = \sum_j \mu_j \nu_j d\hat{N}$$

then for equilibrium  $dG = 0$ ,

$$\boxed{\sum_j \nu_j \mu_j = 0} \quad (120)$$

### 22.1 Equilibrium for Ideal Gases

cf. Subsection Equilibrium for Ideal Gases for Chapter 9: Gibbs Free Energy and Chemical Reactions of Kittel and Kroemer (1980) [2]

From Eq. (6.48) of Kittel and Kroemer (1980) [2],

$$\mu = \tau \left[ \ln \left( \frac{n}{n_Q} \right) - \ln Z_{\text{int}} \right]$$

and so

$$\mu_j = \tau (\ln n_j - \ln c_j)$$

where

$$c_j \equiv n_{Q_j} Z_j(\text{int})$$

where

$$Z_j(\text{int}) = \sum_{\text{int}} \exp \left( \frac{-\epsilon_{\text{int}}}{\tau} \right)$$

Thus

$$\implies \sum_j \nu_j \ln n_j = \sum_j \nu_j \ln c_j$$

and so the **law of mass action** is obtained:

$$\boxed{\prod_j n_j^{\nu_j} = K(\tau) \equiv \prod_j n_{Q_j}^{\nu_j} \exp(-\nu_j F_j(\text{int})/\tau)} \quad (121)$$

where

$$F_j(\text{int}) = -\tau \ln Z_j(\text{int})$$

From Problem 6 **Rotation of diatomic molecules** of Chapter 3: Boltzmann Distribution and Helmholtz Free Energy, Kittel and Kroemer [2],

rotational energy at level  $j \in \mathbb{Z}^+$ ,  $\epsilon(j) = j(j+1)\epsilon_0$

multiplicity  $g(j) = 2j+1$

$$Z_R = \sum_{j=0}^{\infty} (2j+1) \exp\left(\frac{1}{-\tau} j(j+1)\epsilon_0\right) = \frac{-\tau}{\epsilon_0} \sum_{j=0}^{\infty} \frac{d}{dj} \left[ (e^{-\epsilon_0/\tau})^{j^2+j} \right]$$

For  $1 \gg \frac{\epsilon_0}{\tau}$ ,

$$Z_R(\tau) = \frac{-\tau}{\epsilon_0} \int_0^{\infty} \frac{d}{dx} \left[ (e^{-\epsilon_0/\tau})^{x^2+x} \right] = \frac{\tau}{\epsilon_0}$$

So

$$F_R = -\tau \ln Z_R = -\tau \ln \left( \frac{\tau}{\epsilon_0} \right)$$

and

$$\exp(-\nu_j F_j(\text{int})/\tau) = \exp(\nu_j \ln \left( \frac{\tau}{\epsilon_{0;j}} \right)) = \exp(\nu_j \ln \left( \frac{\tau}{\epsilon_{0;j}} \right)) = \left( \frac{\tau}{\epsilon_{0;j}} \right)^{\nu_j}$$

and so

$$K(\tau) = \prod_j (n_{Q_j})^{\nu_j} \left( \frac{\tau}{\epsilon_{0;j}} \right)^{\nu_j} = \prod_j \left[ \left( \frac{M_j \tau}{2\pi \hbar^2} \right)^{d/2} \frac{\tau}{\epsilon_{0;j}} \right]^{\nu_j}$$

2015 manuscript.

## 23 Chemical Reactions

Topics:

- Heat of formation i.e. standard enthalpy of formation

I will follow Chapter 10, “Chemical Reactions”, of Landau and Lifshitz (1980) [7].

From Sec. 101. The condition for chemical equilibrium of Landau and Lifshitz (1980) [7], for a chemical reaction,

$$\sum_i \nu_i A_i = 0 \quad \nu_i \in \mathbb{Z}$$

Denote thermodynamic potential  $\Phi$ .

In equilibrium, say at constant temperature and pressure,

$$d\Phi \left( \frac{\partial}{\partial N_i} \right) = \frac{\partial \Phi}{\partial N_1} + \frac{\partial \Phi}{\partial N_2} \frac{\partial N_2}{\partial N_1} + \frac{\partial \Phi}{\partial N_3} \frac{\partial N_3}{\partial N_1} + \dots = 0 \text{ for}$$

$$d\Phi = \frac{\partial \Phi}{\partial N_1} dN_1 + \frac{\partial \Phi}{\partial N_2} dN_2 + \dots$$

where  $N_1, N_2, \dots$  = number of particles of various substances taking place in reaction.

It's clear that

$$dN_i = \frac{\nu_i}{\nu_1} dN_1$$

so

$$\sum_i \frac{\partial \Phi}{\partial N_i} \frac{\nu_i}{\nu_1} = 0$$

Let  $\mu_i := \frac{\partial \Phi}{\partial N_i}$ .

$$\implies \sum_i \nu_i \mu_i = 0 \quad (122)$$

Here, I will follow Section 3.5 Chemical potential of Le Bellac, Mortessagne, Batrouni (2004) [9]. Recall that

$$G(N, \tau, p) = N\mu(\tau, p)$$

$$G = \sum_j \mu_j N_j$$

from Eq. 84, and Kittel and Kroemer (1980) [2]. Therefore,

$$dG = \sum_j \mu_j dN_j + N_j d\mu_j = -\sigma d\tau + V dp + \sum_j \mu_j dN_j$$

and so the **Gibbs-Duhem** relation is (correctly) obtained:

$$0 = \sigma d\tau - V dp + \sum_j N_j d\mu_j \quad (123)$$

### 23.0.1 Conduction and Convection

In  $\tau d\sigma$ , let's distinguish the *conduction* component and *convection* component.

Consider system at equilibrium, thermal equilibrium with thermostat at temperature  $\tau$   
kept at pressure  $p$   
able to exchange particles with reservoir at chemical potential  $\mu$

Define

$$\hat{\sigma} := \frac{\sigma}{N} \quad \text{entropy per particle}$$

$$\hat{\epsilon} := \frac{E}{N} \quad \text{energy per particle}$$

Then,

$$\boxed{\tau d\sigma = \tau d(\hat{\sigma} N) = \tau N d\hat{\sigma} + \tau \hat{\sigma} dN}$$

where

$\tau N d\hat{\sigma}$  is *entropy change due to change in entropy per particle*, i.e. *conduction term* and

$\tau \hat{\sigma} dN$  is *entropy change due to change in number of particles* i.e. *convection term*.

So

$$dE = Q + W + \mu dN = \tau d\sigma + 0 + \mu dN = \tau N d\hat{\sigma} + (\tau \hat{\sigma} + \mu) dN = \tau N d\hat{\sigma} + \hat{h} dN$$

where

$$\hat{h} := \frac{H}{N} = \hat{\epsilon} + pV \text{ is } \textbf{enthalpy per particle}$$

since

$$H = E + pV$$

$$G = E - \tau\sigma + pV = \mu N = H - \tau\sigma \implies \tau\hat{\sigma} + \mu = \frac{H}{N} = \hat{h}$$

Interpretation of  $\hat{h} dN$ : If  $dN$  particles are transported into a given volume by convection, energy increases by  $\hat{\epsilon} dN$ .

However, to return to initial volume, it's necessary to compress by  $v dN =: \frac{V}{N} dN$  and add work done by pressure  $p v dN$

$$\implies \hat{h} dN = (\hat{\epsilon} + p v) dN$$

An example of this is *Joule-Thomson expansion*.

**Correction (enthalpy per particle).**  $\hat{h} = H/N = \hat{\epsilon} + p v$  with  $v = V/N$ . The display that adds  $pV$  to  $\hat{\epsilon}$  mixes an energy per particle with an energy. See §33.5.

2008 manuscript.

## 24 Phase Transformations

### Vapor Pressure Equation

**Derivation of the Coexistence Curve  $p$  versus  $\tau$**

Suppose  $\mu_g(p_0, \tau_0) = \mu_l(p_0, \tau_0)$  and  $\mu_g(p_0 + dp, \tau_0 + d\tau) = \mu_l(p_0 + dp, \tau_0 + d\tau)$

$$\begin{aligned} \Rightarrow \mu_g(p_0, \tau_0) + \left( \frac{\partial \mu_g}{\partial p} \right)_\tau dp + \left( \frac{\partial \mu_g}{\partial \tau} \right)_p d\tau + \dots &= \mu_l(p_0, \tau_0) + \left( \frac{\partial \mu_l}{\partial p} \right)_\tau dp + \left( \frac{\partial \mu_l}{\partial \tau} \right)_p d\tau + \dots \\ \frac{dp}{d\tau} &= \frac{\left( \frac{\partial \mu_l}{\partial \tau} \right)_p - \left( \frac{\partial \mu_g}{\partial \tau} \right)_p}{\left( \frac{\partial \mu_g}{\partial \tau} \right)_\tau - \left( \frac{\partial \mu_l}{\partial \tau} \right)_\tau} \end{aligned}$$

$$v \equiv V/N$$

$$s \equiv \sigma/N$$

$$\begin{aligned} \frac{1}{N} \left( \frac{\partial G}{\partial p} \right)_{N, \tau} &= v = \left( \frac{\partial \mu}{\partial p} \right)_\tau \\ \frac{1}{N} \left( \frac{\partial G}{\partial \tau} \right)_{N, p} &= \frac{-\sigma}{N} = -s = \left( \frac{\partial \mu}{\partial \tau} \right)_p \end{aligned}$$

$s_g - s_l$  related directly to quantity of heat that must be added to system to transfer 1 molecule reversibly from liquid to gas, while temperature constant. (if heat isn't added to system from outside in the process, temperature will decrease when molecule transferred to gas)

$$L \equiv \tau(s_g - s_l) \quad (124)$$

**latent heat of vaporization**

$$\Delta v = v_g - v_l \quad (125)$$

change of volume when 1 molecule transferred from liquid to gas

$$\frac{dp}{d\tau} = \frac{L}{\tau \Delta v} \quad (126)$$

is the **Clausius-Clapeyron equation** or **vapor pressure equation**.

Assumptions:

- (a) if assume  $v_g \gg v_l$ , then  $\Delta v \cong v_g = V_g/N_g$  at  $p = 1$  atm (at atmospheric pressure),  $v_g/v_l \approx 10^3$
- (b) ideal gas law  $pV_g = N_g\tau \Rightarrow \Delta v \cong \tau/p$

$$\frac{dp}{d\tau} = \frac{L}{\tau^2} p \text{ or } \frac{d}{d\tau} \ln p = \frac{L}{\tau^2}$$

$L \equiv$  latent heat per molecule

If latent heat  $L$  independent of temperature over temperature range of interest,  $L = L_0$

$$\int \frac{dp}{p} = L_0 \int \frac{d\tau}{\tau^2} \text{ whence } \ln p = -L_0/\tau + \text{const or } p(\tau) = p_0 \exp(-L_0/\tau)$$

Consider  $\frac{L_0}{\tau}$ ,  $L_0$ , latent heat of vaporization of 1 molecule.

Now compare it with  $\frac{L_0}{RT}$ ,

where  $R \equiv N_0 k_B \equiv$  gas constant.

$$N_0 = 6.02205 \times 10^{23} \frac{1}{\text{mol}}$$

$$k_B = 1.38066 \times 10^{-23} \text{ J/K (usual Boltzmann constant)}$$

$$R = 8.31441 \text{ J/K} \cdot \text{mol}$$

So for

$$\frac{L_0}{RT} = \frac{L_0}{N_0 k_B T}$$

Then that implies

$$\frac{1}{1 \text{ particle}} \frac{1}{N_0} = \frac{1}{1 \text{ particle}} \frac{1}{\frac{1}{\text{mol}}}$$

which implies that  $L_0$  now is the latent heat of vaporization of a *mole* (*mol*) of the gas.

## Latent heat and enthalpy

crossing the coexistence curve,

$$\tau d\sigma = dU + pdV - (\mu_g - \mu_l)dN$$

if liquid  $\rightarrow$  gas,  $\mu_l > \mu_g$  (or  $0 > \mu_g - \mu_l$ )

on coexistence curve,  $\mu_g = \mu_l$

$$L = \tau d\sigma(\dot{\gamma}) = dU(\dot{\gamma}) + pdV(\dot{\gamma}) = (dU + pdV)(\dot{\gamma}) + 0 = (dU + pdV)(\dot{\gamma}) + Vdp(\dot{\gamma}) = dH(\dot{\gamma}) = H_g - H_l$$

where  $dp(\dot{\gamma}) = 0$  since  $\gamma$  is a path s.t. pressure  $p$  is constant.

Recall that  $H = U + pV$ .

$$dH = dU + pdV + Vdp$$

$$C_p = \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_p = \tau \sigma(\tau, p)(\dot{\gamma}) = (dU(\tau, p)(\dot{\gamma}) + pdV(\tau, p)(\dot{\gamma})) = \left( \frac{\partial H}{\partial \tau} \right)_p$$

$$\text{Then } H = \int C_p d\tau = \int_{\gamma} C_p d\tau$$

**Example: Model system for gas-solid equilibrium** (pp.285 Ch. 10 “Phase transformations”, Kittel and Kroemer (1980) [2])

imagine solid of  $N$  atoms

each atom bound as harmonic oscillator of frequency  $\omega$

binding energy of each atom in ground state  $\epsilon_0$ , i.e. energy of atom in ground state is  $-\epsilon_0$

energy of a single oscillator  $\epsilon = n\hbar\omega - \epsilon_0$ ,  $n \in \mathbb{Z}^+$

$Z_s \equiv$  partition function of a single oscillator in solid,  $\beta \equiv 1/\tau$

$$Z_s = \sum_n \exp[-\beta(n\hbar\omega - \epsilon_0)] = e^{\beta\epsilon_0} \sum_n (e^{-\beta\hbar\omega})^n = \frac{e^{\beta\epsilon_0}}{1 - e^{-\beta\hbar\omega}}$$

Free energy  $F_s = U_s - \tau\sigma_s = -\tau \ln Z_s$

Gibbs free energy in the solid, per atom

$$G_s = U_s - \tau\sigma_s + pv_s = F_s + pV_s = \mu_s$$

Supposing  $v_s \ll v_g$  (volume  $v_s$  per atom in solid phase much smaller than volume  $v_g$  per atom in gas phase)

$$\mu_s = F_s + pv_s \approx F_s$$

$$\lambda_s := \exp(\beta\mu_s) \simeq \exp(\beta F_s) = \exp(-\ln Z_s) = \frac{1}{Z_s} = \exp(-\beta\epsilon_0)(1 - \exp(-\beta\hbar\omega))$$

use ideal gas approximation to describe gas phase, and spin of atom to be 0. From Ch. 6,

$$\lambda_g = \frac{n}{n_Q} = \frac{p}{\tau n_Q} = \frac{p}{\tau} \left( \frac{2\pi\hbar^2}{M\tau} \right)^{3/2}$$

gas in equilibrium with solid with  $\lambda_g = \lambda_s$ , or

$$\frac{p}{\tau n_Q} = \lambda_g = \lambda_s = e^{-\beta\epsilon_0}(1 - e^{-\beta\hbar\omega}) \implies p = \left( \frac{M}{2\pi\hbar^2} \right)^{3/2} \tau^{5/2} \exp(-\epsilon_0/\tau)(1 - \exp(-\hbar\omega/\tau))$$

## Van Der Waals Equation of State

$$F(\text{ideal gas}) = -N\tau[\ln(n_Q/n) + 1]$$

$n := \frac{N}{V} = \frac{N}{V - Nb}$  (hardcore repulsion at short distances)

## Mean Field Method

Let  $\varphi(r) :=$  potential energy of interaction of 2 atoms separated by distance  $r$ .  
 average value of total interaction of all other atoms on atom at  $r = 0$

$$\int_0^\infty dV \varphi(r) n = n \int_b^\infty dV \varphi(r) = -2na \quad (2 \text{ factor is convention})$$

excluded hardcore sphere of volume  $b$  from volume of integration.

assumed mean value of  $n$ , concentration  $n$  constant throughout volume accessible to gas molecules.

ignored increase of concentration in regions of strong attractive potential energy i.e.

mean field method neglects correlations between interacting molecules.

$$\Delta F \simeq \Delta U = \frac{-1}{2}(2Nna) = -N^2a/V$$

where  $\frac{1}{2}N(N-1) \simeq \frac{1}{2}N^2$  ( $N$  large)

$$\Rightarrow F(\text{vdW}) = -N\tau(\ln[n_Q(V-Nb)/N] + 1) - N^2a/V$$

$$p = -\left(\frac{\partial F}{\partial V}\right)_{\tau, N} = \frac{N\tau}{V-Nb} - \frac{N^2a}{V^2} \text{ or } \left(p + \frac{N^2a}{V^2}\right)(V-Nb) = N\tau$$

## Critical Points for the Van der Waals Gas

$$p_c = \frac{a}{27b^2}$$

$$V_c = 3Nb$$

$$\tau_c = \frac{8a}{27b}$$

$$\hat{p} := \frac{p}{p_c}$$

$$\hat{V} := \frac{V}{V_c}$$

$$\hat{\tau} := \frac{\tau}{\tau_c}$$

since

$$\begin{aligned} \left(\frac{p}{p_c} + \frac{3}{(V/V_c)^2}\right) \left(\frac{V}{V_c} - \frac{1}{3}\right) &= \frac{8\tau}{3\tau_c} & \frac{8\tau}{3\tau_c} &= \frac{8\tau}{3\frac{8a}{27b}} = \frac{9\tau}{a/b} \\ \left(\frac{p}{p_c} + \frac{3}{(V/V_c)^2}\right) &= \frac{1}{p_c} \left(p + \frac{3a/27b^2}{V^2/9N^2b^2}\right) = \frac{1}{p_c} \left(p + \frac{N^2a}{V^2}\right) & \frac{1}{p_c V_c} &= \frac{1}{\frac{a}{27b^2} 3Nb} = \frac{9}{a/b} \\ \left(\frac{V}{V_c} - \frac{1}{3}\right) &= \frac{1}{V_0}(V-Nb) \\ \Rightarrow \left(\hat{p} + \frac{3}{\hat{V}^2}\right) \left(\hat{V} - \frac{1}{3}\right) &= \frac{8}{3}\hat{\tau} \text{ or } \hat{p} = \frac{\frac{8}{3}\hat{\tau}}{\hat{V} - \frac{1}{3}} - \frac{3}{\hat{V}^2} \end{aligned}$$

at  $\hat{p} = 1$ ,  $\hat{V} = 1$ ,  $\hat{\tau} = 1$ ,  $\hat{\tau}$  constant,  $\left(\frac{\partial \hat{p}}{\partial \hat{V}}\right)_{\hat{\tau}} = 0$ ,  $\left(\frac{\partial^2 \hat{p}}{\partial \hat{V}^2}\right)_{\hat{\tau}} = 0$

Above  $\tau_c$ , no phase separation exists.

## Gibbs Free Energy of the Van der Waals Gas

$$G = F + pV$$

$$G(\tau, V, N) = \frac{N\tau V}{V-Nb} - \frac{2N^2a}{V} - N\tau(\ln[n_Q(V-Nb)/N] + 1)$$

Recall  $dG = \mu dN - \sigma d\tau + V dp$

$$\Rightarrow \left(\frac{\partial G}{\partial N}\right)_{\tau, p} = \mu = \mu(p, \tau) \text{ or } G(N, p, \tau) = N\mu(p, \tau)$$

cf. (9.13) of Kittel and Kroemer (1980) [2]

$G(\tau, p, N)$  solved numerically.

on p-V diagram,

coexistence line: consider  $dG = -\sigma d\tau + V dp + \mu dN$

at constant  $\tau$ , constant total number of particles,  $N$ ,

$$dG(\dot{\gamma}) = V dp(\dot{\gamma}) = G_g - G_l = \int V dp$$

if  $\int V dp = 0$ ,  $G_p = G_l$ ,  $G_g(\tau, p) = G_l(\tau, p)$ , so  $\mu_g(\tau, p) = \mu_l(\tau, p)$  along horizontal coexistence line

## Nucleation

Let  $\Delta\mu = \mu_g - \mu_l$  chemical potential difference between

vapor surrounding small liquid droplet and

liquid in bulk (infinitely large drop)

surface free energy of liquid drop is positive and tends to increase free energy of liquid

for small drop radii, surface can be dominant and drop can be unstable with respect to gas

$n_l \equiv$  concentration of molecules in liquid

$$\Delta G = G_l - G_g = -\left(\frac{4\pi}{3}\right) R^3 n_l \Delta\mu + 4\pi R^2 \gamma$$

where  $\gamma$  surface free energy per limit area.

$$\frac{d\Delta G}{dR} = 0 = -4\pi R^2 n_l \Delta\mu + 8\pi R \gamma \text{ or } R_c = \frac{2\gamma}{n_l \Delta\mu}$$

$R_c \equiv$  critical radius for nucleation

when  $R < R_c$ ,  $\Delta G > 0$ , so drop will evaporate (to gas) spontaneously

when  $R > R_c$ ,  $\Delta G < 0$ , drop will tend to grow spontaneously

$$(\Delta G)_c = \frac{16\pi}{3} \left[ \frac{\gamma^3}{n_l^2 (\Delta\mu)^2} \right]$$

where  $(\Delta G)_c = \Delta G(R = R_c)$

assume vapor behaves like ideal gas:

$$\Delta\mu = \tau \ln \left( \frac{p}{p_{\text{eq}}} \right)$$

Eq. (12a), chemical potential of the ideal gas,  $\mu = \tau \ln \left( \frac{n}{n_Q} \right) = \tau \ln \left( \frac{p}{p_{\text{eq}}} \right)$

$p \equiv$  vapor pressure in gas phase

$p_{\text{eq}} \equiv$  equilibrium vapor pressure of bulk liquid ( $R \rightarrow \infty$ )

use  $\gamma = 72 \frac{\text{erg}}{\text{cm}^2}$ ,  $p = 1.1 p_{\text{eq}}$ ,  $\tau = 300 \text{ K}$  ( $H_2O$ ),  $R_c = \frac{2\gamma}{n_l \Delta\mu} = 1 \times 10^{-6} \text{ cm}$ .

## Problems

### Problem 1. Entropy, energy, and enthalpy of van der Waals gas.

- (a) Recall the generalized thermodynamic identity; starting with  $\sigma = \sigma(U, V, N) \in C^\infty(\Sigma) = C^\infty((U, V, N))$ , the identity is  $dU = \tau d\sigma - p dV + \mu dN$ .

By definition,  $F = U - \tau\sigma$ , so that

$$dF = dU - d\tau\sigma - \tau d\sigma = -p dV + \mu dN - \sigma d\tau$$

Then clearly,  $\left( \frac{\partial F}{\partial \tau} \right)_{V, N} = -\sigma$ .

Now

$$F(\text{vdW}) = -N\tau(\ln[n_Q(V - Nb)/N] + 1) - N^2 a/V$$

as was postulated about the hardcore repulsion inside a molecule, making  $V$  into  $V - Nb$ .

Remembering that quantum concentration does depend on  $\tau$ :  $n_Q \equiv \left( \frac{M\tau}{2\pi\hbar^2} \right)^{3/2}$

Consider

$$\left( \ln \left( \left( \frac{M\tau}{2\pi\hbar^2} \right)^{3/2} \frac{(V-Nb)}{N} \right) + 1 \right) = \left( \frac{3}{2} \ln \tau + \ln \left( \left( \frac{M}{2\pi\hbar^2} \right)^{3/2} \frac{(V-Nb)}{N} \right) + 1 \right) \xrightarrow{\frac{\partial}{\partial \tau}} \frac{3}{2\tau}$$

Thus

$$\sigma = - \left( \frac{\partial F}{\partial \tau} \right)_{V,N} = N \left( \ln \left( \frac{n_Q(V-Nb)}{N} \right) + 1 \right) + N \frac{3}{2} = N \left( \ln \left( \frac{n_Q(V-Nb)}{N} \right) + \frac{5}{2} \right)$$

(b) Recall that  $F := U - \tau\sigma$  or  $U = F + \tau\sigma = F - \tau \left( \frac{\partial F}{\partial \tau} \right)_{V,N} = -\tau^2 \left( \frac{\partial}{\partial \tau} \left( \frac{F}{\tau} \right) \right)_{V,N}$

$$U = -\tau^2 \frac{3}{2\tau} (-N) + \frac{N^2 a}{V} = \boxed{\frac{3N\tau}{2} + \frac{N^2 a}{V} = U}$$

(c) Give to first order in the van der Waals correction terms  $a, b$ .

$$p = \frac{N\tau}{V-Nb} - \frac{N^2 a}{V^2} = \frac{N\tau}{V(1-Nb/V)} - \frac{N^2 a}{V^2} = \frac{N\tau}{V} \left( 1 + \frac{Nb}{V} \right) - \frac{N^2 a}{V^2}$$

where  $Nb \ll V$ .

Thus,

$$H(\tau, V) := U + pV = \boxed{\frac{5}{2}N\tau + \frac{N^2 b\tau}{V} - \frac{2N^2 a}{V}}$$

## Problem 2. Calculation of $dT/dp$ for water

Using the **Clausius-Clapeyron equation** or **vapor pressure equation**,

$$\frac{dp}{d\tau} = \frac{L}{\tau \Delta v}$$

assume  $\Delta v := v_g - v_l \approx v_g \equiv \frac{V_g}{N_g}$  (i.e. the volume of the gas is much bigger than the volume of the liquid), and  $pV_g = N_g\tau$  or  $\frac{V_g}{N_g} = \frac{\tau}{p}$  (i.e. water vapor acts like an ideal gas)

$$\frac{L}{\tau \Delta v} \approx \frac{L}{\tau^2 p}$$

Looking up fundamental constants and unit conversions *manually* appears, to me at least, in the 21st century, to be an obsolete exercise (“an exercise in insanity?”). I would like to propose presenting the fundamental constants and unit conversions, sourced from the NIST (National Institute of Standards and Technology) website, and using Python library **pandas**, as panda dataframe objects, to make the data accessible and computable, and persisting them or databasing the values as panda dataframe objects.

Open up **thermo.py**, which relies on the **Physique** repository (I’ll put it up on my **github**, **ernestyalumni**).

```
N_A = FundConst[ FundConst["Quantity"].str.contains("Avogadro") ].loc[42,:]
k_BOLTZ = FundConst[ FundConst["Quantity"].str.contains("Boltzmann") ].loc[49,:]
P_fmATM = conv[ conv["Toconvertfrom"].str.contains("atm") ].loc[15,:] # 1 atm to Pascal conversion
for pressure
...
## 2. Calculation of $dT/dp$ for water

L = 2260 # J g^{-1}
boilingwatertemp_K = KCconv.subs(T_C,100).rhs # room temperature in Kelvin
tau_boilingwater = boilingwatertemp_K * k_BOLTZ.Value
P = 1 # atm

dpdtau_1002 = L*18.0153/float(N_A.Value)/( tau_boilingwater**2 )
dtaudp_1002 = 1./ dpdtau_1002 # in J/Pascal
dTdp_1002 = dtaudp_1002 / (k_BOLTZ.Value ) # 28.4348535111262 K/atm
```

$$\boxed{\frac{dT}{dp} = 28.43 K/\text{atm}}$$

### Problem 3. Heat of vaporization of ice

Given

$$P(\text{H}_2\text{O vapor})(T = -2^\circ\text{C}) = 3.88 \text{ mm Hg}$$

$$P(\text{H}_2\text{O vapor})(T = 0^\circ\text{C}) = 4.58 \text{ mm Hg}$$

Use the **Clausius-Clapeyron equation** or **vapor pressure equation** (for coexistence equilibrium).

$$\frac{dp}{d\tau} = \frac{L}{\tau \Delta v} \approx \frac{L}{\tau^2/p}$$

$$\text{with } \frac{dp}{d\tau} \approx \frac{\Delta p}{\Delta \tau} = \frac{p(\tau_2) - p(\tau_1)}{\tau_2 - \tau_1}$$

Now idealizing water vapor as a perfect ideal gas,  $p = \frac{N}{V}\tau$ , so  $p$  linear in  $\tau$ , i.e.  $p = p(\tau)$ .

Note that while  $p$  is linear in  $\tau$ , the given temperature values correspond to  $T$ , in  $K$ , and so one would need this form of the slope equation  $y = mx + b$ , where  $m = \frac{N}{V}$  times a bunch of conversion factors, and one must also include a nonzero  $b$  constant, the result of choice of units (celsius versus energy units).

Thus,  $L = \frac{dp}{d\tau} \frac{\tau^2}{p}$ , with  $L$  being the latent heat of vaporization per particle.

Note, I treat Avogadro's number as a units conversion,  $1 = N_A = 6.02205 \times 10^{23} \frac{\text{particles}}{\text{mol}}$ . So in the end, for the desired  $J/\text{mol}$ , multiply  $L$  by  $N_A$ ,  $LN_A$ .

`L1003 = ( 4.58 - 3.88 ) / ( 0 - (-2.) ) * ((KCconv.subs(T_C,1.) . rhs) ** 2) / ( (4.58 - 3.88) / (0 - (-2)) * ( 1.) + 3.88 ) *`  
`k_BOLTZ.Value*N_A.Value # 51705.6757640485`

$$\boxed{L = 5.17 \times 10^4 J/\text{mol}}$$

### Problem 4. Gas-solid equilibrium

(a) Let's review the quantum harmonic oscillator (qho).

Recall the Hamiltonian in  $d$  dimensions

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

where  $\hat{p} = -i\frac{\partial}{\partial x^i}\mathbf{e}_i$ .

Suppose for the Hilbert space  $\mathcal{H}$ , that  $\mathcal{H} = \mathcal{H}_{x^1} \otimes \mathcal{H}_{x^2} \otimes \mathcal{H}_{x^3}$ ,

$$|\psi\rangle \in \mathcal{H} = \mathcal{H}_{x^1} \otimes \mathcal{H}_{x^2} \otimes \mathcal{H}_{x^3} \implies |\psi\rangle = |\psi_{x^1}\rangle \otimes |\psi_{x^2}\rangle \otimes |\psi_{x^3}\rangle \equiv |\psi_{x^1}\rangle |\psi_{x^2}\rangle |\psi_{x^3}\rangle$$

Now

$$\hat{H}|\psi\rangle = \sum_{i=1}^d \left( \frac{-1}{2m} \left( \frac{\partial}{\partial x^i} \right)^2 + \frac{1}{2}m\omega^2(\hat{x}^i)^2 \right) |\psi\rangle \equiv \left( \frac{-1}{2m} \left( \frac{\partial}{\partial x^i} \right)^2 + \frac{1}{2}m\omega^2(\hat{x}^i)^2 \right) |\psi\rangle$$

The ladder operators make it much more simpler and straightforward than directly solving the differential equation. Let

$$a_i = \sqrt{\frac{m\omega}{2}} \left( \hat{x}^i + \frac{i}{m\omega} \hat{p}_i \right) \quad \hat{x}^i = \sqrt{\frac{1}{2m\omega}} (a_i + a_i^\dagger)$$

$$a_i^\dagger = \sqrt{\frac{m\omega}{2}} \left( \hat{x}^i - \frac{i}{m\omega} \hat{p}_i \right) \quad \hat{p}_i = i\sqrt{\frac{m\omega}{2}} (a_i^\dagger - a_i)$$

Now let

$$N_i = a_i^\dagger a_i = \frac{m\omega}{2} ((\hat{x}^i)^2 + \frac{(\hat{p}_i)^2}{(m\omega)^2}) - \frac{1}{2}$$

where the uncertainty principle commutator relation was used:

$$\hat{x}^i \hat{p}_i - \hat{p}_i \hat{x}^i = [\hat{x}^i, \hat{p}_i] = i$$

Thus

$$\begin{aligned}\hat{H} &= (N_1 + \frac{1}{2})\omega + (N_2 + \frac{1}{2})\omega + (N_3 + \frac{1}{2})\omega \\ \hat{H}|\psi\rangle &= ((n_x + \frac{1}{2})\omega + (n_y + \frac{1}{2})\omega + (n_z + \frac{1}{2})\omega)|\psi\rangle\end{aligned}$$

Consider

$$Z_s = \sum_{n_x, n_y, n_z} \exp \left[ -\beta \left( (n_x + \frac{1}{2})\omega + (n_y + \frac{1}{2})\omega + (n_z + \frac{1}{2})\omega - \epsilon_0 \right) \right] = e^{\beta\epsilon_0} \left( \sum_n (e^{-\beta\omega})^n \right)^d = e^{\beta\epsilon_0} \left( \frac{1}{1 - e^{-\beta\omega}} \right)^d$$

The derivation of  $p = p(\tau)$  goes, as above (with Example: Model system for gas-solid equilibrium (pp.285 Ch. 10 “Phase transformations”, Kittel and Kroemer (1980) [2])); recall

$$\lambda_g = \lambda_s, \lambda_g = \frac{n}{n_Q} = \frac{p}{\tau n_Q}$$

and so for  $\tau \gg \hbar\omega$  or  $1 \gg \beta\omega$ ,

$$p = \left( \frac{M}{2\pi\hbar^2} \right)^{3/2} \tau^{5/2} e^{-\beta\epsilon_0} (1 - e^{-\beta\omega})^d \cong \left( \frac{M}{2\pi} \right)^{3/2} \frac{1}{\hbar^3} \tau^{5/2} e^{-\beta\epsilon_0} \beta^d \omega^d$$

For  $d = 3$ ,

$$p = \left( \frac{M}{2\pi} \right)^{3/2} \frac{\omega^3}{\tau^{1/2}} e^{-\beta\epsilon_0}$$

(b) From above part (a),

$$\frac{dp}{d\tau} \frac{1}{p} = \frac{-1}{2} \tau^{-1} + (-\epsilon_0) \left( \frac{-1}{\tau^2} \right)$$

Now using the approximation that  $\Delta v \approx v_g$ , and the ideal gas idealization,

$$L = \frac{dp}{d\tau} \frac{\tau^2}{p} = \epsilon_0 - \frac{\tau}{2}$$

My explanation: we must excite the atom by its (bounded) ground state energy,  $\epsilon_0$ . On the other hand, thermal energy of gas, with increasing linear temperature, helps to remove atom from solid to gas.

### Problem 5. Gas-solid equilibrium

(a) Now

$$F_s = -\tau \ln \frac{Z_s^{N_s}}{N_s!} = -\tau (N_s \ln Z_s - \ln N_s!) \simeq -\tau (N_s \ln Z_s - N_s \ln N_s + N_s)$$

where Stirling's approximation was used and  $(1/N_s!)$  factor is for indistinguishability (from quantum mechanics).

Recall

$$Z_s = \frac{e^{\beta\epsilon_0}}{(1 - e^{-\beta\omega})^d} \text{ so } \ln Z_s = \beta\epsilon_0 - d \ln (1 - e^{-\beta\omega})$$

from Example: Model system for gas-solid equilibrium (pp.285 Ch. 10 “Phase transformations”, Kittel and Kroemer (1980) [2]).

Then

$$F_s = -N_s \epsilon_0 + N_s \tau d \ln (1 - e^{-\beta\omega}) + \tau N_s \ln N_s - N_s \tau \quad (127)$$

Recall this useful relation (it's easy to calculate  $\sigma$  from Helmholtz free energy  $F$ ):

$$\left( \frac{\partial F}{\partial \tau} \right)_V = \sigma$$

**Correction (Helmholtz derivative).** Differentiating  $F = U - \tau\sigma$  gives  $(\partial F/\partial\tau)_V = -\sigma$ . The minimization in  $N_g$  below does not use this line. The boxed  $N_g$  and  $p_g$  are unchanged. See §33.4.

and so

$$\begin{aligned}\frac{\partial F_s}{\partial\tau} &= N_s d \ln(1 - e^{-\omega/\tau}) + N_s \ln N_s - N_s + N_s \tau d \left( \frac{1}{1 - e^{-\beta\omega}} \right) (-e^{-\beta\omega})(-\omega) \left( \frac{-1}{\tau} \right) = \\ &= N_s d \ln(1 - e^{-\beta\omega}) + N_s \ln N_s - N_s - N_s d \beta \omega \frac{e^{-\beta\omega}}{1 - e^{-\beta\omega}} = \sigma\end{aligned}\tag{128}$$

Suppose  $\tau \gg \omega$  or  $1 \gg \beta\omega$ . Then  $\sigma = N_s d \ln(1 - e^{-\beta\omega}) + N_s \ln N_s - N_s(1 + d)$ .

So take the (given) “extreme assumption that the entropy of the solid maybe neglected over the temperature range of interest.” [2]

Then comparing  $\sigma$  from Eq. 128 and  $F_s$  from Eq. 127, then

$$F_s \simeq -N_s \epsilon_0$$

$$F = F_s + F_g = -N_s \epsilon_0 + N\tau \left[ \ln \left( \frac{N_g}{V n_Q} \right) - 1 \right]$$

(b) Now

$$F = -N_s \epsilon_0 + N_g \tau \left[ \ln \left( \frac{N_g}{V n_Q} \right) - 1 \right] = -\epsilon_0(N - N_g) + N_g \tau \left( \ln \left( \frac{N_g}{V n_Q} \right) - 1 \right)$$

Now, taking the partial derivative,

$$\frac{\partial F}{\partial N_g} = \epsilon_0 + \tau \left( \ln \left( \frac{N_g}{V n_Q} \right) - 1 \right) + N_g \tau \frac{1}{N_g}$$

Taking the minimum free energy with respect to  $N_g$ , i.e. set the partial derivative to 0,

$$\frac{\partial F}{\partial N_g} = 0 = \epsilon_0 + \tau \ln \left( \frac{N_g}{V n_Q} \right) \text{ or } N_g = n_Q V \exp \left( \frac{-\epsilon_0}{\tau} \right)$$

(c) Idealize the gas with the ideal gas law.

$$pV = N_g \tau \implies p_g = n_Q \tau \exp \left( \frac{-\epsilon_0}{\tau} \right) = \left( \frac{M}{2\pi\hbar^2} \right)^{3/2} \tau^{5/2} \exp \left( \frac{-\epsilon_0}{\tau} \right) = p_g$$

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## 25 Phase Equilibrium

### 25.1 Condition of coexistence; Coexistence of phases

cf. Chapter 10: Phase Transformation, earlier sections of Kittel and Kroemer (1980) [2]; Subsection 3.5.2. “Coexistence of phases” of Le Bellac, Mortessagne, Batrouni (2004) [9]

The *condition of coexistence* is

$$\mu_1(\tau, p) = \mu_2(\tau, p)$$

For example 1 = g = gas

2 = l = liquid

Then Taylor expand:

$$\begin{aligned}\mu_1(\tau_0, p_0) &= \mu_2(\tau_0, p_0) = \mu_1(\tau_0, p_0) + \left(\frac{\partial \mu_1}{\partial p}\right)_\tau dp + \left(\frac{\partial \mu_1}{\partial \tau}\right)_p d\tau + \dots = \mu_2(\tau_0, p_0) + \left(\frac{\partial \mu_2}{\partial p}\right)_\tau dp + \left(\frac{\partial \mu_2}{\partial \tau}\right)_p d\tau + \dots \\ \implies \frac{dp}{d\tau} &= \frac{\left(\frac{\partial \mu_1}{\partial \tau}\right)_p - \left(\frac{\partial \mu_2}{\partial \tau}\right)_p}{\left(\frac{\partial \mu_2}{\partial p}\right)_\tau - \left(\frac{\partial \mu_1}{\partial p}\right)_\tau}\end{aligned}$$

Now

$$G = N\mu(\tau, p)$$

and so define

$$\begin{aligned}\frac{1}{N} \left(\frac{\partial G}{\partial \tau}\right)_{N,p} &= \frac{-\sigma}{N} \equiv -\hat{\sigma} = \left(\frac{\partial \mu}{\partial \tau}\right)_{N,p} \\ \frac{1}{N} \left(\frac{\partial G}{\partial p}\right)_{N,\tau} &= \frac{V}{N} \equiv v = \left(\frac{\partial \mu}{\partial p}\right)_{N,\tau}\end{aligned}$$

and so

$$\implies \frac{dp}{d\tau} = \frac{\hat{\sigma}_2 - \hat{\sigma}_1}{v_2 - v_1}$$

Equation 11 in Chapter 10 of Kittel and Kroemer (1980) [2] and Eq. 3.96 of Le Bellac, Mortessagne, Batrouni (2004) [9] agree.

Now consider  $\tau N d\hat{\sigma}$  for constant  $\tau$ ,

$$Q = \tau N d\hat{\sigma} \implies \int Q = \tau N (\hat{\sigma}_2 - \hat{\sigma}_1) \text{ or } \frac{\int Q}{N} = \tau (\hat{\sigma}_2 - \hat{\sigma}_1)$$

which is the heat added in the transfer.

Define the **latent heat of vaporization**

$$L := \tau (\hat{\sigma}_2 - \hat{\sigma}_1) \quad (\text{latent heat of vaporization}) \quad (129)$$

For the change of volume when 1 molecule is transferred from liquid to gas or 1 to 2,  $\Delta v$ ,  $\Delta v \equiv v_2 - v_1$ , and so

$$\left. \frac{dp}{d\tau} \right|_{\text{coexist}} = \frac{L}{\Delta v \tau} \quad (130)$$

## 25.2 Latent heat

$p, T$  const., liquid  $\rightarrow$  gas, e.g.  $p = 1$  atm

$$T = 373 \text{ K}$$

latent heat of vaporization  $L_{\text{vap}}$  (by def.) amount of energy supplied by heating.

$$Q = dU + W$$

$\epsilon := U/N$  = average (internal) energy per molecule

$v := V/N$  = volume per molecule

$$L_{\text{vap}} = d\epsilon + p dV$$

$$d\epsilon(\dot{\gamma}) = \epsilon_{\text{vap}} - \epsilon_{\text{liq}}$$

$$dv(\dot{\gamma}) = v_{\text{vap}} - v_{\text{liq}} > 0$$

if  $p$  const.,  $L_{\text{vap}} = d(\epsilon + pV) = dh$ ,  $h := H/N = \frac{U+pV}{N}$

EY : 20151031 Another way to think about it is this: recall that

$$Q = dU + W = dU + p dV = \tau d\sigma$$

$$H = U + pV \text{ so } dH = dU + p dV + V dp = Q + V dp$$

Consider a path  $\gamma \in M$  s.t.  $d\tau(\dot{\gamma}) = 0$  (constant  $\tau, p$ )

$$dp(\dot{\gamma}) = 0$$

$$Q(\dot{\gamma}) = dH(\dot{\gamma}) - Vdp(\dot{\gamma}) = dH(\dot{\gamma}) - 0 = dH(\dot{\gamma}) \quad Q(\dot{\gamma}) = \tau d\sigma(\dot{\gamma})$$

$$\int Q(\dot{\gamma}) = \int \tau d\sigma(\dot{\gamma}) = \tau(\sigma_g - \sigma_l)$$

Thus

$$\frac{\int Q(\dot{\gamma})}{N} = \tau(s_g - s_l) = \frac{H_g}{N} - \frac{H_l}{N}$$

$$L \equiv \tau(s_g - s_l) = \frac{\int Q(\dot{\gamma})}{N} = \frac{1}{N}(H_g - H_l)$$

### Latent heat versus heat capacity

Take slow, reversible process.

Now

$$C_V := \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_V$$

$$C_P := \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_P$$

It's stated in Kittel and Kroemer (1980), pp. 166, Equation (37), Chapter 6: Ideal Gas, Subsection "Heat capacity" [2], that

$$C_P = \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_P = \left( \frac{\partial U}{\partial \tau} \right)_P + p \left( \frac{\partial V}{\partial \tau} \right)_P \quad (131)$$

Suppose  $\sigma = \sigma(\tau, V)$ . With  $\tau d\sigma = dU + W = \tau d\sigma = dU + pdV$ ,

$$d\sigma = \frac{\partial \sigma}{\partial \tau} d\tau + \frac{\partial \sigma}{\partial V} dV = \frac{dU}{\tau} + \frac{p}{\tau} dV$$

$$d\sigma(\dot{\gamma}) = \left( \frac{\partial \sigma}{\partial \tau} \right)_V + 0 = \frac{1}{\tau} dU(\dot{\gamma}) = \frac{1}{\tau} \left( \frac{\partial U}{\partial \tau} \right)_V$$

Then it's clear that

$$\tau \left( \frac{\partial \sigma}{\partial \tau} \right)_V = \left( \frac{\partial U}{\partial \tau} \right)_V = C_V$$

Now suppose  $\sigma = \sigma(\tau, p)$ .

Consider also the enthalpy,  $H = U + pV$ , and so

$$dH = dU + Vdp + pdV = \tau d\sigma + Vdp$$

Now for  $\sigma = \sigma(\tau, p)$ ,

$$\sigma = \sigma(\tau, p)$$

$$d\sigma = \frac{\partial \sigma}{\partial \tau} d\tau + \frac{\partial \sigma}{\partial p} dp = \frac{dH}{\tau} - \frac{V}{\tau} dp \implies d\sigma(\dot{\gamma}) = \left( \frac{\partial \sigma}{\partial \tau} \right)_p + 0 = \frac{1}{\tau} \left( \frac{\partial H}{\partial \tau} \right)_p - 0$$

So

$$C_p := \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_p = \left( \frac{\partial H}{\partial \tau} \right)_p$$

Now

$$dH(\dot{\gamma}) = dU(\dot{\gamma}) + Vdp(\dot{\gamma}) + pdV(\dot{\gamma}) = \left( \frac{\partial U}{\partial \tau} \right)_p + 0 + p \left( \frac{\partial V}{\partial \tau} \right)_p = \left( \frac{\partial H}{\partial \tau} \right)_p \text{ so}$$

$$C_p = \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_p = \left( \frac{\partial U}{\partial \tau} \right)_p + p \left( \frac{\partial V}{\partial \tau} \right)_p$$

Kittel and Kroemer (1980) [2] argues that for ideal gas,

$$\left( \frac{\partial U}{\partial \tau} \right)_p = \left( \frac{\partial U}{\partial \tau} \right)_V$$

since  $U = U(\tau)$ .

## 25.3 Conditions for coexistence

Sec. 12.3 of Baierlein (1999) [1].

Recall

$$G = F + pV = U + pV - \tau\sigma = G(\tau, p, N)$$

Now

$$\begin{aligned} G = G(\tau, p, N_{\text{vap}}, N_{\text{liq}}) \implies dG &= \frac{\partial G}{\partial N_{\text{vap}}} dN_{\text{vap}} + \frac{\partial G}{\partial N_{\text{liq}}} dN_{\text{liq}} = \mu_{\text{vap}} 1 + \mu_{\text{liq}} (-1) = 0 \\ \implies \mu_{\text{vap}}(\tau, p) &= \mu_{\text{liq}}(\tau, p) \end{aligned}$$

2015 manuscript.

[6]

## 26 Properties of a Pure Substance

EY : 20151030 Is the word vapor the same as gas? vapour, gaz, gas, vapore

For coexistence equilibrium,

$$\begin{aligned} \mu_g(p_0, \tau_0) &= \mu_l(p_0, \tau_0) \text{ and} \\ \mu_g(p_0 + dp, \tau_0 + d\tau) &= \mu_l(p_0 + dp, \tau_0 + d\tau) \end{aligned}$$

and so

$$\frac{dp}{d\tau} = \frac{L}{\tau \Delta v} \quad (132)$$

where  $v \equiv \frac{V}{N}$ , the so-called **vapor pressure equation** or **Clausius-Clapeyron** equation.

For (2) approximations,  $\Delta v = v_g - v_l \approx v_g = \frac{V_g}{N_g}$  and idealize vapor as ideal gas,  $pV = N\tau$  so  $\frac{dp}{d\tau} = \frac{L}{\tau^2/p}$ .

Second, if  $L$  constant,

$$p(\tau) = p_0 \exp(-L_0/\tau) \text{ or } \ln\left(\frac{p(\tau)}{p_0}\right) = \frac{-L_0}{\tau} \quad (133)$$

cf. Ch. 10 Phase Transformations, pp. 278-284, “Derivation of the Coexistence Curve,  $p$  Versus  $\tau$ ” of Kittel and Kroemer (1980) [2].

Eq. 133 explains the shape of the coexistence curve between solid and gas (vapor) (sublimation) and liquid and gas (vapor) (vaporization; vaporization curve).

**Saturation** is this  $p = p(\tau)$  coexistence curve.

**Isotherms, Isothermals**

Recall  $G = F + pV$  and so  $dG = dF + Vdp + pdV = dU - \tau d\sigma - \sigma d\tau + pdV + Vdp = -\sigma d\tau + Vdp$

$$F = U - \tau\sigma \quad dG = -\sigma d\tau + Vdp$$

So then  $G = G(\tau, p)$

$$G = G(\tau, p, N)$$

where the latter statement is when we include particle transfer, so that

$$dG = -\sigma d\tau + Vdp + \mu dN \text{ for } \mu = \mu(\tau, p)$$

For the *ideal gas*:

$$F(\tau, V) = F = -N\tau \left( \ln \left( \frac{n_Q}{n} \right) + 1 \right)$$

$$\text{where } n_Q = \left( \frac{M\tau}{2\pi\hbar^2} \right)^{3/2}$$

$$n \equiv N/V = \frac{p}{\tau}$$

$$G(\tau, p, N) = -N\tau \left( \ln \left( \frac{n_Q}{n} \right) + 1 \right) + N\tau = -N\tau \left( \ln \left( \frac{n_Q}{n} \right) \right) = -N\tau \ln \left( \left( \frac{M\tau}{2\pi\hbar^2} \right)^{3/2} \frac{\tau}{p} \right)$$

$$\text{so then } \left( \frac{\partial G}{\partial N} \right)_{\tau, p} = \mu = -\tau \ln \left( \left( \frac{M\tau}{2\pi\hbar^2} \right)^{3/2} \frac{\tau}{p} \right)$$

For the *Van der Waals gas*

$$F(vdW) = -N\tau \left( \ln \left( \frac{n_Q(V - Nb)}{N} \right) + 1 \right) - \frac{N^2 a}{V}$$

$$p = - \left( \frac{\partial F}{\partial V} \right)_{\tau, N} = \frac{N\tau}{V - Nb} - \frac{N^2 a}{V^2}$$

$$G(\tau, V, N) = \frac{N\tau V}{V - Nb} - \frac{2N^2 a}{V} - N\tau \left( \ln \left( \frac{n_Q(V - Nb)}{N} \right) + 1 \right)$$

Nevertheless, consider, when considering isotherms, isothermals,

$$dG = -\sigma d\tau + V dp + \mu dN$$

Consider a path  $\gamma$  on constant  $\tau$ , constant total number of particles  $N$ , and so

$$dG(\dot{\gamma}) = V dp(\dot{\gamma}) = G_g - G_l = \int_{\gamma} V dp$$

*2015 manuscript.*

## 27 The coexistence curve

*Manuscript section title: Phase Equilibrium.*

Phase Equilibrium

Phase Diagram

(Phase) Coexistence Curve

Antoine Equation (parameters)

For coexistence equilibrium,  $\mu_g(p_0, \tau_0) = \mu_l(p_0, \tau_0)$  and  $\mu_g(p_0 + dp, \tau_0 + d\tau) = \mu_l(p_0 + dp, \tau_0 + d\tau)$ , and so  $\frac{dp}{d\tau} = \frac{L}{\tau \Delta v}$  with  $v \equiv \frac{V}{N}$ .

Make the approximation that  $\Delta v \equiv v_g - v_l \approx v_g = \frac{V_g}{N_g}$  and idealize vapor as ideal gas,  $pV = N\tau$ , so  $\frac{dp}{d\tau} = \frac{L}{\tau^2/p}$ . If  $L$  constant,

$$p(\tau) = p_0 \exp(-L_0/\tau) \text{ or } \ln \left( \frac{p(\tau)}{p_0} \right) = \frac{-L_0}{\tau} \quad (134)$$

Consider the Antoine Equation Parameters given in the NIST (National Institute of Standards and Technology) Chemistry WebBook <sup>5</sup>

$$\log_{10}(P) = A - (B/(T + C)) \quad (135)$$

---

<sup>5</sup>Phase change data for Oxygen

Then

$$P = 10^A = \frac{B}{T+C} = 10^A 10^{-\frac{B}{T+C}} = 10^A \exp\left(\frac{-B}{T+C} \ln 10\right)$$

Now  $\frac{D}{T+C} = \frac{D}{T} - \frac{DC}{T(T+C)}$ , and so

$$P = 10^A \exp\left(\frac{(B \ln 10)C}{T(T+C)}\right) \exp\left(\frac{-B \ln 10}{T}\right)$$

Consider how much the pressure is changed due to this  $C$  parameter. Consider  $P_0$ ,  $P_1$ , defined as such:

$$P_0 = 10^A \exp\left(\frac{-B \ln 10}{T}\right) \quad P_1 = 10^A \exp\left(\frac{(B \ln 10)C}{T(T+C)}\right) \exp\left(\frac{-B \ln 10}{T}\right) \Rightarrow \frac{P_0}{P_1} = \exp\left(\frac{(B \ln 10)C}{T(T+C)}\right)$$

So a deviation can be estimated from  $1 - \frac{P_0}{P_1}$ .

Open up `thermochem.py`. The *saturation curve* or *coexistence curve* for, for example, oxygen, from liquid to gas, can be reproduced. One needs to input in the Antoine parameters from the NIST website <sup>6</sup>

```
>>> O2coec = CoexistCurve(3.85845, 325.675, -5.667)
>>> plot(O2coec.curveSI.rhs, (T, 54.36, 100.16))

>>> N(O2coec.curveSI.rhs.subs(T, 60.))
731.804072053687
>>> N(O2coec.curveSI.rhs.subs(T, 70.))
6253.42680398774
>>> N(O2coec.curveSI.rhs.subs(T, 75.))
14494.1195824433
```

---

2015 manuscript.

## 28 Joule-Thomson Effect

cf. Problem 1.7.6 of Le Bellac, Mortessagne, Batrouni (2004) [9], Lecture 18, Sections “Throttling Processes”, “The Joule-Thomson Effect” of Groth, Phys301 Fall 1999 <sup>7</sup>

From Prob. 1.7.6 of Le Bellac, Mortessagne, Batrouni (2004) [9], their setup is the following: gas initially confined in fixed volume escapes freely without exchanging heat with its environment,

- initially, gas in compartment and  $(T_1, V_1, p_1)$
- expands into container of volume  $V_2$  under constant pressure  $p_2$  ( $p_2 < p_1$ )
- 2 compartments are thermally isolated; connected by porous plug to maintain pressure difference
- adiabatic (but not quasi-static) expansion.

$Q = 0$ . So  $Q = 0 = dU - W$ . Then

$$\int dU - \int W = \Delta U - \int W = 0$$

Now

$$\Delta H = H_2 - H_1 = u_2 + p_2 V_2 - U_1 - p_1 V_1 = \Delta U + p_2 V_2 - p_1 V_1$$

Treat the process of gas evacuating, or going from compartment 1 to container 2 as 2 processes:  $\int W = W_1 + W_2$  where

$W_1 = -p_1(0 - V_1) = p_1 V_1 \equiv$  work done on gas as the gas evacuates the compartment.

$W_2 = -p_2(V_2 - 0) = -p_2 V_2 \equiv$  work done by gas as gas expands into container at constant pressure  $p_2$ .

Thus  $\int W = W_1 + W_2 = p_1 V_1 - p_2 V_2$  and so

$\Delta H = 0$

---

<sup>6</sup><http://webbook.nist.gov/cgi/cbook.cgi?ID=C7782447&Mask=4#Thermo-Phase>

<sup>7</sup>Edward J. Groth, 1999, Fall 1999 Physics 301, [Phys301 23-Nov-1999 18-10](#)

The enthalpy is constant for this expansion.

For an ideal gas,  $H = U + pV = \frac{\gamma}{\gamma-1}N\tau$ , for an ideal gas has  $U = \frac{1}{\gamma-1}N\tau$ .  $\gamma = \frac{5}{3}$  and  $\frac{\gamma}{\gamma-1} = \frac{5/3}{2/3} = 5/2$ . Nevertheless, since  $H$  is constant,  $T$  is constant, and so there's no cooling of an ideal gas in a Joule-Thomson apparatus! But an ideal gas has no interaction energy, and it can be argued that one has to consider interaction energy between molecules for liquefaction (cf. Groth (1999) [10], [Lecture 18 Physics 301 Fall 1999](#)).

From the “Derivation of the Joule-Thomson (Kelvin) coefficient” section of the *Joule-Thomson effect* article in [Wikipedia](#), recall the definition of enthalpy  $H := U + pV$ , and so that  $dH = dU + pdV + Vdp = \tau d\sigma + Vdp$ .

Let  $\sigma = \sigma(\tau, p)$ .

$$dH = \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_p d\tau + (V + \tau \left( \frac{\partial \sigma}{\partial p} \right)_\tau) dp = C_p d\tau + (V + \tau \left( \frac{\partial \sigma}{\partial p} \right)_\tau) dp$$

since

$$Q = Q(\tau, p) = \frac{\partial Q}{\partial \tau} d\tau + \frac{\partial Q}{\partial p} dp = C_p d\tau + \frac{\partial Q}{\partial p} dp$$

and so for a thermodynamic process  $\gamma$  such that  $dp(\dot{\gamma}) = 0$  (i.e. constant pressure), then we can say

$$\frac{C_p}{\tau} d\tau = d\sigma \xrightarrow{\gamma} \int_\gamma \frac{C_p}{\tau} d\tau = \Delta\sigma \implies \tau \left( \frac{\partial \sigma}{\partial \tau} \right)_p = C_p$$

And now

$$dG = -\sigma d\tau + V dp = \left( \frac{\partial G}{\partial \tau} \right)_p d\tau + \left( \frac{\partial G}{\partial p} \right)_\tau dp$$

and since  $\frac{\partial^2 G}{\partial \tau \partial p} = \frac{\partial^2 G}{\partial p \partial \tau}$ ,

$$\left( \frac{\partial \sigma}{\partial p} \right)_\tau = - \left( \frac{\partial V}{\partial \tau} \right)_p = -V\alpha$$

Then

$$dH = C_p d\tau + V(1 - \tau\alpha) dp$$

and so

$$\left( \frac{\partial \tau}{\partial p} \right)_H = \frac{V(1 - \tau\alpha)}{-C_p} \tag{136}$$

is the **Joule-Thomson coefficient**, where  $\alpha$  is the coefficient of thermal expansion.

For an ideal gas,

$$\begin{aligned} pV &= N\tau \\ V &= \frac{N\tau}{p} \implies \frac{\partial V}{\partial \tau} = \frac{N}{p} = \frac{V}{\tau} \end{aligned}$$

so

$$dH = C_p d\tau + (V + \tau \left( \frac{-V}{\tau} \right)) dp = C_p d\tau$$

so that

$$\left( \frac{\partial \tau}{\partial p} \right)_H = 0$$

for an ideal gas.

For every real gas, every real gas has an *inversion temperature* such that

below the inversion temperature,  $\left( \frac{\partial T}{\partial p} \right)_H$  is positive, and so  $dp < 0$  always in an expansion, and so  $dT < 0$  and the gas cools, and

above the inversion temperature,  $\left( \frac{\partial T}{\partial p} \right)_H$  is negative, and so  $dp < 0$  always in an expansion, and so  $dT < 0$  and the gas warms.

**Correction (Joule-Thomson warming).** Above inversion the coefficient is negative and  $dp < 0$ , so  $d\tau > 0$ . The gas warms. The clause  $dT < 0$  in the sentence has the wrong sign. See §33.6.

An inversion temperature must exist because when the molecules are close enough, there are intermolecular forces that'll make them attract each other (such as temporary configurations so a slightly (electrically) negative cloud of electrons attract slightly positively charged part of another molecule) so the interaction energy is negative.

However, it appears that this inversion temperature is not found from the Joule-Thomson coefficient expression, Eq. 136, but from, possibly, the Van der Waals expression or another model (cf. Lecture 18 of Groth (1999) [10]).

Looking at Figure 1.16 of Le Bellac, Mortessagne, Batrouni (2004) [9], for isenthalpy curves ( $H = \text{const.}$ ), in  $(T, p)$  plane,  $T$  vs.  $p$ , the dashed line intersects the points on each isenthalpy curves where  $\left(\frac{\partial T}{\partial p}\right)_H = 0$  and for lower pressures ( $dp < 0$ , gas expands), cooling occurs.

There's a maximum temperature, about  $300^\circ C$  for nitrogen, where the dashed line meets the  $p = 0$  atm axis, meaning  $\left(\frac{\partial \tau}{\partial p}\right)_H < 0 \quad \forall p > 0$ , so no cooling occurs.

2008 manuscript.

## 29 Kinetic Theory

The wall-collision argument for  $P = N\tau/V$  is in §4.2.

### Kinetic Theory of the Ideal Gas Law

Consider molecule strike unit area of wall.

Let  $v_z \equiv$  velocity component normal to plane of wall.

Suppose molecules, of mass  $M$ , reflected specularly (mirror-like) from wall,

$$\Delta p_z = -2M|v_z|$$

Let  $a(v_z)dv_z$ , number of molecules per unit volume with  $z$ -component of velocity between  $v_z$  and  $v_z + dv_z$ .

$$\int a(v_z)dv_z = \frac{N}{V} = n$$

$a(v_z)v_zdv_z$  number of molecules in  $(v_z, v_z + dv_z)$  velocity range that strike unit area of wall in (per) unit time

$$\text{pressure } p = \int_0^\infty 2Mv_z a(v_z)v_z dv_z = M \int_{-\infty}^\infty v_z^2 a(v_z)dv_z = Mn\langle v_z^2 \rangle$$

$\frac{1}{2}M\langle v_z^2 \rangle = \frac{1}{2}\tau$  by equipartition of energy (Ch.3)

$$p = nM\langle v_z^2 \rangle = n\tau = \frac{N\tau}{V}$$

### Maxwell Distribution of Velocities

cf. Ch.6. distribution function of ideal gas  $f(\epsilon_n) = \lambda \exp\left(\frac{-\epsilon_n}{\tau}\right)$

Recall, Ch. 6, Sec. "Classical Limit" of Kittel and Kroemer [2], **an ideal gas is defined as a system of free noninteracting particles in the classical regime.**

$f(\epsilon) \equiv$  average occupancy of an orbital at energy  $\epsilon$

$\epsilon \equiv$  energy of orbital occupied by 1 particle; not energy of system of  $N$  particles

Fermi-Dirac and Bose-Einstein distribution  $f(\epsilon) = \frac{1}{\exp[(\epsilon - \mu)/\tau] \pm 1}$

In order for  $f(\epsilon) \ll 1 \quad \forall$  orbitals,  $\exp[(\epsilon - \mu)/\tau] \gg 1 \quad \forall \epsilon$ .

$$\Rightarrow f(\epsilon) \simeq \exp[(\mu - \epsilon)/\tau] = \lambda \exp(-\epsilon/\tau) \quad \lambda \equiv \exp\left(\frac{\mu}{\tau}\right)$$

$f(\epsilon)$ , average occupancy of orbital of energy  $\epsilon$ , is classical distribution function.

particle in a box:  $\epsilon_n = \frac{1}{2M} \left(\frac{\pi n}{L}\right)^2$  (for, recall  $\frac{1}{2M} \left(\frac{1}{i}\partial\right)^2 \psi = \frac{1}{2M} \partial^2 \psi = E\psi$ )

number of orbitals in range of quantum number  $(n, n + dn)$ , probability such orbital is occupied

$$\left(\frac{1}{2}\pi n^2 dn\right)f(\epsilon_n) = \frac{1}{2}\pi n^2 \lambda \exp(-\epsilon_n/\tau) dn$$

$$\frac{1}{2}Mv^2 = \frac{1}{2M} \left( \frac{\pi n}{L} \right)^2 \text{ or } n^2 = \frac{(ML)^2}{\pi^2} v^2 \text{ or } n = \frac{MLv}{\pi}$$

Consider system of  $N$  particles in volume  $V$ .

Let  $NP(v)dv$  number of atoms with velocity magnitude in range  $dv$  at  $v$

$$NP(v)dv = \frac{1}{2}\pi n^2 \lambda \exp(-\epsilon_n/\tau) \frac{dn}{dv} dv = \frac{1}{2}\pi \lambda \left( \frac{ML}{\pi} \right)^3 v^2 \exp\left(\frac{-Mv^2}{2\tau}\right) dv$$

cf. Ch. 6,  $\lambda = \frac{n}{n_Q} = \frac{N}{L^3} \left( \frac{2\pi\hbar^2}{M\tau} \right)^{3/2}$

$$\frac{1}{2}\pi \frac{N}{L^3} \left( \frac{2\pi}{M\tau} \right)^{3/2} \left( \frac{ML}{\pi} \right)^3 = 4\pi N \left( \frac{N}{2\pi\tau} \right)^{3/2}$$

**Correction (Maxwell prefactor).** The right-hand side above is not the prefactor of the speed density in the next display. That density is normalized, as checked in §33.8.

$$\implies P(v) = 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} v^2 \exp\left(\frac{-Mv^2}{2\tau}\right) \quad (137)$$

$P(v)$  is **Maxwell velocity distribution**,  $P(v)dv$  is probability particle has speed in  $dv$  at  $v$

### Experimental verification

velocity distribution of atoms which exit from slit of oven.

exit beam weighted in favor of atoms of high velocity at expense of those at low velocity.

weight factor is velocity component  $v \cos \theta$  normal to plane of hole.

$$\begin{aligned} \int (\cos \theta) drrd\varphi r \sin \theta d\theta &= \left( \frac{1}{3} (2\pi) R^3 \right) \int \cos \theta \sin \theta d\theta = \left( \frac{2\pi}{3} R^3 \right) \int \frac{\sin 2\theta d\theta}{2} = \frac{2\pi R^3}{3} \left( \frac{-\cos 2\theta}{4} \right) \Big|_0^{\pi/2} = \\ &= \frac{2\pi R^3}{3} \left( \frac{1+1}{4} \right) = \frac{2\pi R^3}{3} \left( \frac{1}{2} \right) \end{aligned}$$

Probability atom leaves hole will have velocity in  $(v, v + dv)$  :  $P_{\text{beam}}(v)dv$

$$P_{\text{beam}}(v) \propto v P_{\text{Maxwell}} \propto v^3 \exp\left(\frac{-Mv^2}{2\tau}\right)$$

with, recall  $P_{\text{Maxwell}} = 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} v^2 \exp\left(\frac{-Mv^2}{2\tau}\right)$

$P_{\text{beam}}$  distribution of transmission through a hole is Maxwell transmission distribution.

$$\begin{aligned} \langle v_{\text{out}} \rangle &= \int \int_0^{\pi/2} v \cos \theta \sin \theta d\theta P_{\text{Maxwell}}(v) dv = 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} \frac{1}{2} \int_0^\infty v^3 \exp\left(\frac{-Mv^2}{2\tau}\right) dv = \\ &= 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} \frac{1}{2} \left( \frac{1}{-2\alpha} \right) \left[ 0 - \frac{1}{\alpha} \right] = 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} \frac{1}{4} \left( \frac{4\tau^2}{M^2} \right) = \boxed{\frac{2^{1/2}}{\pi^{1/2}} \left( \frac{\tau}{M} \right)^{1/2}} \end{aligned}$$

for (doing the integration by hand)

$$\begin{aligned} (e^{-\alpha v^2})' &= -2\alpha v e^{-\alpha v^2} \\ (v^2 e^{-\alpha v^2})' &= -2\alpha v^3 e^{-\alpha v^2} + 2v e^{-\alpha v^2} \\ (v^2 e^{-\alpha v^2} + \frac{e^{-\alpha v^2}}{\alpha})' &= -2\alpha v^3 e^{-\alpha v^2} \end{aligned}$$

## Particle Diffusion

pp. 399 of Kittel and Kroemer [2]

Consider a system.

One end in diffusive contact with reservoir at chemical potential  $\mu_1$

Other end in diffusive contact with reservoir at chemical potential  $\mu_2$

Constant temperature  $\tau$ .

If  $\mu_1 > \mu_2$ , particle flow through system from reservoir 1 to reservoir 2;  $1 \rightarrow 2$ .

$n_i \equiv$  particle concentration in  $i$

Take

$$\mathbf{j}_n = -D \text{grad} n \quad (138)$$

which is **Fick's law**, and where  $D \equiv$  particle diffusion constant or **diffusivity**.

Mean free path  $l$ . Particles freely travel over  $l$ .

Assume in a collision at  $z$ , particles come into local equilibrium at local chemical potential  $\mu(z)$ , local concentration  $n(z)$ .

At  $z$ , particle flux density in positive  $z$  direction  $\frac{1}{2}n(z-l_z)\bar{c}_z$

particle flux density in negative  $z$  direction  $-\frac{1}{2}n(z+l_z)\bar{c}_z$

Note  $n(z-l_z)$  is particle concentration at  $z-l_z$

$$J_n^z = \frac{1}{2} [n(z-l_z) - n(z+l_z)] \bar{c}_z = -\frac{dn}{dz} \bar{c}_z l_z$$

where  $\bar{c}_z = \bar{c} \cos \theta$

$$\bar{l}_z = \bar{l} \cos \theta$$

$$\langle \bar{c}_z l_z \rangle = \bar{c} \bar{l} \frac{\int_{\text{hemisphere}} \cos^2 \theta dS}{\int_{\text{hemisphere}} dS} = \bar{c} \bar{l} \frac{\int_0^{\pi/2} d\theta \int_0^{2\pi} d\varphi \cos^2 \theta \sin \theta d\theta}{\int_0^{\pi/2} d\theta \sin \theta \int_0^{2\pi} d\varphi} = \bar{c} \bar{l} \frac{1}{3}$$

Comparing with Fick's law,

$$J_n^z = \frac{-1}{3} \bar{c} \bar{l} \frac{dn}{dz} \text{ or } \mathbf{J}_N = -\frac{1}{3} \bar{c} \bar{l} \nabla n$$

For diffusivity  $D$  is then  $D = \frac{1}{3} \bar{c} \bar{l}$ .

Now recall that  $\bar{l}$ , the *mean free path*, was derived from kinetic theory:

$$l = \frac{1}{n\pi d^2}$$

where  $d$  is the diameter of the particle.

The mean thermal velocity was derived from the Maxwell distribution:

$$\bar{c} = \left( \frac{8\tau}{M\pi} \right)^{1/2}$$

## Problems

### Problem 1. Mean speeds in a Maxwellian distribution.

$$(a) P(v) = 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} v^2 \exp \left( \frac{-Mv^2}{2\tau} \right)$$

Now

$$\begin{aligned} \int_0^\infty dv v^4 \exp(-\alpha v^2) &= \int_0^\infty v^3 \left( \frac{\exp(-\alpha v^2)}{-2\alpha} \right)' = 0 - \int 3v^2 \frac{\exp(-\alpha v^2)}{-2\alpha} = \int_0^\infty \frac{3}{2\alpha} v^2 \exp(-\alpha v^2) = \\ &= \frac{3}{2\alpha} \int v \left( \frac{e^{-\alpha v^2}}{-2\alpha} \right)' = \frac{3}{2\alpha} \left[ 0 - \int \frac{e^{-\alpha v^2}}{-2\alpha} \right] = \frac{3}{4\alpha^2} \frac{\sqrt{\pi}}{2\sqrt{\alpha}} \end{aligned}$$

So

$$\langle v^2 \rangle = \int_0^\infty v^2 P(v) = 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} \frac{3}{4 \left( \frac{M}{2\tau} \right)^2} \frac{\sqrt{\pi}}{\sqrt{\frac{M}{2\tau}}} \frac{1}{2} = \frac{3}{\frac{M}{2\tau}} \frac{1}{2} = \frac{3\tau}{M}$$

so

$$v_{rms} = \left( \frac{3\tau}{M} \right)^{1/2}$$

Now

$$\langle v^2 \rangle = 3\langle v_x^2 \rangle = \frac{3\tau}{M} \text{ so } \langle v_x^2 \rangle^{1/2} = \left( \frac{\tau}{M} \right)^{1/2}$$

(b)

$$\frac{\partial P(v)}{\partial v} = \frac{2P(v)}{v} + P(v) \left( \frac{-M}{\tau} \right) v = P(v) \left[ \frac{2}{v} - \frac{M}{\tau} v \right] = 0 \text{ if } v_{mp} = 0 \text{ or } v_{mp} = \boxed{\sqrt{\frac{2\tau}{M}}}$$

where  $mp$  stands for most probable.

$$v_{mp} = \sqrt{\frac{2\tau}{M}} < \sqrt{\frac{3\tau}{M}} = v_{rms}$$

(c)

$$\begin{aligned} \bar{c} &= \int_0^\infty dv v P(v) =: \langle v \rangle = \int_0^\infty dv 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} v^3 \exp \left( \frac{-Mv^2}{2\tau} \right) = 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} \int_0^\infty dv v^2 \left( \frac{e^{-\frac{Mv^2}{2\tau}}}{-\frac{M}{\tau}} \right)' \\ &= 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} \left[ 0 - \int_0^\infty dv 2v \frac{e^{-\frac{Mv^2}{2\tau}}}{-\frac{M}{\tau}} \right] = 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} \left[ \frac{2\tau}{M} \left( \frac{e^{-\frac{Mv^2}{2\tau}}}{-\frac{M}{\tau}} \right) \right]_0^\infty = 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} \frac{2\tau^2}{M^2} = \\ &= \frac{1}{(2\pi)^{3/2}} 8\pi \sqrt{\frac{\tau}{M}} = \boxed{\sqrt{\frac{8\tau}{\pi M}}} \end{aligned}$$

Note

$$\frac{v_{rms}}{\bar{c}} = \frac{\left( \frac{3\tau}{M} \right)^{1/2}}{\left( \frac{8\tau}{\pi M} \right)^{1/2}} = \left( \frac{3\pi}{8} \right)^{1/2}$$

(d)  $|v_z| = |v \cos \theta| = v |\cos \theta|$ . For a fixed  $\theta$ ,  $v \cos \theta$  is 1 point on a sphere, whereas  $v |\cos \theta|$  could be 2 pts on a sphere.

Now

$$\begin{aligned} \int_0^\pi |\cos \theta| \sin \theta d\theta &= \int_0^{\pi/2} \cos \theta \sin \theta d\theta + \int_{\pi/2}^\pi -\cos \theta \sin \theta d\theta = \int_0^{\pi/2} \frac{\sin 2\theta d\theta}{2} + \int_{\pi/2}^\pi \frac{-\sin 2\theta d\theta}{2} = \\ &= \left( \frac{\cos 2\theta d\theta}{-4} \right) \Big|_0^{\pi/2} + \left( \frac{\cos 2\theta}{4} \right) \Big|_{\pi/2}^\pi = \frac{1}{2} + \frac{1+1}{4} = 1 \end{aligned}$$

$P(v)dv$  is probability that particle has speed or velocity magnitude in  $(v, v + dv)$ .

It counts the event of  $v \cos \theta$  and  $v \cos(\pi - \theta)$ , so it counts  $v |\cos \theta|$  twice.  $P(v)dv = \int_0^{\pi/2} d\theta 2P(v |\cos \theta|)$

$$\Rightarrow \bar{c}_z = \langle |v_z| \rangle = \langle |v \cos \theta| \rangle = \langle v |\cos \theta| \rangle = \int_0^\pi \int_0^\infty v \frac{P(v)dv}{2} |\cos \theta| d\theta = \left( \frac{2\tau}{\pi M} \right)^{1/2}$$

## 30 Propagation

### Heat Conduction Equation

nonrelativistic case:

Let manifold  $N = \mathbb{R} \times M$ , with  $\dim M = n$ .

Let  $\mathbf{J} \in \mathfrak{X}(N) = \mathfrak{X}(\mathbb{R} \times M)$  be a vector field in  $N$ .

Let  $\rho \in C^\infty(N)$  be a smooth function on  $N$ .

$\rho = \rho(t, x)$  locally

Let  $J \in \Omega^1(N)$  be a 1-form on  $N$  that is isomorphic to  $\mathbf{J}$  (Tangent-Cotangent isomorphism theorem), i.e.

$$J = \mathbf{J}^\flat$$

$$J_i = g_{ij} \mathbf{J}^j$$

with  $g_{ij}$  being the metric on  $N$  (not just  $M$ !)

Note that as  $N = \mathbb{R} \times M$ ,  $g_{0j} = \delta_{0j}$

The local form of  $\mathbf{J}$  is the following:

$$\mathbf{J} = \rho \frac{\partial}{\partial t} + \mathbf{j}^i \frac{\partial}{\partial x^i} \quad i = 1 \dots n$$

So

$$J = J_i dx^i = g_{ij} \mathbf{J}^j dx^i = \rho dt + j_k dx^k \quad k = 1 \dots n$$

Thus

$$J = \rho dt + j_k dx^k \quad k = 1 \dots n \quad (139)$$

Now do the Hodge star operator, resulting in a  $n$ -form

$$*J \in \Omega^n(N)$$

and so

$$*J = \rho * dt + j_k * dx^k = \rho \text{vol}^n + j_j \text{vol}^{n+1}$$

as

$$\rho * dt = \rho \frac{\sqrt{g}}{n!} \epsilon^0_{i_1 \dots i_n} dx^1 \wedge \dots \wedge dx^n \quad i_1 \dots i_n \in \{1 \dots n\}$$

$$j_k * dx^k = g_{kl} j^l \frac{\sqrt{g}}{(n+1)!} \epsilon^k_{i_1 \dots i_n} dx^{i_1} \wedge \dots \wedge dx^{i_n} = j_j \text{vol}^{n+1}$$

so thus

$$*J = \rho \text{vol}^n + j_j \text{vol}^{n+1}$$

Hence

$$d * J = \frac{\partial}{\partial t} (\rho \sqrt{g}) \frac{1}{\sqrt{g}} \text{vol}^{n+1} + \frac{\partial}{\partial x^k} (\sqrt{g} j_k) \frac{1}{\sqrt{g}} \text{vol}^{n+1} = d(\rho \text{vol}^n) + d j_j \text{vol}^{n+1} \quad (140)$$

Special case:  $\frac{\partial \sqrt{g}}{\partial t} = 0$

$$d * J = \frac{\partial \rho}{\partial t} \text{vol}^{n+1} + \left( \frac{\partial}{\partial x^k} \ln \sqrt{g} \right) j_k \text{vol}^{n+1} + \frac{\partial j_k}{\partial x^k} \text{vol}^{n+1} = 0$$

$$\implies \frac{\partial \rho}{\partial t} + \left( \frac{\partial}{\partial x^k} \ln \sqrt{g} \right) j_k + \frac{\partial j_k}{\partial x^k} = 0$$

Special case: if  $\sqrt{g}$  constant,  $\frac{\partial \rho}{\partial t} + \frac{\partial j_k}{\partial x^k} = 0$

Let  $j \equiv -D \mathbf{d}\rho$  ( $j$  is a closed form on  $M$ ) where  $\mathbf{d}\rho = \frac{\partial \rho}{\partial x^i} dx^i \quad i = 1 \dots n$ ,  $D$  constant

$$d * J = d\rho \text{vol}^n + -D d * \mathbf{d}\rho = 0$$

For flat metric,  $\sqrt{g}$  constant,

$$\frac{\partial \rho}{\partial t} = D \frac{\partial^2 \rho}{(\partial x^k)^2}$$

For relativistic case,

Consider manifold  $M$ , with dimension  $\dim M = n + 1$

$$\mathbf{J} = \rho \frac{\partial}{\partial t} + \mathbf{j}^i \frac{\partial}{\partial x^i}$$

Let  $J = \mathbf{J}^\flat$ .  $J_\mu = g_{\mu\nu} \mathbf{J}^\nu$

For special case of flat Minkowski space,

$$J = -\rho dt + j_i dx^i$$

$$\begin{aligned} *J &= -\rho *dt + j_i *dx^i \\ &= -\rho \frac{\sqrt{g}}{(n+1)!} \epsilon_{i_1 \dots i_n} dx^{i_1} \wedge \dots \wedge dx^{i_n} + j_i \frac{\sqrt{g}}{(n+1)!} \epsilon^i_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_n} \\ d *J &= -\frac{\partial(\rho\sqrt{g})}{\partial t} \frac{1}{\sqrt{g}} \text{vol}^{n+1} + \frac{\partial(j_i \sqrt{g})}{\partial x^i} \frac{1}{\sqrt{g}} \text{vol}^{n+1} = 0 \\ &\implies -\frac{\partial(\rho\sqrt{g})}{\partial t} + \frac{\partial(j_i \sqrt{g})}{\partial x^i} = 0 \end{aligned}$$

Fick law (14.19) for particle flux density,  $j = -D_n \mathbf{d}n$  where  $D_n$  particle diffusivity constant  
 $n$  particle concentration

$$J = n dt + j$$

thermal conductivity; homogeneous medium  $\hat{C}$  heat capacity per unit volume.  $j_u = -K \mathbf{d}\tau$

$$J = \hat{C} \tau dt + j_u$$

$$\hat{C} \frac{\partial \tau}{\partial t} + \frac{\partial (j_u)_k}{\partial x^k} = 0 \quad (5)$$

$$\frac{\partial \tau}{\partial t} = D_\tau \frac{\partial^2 \tau}{(\partial x^k)^2} \quad D_\tau \equiv K/\hat{C} \quad (6)$$

## Propagation of Sound Waves in Gases

pressure associated with sound wave

$$\delta p = \delta p_0 \exp[i(kx - \omega t)] \quad (27)$$

Suppose ideal gas:

$$pV = N\tau \text{ or } p = \rho\tau/M \quad (28)$$

Consider “solid ball” or “billiard ball” particle (extended particle, not pt. particle, but no internal structure)

$$\rho = \frac{NM}{V}$$

Force on particle

$$F = \frac{dP}{dt} = \frac{d}{dt} \int \rho \text{vol}^n u = \frac{M}{V} \int \mathcal{L}_{\frac{\partial}{\partial t} + \mathbf{u}} \mathbf{u} = \frac{M}{V} \int \frac{\partial u}{\partial t} + [u, u]$$

Suppose  $[u, u] = 0$  (certainly for flat spaces; what about for curved spaces?  $[u, u] \neq 0$ ? Possibly? I don't know.  
 EY: 20150317

$$dU + p dV = \tau d\sigma \quad (145)$$

define fractional deviations  $s, \theta$

$$\begin{aligned}\rho &= \rho_0(1 + s) \\ \tau &= \tau_0(1 + \theta)\end{aligned}\quad (5) \quad (146)$$

where  $\rho_0, \tau_0$  are density and temperature in absence of sound wave.  
assume  $u, s, \theta$  have form of traveling  $\exp[i(kx - \omega t)]$

$$\omega u - \left(\frac{k\tau}{M}\right)s - \left(\frac{k\tau}{M}\right)\theta = 0 \quad (39) \quad (147)$$

$$\omega s - ku = 0 \quad (40) \quad (148)$$

$$\tau \hat{C}_V \theta - ps = 0 \text{ or } \hat{C}_V \theta - ns = 0 \quad (41) \quad (149)$$

$$\begin{aligned}\omega u &= \frac{k\tau}{M} \left(1 + \frac{n}{\hat{C}_V}\right)s \\ \omega &= \frac{k\tau}{M} \left(1 + \frac{n}{\hat{C}_V}\right) \frac{k}{\omega}\end{aligned}$$

So

$$\omega = \left(\frac{\gamma\tau}{M}\right)^{1/2} k \quad (42) \quad (150)$$

$$\begin{aligned}\gamma &= \frac{\hat{C}_V + n}{\hat{C}_V} = \frac{\hat{C}_p}{\hat{C}_V} \\ v_s &= \frac{\partial \omega}{\partial k} = \left(\frac{\gamma\tau}{M}\right)^{1/2}\end{aligned}$$

## Problems

### Problem 1. Fourier analysis of pulse

$t = 0$

$$\theta(x, 0) = \delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dk \exp(ikx) \quad (58) \quad (151)$$

$$\theta(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dk \exp[i(kx - \omega t)] \quad (59) \quad (152)$$

Given a dispersion relation at this form:

$$Dk^2 = i\omega \quad (10) \quad (153)$$

$$\theta(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dk \exp[ikx - Dk^2 t] \quad (60) \quad (154)$$

and so, doing the Gaussian integral,

$$\theta(x, t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(\frac{-x^2}{4Dt}\right) \quad (14) \quad (155)$$

### Problem Diffusion in two and three dimensions.

(a)

$$\begin{aligned}\frac{\partial \theta_2}{\partial t} &= -\frac{\theta_2}{t} + \frac{r^2}{4} \frac{\theta_2}{Dt^2} \\ \frac{\partial^2 \theta_2}{(\partial x^i)^2} &= -\frac{1}{2} \frac{\theta_2}{Dt} - \frac{1}{2} \frac{\theta_2}{Dt} + \frac{1}{4} \frac{\theta_2}{D^2 t^2} (x^2 + y^2) \\ \implies \frac{\partial \theta_2}{\partial t} &= D \frac{\partial^2 \theta_2}{(\partial x^i)^2}\end{aligned}$$

(b)

(c)

2015 manuscript.

## 31 Hydrodynamics

Another example: hydrodynamics; Eulerian description in fixed volume.

Define  $\epsilon := \frac{E}{V} = \hat{\epsilon} \cdot \left(\frac{N}{V}\right) \equiv \hat{\epsilon}n$ .

$$dE = \tau Nd\hat{\sigma} + \hat{h}dN \xrightarrow{\frac{1}{V}} d\epsilon = \tau nd\hat{\sigma} + \hat{h}dn$$

in isentropic process,  $(\frac{1}{2}\rho u^2 + h)$   $\mathbf{u}$  is energy current;  $h = \frac{H}{V}$ ,  $\mathbf{u}$  velocity of fluid  
2015 manuscript.

## Part II

# Nonequilibrium

## 32 Irreversible processes: macroscopic theory

Chapter 6 of Le Bellac, Mortessagne, Batrouni (2004) [9]

### 32.1 Flux, affinities, transport coefficients

Section 6.1 of Le Bellac, Mortessagne, Batrouni (2004) [9].

Assume system composed of homogeneous cells, small on macroscopic scale, but large on microscopic, labelled  $(a, b, \dots)$ .

Assume cells weakly interacting, so each cell independently attains local equilibrium with microscopic relaxation time  $\tau_{\text{micro}}$

Assume  $\tau_{\text{micro}} \ll \tau_{\text{macro}}$  where

$\tau_{\text{macro}}$  macroscopic relaxation time, for global equilibrium

#### 32.1.1 Affinities and transport coefficients

Suppose neighboring cells  $a, b$  have different intensive variables  $\gamma_i$ , an exchange of  $A_i$  will take place between them.

Recall, from the Gibbs-Duhem relation, 123,

$$Nd\mu + \sigma d\tau - Vdp = 0$$

For the  $d\tau = 0$  isothermal case, define

$$n := \frac{N}{V} \equiv \text{particle density} \quad (156)$$

Then

$$\begin{aligned} dp &= nd\mu \\ \Rightarrow \left( \frac{\partial p}{\partial \mu} \right)_{\tau} &= n \end{aligned} \quad (157)$$

Recall

### 32.1.2 Dissipation and Entropy Production

cf. Subsection 6.1.5 of Le Bellac, Mortessagne, Batrouni (2004) [9].

$$\frac{dS(a)}{dt} + \sum_{b \neq a} \Phi_S(a \rightarrow b) = \frac{1}{2} \sum_{i, b \neq a} \Gamma_i(a, b) \Phi_i(a \rightarrow b) \quad (158)$$

since

$$\frac{dS(a)}{dt} = - \sum_{i, b \neq a} \frac{1}{2} (\gamma_i(a) + \gamma_i(b)) \Phi_i(a \rightarrow b) + \frac{1}{2} \sum_{i, b \neq a} \Gamma_i(a, b) \Phi_i(a \rightarrow b)$$

and since

$$\Phi_S(a \rightarrow b) = \frac{1}{2} \sum_i (\gamma_i(a) + \gamma_i(b)) \Phi_i(a \rightarrow b) = -\Phi_S(b \rightarrow a) \quad (159)$$

Calculate  $\frac{dS_{\text{tot}}}{dt}$ , evolution of total entropy.

$$\frac{dS_{\text{tot}}}{dt} = \frac{d}{dt} \sum_a S(a) = \sum_a \frac{dS(a)}{dt}$$

Note that

$$\Phi_S(b \rightarrow a) + \Phi_S(a \rightarrow b) = 0$$

by antisymmetry of  $\Phi_S(a \rightarrow b)$ .

The  $\sum_a \sum_{b \neq a}$  is the summation of all the ways of making a size 2 ordered choice of 2 elements out of the possible cells  $\{a, b, \dots\}$ ; if there are  $n$  total cells, then there are  $n(n-1)$  ways of picking out 2, *in order*. Note that  $n(n-1)$  *must* be even. Then

$$\sum_a \sum_{b \neq a} \Phi_S(a \rightarrow b) = \frac{1}{2} \sum_a \sum_{b \neq a} \Phi_S(a \rightarrow b) + \frac{1}{2} \sum_b \sum_{a \neq b} \Phi_S(b \rightarrow a) = \frac{1}{2} \sum_a \sum_{b \neq a} \Phi_S(a \rightarrow b) + \frac{1}{2} \sum_b \sum_{a \neq b} -\Phi_S(a \rightarrow b) = 0$$

So  $\Phi_S$  doesn't contribute to  $\frac{dS_{\text{tot}}}{dt}$ :  $\implies$  *reversible exchange*

$$\frac{dS_{\text{tot}}}{dt} = \frac{1}{2} \sum_a \sum_{i, b \neq a} \Gamma_i(a, b) \Phi_i(a \rightarrow b)$$

entropy production, called *dissipation*; all physical phenomena accompanied by entropy production called *dissipative*.

### 32.1.3 Particle Diffusion

cf. pp. 399 of Kittel and Kroemer [2]

Consider a system.

One end in diffusive contact with reservoir at chemical potential  $\mu_1$

Other end in diffusive contact with reservoir at chemical potential  $\mu_2$

Constant temperature  $\tau$ .

If  $\mu_1 > \mu_2$ , particle flow through system from reservoir 1 to reservoir 2;  $1 \rightarrow 2$ .

$n_i \equiv$  particle concentration in  $i$

Take

$$\mathbf{j}_n = -D \text{grad} n \quad (160)$$

which is **Fick's law**, and where  $D \equiv$  particle diffusion constant or **diffusivity**.

Mean free path  $l$ . Particles freely travel over  $l$ .

Assume in a collision at  $z$ , particles come into local equilibrium at local chemical potential  $\mu(z)$ , local concentration  $n(z)$ .

At  $z$ , particle flux density in positive  $z$  direction  $\frac{1}{2} n(z - l_z) \bar{c}_z$

particle flux density in negative  $z$  direction  $-\frac{1}{2} n(z + l_z) \bar{c}_z$

Note  $n(z - l_z)$  is particle concentration at  $z - l_z$

$$J_n^z = \frac{1}{2} [n(z - l_z) - n(z + l_z)] \bar{c}_z = -\frac{dn}{dz} \bar{c}_z l_z$$

where  $\bar{c}_z = \bar{c} \cos \theta$

$$\bar{l}_z = \bar{l} \cos \theta$$

$$\langle \bar{c}_z l_z \rangle = \bar{c} \bar{l} \frac{\int_{\text{hemisphere}} \cos^2 \theta dS}{\int_{\text{hemisphere}} dS} = \bar{c} \bar{l} \frac{\int_0^{\pi/2} d\theta \int_0^{2\pi} d\varphi \cos^2 \theta \sin \theta d\theta}{\int_0^{\pi/2} d\theta \sin \theta \int_0^{2\pi} d\varphi} = \bar{c} \bar{l} \frac{1}{3}$$

Comparing with Fick's law,

$$J_n^z = -\frac{1}{3} \bar{c} \bar{l} \frac{dn}{dz} \text{ or } \mathbf{J}_N = -\frac{1}{3} \bar{c} \bar{l} \nabla n$$

For diffusivity  $D$  is then  $D = \frac{1}{3} \bar{c} \bar{l}$ .

Now recall that  $\bar{l}$ , the *mean free path*, was derived from kinetic theory:

$$l = \frac{1}{n\pi d^2}$$

where  $d$  is the diameter of the particle.

The mean thermal velocity was derived from the Maxwell distribution:

$$\bar{c} = \left( \frac{8\tau}{M\pi} \right)^{1/2}$$

### 32.1.4 Momentum conservation

Introduce *stress tensor*  $\mathcal{P}_{\alpha\beta}$ ,  $-\mathcal{P}_{\alpha\beta}$  is  $\alpha$  component of force per unit area applied by fluid outside parallelepiped on surface  $\mathcal{S}_\beta$ .

### 32.1.5 Hydrodynamics of the perfect fluid

cf. Subsection 6.5.2 Hydrodynamics of the perfect fluid, of Le Bellac, Mortessagne, Batrouni (2004) [9].

Consider a fluid where the only internal force is pressure. Assume there's no thermal conduction. In such a fluid, dissipation is absent. The fluid is a so-called "perfect fluid." Then right hand side for these transport equations (Eq. (6.87), (6.88) of Le Bellac, Mortessagne, Batrouni (2004) [9] is 0:

$$\mathbf{j}'_E = L_{EE} \nabla \frac{1}{T}$$

$$\mathcal{P}_{\alpha\beta} - \delta_{\alpha\beta} \mathcal{P} = -\xi \delta_{\alpha\beta} (\partial_\gamma u_\gamma) - 2\eta \Delta_{\alpha\beta}$$

1. Consider the force acting on a fluid volume element  $\text{vol}^n \in \Omega^n(N)$ , where spatial manifold has dimension  $n$ , i.e.  $\dim N = n$ .

The **total force** applied by rest of fluid in fluid enclosed in parallelepiped (generalized to  $V$ ), integrate over  $\partial V$  (all forces)

$$-\int_{\partial V} \mathcal{P}_{\alpha\beta} dx^\alpha \otimes dS^\beta = -\int_{\partial V} \mathcal{P}_{\alpha\beta} dS^\beta \otimes dx^\alpha = -\int_V d(\mathcal{P}_{\alpha\beta} \sqrt{g}) \otimes dx^\alpha = -\int_V \frac{\partial(\mathcal{P}_{\alpha\beta} \sqrt{g})}{\partial x^\beta} \frac{1}{\sqrt{g}} \text{vol}^n \otimes dx^\alpha$$

Now consider the "left-hand side" which describes the dynamics (or kinematics?). For *total* momentum  $\Pi$

$$\Pi = \int_V \rho u_i dx^i \otimes \text{vol}^n$$

$$\frac{d\Pi}{dt} = \dot{\Pi} = \frac{d}{dt} \int_V \rho u_i \text{vol}^n \otimes dx^i = \int_V \frac{\partial(\rho u_i)}{\partial t} \text{vol}^n \otimes dx^i + \int_V \mathcal{L}_{\mathbf{u}} \rho u_i \text{vol}^n \otimes dx^i = \int_V \frac{\partial(\rho u_i)}{\partial t} \text{vol}^n \otimes dx^i + \int_V \mathbf{d}(\rho u_i u^j dS_j)$$

Now

$$\mathbf{d}(\rho u_i u^j dS_j) = \frac{\partial \rho}{\partial x^j} u^j u_i \text{vol}^n + \rho \left[ \frac{\partial u_i}{\partial x^j} u^j \text{vol}^n + u_i \frac{\partial u^j}{\partial x^j} \text{vol}^n \right]$$

where, recalling

$$dS_j = \frac{\sqrt{g}}{(n-1)!} \epsilon_{ji_2 \dots i_n} dx^{i_2} \wedge \dots \wedge dx^{i_n}$$

then

$$dx^k \wedge dS_j = \frac{\sqrt{g}}{(n-1)!} \epsilon_{ji_2 \dots i_n} dx^k \wedge dx^{i_2} \wedge \dots \wedge dx^{i_n} = \frac{\sqrt{g}}{(n-1)!} \epsilon_{ji_2 \dots i_n} \epsilon_{12 \dots n}^{ki_2 \dots i_n} \frac{dx^1 \wedge \dots \wedge dx^n}{n} = \delta_j^k \text{vol}^n$$

2. Consider the *lab frame* (with unprimed notation).

Consider the frame *at rest in the fluid's frame* (with primed notation).

Note that

$$\frac{d}{dt} \int_V \text{vol}^n = \int_V \mathcal{L}_{\frac{\partial}{\partial t} + \mathbf{u}} \text{vol}^n = \int_V 0 + \mathbf{d}i_{\mathbf{u}} \text{vol}^n + 0 = \int_{\partial V} i_{\mathbf{u}} \text{vol}^n = \int_{\partial V} u^i dS_i$$

Let  $s \equiv S/V \equiv \bar{\sigma} \equiv \frac{\sigma}{V}$  entropy per unit volume in fluid's frame

$h' \equiv H/V$  enthalpy per unit volume in *fluid's frame*.

Note that  $s = s'$  since entropy is Galilean invariant.

Now

$$\check{s} \equiv S/N \equiv \check{\sigma} \equiv \sigma/N$$

$$\check{h}' \equiv H/N$$

Let's review Subsubsection 23.0.1, which is on pp.137 at the end of Section 3.5.1. Basic formulae of Le Bellac, Mortessagne, Batrouni (2004) [9].

### 32.1.6 Conduction and Convection

Consider system at equilibrium, with thermostat at temperature  $T$ , kept at pressure  $p$ , able to exchange particles, with reservoir at chemical potential  $\mu$ .

For 1 kind of particle, let

$$\check{s} \equiv S/N \text{ entropy per particle}$$

$$\check{\epsilon} \equiv E/N \text{ energy per particle}$$

Thus

$$\tau d\sigma = \tau d(N\check{\sigma}) = \tau N d\check{\sigma} + \tau \check{\sigma} dN$$

where

$\tau N d\check{\sigma}$  is entropy change due to change in entropy per particle, i.e. **conduction term**

$\tau \check{\sigma} dN$  is entropy change due to change in number of particles, i.e. **convection term**

For constant volume,

$$dE = TN d\check{\sigma} + (T\check{\sigma} + \mu) dN = TN d\check{\sigma} + \check{h} dN$$

where  $\check{h} = \bar{H}/N = \check{\epsilon} + Pv$  is enthalpy per particle, since, recall,

$$\bar{H}(S, P) = E + PV$$

$$\mu N = E - TS + PV = G$$

*physical interpretation* of  $\check{h} dN$ : if  $dN$  particles transported into a given volume by convection, energy increases by  $\check{\epsilon} dN$ .

But to return to initial volume, it's necessary to compress by  $vdN$  and so add energy to do work  $PvdN$  by pressure.

Example: hydrodynamics, where 1 uses Eulerian description in fixed volume.

Define energy density  $\epsilon = \frac{E}{V} = \check{\epsilon} n$ .

$$d\epsilon = Tnd\check{s} + \check{h}dn$$

separates conduction and convection terms.

To reiterate, in my notation,

$$\begin{aligned}
H &= E + pV \xrightarrow{1/N} \check{h} = \check{\epsilon} + pV \\
G &= F + pV = E - \tau\sigma + pV = \mu N = H - \tau\sigma \implies \mu = \check{h} - \tau\check{\sigma} \\
dE &= Q + W + \mu dN \implies dE = \tau d\sigma - p dV + \mu dN \\
&= \tau N d\check{\sigma} + \tau \check{\sigma} dN - p dV + \mu dN = \tau N d\check{\sigma} + (\tau \check{\sigma} + \mu) dN = \tau N d\check{\sigma} + \check{h} dN
\end{aligned}$$

where the process considered was for constant volume, and so  $dV = 0$ , and one must keep in mind that the general form of the (internal) energy is as a function of  $\sigma, V, \{N_i\}$ , so that

**Correction (enthalpy per particle, again).** The same per-particle enthalpy is  $\check{\epsilon} + pv$ , with  $v = V/N$ .

$$dE = \tau d\sigma - p dV + \sum_i \mu_i dN_i$$

## 33 Resolutions

Each item states the correct relation, the hypotheses, and the manuscript line it replaces. The manuscript text is still in the preceding sections.

### 33.1 Ideal-gas internal energy, and the adiabatic path

The 2015 notes ask why, on an adiabatic path, one may write  $dU = C_V d\tau$  and drop  $(\partial U/\partial V)_\tau dV$ . They also write, in that question,  $dU = C_V \tau$ . The differential is  $C_V d\tau$ . The product  $C_V \tau$  is not that differential.

Start from (2) at fixed  $N$ ,

$$dU = \tau d\sigma - p dV. \quad (161)$$

Take  $U = U(\tau, V)$  on the equilibrium surface, so

$$dU = \left( \frac{\partial U}{\partial \tau} \right)_V d\tau + \left( \frac{\partial U}{\partial V} \right)_\tau dV. \quad (162)$$

The definition  $C_V = (\partial U/\partial \tau)_V$  turns the first coefficient into  $C_V$ . It remains to compute the second. From  $F = U - \tau\sigma$  and  $dF = -\sigma d\tau - p dV$ ,

$$\left( \frac{\partial \sigma}{\partial V} \right)_\tau = \left( \frac{\partial p}{\partial \tau} \right)_V. \quad (163)$$

Hold  $\tau$  fixed in (161) and solve for the volume derivative:

$$\left( \frac{\partial U}{\partial V} \right)_\tau = \tau \left( \frac{\partial \sigma}{\partial V} \right)_\tau - p = \tau \left( \frac{\partial p}{\partial \tau} \right)_V - p. \quad (164)$$

The second equality is (163). This identity is general at fixed  $N$ . It has not used an ideal-gas law and it has not used  $Q = 0$ .

Now restrict to a classical ideal gas,

$$p = \frac{N\tau}{V}. \quad (165)$$

Then  $(\partial p/\partial \tau)_V = N/V = p/\tau$ , and (164) collapses:

$$\left( \frac{\partial U}{\partial V} \right)_\tau = \tau \cdot \frac{p}{\tau} - p = 0. \quad (166)$$

So  $U = U(\tau)$  only. Equation (162) becomes

$$dU = C_V(\tau) d\tau \quad (167)$$

on every path on the ideal-gas surface, adiabatic or not. The term that the 2015 question could not place is zero because of (165), before  $Q = 0$  is imposed.

The reversible heat is then

$$Q = dU + p dV = C_V d\tau + \frac{N\tau}{V} dV. \quad (168)$$

An adiabatic reversible path is  $Q = 0$ . If  $C_V$  is independent of  $\tau$ , which is the classical gas with a temperature-independent heat capacity,

$$\frac{d\tau}{\tau} + \frac{N}{C_V} \frac{dV}{V} = 0. \quad (169)$$

Integrate from an initial state to a final state:

$$\ln \frac{\tau_f}{\tau_i} + \frac{N}{C_V} \ln \frac{V_f}{V_i} = 0, \quad \tau V^{N/C_V} = \text{constant}. \quad (170)$$

The difference  $C_p - C_V$  uses the same two facts,  $U = U(\tau)$  and (165), and does not need the monatomic value of  $C_V$ . The enthalpy is  $H = U + pV = U + N\tau$ , so

$$C_p = \left( \frac{\partial H}{\partial \tau} \right)_p = \left( \frac{\partial U}{\partial \tau} \right)_p + N = C_V + N, \quad (171)$$

where  $(\partial U / \partial \tau)_p = (\partial U / \partial \tau)_V$  because  $U$  does not depend on  $V$  or on  $p$ . Therefore

$$\frac{N}{C_V} = \frac{C_p - C_V}{C_V} = \gamma - 1, \quad \gamma = \frac{C_p}{C_V}, \quad (172)$$

and  $\tau V^{\gamma-1}$  is constant. For a monatomic gas in three dimensions, equipartition gives  $C_V = 3N/2$  and  $\gamma = 5/3$ . For a diatomic gas with  $C_V = 5N/2$ , the same  $C_p - C_V = N$  holds and  $\gamma = 7/5$ . The 2015 question about molecules other than monatomic ones is this split:  $C_p - C_V = N$  is the ideal-gas law, while the value of  $C_V$  is the internal spectrum.

The Carnot-cycle notes ask why  $C_V$ , whose definition holds  $V$  fixed, may be used on the adiabatic leg  $2 \rightarrow 3$ , where  $V$  changes. The subscript  $V$  names the variable held fixed in the partial derivative that defines the state function  $C_V(\tau)$ . It does not name a constraint on the path. Once (166) is in force, (167) is the differential of  $U$  along the adiabatic leg and along every other leg.

### 33.2 Path dependence is the failure of $dQ = 0$

The 2008 notes say that a reversible process is one whose 1-forms are exact, and later that  $Q$  and  $W$  are not necessarily exact, so the state space would have to be topologically nontrivial. Those two sentences cannot stand together with  $Q = \tau d\sigma$ .

The energy  $U$  is a state function, so  $dU$  is exact. On a reversible path the heat 1-form is defined to be

$$Q = \tau d\sigma. \quad (173)$$

Its exterior derivative, with  $\tau$  and  $\sigma$  treated as coordinates, or as functions whose differentials are known, is

$$dQ = d(\tau d\sigma) = d\tau \wedge d\sigma. \quad (174)$$

The 2-form  $d\tau \wedge d\sigma$  vanishes when  $\tau$  is a function of  $\sigma$  alone. In the  $(\tau, V)$  chart it does not vanish:  $\sigma = \sigma(\tau, V)$  and  $C_V$  is not the whole story of  $Q$ , because of the second term in (168). So  $Q$  fails to be closed. On an open set of  $\mathbb{R}^2$ , which is contractible, the Poincaré lemma says that a closed 1-form is exact. The contrapositive used here is local: a 1-form that is not closed is not exact, on a domain with no interesting topology. Path dependence of the reversible heat is  $dQ \neq 0$ . It is already visible for an ideal gas in the  $(\tau, V)$  plane.

The work 1-form  $W_{\text{by}} = p dV$  is likewise not closed once  $p = p(\tau, V)$ . Their sum  $Q - W_{\text{by}} = dU$  is exact. Exactness belongs to  $dU$ , including on irreversible processes when  $U$  is evaluated between equilibrium endpoints. It is not the definition of a reversible path.

### 33.3 Work sign in the heat-and-work notes

One line in the heat-and-work notes writes  $dU = Q + W = \tau d\sigma + p dV$ . The identity in force everywhere else in that discussion is (161), and the Carnot calculation a few lines later uses  $\oint dU = 0 = \oint \tau d\sigma - \oint p dV$ . The plus sign on  $p dV$  in that one line is the slip. The Carnot algebra that follows it uses the minus sign and is not changed by the slip.

### 33.4 Derivative of the Helmholtz free energy

From  $F = U - \tau\sigma$  and (161),

$$dF = -\sigma d\tau - p dV \quad (175)$$

at fixed  $N$ , so

$$\left(\frac{\partial F}{\partial \tau}\right)_V = -\sigma. \quad (176)$$

The heat-and-work notes record (176). The gas–solid solution in the phase chapter states the opposite,  $(\partial F/\partial \tau)_V = \sigma$ , and then differentiates  $F_s$ . The entropy computed there is  $-\sigma$  under (176). That solution then drops the solid entropy by the assumption stated in the problem, and the boxed  $N_g$  and  $p_g$  come from minimizing  $F$  in  $N_g$ :

$$F = -\epsilon_0(N - N_g) + N_g\tau \left[ \ln\left(\frac{N_g}{V n_Q}\right) - 1 \right],$$

$$\left(\frac{\partial F}{\partial N_g}\right) = \epsilon_0 + \tau \ln\left(\frac{N_g}{V n_Q}\right). \quad (177)$$

The  $\tau$  terms  $-\tau + \tau$  from the product rule cancelled. Setting the derivative to zero gives  $N_g = V n_Q e^{-\epsilon_0/\tau}$ , and  $p_g = N_g\tau/V$  gives the boxed pressure. Those two boxed results do not use the flipped entropy identity.

### 33.5 Enthalpy per particle

The extensive enthalpy is  $H = U + pV$ . Divide by  $N$ :

$$\hat{h} = \frac{H}{N} = \hat{\epsilon} + pv, \quad \hat{\epsilon} = \frac{U}{N}, \quad v = \frac{V}{N}. \quad (178)$$

The sum  $\hat{\epsilon} + pV$  adds an energy per particle to an energy. The 2015 definition writes that sum, and the later reminder writes it again as  $\hat{h} = \check{\epsilon} + pV$ . The line  $\hat{h} dN = (\hat{\epsilon} + pv) dN$ , in the same subsection, has the volume right. Dimensions select (178). The conduction term  $\tau N d\hat{\sigma}$  and the convection term  $\hat{h} dN$  then add as energies. This is the comparison the 2019 note in the differential-geometry dump asked for.

### 33.6 Joule–Thomson sign

The 2015 derivation of the coefficient is kept. With  $\alpha = V^{-1}(\partial V/\partial \tau)_p$  and  $(\partial \sigma/\partial p)_\tau = -(\partial V/\partial \tau)_p$  from  $dG = -\sigma d\tau + V dp$ ,

$$dH = C_p d\tau + V(1 - \tau\alpha) dp. \quad (179)$$

At fixed  $H$ ,

$$\left(\frac{\partial \tau}{\partial p}\right)_H = -\frac{V(1 - \tau\alpha)}{C_p}. \quad (180)$$

For the ideal gas,  $\alpha = 1/\tau$ , so the coefficient vanishes and an isenthalpic expansion does not change  $\tau$ .

Below the inversion temperature one has  $\tau\alpha > 1$ , so (180) is positive. An expansion has  $dp < 0$ , and therefore  $d\tau < 0$ : the gas cools. The 2015 sentence for that side agrees.

Above the inversion temperature,  $\tau\alpha < 1$ , so (180) is negative. The same expansion has  $dp < 0$ , and the product is

$$d\tau = \left(\frac{\partial \tau}{\partial p}\right)_H dp > 0. \quad (181)$$

The gas warms. The 2015 sentence for that side says both  $dT < 0$  and that the gas warms. The warming conclusion matches (181). The clause  $dT < 0$  does not.

The inversion curve itself, where the coefficient vanishes, is  $\tau\alpha = 1$ . For a van der Waals gas that locus is a curve in the  $(\tau, p)$  plane. Equation (180) does not by itself produce the numerical inversion temperature; a model for  $\alpha(\tau, p)$  does. The 2015 note already says that.

### 33.7 Rotational factor in the ideal-gas summary

In the 2015 summary the energy of translation plus rotation is written correctly as

$$\epsilon_{\mathbf{n},j} = \frac{\hbar^2}{2M} \left( \frac{\pi}{L} \right)^2 \mathbf{n}^2 + j(j+1)\epsilon_0, \quad (182)$$

and then the Boltzmann factor is written with  $2\pi$  in place of  $2M$  in the translational piece and with  $e^{j(j+1)\epsilon_0}$  in place of the rotational factor. The factor required by (182) is

$$\exp(-\epsilon_{\mathbf{n},j}/\tau) = \exp\left(-\frac{\hbar^2\pi^2\mathbf{n}^2}{2ML^2\tau}\right) \exp\left(-\frac{j(j+1)\epsilon_0}{\tau}\right). \quad (183)$$

One line in that summary also sets the power on  $M\tau/(2\pi\hbar^2)$  equal to 2. The quantum concentration in three dimensions is the power 3/2, as in (104).

The later replacement of the rotational sum by  $\tau/\epsilon_0$  is the high-temperature integral, and it is consistent with the minus sign in (183). For a linear rotor,  $g(j) = 2j + 1$ . When  $\epsilon_0 \ll \tau$  the sum over  $j$  is the integral

$$\sum_{j=0}^{\infty} (2j+1) \exp\left(-\frac{j(j+1)\epsilon_0}{\tau}\right) \approx \int_0^{\infty} (2j+1) \exp\left(-\frac{j(j+1)\epsilon_0}{\tau}\right) dj. \quad (184)$$

Set  $y = j(j+1)$ . Then  $dy = (2j+1) dj$ , and

$$\int_0^{\infty} e^{-y\epsilon_0/\tau} dy = \frac{\tau}{\epsilon_0}. \quad (185)$$

The hypothesis  $\epsilon_0 \ll \tau$  is what licenses the integral. The factorization  $Z_1 = n_Q V Z_{\text{int}}$  in that summary survives; the displayed exponential between (182) and (185) is the line to replace by (183).

### 33.8 Maxwell speed distribution

The kinetic-theory notes contain an intermediate reduction whose right-hand side is  $4\pi N(N/(2\pi\tau))^{3/2}$ . That expression does not have the dimension of the prefactor of a speed distribution, and it is not the prefactor that appears in the final formula. The final formula is the Maxwell speed density

$$P(v) = 4\pi \left( \frac{M}{2\pi\tau} \right)^{3/2} v^2 \exp\left(-\frac{Mv^2}{2\tau}\right), \quad v \geq 0. \quad (186)$$

It is normalized. Set  $\alpha = M/(2\tau)$ . The Gaussian integrals

$$\int_{-\infty}^{\infty} e^{-\alpha v^2} dv = \sqrt{\frac{\pi}{\alpha}}, \quad \int_{-\infty}^{\infty} v^2 e^{-\alpha v^2} dv = \frac{1}{2} \sqrt{\frac{\pi}{\alpha^3}} \quad (187)$$

give, because  $v^2 e^{-\alpha v^2}$  is even,

$$\int_0^{\infty} v^2 e^{-\alpha v^2} dv = \frac{1}{4} \sqrt{\frac{\pi}{\alpha^3}}. \quad (188)$$

Then

$$\begin{aligned} \int_0^{\infty} P(v) dv &= 4\pi \left( \frac{\alpha}{\pi} \right)^{3/2} \cdot \frac{1}{4} \sqrt{\frac{\pi}{\alpha^3}} \\ &= \pi \cdot \frac{\alpha^{3/2}}{\pi^{3/2}} \cdot \frac{\pi^{1/2}}{\alpha^{3/2}} = 1. \end{aligned} \quad (189)$$

The mean-speed solution that follows (186) in the manuscript is using this normalized density. The intermediate prefactor line is not an input to that solution.

### 33.9 Donor arithmetic

The preserved donor solution reports  $n_Q = 1263 \text{ cm}^{-3}$  and  $n = 2 \times 10^{-18} \text{ cm}^{-3}$  at  $T = 100 \text{ K}$ ,  $m^* = 0.3 m_e$ , and  $N_d = 10^{17} \text{ cm}^{-3}$ . Section 21 evaluates (105) and the hydrogenic binding energy (115) at  $\epsilon_r = 11.7$ , and it solves the quadratic (112). The result is  $n = 2.965 \times 10^{16} \text{ cm}^{-3}$ .

### 33.10 A numerical check that already matched

The adiabatic compression in the 2015 background uses  $\gamma = 1.4$ ,  $V_i/V_f = 15$ , and  $\tau_i = 293.15 \text{ K}$ , which is  $20^\circ\text{C}$  on the Celsius shift used in `Propulsion/thermo.py`. Equation (170) gives

$$\tau_f = \tau_i \left( \frac{V_i}{V_f} \right)^{\gamma-1} = 293.15 \times 15^{0.4} = 866.017 \text{ K}. \quad (190)$$

The pressure ratio at fixed  $N$  is

$$\frac{p_f}{p_i} = \frac{V_i}{V_f} \frac{\tau_f}{\tau_i} = 15^{1.4} = 44.313. \quad (191)$$

Both match the values written in that solution, 866.01 K and 44.31.

### 33.11 Left blank on purpose

Radiation solution 16 is absent from the 2008 file. Parts (b) and (c) of the last propagation problem are empty and the problem statement is not in the file. Those three slots were not filled with a guessed question. Entropy of mixing, which was an empty heading, is §14.

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